

Counterexamples to List and Line Decodability Bounds over Prime Fields

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Abstract

We give counterexamples to proposed list-decoding and proximity-gap bounds for Reed–Solomon codes over prime fields. The main line-level phenomenon is simple: even strictly below the characteristic-based Elias radius, two far words can have every other affine combination close to the code.

At every fixed rational rate $\rho \in (0, 1)$, write $\theta = 1 - \rho - \eta$ for the tested radius. We construct two words at distance $\theta + (1/2 - o(1))\eta$ whose other $p - 2$ affine combinations all have distance exactly θ . The underlying construction also gives a line nearby at all $p - 1$ nonzero parameters, while zero has distance $1 - \rho - 1/n$, only one coordinate short of the maximum possible distance. Here $n = \Theta_\rho(\log p \log \log p)$ and $\eta \sim H_2(\rho)/\log_2 p$. The actual nearby fraction tends to one while the proposed numerical fraction tends to zero. The nearby polynomials change with the parameter: their decoding lists are pairwise disjoint. The same geometry gives failures for batches consisting entirely of far inputs, including multilinear mixtures.

Decoding ambiguity is unnecessary. Separate constructions have exponentially more nearby parameters than prescribed, each with a unique witness. Almost full-gap separation persists after doubling the proposed exponent constant. We give efficient witness recovery over sufficiently large splitting primes, and logarithmic-length existence without that recovery guarantee. A different construction gives complete nonzero coverage against arbitrary fixed numerical constants.

For lists, integer-interval domains give $\log L = \Omega_\rho(\eta^{-2}/\log(1/\eta))$, ruling out every uniform $\exp(O_\rho(1/\eta))$ prescription. Over every sufficiently large prime, with $b = \log_2 p$, we obtain $\log_2 L \geq (1/2 - o(1))b^2/\log_2 b$ below Elias. We also give fixed-gap counterexamples to proposed numerical line coefficients. Separately, at gap $1/n$, codes with global maximum list size exactly $n - k - 1$ have $\binom{n}{k+1}$ uniquely nearby points on one line, attaining a general puncturing bound.

We apply these results and finite circle-code constructions to the quantitative premises in the S-two whitepaper. The shrinking-gap and fixed-gap quantifiers, numerical prescriptions and actual-list-size statements, and coding premises and complete-protocol conclusions are distinguished throughout.

1 Introduction

Many hash-based SNARKs ask the prover to commit to evaluations of low-degree polynomials. The verifier then checks a small number of positions and random linear combinations of the committed words. Analyzing these checks requires quantitative information about the underlying code. How many low-degree polynomials can be close to one word? How many points on an affine line can be close to the code? When do the nearby polynomials imply a common agreement set for the original words?

These are coding-theoretic questions, but the numerical answers enter soundness calculations. A bound that is polynomial in the block length may be adequate for an asymptotic theorem and

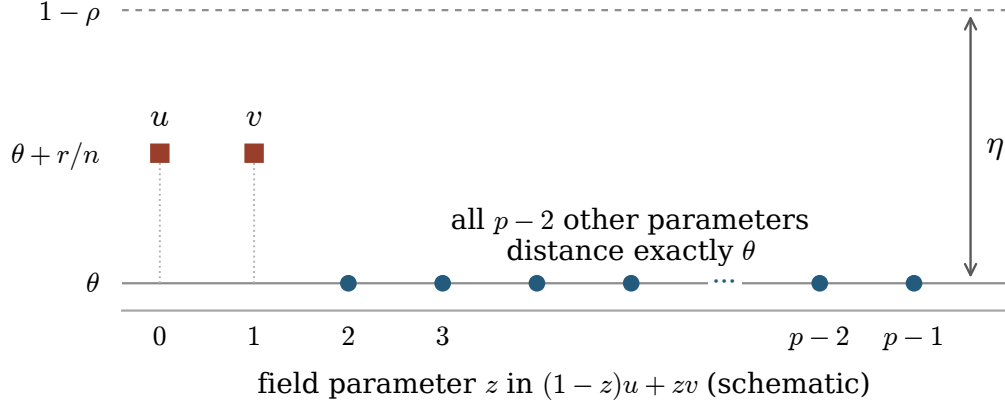


Figure 1: Two far inputs, but all $p - 2$ other affine combinations are nearby (Corollary 4.16). The endpoint separation r/n approaches half the capacity gap $\eta = (2r + 1)/n$. The diagram suppresses most field parameters.

inadequate for a claimed concrete security level. Likewise, a bound proved for random evaluation domains does not automatically apply to a prescribed subgroup. The field size, domain, and order of quantifiers must be explicit.

This paper gives counterexamples to quantitative list and line bounds over prime fields. The main results are independent of any particular proof system. We first prove the general statements, then give finite circle-code examples and apply them to the two conjectures underlying the S-two security analysis.

A useful starting point is the root bound. A nonzero polynomial of degree less than k has fewer than k roots, so two distinct Reed–Solomon codewords cannot agree in k positions. This proves a minimum-distance guarantee. It does not answer the first question above: many polynomials may each agree with an arbitrary received word in more than k positions, using different agreement sets.

Write n for the number of evaluation points and $\rho = k/n$ for the rate. We consider agreement on at least $(\rho + \eta)n$ positions, equivalently relative distance at most $1 - \rho - \eta$. The gap $\eta > 0$ is the excess agreement over the rate. Throughout the paper, we distinguish the largest list at one received word, the number of nearby parameters on an affine line, and correlated agreement across that line. None of these quantities can be substituted for another without a theorem relating them.

The main quantitative message is that the proposed dependence on η fails even when $n = o(p)$. At any fixed rational rate, the proposed list bound has $\log L = O_\rho(1/\eta)$, whereas our short-domain examples require

$$\log L = \Omega_\rho\left(\frac{1}{\eta^2 \log(1/\eta)}\right).$$

Thus the exponent must grow almost quadratically in the reciprocal gap, rather than linearly. The examples remain strictly below the characteristic-based Elias radius.

The clearest line-level message is that *even below Elias, two far words can have every affine combination other than the two endpoints close to the code*. Corollary 4.16 places both endpoints almost half a gap outside the tested radius, while a random mixture is nearby with probability $1 - 2/p$. More generally, for any fixed batch size t , all t inputs can lie almost $(1 - 1/t)\eta$ outside the radius while a random affine combination is nearby with probability $1 - O(t/p)$. The same obstruction holds for unrestricted linear combinations and for multilinear mixtures (Corollary 4.17).

The underlying construction gives an even sharper exceptional point: *a line can be close to the code everywhere except at one point, and that point can be only one coordinate short of being maximally far*. At every fixed rational rate, Corollary 4.15 gives prime-field codes for which all $p - 1$ nonzero parameters are nearby, but parameter zero has distance $1 - \rho - 1/n$. Every word is within $1 - \rho$ of a Reed–Solomon code, so this is almost the largest distance possible. Its separation from the tested radius is $(1 - o(1))\eta$, almost the entire capacity gap. The radius remains strictly below Elias. Every nonzero point has exactly the tested distance, and no codeword is nearby at two distinct nonzero parameters. The line changes only $2r$ coordinates to improve the distance by exactly $2r/n$, attaining the Hamming-distance bound. No pair of codewords explains the line on the required common coordinate set.

The obstruction is that the nearby polynomial changes with the mixing coefficient. In these complete-coverage examples, no polynomial is close at two distinct nearby parameters. A root bound controls agreements between fixed polynomials; it does not force the polynomials selected at different line parameters to fit together. The construction realizes this behavior through products of locator values on the evaluation domain, without any proper subfield.

The $c_1 = c_2 = 1$ numerical prescription predicts a nearby fraction tending to zero; the actual fraction is $1 - 1/p$. Thus testing a random linear combination can find a nearby word with probability arbitrarily close to one despite an almost maximally far input. Theorem 4.41 separately defeats any fixed constants c_1, c_2 , with a fixed positive separation fraction depending on c_2 .

Decoding ambiguity is unnecessary for numerical failure. Separately, Theorem 4.24 gives explicit powers-of-two domains where every nearby point has a unique codeword, recoverable in polynomial time in $\log p$, and the number of nearby parameters exceeds the numerical prescription exponentially. The exceptional point is again one coordinate short of maximum distance. At half rate these examples can have only two distance levels: every parameter outside the nearby bank is at the far level, including both displayed endpoints. Thus the obstruction persists even when proximity along the line is easy to decide and every accepted parameter has an unambiguous witness. Corollary 4.28 strengthens this conclusion: for every fixed $\varepsilon > 0$, suitable infinitely many prime fields admit unique-witness examples whose far point lies at least $(1 - \varepsilon)\eta$ outside the nearby radius. These examples use scaled prime-order orbits and retain efficient witness recovery. They have a sparse nearby set; they are separate from the complete-coverage construction. Theorem 4.30 obtains the same relative separation and uniqueness on domains of length $\Theta(\log p)$ by random sampling, with a polynomial-in- p violation of the numerical bound, but without a witness-recovery guarantee. Theorem 4.32 goes further: two padding orbits preserve uniqueness and almost full-gap separation even when the proposed exponent constant is doubled to $c_2 = 2$. It provides efficient witness recovery over sufficiently large splitting primes. Corollary 4.34 obtains the same numerical strengthening at length $\Theta(\log p)$ without that recovery guarantee; its finite half-rate certificates include separation $38\eta/39$ and a violation factor greater than 2^9 over $\mathbb{F}_{2^{44497-1}}$. A dimension tradeoff preserves uniqueness for every fixed $c_2 < 3$, with separation greater than two thirds of the gap (Corollary 4.35).

In the full-coverage examples the gap shrinks as $\Theta(1/\log p)$, and the strongest relative separation approaches the Elias boundary. These results do not establish superlinear growth in n at one fixed positive gap on short prime-field domains; that stronger question remains open. Separate interval-padding constructions give explicit density/separation tradeoffs and can make the number of missing coordinates grow.

In the fixed-gap line examples, every nearby word can have a unique nearby codeword. Thus their many nearby parameters do not arise from decoding ambiguity at individual points of the line. This uniqueness is a statement about the displayed line, not about all received words for the code.

A separate construction keeps the code’s global maximum list size linear in n , yet has exponentially many uniquely nearby points on a line; every other point on that line is as far from the code as any

word can be (Theorem 5.1).

1.1 The list-size obstruction

Our first result concerns a particularly simple evaluation domain: $0, 1, \dots, n-1$ in a prime field.

Theorem 1.1 (Exponential lists on integer intervals, informal). *Fix any rational rate $0 < \rho < 1$. For arbitrarily large n , there is a prime p , a rate- ρ Reed–Solomon code over $\mathbb{F}_p$, and a received word with $2^{\Omega_\rho(n)}$ nearby codewords, where*

$$\eta = \Theta_\rho\left(\frac{1}{\sqrt{n \log n}}\right), \quad \log_2 p = \Theta_\rho(\sqrt{n \log n}).$$

Every listed codeword agrees with the received word on exactly $(\rho + \eta)n$ positions. The associated distance is strictly below the characteristic-based Elias radius.

The field is large, but its logarithm is sublinear in the block length. The list is much larger than any bound of the form $c_1 2^{c_2 H_2(\rho)/\eta}$ with constants depending only on the rate: the logarithm of that proposed bound is $O_\rho(\sqrt{n \log n})$, whereas the logarithm of our list size is $\Omega_\rho(n)$. Here H_2 is binary entropy. For a fixed rate it is simply a positive constant.

The qualification about the field and domain matters. This is a statement about what can be guaranteed for *every* prime-field Reed–Solomon code. It does not give exponential lists at a fixed positive gap, or when the prime is required to be polynomial in n .

Finite and uniform refinements. The moment count also gives explicit certificates. At length 64 and dimension 32, the exact enumeration in Section 7.1 proves a list of 5,552,914,238,035 codewords with 34 agreements. Over the M31 prime, a further interval example at length 157 and dimension 63 has at least 28,169,451,256,663,519,418 codewords with 68 agreements, exceeding the finite prescription with $c_1 = c_2 = 1$ by more than $2^{34.09967}$ (Section 7.2). Proposition 2.12 gives an analytic refinement: for each fixed number m of canceled coefficients and fixed limiting rate, concentration improves the full-range counting guarantee by a factor of order $n^{m/2}$. Classical Gram orthogonalization gives explicit constants; Proposition 2.13 strengthens the finite bound, and Section 2.7 obtains the Gaussian central-density constant when m is fixed. These refinements retain the exponent above. The exact checks certify the finite arithmetic; the proofs establish the asymptotic statements.

The same construction has a stronger formulation in terms of the gap. Corollary 2.3 rules out every uniform bound whose logarithm is $o_\rho(\eta^{-2}/\log(1/\eta))$. Moreover, over every sufficiently large prime field, with $b = \log_2 p$, it gives a below-Elias list with

$$\log_2 L \geq (1/2 - o(1))b^2/\log_2 b.$$

Here $n = (H_2(\rho)^{-1} + o(1))b^2/\log_2 b$ and $\eta = (H_2(\rho) + o(1))/b$. These sharper parameter statements follow by optimizing the length in the existing moment construction.

1.2 Why the construction works

There is a useful way to reverse the usual root-counting argument. Choose a set A of $r = k + s$ evaluation points, and form its root polynomial

$$F_A(X) = \prod_{a \in A} (X - a).$$

Suppose many choices of A give the same coefficients in degrees $r, r-1, \dots, k$. Let W be this common high-degree part. Then $G_A = W - F_A$ has degree less than k , and it agrees with W at every point of A . Thus each such set produces a member of one decoding list. The problem has become a counting question about leading coefficients.

Those coefficients are determined by the first s power sums of A . When $A \subseteq \{0, \dots, n-1\}$, each power sum is a relatively small *integer*. There are $\binom{n}{r}$ possible sets, but far fewer possible vectors of s power sums. Some vector must occur many times. The crucial point is that we count these vectors over the integers *before choosing the prime*. The count therefore does not grow with the field size.

For a small example, take

$$A = \{0, 1, 6, 7\}, \quad B = \{0, 2, 5, 7\}.$$

Both sets have size four and sum fourteen. Their root polynomials therefore begin with $X^4 - 14X^3$. Subtracting either polynomial from $W = X^4 - 14X^3$ leaves a polynomial of degree at most two. Over any prime larger than seven, on the domain $\{0, \dots, 7\}$ each agrees with W at four positions. The general construction repeats this idea with many more sets and more moments. This example illustrates the mechanism; its two codewords are not the quantitative lower bound of Theorem 1.1.

1.3 Lines of received words

Proof systems often test a random linear combination of several words. This leads to a related question. Suppose many words on an affine line $f_z = f_0 + zf_1$ are close to the code. Must the nearby codewords fit together into a single affine line of codewords? Must f_0 and f_1 agree with two codewords on a large common set of coordinates?

We give three constructions, because these questions have different quantifiers.

Many nearby points without correlated agreement. Corollary 4.15 gives complete nonzero coverage with an almost maximally far exception. On a paired core, even locator polynomials leave codeword candidates after their common leading term cancels. A padding block supplies several pairs of new agreements at once. Multiplicative-character estimates show that the candidates' joint evaluation images fill almost all possible vectors, so random padding directions cover every nonzero line parameter.

Theorem 4.41 gives complete coverage against arbitrary fixed numerical constants. Matching three times as many moments permits a cubic change of the seed domain without losing the required common locator prefix. A geometric irreducibility argument and a finite-field point-count estimate then control outside collisions.

The interval constructions also give elementary alternatives with quantified nearby density. Division by $X - \tau$ yields $\Omega(p/\log p)$ nearby labels at logarithmic length (Theorem 3.4); padding gives the stronger density bounds in Theorems 4.43 and 4.45. Corollary 4.48 makes the number of coordinates outside the radius grow while retaining density tending to one. These variants use padded interval domains. The complete-coverage constructions use random paired domains or cubic images of intervals with added coordinates.

An obstruction that persists at a fixed gap. The gap in Theorem 1.1 shrinks with n . To address asymptotic claims at a fixed gap, we start with one fixed large list and replace each evaluation point by B points on which X^B has the same value. Polynomial composition preserves the rate, relative agreement, and list size. Suitable prime fields exist for every B .

An anchored version removes one common agreement point and divides by its linear factor. Padding then produces many nearby challenges while keeping every common coefficient agreement

below the threshold. Theorem 4.6 gives actual failure of correlated agreement at a fixed rate and gap, with a linear count whose coefficient has logarithm of order $\eta^{-2}/\log(1/\eta)$. Every nearby word on these lines has exactly one nearby codeword, while the parameter-zero word is one coordinate beyond the radius. The obstruction persists without local list ambiguity. The same exceptional-count lower bound also has realizations with $p = n^{O_{\rho,\eta}(1)}$ (Corollary 4.7); uniqueness is not asserted for those smaller-field realizations. No remainder that is $o(n)$ along this family can absorb a smaller linear coefficient. The simpler selection argument in Theorem 4.4 is retained as a preliminary construction.

Small lists with many isolated nearby line points. Large lists are not the only obstruction. At every fixed rate $\rho = k/n$, we also construct codes whose maximum list size is exactly $n - k - 1$, together with a line having $\binom{n}{k+1}$ nearby parameters. Each nearby word has a unique nearby codeword, yet no parametrized codeword line contains more than $n - k$ of those witnesses. The gap is $1/n$. At rate one-half these bounds are $n/2 - 1$, $\binom{n}{n/2+1}$, and $n/2$, respectively. The construction combines an elementary line spanned by consecutive powers with a domain chosen using a known list-decoding theorem for random Reed–Solomon codes. Proposition 5.4 shows that its nearby-parameter count is optimal among lines without correlated agreement at this radius.

What remains open at a fixed positive gap. Our fixed-gap lower bound forces a large coefficient of n , but does not force a superlinear exponent. Proposition 4.9 identifies a sufficient route: a source list of size L at length N gives $\Omega(\min\{NL, p\})$ full-support MCA exceptions over the same prime field, after exactly halving both rate and gap. Thus lists of size N^c over superpolynomial prime fields would force exceptions of order N^{c+1} . The missing ingredient is such a growing fixed-gap list; full-length lists with $N = p - 1$ cannot give superlinear scalar-label counts. The factor-of-two parameter change is not essential: Corollary 4.10 uses a small positive padding fraction to preserve any strict margin above a continuous sufficient agreement curve. Thus the same exponent transfer can stay inside an open regime where a sharper upper bound is available.

Sharp algebraic counts need not give sharp proximity gaps. For a differential equation of fixed order d , fixed jet degree, and fixed challenge degree, Appendix I proves that the actual regular polynomial–challenge locus has at most $O(D^{d+1})$ isolated points for message degree D . An explicit Wronskian family attains $\binom{D+d+1}{d+1}$ such points, all regular, with distinct labels over suitable prime fields. Nevertheless, that family has only constantly many nearby labels on any received line at a fixed positive agreement gap. At every fixed order d , Appendix J also bounds the degree of each positive-dimensional actual stratum by $O(D^d)$, and gives examples attaining this exponent. The first-order case in Appendix H is linear in D . These results distinguish sharpness of algebraic candidate counts from the still-open fixed-gap superlinear proximity-gap lower bound over sufficiently large prime fields. Proposition I.4 extends this obstruction to arbitrary candidate families with bounded-degree rational residuals after a common normalization: their full-support exceptional count is linear, regardless of how many algebraic candidates they contain. Theorem H.5 gives a further first-order restriction: bad labels with a positive fraction of nonsingular agreement coordinates number only $O(n)$ at a fixed positive gap. For $R(X, z)P' = \mathcal{A}(X, z, P)$ this bounds all bad labels linearly when $\deg_X R$ also stays a fixed positive fraction below the agreement threshold. A nonlinear two-solution family attains $n/2$ bad labels at fixed rate and gap. The general first-order conjecture still requires controlling singular agreement coordinates. Appendix M removes that restriction for a fixed nonlinear Riccati equation, proving a linear full-support bound and a sharp ordinary-list bound with prime-field equality families.

1.4 Finite examples on circle codes

We also study a different regime: the fixed prime $p = 2^{31} - 1$ and domains on the norm-one circle. These are structured evaluation domains used by Circle STARKs. At the common parameters

$$n = 2^{26}, \quad \rho = \frac{1}{2}, \quad \eta = \frac{33}{1024} - \frac{2}{n},$$

we obtain the following finite lower bounds:

Object constructed	Guaranteed size
One base-field decoding list	$> 2^{40}$
Distinct nearby parameters on an extension-field line	$> 2^{74}$

The line has no correlated agreement at the specified threshold. The constructions give existence with explicit numerical certificates; they do not enumerate these very large lists. Section 6 explains the idea, and Appendix A gives the coefficient calculations.

1.5 Context and organization

Proximity gaps are established tools in the analysis of low-degree tests [8, 18]. In particular, existing theorems give proximity guarantees in specified radius regimes. Our fixed-rate asymptotic examples have $\eta \rightarrow 0$ and eventually lie beyond the Johnson radius $1 - \sqrt{\rho}$, although they remain below the characteristic-based Elias radius. The distinction between these radii is essential when comparing the results with proved soundness bounds for a low-degree test.

The connection between equal subset moments, elementary symmetric functions, and Reed–Solomon decoding is established prior work; Gandikota, Ghazi, and Grigorescu use this connection in decoding hardness results [13]. The results here use quantitative counts on integer intervals to obtain the stated list and line separations. The number-theoretic splitting theorem used in the lift is also classical.

Prime-field list constructions using multiplicative cosets also appear in work of Guruswami and Rudra, presented in Rudra’s thesis [14, Section 6.4.3]. Those full-field examples already use cancellation of leading coefficients through unions of multiplicative orbits. Their displayed coset bank has logarithmic size $O(1/\eta)$; our interval construction gives the larger, almost-quadratic reciprocal-gap exponent above. Neither comparison supplies growing lists at one fixed positive gap.

The generic list-profile upper bound is due to Brakensiek, Gopi, and Makam [11]. At every fixed positive rate, Roth’s ordinary list-decoding bound [12] gives the matching lower bound for all sufficiently large lengths. We include an explicit construction that also covers the remaining finite parameters $k \geq 2$, without claiming priority for that construction. The separation here combines these list bounds with a line containing many isolated nearby points.

Goyal, Guruswami, Sun, and Wootters [19] give further recent proximity-gap and correlated-agreement bounds for random Reed–Solomon domains. Their results allow general received curves and retain explicit slack and field-size hypotheses. They provide useful context for the shrinking-gap regime studied here, but do not subsume the specific actual-list-size separation in Theorem 5.1.

Krachun, Kazanin, and Haböck [17] construct large lists and affine-line obstructions on multiplicative subgroups of prime fields, with a gap of order $1/\log n$. Our interval construction addresses a different family of domains and gives lists larger than every fixed exponential in the reciprocal gap. The multiplicative structure of their examples remains a separate advantage. Our finite circle examples concern the evaluation spaces of Circle STARKs [21].

The Proximity Prize’s orbit-pencil upper certificate also uses multiplicative fibers and a fixed core [22]. Our circle construction expresses the analogous core through a common anchor and division at two circle points. Appendix A.6 makes this change of presentation explicit. The contest’s field, code, and score are different from the circle specialization considered here.

The distinction between fixed and shrinking gap separates our results from the fixed-gap polynomial-in- n list bound stated in the recent preprint of Jeronimo [15]. Our exponential-list family has a shrinking gap. Our fixed-gap family preserves a fixed finite list of witnesses as n grows; it does not bound the full decoding list by that constant. Neither construction contradicts such a polynomial bound.

The current Dao–Kominers–Thaler paper [34] gives, above its first-order curve by margin η_1 , list bound $O_\rho(n/\eta_1^2)$ and full agreement-set MCA exceptional count $O_\rho(n^2/\eta_1^4)$ in sufficiently large characteristic. Its capacity bounds have exponents depending on the fixed capacity gap. There are two qualitative intrinsic lower bounds: Proposition 2.4 gives $n/2$ list elements over prime fields at fixed rate $1/4$ and gap $1/8$; Corollary 2.6 gives $\Omega(n^2)$ full agreement-set MCA exceptions over quadratic extension fields of growing prime characteristic, at fixed rate $1/8$ and gap $1/16$. The latter rules out a universal linear capacity MCA bound in that larger field class. Theorem N.1 also gives $\Omega_\varepsilon(n^{5/4-\varepsilon})$ nearby labels with no ordinary correlated agreement, over quadratic extensions at exact rate $3/13$ and capacity gaps bounded below by $3/26$ (not fixed exactly). These results do not establish first-order-regime tightness or force an exponent growing as the gap shrinks. Over prime ambient fields, even a superlinear exceptional-count lower bound at one fixed positive gap remains open here. The large gap-dependent coefficient in our linear lower bound must not be confused with a growing exponent of n . Likewise, optimality of a specified interpolation dimension count is a statement about that proof method, not an intrinsic list or MCA lower bound. The capacity gap and the margin above the first-order curve are different parameters.

Sections 2–5 prove the general list and line results. Section 6 gives finite circle examples. Section 8 explains what these results imply for hash-based SNARK analyses, and Section 9 applies them to S-two. Section 10 states the remaining coding-theoretic questions. The first appendices contain the circle proofs, quotient calculations, and finite certificates. The later appendices study restrictions on possible constructions: rational fibers, binary-affine locator families, and polynomial differential equations. These additional results address the open fixed-gap problem; the main list and line counterexamples do not depend on them.

2 Large lists from subsets with equal moments

The construction starts with a simple observation: if two polynomials differ by a polynomial whose roots are a set A , then they agree at every point of A . We will choose many sets A whose vanishing polynomials have the same leading coefficients. Subtracting those common coefficients will give many low-degree polynomials close to one received word.

Notation. For a field F and a set $D \subset F$ of n distinct points, write

$$\text{RS}(F, D, k) = \{(P(x))_{x \in D} : \deg P < k\}.$$

This code has dimension k and rate $\rho = k/n$, provided $k \leq n$. The relative Hamming distance between two words is the fraction of coordinates on which they differ. A list at radius θ consists of all codewords within that distance of a received word. Write $\ell_C(\theta)$ for the largest such list over all received words. We parameterize the radius by

$$\theta = 1 - \rho - \eta.$$

Thus a nearby codeword agrees with the received word on at least a $\rho + \eta$ fraction of the coordinates. We call $\eta > 0$ the gap from $1 - \rho$.

All logarithms are base two, and $H_2(t) = -t \log_2 t - (1 - t) \log_2(1 - t)$ is binary entropy. For characteristic p , define

$$H_p(t) = \frac{H_2(t) + t \log_2(p - 1)}{\log_2 p}.$$

The characteristic-based Elias radius is the solution of $H_p(r_E) = 1 - \rho$ on the increasing branch $[0, 1 - 1/p]$. We will keep our examples strictly below this radius. The simple estimate

$$H_p(\theta) \leq \theta + \frac{1}{\log_2 p} \quad (1)$$

shows that it suffices to have $\eta \log_2 p > 1$. In our prime-field examples $p > n$ and $\rho \geq 1/n$, so $0 \leq \theta < 1 - 1/p$ as required.

2.1 Why matching moments produces a list

Consider the 34-element subsets A of $\{0, \dots, 63\}$. Their vanishing polynomials begin

$$F_A(X) = \prod_{a \in A} (X - a) = X^{34} - \left(\sum_{a \in A} a \right) X^{33} + \left(\sum_{a < b \in A} ab \right) X^{32} + \dots$$

Subsets with the same sum and the same sum of squares have the same three leading coefficients, because

$$\sum_{a < b \in A} ab = \frac{(\sum_{a \in A} a)^2 - \sum_{a \in A} a^2}{2}.$$

Let W be those common leading terms. Each polynomial $W - F_A$ has degree less than 32 and agrees with W at exactly the 34 points of A . Consequently, a large class of subsets sharing these two moments gives a large list in a rate-one-half Reed–Solomon code.

For the general construction we fix the first s moments. It is convenient to use *binomial moments*, $\sum_{a \in A} \binom{a}{j}$, because their integer ranges can be counted exactly. They determine the usual power sums and hence the leading coefficients in exactly the same way.

Lemma 2.1 (A finite list bound). *Let $n \geq 2$, $k \geq 1$, $s \geq 1$, and $t = k + s < n$. Let $p > n$ be prime and $D = \{0, \dots, n - 1\} \subset \mathbb{F}_p$. Put*

$$R_j = \binom{n}{j+1} - \binom{n-t}{j+1} - \binom{t}{j+1} + 1 \quad (1 \leq j \leq s), \quad L = \left\lceil \frac{\binom{n}{t}}{\prod_{j=1}^s R_j} \right\rceil.$$

There is a received word with at least L distinct codewords of $\text{RS}(\mathbb{F}_p, D, k)$ at distance exactly $1 - t/n$. If $p > \max_j (R_j - 1)$, its entire list at radius $1 - t/n$ corresponds exactly to one class of subsets with equal integer binomial moments.

Proof. For each t -subset $A \subset D$, record $T_j(A) = \sum_{a \in A} \binom{a}{j}$ for $1 \leq j \leq s$. The minimum and maximum values of T_j occur on the first and last t integers in D . The identity $\sum_{a=0}^{b-1} \binom{a}{j} = \binom{b}{j+1}$ gives their values as

$$\binom{t}{j+1} \quad \text{and} \quad \binom{n}{j+1} - \binom{n-t}{j+1}.$$

There are therefore at most $\prod_j R_j$ moment vectors. By pigeonhole, at least L subsets share one vector.

Equal binomial moments imply equal power sums through degree s : each X^j is an integer linear combination of $\binom{X}{0}, \dots, \binom{X}{j}$, and all subsets have size t . Newton identities then imply equal elementary symmetric functions through degree s . These identities are valid modulo p since $p > n > s$. Thus the monic polynomials $F_A(X) = \prod_{a \in A} (X - a)$ have the same coefficients in degrees $t, t-1, \dots, t-s = k$. Let W be their common degree- $\geq k$ part. Then $P_A = W - F_A$ has degree less than k and agrees with W exactly on A : outside A , each factor of $F_A(x) = \prod_{a \in A} (x - a)$ is nonzero in the field because $p > n$ makes the evaluation points distinct. Different subsets give different codewords, since these polynomials have degree less than n .

For the last assertion, any codeword P in the list makes $W - P$ a monic degree- t polynomial with at least t roots in D . It must therefore be F_A for a t -subset A . Its first s binomial moments are congruent to those of the selected class modulo p . Each lies in an integer interval of width $R_j - 1 < p$, so the congruences are equalities. $\square$

2.2 An exponential list and a subexponential proposed bound

There are exponentially many subsets of a fixed density. Fixing a modest number of moments divides them into far fewer classes. The following choice of parameters makes this discrepancy precise.

Theorem 2.2 (No uniform $\exp(O_\rho(1/\eta))$ list bound). *Fix a rational rate $\rho \in (0, 1)$. For arbitrarily large n , there are prime-field Reed–Solomon codes of exact rate ρ , on $D = \{0, \dots, n-1\}$, with*

$$\eta = \Theta_\rho\left(\frac{1}{\sqrt{n \log n}}\right), \quad \log_2 p = \Theta_\rho(\sqrt{n \log n}), \quad \log_2 \ell_C(1 - \rho - \eta) = \Omega_\rho(n).$$

The radius lies strictly below the characteristic-based Elias radius. Consequently no finite constants $c_1, c_2 > 0$ make

$$\ell_C(1 - \rho - \eta) \leq c_1 2^{c_2 H_2(\rho)/\eta} \tag{2}$$

hold uniformly over prime fields, evaluation domains, and these radii. The same sequence works for every fixed pair c_1, c_2 .

Proof. Let n grow through multiples of the denominator of ρ and set $k = \rho n$. Choose a constant $c > 0$ with $c^2 < 4H_2(\rho)$, and put

$$s = \left\lfloor c \sqrt{\frac{n}{\log_2 n}} \right\rfloor, \quad t = k + s, \quad \eta = s/n.$$

For large n , the hypotheses of Lemma 2.1 hold. Retaining the factorials in the binomial coefficients gives $R_j \leq 2 \binom{n}{j+1} \leq 2(3n/(j+1))^{j+1}$. Hence

$$\log_2 \prod_{j=1}^s R_j \leq s + \sum_{j=1}^s (j+1) \log_2 \frac{3n}{j+1} = \frac{s^2}{2} \log_2 \frac{n}{s} + O(s^2 + s \log n) = \left(\frac{c^2}{4} + o(1) \right) n.$$

The middle equality follows by summing $j \log j$, or by integral comparison. The final equality uses $\log_2(n/s) = \frac{1}{2} \log_2 n + O(\log \log n)$. On the other hand, $\log_2 \binom{n}{t} = H_2(\rho)n + o(n)$. The lemma gives

$$\log_2 L \geq \left(H_2(\rho) - \frac{c^2}{4} - o(1) \right) n = \Omega_\rho(n). \tag{3}$$

Choose any constant $A > H_2(\rho)/c$. Bertrand's postulate supplies a prime with $\log_2 p = A\sqrt{n \log_2 n} + O(1)$. Eventually $p > n$ and $\eta \log_2 p \rightarrow cA > H_2(\rho)$. Since $H_2(1 - \rho - \eta) \rightarrow H_2(\rho)$, the sharper

entropy estimate $H_p(\theta) \leq \theta + H_2(\theta)/\log_2 p$ proves the Elias condition. In contrast, the logarithm of the right-hand side of (2) is

$$\log_2 c_1 + c_2 H_2(\rho) \frac{n}{s} = O_{\rho, c_1, c_2}(\sqrt{n \log n}) = o(n).$$

It is eventually smaller than the logarithm of the constructed list. $\square$

The rate is fixed, while the gap decreases and the prime grows. The theorem leaves open stronger bounds under restrictions on the domains or on the relation between field size and block length.

Corollary 2.3 (The obstruction as a function of gap and field size). *Fix a rational rate $\rho \in (0, 1)$ and write $H = H_2(\rho)$. For every sufficiently small real $\varepsilon > 0$, there is a length*

$$n = (H + o(1))/(\varepsilon^2 \log_2(1/\varepsilon))$$

such that, for every prime $p > n$, the interval-domain code $C = \text{RS}(\mathbb{F}_p, \{0, \dots, n-1\}, \rho n)$ satisfies

$$\log_2 \ell_C(1 - \rho - \varepsilon) \geq \frac{H^2/2 - o(1)}{\varepsilon^2 \log_2(1/\varepsilon)}. \quad (4)$$

This radius is strictly below the characteristic-based Elias radius whenever $\varepsilon \log_2 p > H_2(1 - \rho - \varepsilon)$.

Moreover, for every sufficiently large prime p , writing $b = \log_2 p$, there is such a code of length $n = (H^{-1} + o(1))b^2/\log_2 b$ and a radius strictly below Elias with

$$\log_2 \ell_C(1 - \rho - \varepsilon) \geq (1/2 - o(1)) \frac{b^2}{\log_2 b}, \quad \varepsilon = (H + o(1))/b. \quad (5)$$

All asymptotics fix ρ . These are consequences of the preceding interval construction.

Proof. Choose n to be a nearest multiple of the denominator of ρ to $H/(\varepsilon^2 \log_2(1/\varepsilon))$, and put $k = \rho n$, $m = \lceil \varepsilon n \rceil$, and $t = k + m$. For sufficiently small ε , we have $1 \leq m < t < n$. The range estimate in the proof of Theorem 2.2, applied to Lemma 2.1, gives

$$\log_2 \ell_C(1 - t/n) \geq Hn - \frac{m^2}{2} \log_2(n/m) - O_\rho(m^2 + m \log n + \log n).$$

Here $\log_2 \binom{n}{t} = Hn + O_\rho(m + \log n)$, and all error terms are $o(n)$ for the chosen parameters. Substitution gives (4): its two leading terms are respectively $H^2/(\varepsilon^2 \log_2(1/\varepsilon))$ and half that quantity. Since $m/n \geq \varepsilon$, the constructed radius is at most the claimed radius; list-size monotonicity handles every real prescribed gap, including the rounding of m .

The entropy estimate $H_p(\theta) \leq \theta + H_2(\theta)/\log_2 p$ proves the stated sufficient condition for strict distance below Elias. For the final claim, start with any sufficiently large prime, put $A_b = H + b^{-1/2}$, and take $\varepsilon = A_b/b$ in the first construction. Then $p > n$ eventually, $H_2(1 - \rho - \varepsilon) = H + O_\rho(1/b) < A_b$, and $\log_2(1/\varepsilon) = \log_2 b + O_\rho(1)$. Equation (4) now gives (5) and the asserted length. The fields grow and the gap shrinks; these statements impose no conclusion at a fixed field or fixed gap. $\square$

In particular, a bound with $\log_2 \ell_C(1 - \rho - \varepsilon) = o_\rho(\varepsilon^{-2}/\log(1/\varepsilon))$ cannot hold uniformly over prime fields and evaluation domains, even below Elias. In terms of field size, the exhibited lists are superpolynomial in p .

Full-length comparison at a fixed gap. The short-domain condition is a substantive distinction. The following elementary comparison already rules out a list bound depending only on rate and gap when the domain occupies almost the whole prime field. We make no novelty claim for this full-length observation.

Proposition 2.4. *Let $p \equiv 1 \pmod{8}$ be prime, $n = p - 1$, and $k = n/4$. On $\mathbb{F}_p^\times$, there is a received word with $n/2$ distinct degree- $(k - 1)$ polynomials, each agreeing on exactly $3n/8$ coordinates. Thus the rate is $1/4$ and the gap is $1/8$. For $p > 256$, the radius $5/8$ is strictly below the characteristic-based Elias radius.*

Proof. Let χ be the quadratic character, with $\chi(0) = 0$, and put $e = (p + 1)/2 = 2k + 1$. For $a \neq 0$ define

$$G_a(X) = \sum_{j=0}^k \binom{e}{2j+1} a^{e-2j-1} X^j, \quad P_a(X) = G_a(X) - X^k, \quad w(x) = \frac{1 + \chi(x)}{2} - x^k.$$

The degree- k terms cancel, and $[X^{k-1}]P_a = -a^2/8$. All powers of a are even, so the parameters a modulo sign give exactly $n/2$ distinct polynomials of degree $k - 1$. The binomial identity

$$G_a(s^2) = \frac{(a + s)^e - (a - s)^e}{2s}$$

counts their agreements. If $x = s^2$ is nonsquare, take s in $\mathbb{F}_{p^2}$. Then $s^p = -s$, and $u = (a + s)^e$ satisfies $u^2 = a^2 - x$ and $(a - s)^e = u^p$. Hence $G_a(x) = 0$ exactly when $\chi(a^2 - x) = 1$. There are $(p - 1)/4$ such nonsquare x .

For square $x = s^2$ with $s \in \mathbb{F}_p^\times$, Euler's criterion gives

$$G_a(s^2) = \frac{(a + s)\chi(a + s) - (a - s)\chi(a - s)}{2s}.$$

Away from $s = \pm a$, this equals one exactly when both characters are one. The weighted count $\frac{1}{4} \sum_s (1 + \chi(a + s))(1 + \chi(a - s)) = (p - 1)/4$ uses $\sum_s \chi(a^2 - s^2) = -1$. Removing $s = 0$ and correcting the half-weights at $s = \pm a$ cancel: at the endpoints the actual value is $\chi(2a) = \chi(a)$. Dividing by two for s and $-s$ gives $(p - 1)/8$ square agreements. The nonsquare count above follows from the same quadratic-character identity, applied to $(1 - \chi(x))(1 + \chi(a^2 - x))/4$ with its endpoints removed. Thus the total is $3n/8$. Finally, $H_p(5/8) \leq 5/8 + 1/\log_2 p < 3/4$ for $p > 256$. $\square$

Here $n/p \rightarrow 1$. In contrast, Theorem 2.2 and Corollary 2.3 give exponentially large lists on short integer intervals, with the stated quantitative dependence on the gap. The full-length example does not settle the fixed-gap, short-domain proximity-gap question. Exact checks of the displayed agreement counts are in [research/dickson_fixed_gap/](#).

Lemma 2.5 (Splitting of pairwise differences). *For the polynomials above, if $a^2 \neq b^2$, every root of $G_a - G_b$ belongs to $\mathbb{F}_p$.*

Proof. Let t^2 be a root in an algebraic closure, with $y = G_a(t^2) = G_b(t^2)$. The case $t = 0$ is immediate. For $c \in \{a, b\}$, put $U_c = (c + t)^e$ and $V_c = (c - t)^e$. Then

$$U_c - V_c = 2ty, \quad U_c^2 - V_c^2 = 2c(t + t^p), \quad U_c^2 + V_c^2 = 2c^2 + 2t^{p+1}.$$

If $y = 0$, then $t^p = -t$, so $t^2 \in \mathbb{F}_p$. Otherwise put $H = (t + t^p)/(ty)$. We have $U_c + V_c = cH$, whence

$$c^2 H^2 + 4t^2 y^2 = 4c^2 + 4t^{p+1}.$$

Subtracting the equations for a and b gives $H^2 = 4$, and then $t^2 y^2 = t^{p+1}$. Thus $(t + t^p)^2 = 4t^{p+1}$, or $(t^p - t)^2 = 0$. Again $t^2 \in \mathbb{F}_p$. $\square$

Corollary 2.6 (Quadratic exceptions in growing prime characteristic). *For every prime $p \equiv 1 \pmod{8}$, $p \geq 17$, put $n = 2(p-1)$. Over $\mathbb{F}_{p^2}$ there is a Reed–Solomon code of length n and dimension $n/8$, and a received line with at least $\lceil n^2/20 \rceil$ full agreement-set MCA exceptional labels at agreement $3n/16$. Thus the capacity gap is exactly $1/16$, and $p = n/2 + 1 > n/8 - 1$.*

Proof. Set $N = p - 1$. Proposition 2.4 supplies $L = N/2$ candidates at agreement $A = 3N/8$, with dimension $k = N/4$. Choose any nonsquare anchor. Exactly $\ell = N/4$ candidates agree there: indeed $\sum_{a \in \mathbb{F}_p} \chi(a^2 - x) = -1$ for nonsquare x , none of the summands is zero, and the $a = 0$ summand is -1 . There are therefore $N/2$ admissible nonzero parameters, or $N/4$ modulo sign. As in Proposition 4.9, divide each anchored candidate minus the received anchor value by the anchor’s linear factor. The resulting quotients have degree at most $k - 2$. By Lemma 2.5, their evaluations are pairwise distinct at every point outside $\mathbb{F}_p$.

Append any $N + 1 = p$ distinct points of $\mathbb{F}_{p^2} \setminus \mathbb{F}_p$, with direction one, and retain direction zero on the $N - 1$ old coordinates. Independent uniform translations of the padding value sets yield an expected number of counted labels equal to

$$p^2[1 - (1 - \ell/p^2)^p] \geq p^2[1 - \exp(-N/(4p))] \geq Np[1 - \exp(-1/4)] > N^2/5 = n^2/20.$$

The second inequality uses concavity of $1 - \exp(-u/4)$ on $[0, 1]$; the last uses $1 - e^{-1/4} \geq 1/4 - 1/32 = 7/32 > 1/5$. Every counted label has a selected quotient with at least $A - 1 \geq k$ old agreements and one new agreement. A degree- $< k$ direction polynomial matching this full support would vanish on the old agreements, hence identically, but would equal one on the new agreement. This is impossible. The dimension and agreement threshold remain k and A . $\square$

This elementary consequence rules out a universal linear capacity MCA bound in the large-characteristic field class of [34]. Its ambient field is a quadratic extension, not a prime field, and its agreement lies below the first-order regime. It therefore neither proves tightness of the first-order quadratic upper bound nor resolves the prime-field superlinear-exception target. It concerns full agreement-set MCA; absence of ordinary correlated agreement is not asserted. Exact extension-field replays at $p = 17, 41$ are recorded in `research/prime_field_tightness/quadratic_extension_verification.json`.

Finite examples. At $p = 2^{127} - 1$, $n = 64$, $k = 32$, $s = 2$, Lemma 2.1 gives

$$L > 2^{35}.$$

The exact moment-class enumeration in Section 7.1 strengthens this to $L \geq 5552914238035 > 2^{42}$. Here $\eta = 1/32$, so the choice $c_1 = c_2 = 1$ in (2) gives only 2^{32} ; the improved ratio exceeds 1292. The exact inequality $p^2 > 2^{64}$ verifies the sufficient Elias condition. At $p = 2^{521} - 1$, $n = 4096$, $k = 2048$, $s = 8$, the same formula gives $L > 2^{3600}$, compared with 2^{512} . The exact inequality $p^8 > 2^{4096}$ verifies Elias. At $c_1 = 1$, this example forces $c_2 > 7$. A larger example over the same prime, using Gram concentration, forces $c_2 > 27.31865$. Already over $p = 2^{31} - 1$, an interval-domain example forces $c_2 > 2.00475$ at rate one-half. Appendix D.1 records the exact parameters and counts. As a small check of the algebra, among the 12,870 eight-subsets of $\{0, \dots, 15\}$, the binomial-moment class $(56, 252)$ has exactly 22 members. Over $p = 2^{31} - 1$, they give 22 distinct degree- < 6 polynomials, each agreeing with the common word at exactly eight positions.

2.3 Sharper finite bounds from conditional moments

The product of moment ranges in Lemma 2.1 counts many combinations without using how the moments depend on each other. For two moments, one can first fix the sum of the selected integers, then bound the spread of the second binomial moment within that class. Both steps can be computed exactly without enumerating the subsets.

Proposition 2.7 (Conditional range and dispersion bounds). *Let $3 \leq t < n$, $k = t - 2$, and $p > n$ be prime. For an integer q , let $\mathcal{A}_q$ be the family of t -subsets of $\{0, \dots, n - 1\}$ with sum q , and put*

$$C_q = |\mathcal{A}_q|, \quad Y_A = \sum_{a \in A} \binom{a}{2}.$$

For $C_q > 0$, let l_q and u_q be the minimum and maximum of Y_A in this family. There is a received word for $\text{RS}(\mathbb{F}_p, \{0, \dots, n - 1\}, k)$ whose list at radius $1 - t/n$ has size at least

$$\left\lceil \frac{C_q}{u_q - l_q + 1} \right\rceil. \quad (6)$$

More generally, choose an integer $c \in [l_q, u_q]$ and an integer $r \geq 0$. Define

$$D_q(c) = \sum_{A \in \mathcal{A}_q} (Y_A - c)^2, \quad W_q(c, r) = \min(u_q, c + r) - \max(l_q, c - r) + 1.$$

Then a possibly different received word has a list of size at least

$$\left\lceil \frac{\max\{0, C_q - \lfloor D_q(c)/(r + 1)^2 \rfloor\}}{W_q(c, r)} \right\rceil. \quad (7)$$

Every codeword produced by these bounds agrees in exactly t positions.

Proof. Pigeonhole over the integer values of Y_A proves (6). For the second bound, every subset with $|Y_A - c| > r$ contributes at least $(r + 1)^2$ to $D_q(c)$. Thus at most $\lfloor D_q(c)/(r + 1)^2 \rfloor$ subsets lie outside the specified interval. The remaining subsets occupy at most $W_q(c, r)$ integer values, proving (7) by pigeonhole. In either case, a class shares its first two binomial moments. The polynomial construction in Lemma 2.1 applies. $\square$

Exact computation. Process the integers $0, \dots, n - 1$ in order. For each subset size and first-moment sum, store its count, the minimum and maximum of Y , and the two totals $S = \sum Y$ and $Q = \sum Y^2$. On adding the integer j , put $b = \binom{j}{2}$. A predecessor record with count C contributes

$$C, \quad S + bC, \quad Q + 2bS + b^2C$$

to the new count and totals; its minimum and maximum each increase by b . Merge the contributions from taking and omitting j . The desired squared deviation is then $D_q(c) = Q - 2cS + c^2C$. There are $O(nt^2)$ possible size–sum states and n stages, giving $O(n^2t^2)$ integer operations and $O(nt^2)$ records. All stored integers have $O(n + \log n)$ bits. No assumption that the second moment fills its entire range is needed.

A stronger length-64 certificate. For $n = 64$ and $t = 34$, conditioning on $q = 1061$ gives $C_q = 8558724088132137$, $l_q = 17784$, and $u_q = 25714$. The range bound alone gives 1079148163931 codewords. The dispersion bound is stronger. For $q = 1071$, take $c = 22138$ and $r = 1076$. The exact counts are

$$\begin{aligned} C_q &= 8634334451040106, \\ D_q(c) &= 3337241714848691877182, \\ C_q - \left\lfloor \frac{D_q(c)}{1077^2} \right\rfloor &= 5757225839350346, \quad W_q(c, r) = 2153. \end{aligned}$$

Consequently,

$$\ell_C(15/32) \geq \left\lceil \frac{5757225839350346}{2153} \right\rceil = 2674048230075 > 2^{41}, \quad C = \text{RS}(\mathbb{F}_p, \{0, \dots, 63\}, 32). \quad (8)$$

This holds for every prime $p > 64$. At $p = 2^{127} - 1$, the original certificate $p^2 > 2^{64}$ keeps the radius strictly below Elias. The dispersion bound exceeds the product-range bound by a factor of 53, and exceeds $2^{1/\eta} = 2^{32}$ by a factor of 622. These are certified list lower bounds; neither is an exact maximum list size or an enumeration of the large moment class.

The script `checks/verify_conditional_moments.py` checks all counts, extrema, and moment totals against exhaustive enumeration for 66 smaller parameter pairs, covering 32,180 subsets. It independently verifies both displayed conditional counts using the coefficient identity

$$[y^t] \prod_{j=0}^{n-1} (1 + yx^j) = x^{\binom{t}{2}} \prod_{j=1}^t \frac{1 - x^{n-t+j}}{1 - x^j}.$$

Summing the conditional moment totals also agrees with the direct one-element and two-element inclusion counts over all t -subsets.

Signed polynomial weights. The interval argument can be sharpened further: instead of charging every retained moment value equally, assign a polynomial weight to each value. Negative weights outside favored ranges are allowed and improve the bound.

Proposition 2.8 (Polynomial-weight refinement). *Use the notation of Proposition 2.7, fix an integer c , and put*

$$M_j = \sum_{A \in \mathcal{A}_q} (Y_A - c)^j.$$

For any polynomial $P(Z) = \sum_{j=0}^d b_j Z^j$ with rational coefficients, define

$$S_P = \sum_{y=l_q}^{u_q} \max\{P(y - c), 0\}.$$

If $S_P > 0$, there is a received word whose list at radius $1 - t/n$ has size at least

$$\max \left\{ 0, \left\lceil \frac{\sum_{j=0}^d b_j M_j}{S_P} \right\rceil \right\}. \quad (9)$$

Each resulting codeword agrees in exactly t positions.

Proof. Let N_y count the members of $\mathcal{A}_q$ with second moment y , and let $L = \max_y N_y$. Since $0 \leq N_y \leq L$,

$$\sum_j b_j M_j = \sum_y N_y P(y - c) \leq L \sum_y \max\{P(y - c), 0\} = L S_P.$$

The equal-moment construction then gives the asserted codeword list. No positivity assumption on P over the entire moment range is needed. $\square$

Taking $P(Z) = h^2 - Z^2$ gives the particularly simple certificate

$$L \geq \left\lceil \frac{C_q h^2 - D_q(c)}{h(4h^2 - 1)/3} \right\rceil \quad \text{when } C_q h^2 > D_q(c).$$

Here the denominator sums positive weights over all integers, so it remains a valid upper bound if the actual moment range is shorter. The preceding $q = 1071$, $c = 22138$ data with $h = 1077$ already give $L \geq 4009211058641$ without computing any further moments.

An exact length-64 certificate. Continue with $n = 64$, $t = 34$, $q = 1071$, and $c = 22138$. The exact second-moment range is $[17969, 26129]$. Define the monic degree-20 polynomial $P(Z) = \prod_{r \in R} (Z - r)$, where

$$R = \{-4169, -2688, -2687, -2062, -2059, -1507, -1472, \\ -1038, -868, -643, 647, 870, 1039, 1464, 1498, \\ 2035, 2038, 2637, 2638, 3991\}.$$

Exact evaluation of (9) yields

$$\ell_C(15/32) \geq 5074503250115 > 2^{42}, \quad C = \text{RS}(\mathbb{F}_p, \{0, \dots, 63\}, 32), \quad p > 64. \quad (10)$$

This exceeds the original product-range certificate by a factor of 101, the preceding dispersion certificate by a factor of 1.89, and 2^{32} by a factor of 1181. At $p = 2^{127} - 1$, the same below-Elias certificate applies. These remain list lower bounds, not exact maximum list sizes.

For reproducibility, extend the count recurrence to raw moment totals $T_r = \sum Y_A^r$ for $0 \leq r \leq 20$. Adding an integer a with $b = \binom{a}{2}$ contributes

$$T'_r = \sum_{j=0}^r \binom{r}{j} b^{r-j} T_j, \quad M_r = \sum_{j=0}^r \binom{r}{j} (-c)^{r-j} T_j.$$

The positive-weight denominator is an exact sum over 8161 integers. The files in `moment_certificates/` record all raw and centered moments, polynomial coefficients, numerator, and denominator. The polynomial was discovered numerically, then replaced by the displayed integer-root polynomial. Its proof and certificate verification use only integer arithmetic; numerical optimizer tolerances play no role. The checker compares every moment through degree 20 with exhaustive small-subset enumeration and independently checks the length-64 moments using complementary 30-subsets.

A degree-40 refinement. At the same n, t, q, c , take the monic polynomial with roots

$$R_{40} = \{-4169, -3591, -3590, -3214, -3213, -2844, -2843, -2475, -2474, -2107, \\ -2106, -1742, -1735, -1391, -1358, -1065, -968, -765, -565, -476, \\ 482, 571, 769, 969, 1065, 1352, 1385, 1721, 1729, 2081, \\ 2082, 2436, 2437, 2787, 2788, 3135, 3136, 3484, 3485, 4184\}.$$

The exact numerator and positive-weight denominator in (9) give the stronger certificate

$$\ell_C(15/32) \geq 5285900426578, \quad C = \text{RS}(\mathbb{F}_p, \{0, \dots, 63\}, 32), \quad p > 64. \quad (11)$$

The radius and below-Elias qualification are unchanged. This improves (10) by more than 4.16% and exceeds the original product-range certificate by a factor of 105. The files in `research/finite_weights/` record the integer roots, expanded coefficients, exact moments, numerator, and denominator. Independent checks compare all moments through degree 40 with the complementary-subset identity and exhaustive smaller examples; the support extrema are recomputed separately. Earlier degree-24 and 28 certificates are retained for comparison. Numerical optimization discovers candidate weights; all claimed lower bounds and the final integer-root refinement use exact integer arithmetic.

The weighted inequality applies to any finite signature map, but its application to a prescribed evaluation domain requires statistics for that domain's coefficient signatures. The interval moments above do not by themselves supply such statistics for a multiplicative NTT subgroup.

Why an exponent substitution does not transfer the certificate. On a multiplicative subgroup generated by g , equal integer moments of exponent indices need not imply equal coefficient signatures. For $n \geq 2^{m+1}$, the positive and negative coefficient supports of $Q(X) = \prod_{j=0}^m (1 - X^{2^j})$ have equal integer moments through degree m , because Q has a root of multiplicity $m+1$ at 1. But their first field moments differ by $Q(g) \neq 0$ when g has order n . Thus replacing each index a by g^a loses the required coefficient equality.

Nor does the bare field-signature pigeonhole estimate certify a large list below Elias. For $p > n$, $t = k + m < n$, and $k, m \geq 1$, it gives only $\lceil \binom{n}{t} / p^m \rceil$. Writing $\theta = 1 - t/n$, the strict Elias inequality $H_p(\theta) < 1 - k/n$ implies

$$\binom{n}{t} / p^m < (p/(p-1))^{n\theta} < e,$$

so this guarantee is at most three. These observations limit the two transfer arguments, rather than the true list size on a subgroup. The exact checks in `research/structured_domains/` accompany these elementary obstructions and a separate complete-fiber transfer lemma.

What full polynomial fibers can accomplish. The following elementary composition argument explains why replacing a power map by another full balanced polynomial map gives no additional fiber partitions on a multiplicative subgroup.

Lemma 2.9 (Full polynomial fibers on a subgroup). *Let F be a field and $n \geq 1$. Suppose $D = \mu_n \subset F$ has n distinct points, and let $R \in F[X]$ have degree $B \geq 1$. If every fiber of $R|_D$ has exactly B points, then $B \mid n$ and $R(X) = aX^B + b$, with $a \neq 0$. Conversely, every such polynomial has B points per fiber when $B \mid n$.*

Proof. Put $M = |R(D)| = n/B$, $a = \text{lc}(R)$, and $P(Y) = \prod_{y \in R(D)} (Y - y)$. Comparing roots and leading coefficients gives

$$P(R(X)) = a^M (X^n - 1).$$

Write $b = R(0)$, $T = R - b$, and $Q(Y) = P(Y + b)$. If T has a nonzero coefficient c_j below degree B , choose its largest positive index j . The coefficient of X^{n-B+j} in T^M is $Ma^{M-1}c_j \neq 0$, since the distinct-root hypothesis implies $\text{char } F \nmid n$. Every lower power of T in $Q(T)$ has degree at most $n - B$. This contradicts $Q(T) = a^M (X^n - 1)$, so $T = aX^B$. The converse follows from the kernel size of $x \mapsto x^B$ on μ_n . $\square$

Rational maps permit different fibers. The lemma exhausts full polynomial fibers, but rational maps admit other partitions. If $2d \mid n$, choose a nonsquare $c \in \mu_{n/d}$. The rational map

$$R(x) = x^d + c/x^d$$

has degree $B = 2d$ and no pole on μ_n . After setting $u = x^d$, its fibers pair u with c/u : equality of two outputs is equivalent to $(u - v)(uv - c) = 0$. These pairs have no fixed points, so every fiber has exactly $2d$ elements.

Rational substitution also preserves a useful polynomial degree bound. In general, write $R = A/V$ in reduced form with $\max\{\deg A, \deg V\} = B$. Suppose it has no domain poles and exactly B points over each of its N image labels. For $1 \leq k \leq N$, replace a polynomial f of degree less than k by

$$V^{k-1}f(A/V),$$

whose degree is at most $B(k-1)$. Apply the same nonzero coordinate scaling to the composed received word. Each agreement repeats exactly B times, preserving relative Hamming distance and distinctness. The images lie in a Reed–Solomon code of dimension $B(k-1) + 1$. A large list still requires a separate certificate on the image domain.

Corollary 2.10 (Balanced fibers of Galois rational maps). *Let F be a finite field of characteristic $p > n \geq 3$, with $\mu_n \subset F$. Suppose $R \in F(X)$ has degree $B \geq 1$, has no poles on μ_n , and has exactly B points in every fiber of $R|_{\mu_n}$. If $\overline{F}(X)/\overline{F}(R)$ is Galois, then $F(X)/F(R)$ is Galois and, for some $h \in \text{PGL}_2(F)$, either*

$$R(X) = h(X^B), \quad B \mid n,$$

or

$$R(X) = h(X^d + c/X^d), \quad B = 2d \mid n,$$

where c is a nonsquare in $\mu_{n/d}$. Here h has no pole on the corresponding image of μ_n .

Proof. Each domain fiber already contains the full B distinct geometric preimages of its image. The geometric deck group G , of order B , therefore preserves μ_n and acts freely there.

A Möbius map $(aX + b)/(cX + d)$ preserving μ_n satisfies $(aX + b)^n - (cX + d)^n = \lambda(X^n - 1)$ with $\lambda \neq 0$; its intermediate coefficients give $a^i b^{n-i} = c^i d^{n-i}$ for $1 \leq i < n$. Comparing $i = 1, 2$ rules out four nonzero entries, and the remaining cases give precisely $X \mapsto \zeta X$ and $X \mapsto \zeta/X$, with $\zeta \in \mu_n$. These maps are defined over F , proving arithmetic descent.

If G consists of rotations, its invariant field is $F(X^B)$. Otherwise its rotations have order d , and a reflection $X \mapsto \alpha/X$ gives the invariant $X^d + \alpha^d/X^d$, of degree $2d = |G|$. Writing $c = \alpha^d$, freeness is equivalent to the absence of $u \in \mu_{n/d}$ with $u^2 = c$: such a u lifts to a fixed point of some reflection. The displayed invariant generates $F(X)^G$; any other generator R differs by an output Möbius map over F . $\square$

Thus this Galois subclass reduces to power and twisted-inversion quotients. Full balanced fibers alone do not imply the Galois hypothesis: the explicit cubic over $\mathbb{F}_{13}$ on μ_6 in `research/gram_norm/BALANCED_FIBER_REVIEW.md` has four simple critical points and is not a Galois cover.

For the large-index congruence classes below, full balance forces the Galois hypothesis. In fact, a small number of uncovered domain points is allowed. This includes the field and subgroup in the pinned `better.codes` instance.

Theorem 2.11 (Nearly complete rational fibers at large Frobenius index). *Let F be a finite field of odd characteristic p , containing μ_n , and suppose*

$$p = \ell n + \epsilon, \quad \epsilon \in \{1, -1\}, \quad \ell \geq 6.$$

Let $R \in F(X)$ have degree $B \geq 2$ and no poles on μ_n . Suppose $T \subseteq \mu_n$ is a union of complete geometric fibers, each consisting of B distinct points, and $|\mu_n \setminus T| \leq c$. If $n > 6(B - 1 + c)$, then $\overline{F}(X)/\overline{F}(R)$ is Galois, and all its deck transformations have the form $X \mapsto \zeta X$ or $X \mapsto \zeta/X$, with $\zeta \in \mu_n$. Consequently $B \mid 2n$; when n is a power of two, $B \mid n$. If also $c < B/2$, then the deck group acts freely on μ_n , every fiber of $R|_{\mu_n}$ has B points, and $B \mid n$. In this case R has one of the two forms in Corollary 2.10.

The proof, in Appendix E, uses the four-term unit equation on each collision component and a Frobenius-index argument establishing the required linear independence. At $p = 2130706433$ and $n = 262144$, one has $\ell = 8128$ and $\epsilon = 1$. For full balance ($c = 0$), the theorem covers every integer $B \leq 43691$. Since full balance forces $B \mid n$, it covers every possible degree through 32768. The remaining degrees have at most four fibers and therefore at most sixteen unions of complete fibers. This classifies the small-degree fixed-map full-fiber partitions at these parameters; it gives neither a new numerical better.codes bound nor a restriction on constructions using partial fibers or varying maps. The extension also excludes degrees 513, 514, 516 with complete-fiber coverage defects at most 1, 4, 16, respectively: each satisfies both inequalities, but none divides 262144. These are exclusions of proposed packet maps, not attained improvements in agreement or score.

On this dyadic domain the divisibility conclusion does not need $c < B/2$. For any non-dyadic degree B , the complete-fiber count is therefore at most

$$\min \left\{ \left\lfloor \frac{262144}{B} \right\rfloor, 1 + \left\lfloor \frac{218452}{B} \right\rfloor \right\}.$$

Indeed, the uncovered count must satisfy $c \geq 43692 - B$, and the covered count is a multiple of B . At degrees 513, 514, 516, this forces at least 43606, 43180, 43360 uncovered points, respectively.

2.4 A uniform concentration bound for several moments

The improvement is not limited to a computed length-64 example. Concentrating the moment signatures near their means saves a polynomial factor for every fixed number of canceled coefficients.

Proposition 2.12 (Concentration of triangular moment signatures). *Let $1 \leq m < t < n$, $k = t - m$, and let $p > n$ be prime. Choose rational coefficients c_{ji} and functions*

$$b_j(a) = \binom{a}{j} + \sum_{i=1}^{j-1} c_{ji} \binom{a}{i}, \quad 1 \leq j \leq m, \quad 0 \leq a < n.$$

For a uniformly chosen t -subset A of $\{0, \dots, n-1\}$, write $B_j(A) = \sum_{a \in A} b_j(a)$ and define

$$\bar{b}_j = \frac{1}{n} \sum_{a=0}^{n-1} b_j(a), \quad V_j = \frac{t(n-t)}{n-1} \left(\frac{1}{n} \sum_{a=0}^{n-1} b_j(a)^2 - \bar{b}_j^2 \right), \quad h_j = \left\lceil \sqrt{(m+2)V_j} \right\rceil.$$

Then $C = \text{RS}(\mathbb{F}_p, \{0, \dots, n-1\}, k)$ satisfies

$$\ell_C(1 - t/n) \geq \left\lceil \frac{2 \binom{n}{t}}{(m+2) \prod_{j=1}^m (2h_j + 1)} \right\rceil. \quad (12)$$

The constructed codewords agree in exactly t positions.

Proof. The inclusion probabilities t/n and $t(t-1)/(n(n-1))$ give $\mathbb{E}B_j = t\bar{b}_j$ and $\text{Var}(B_j) = V_j$. Here $V_j > 0$: b_j is a nonconstant polynomial of degree $j < n$ on n distinct integers, and $0 < t < n$. Chebyshev's inequality and a union bound show that at least a $2/(m+2)$ fraction of all subsets satisfy $|B_j - t\bar{b}_j| \leq h_j$ for every j .

For the first coordinate there are at most $2h_1 + 1$ possible integer values. Given the earlier coordinates, triangular inversion fixes the earlier binomial moments, and B_j differs from the integer $\sum_{a \in A} \binom{a}{j}$ by a fixed rational number. Hence its interval contains at most $2h_j + 1$ possible values. There are at most $\prod_j (2h_j + 1)$ retained signatures. Pigeonhole and triangular inversion give a class sharing its first m integer binomial moments. Lemma 2.1 applies. Rational changes of coordinates are used only for counting; no reduction of their denominators modulo p is required. $\square$

For fixed m and $t/n \rightarrow \rho \in (0, 1)$, choosing all $c_{ji} = 0$ gives $V_j = \Theta_{\rho, m}(n^{2j+1})$ and therefore

$$\ell_C(1 - t/n) = \Omega_{\rho, m} \left(\frac{\binom{n}{t}}{n^{m(m+2)/2}} \right).$$

The product of complete moment ranges instead has order $n^{m(m+3)/2}$ in its denominator. Thus concentration improves that guarantee by a factor of order $n^{m/2}$. This comparison fixes m ; it does not replace the separate estimates when m grows with n .

Two moments with explicit constants. For $m = 2$, take $b_1(a) = a$ and $b_2(a) = \binom{a}{2} - (n-2)a/2$. These centered population functions are uncorrelated, and their subset-sum variances are

$$V_1 = \frac{t(n-t)(n+1)}{12}, \quad V_2 = \frac{t(n-t)(n+1)(n^2-4)}{720}.$$

Consequently (12) implies

$$\ell_C(1 - t/n) \geq \left(\frac{3\sqrt{60}}{8\rho(1-\rho)} + o(1) \right) \frac{\binom{n}{t}}{n^4} \quad \text{if } t/n \rightarrow \rho \in (0, 1).$$

At $n = 64, t = 34$, the exact variances are 5525 and 376805, and $h_1 = 149, h_2 = 1228$. This gives 1102772374154 codewords without the conditional dynamic program. The degree-20 certificate is stronger at that length. The script `moment_certificates/verify_concentration.py` checks the variance identities, triangular signature counts, and finite bounds against exhaustive small-subset enumeration.

2.5 Gram moments and an ellipsoid concentration bound

Orthogonalizing the binomial moments gives explicit variances, and a signed quadratic weight improves the elementary concentration constant. Let $1 \leq m < t < n$. For $1 \leq j \leq m$, let b_j be the degree- j polynomial with leading coefficient $1/j!$ orthogonal to every lower-degree polynomial on $\{0, \dots, n-1\}$ with uniform weight. These are the classical Gram polynomials, with the normalization $b_j(x) = \binom{x}{j} + \sum_{i < j} c_{ji} \binom{x}{i}$. Their classical squared norms [27, eqs. (1.1)–(1.2)] and the sampling-without-replacement variance identity give, for a uniform t -subset A and $B_j(A) = \sum_{a \in A} b_j(a)$,

$$\mathbb{E}B_j = 0, \quad V_j := \text{Var}(B_j) = \frac{t(n-t)}{n-1} \frac{\prod_{i=1}^j (n^2 - i^2)}{(2j+1) \binom{2j}{j}^2 (j!)^2}. \quad (13)$$

The coordinates B_j are pairwise uncorrelated. Put $\kappa_m = \pi^{m/2}/\Gamma(m/2 + 1)$, the volume of the unit ball. The continuous inequality below is the classical maximum-density/covariance bound [30, Proposition III.1 and proof of Corollary III.2]; unit-cube smoothing gives the exact variance correction $1/12$. No log-concavity assumption is needed.

Proposition 2.13 (Finite smoothed ellipsoid bound). *For every prime $p > n$, the code $C = \text{RS}(\mathbb{F}_p, \{0, \dots, n-1\}, t-m)$ satisfies*

$$\ell_C(1 - t/n) \geq \left\lceil \frac{\binom{n}{t}}{\kappa_m(m+2)^{m/2} \prod_{j=1}^m \sqrt{V_j + 1/12}} \right\rceil. \quad (14)$$

The constructed codewords agree in exactly t coordinates.

Proof. Write $N = \binom{n}{t}$, and let L be the largest class sharing its vector $B = (B_1, \dots, B_m)$. Because the zeroth moment is fixed, these vectors lie in an affine lattice $b_0 + T\mathbb{Z}^m$, with T lower triangular and diagonal entries one. The translates of $[-1/2, 1/2]^m$ centered at its lattice points tile $\mathbb{R}^m$: conditional rounding determines the first integer coordinate, then the second, and so on, uniquely.

Add to B an independent uniform vector U from this unit cube. The resulting vector $Y = B + U$ has density at most L/N , mean zero, and coordinate variances $W_j = V_j + 1/12$. Thus $Z_j = Y_j/\sqrt{W_j}$ satisfies $\mathbb{E}\|Z\|^2 = m$ and has density at most $H = (L/N) \prod_j \sqrt{W_j}$. The signed quadratic weight gives

$$2 = \mathbb{E}(m + 2 - \|Z\|^2) \leq H \int_{\mathbb{R}^m} (m + 2 - \|z\|^2)_+ dz = 2H\kappa_m(m+2)^{m/2}.$$

Rearranging bounds L as in (14). Triangular inversion makes these classes exactly the classes sharing their first m integer binomial moments. The construction in Lemma 2.1 converts a class into the stated list. The rational changes of coordinates are used only for real-valued counting, so their denominators impose no further condition on p . $\square$

Asymptotics for a fixed number of moments. For fixed m and $t/n \rightarrow \rho \in (0, 1)$, (14) gives

$$\ell_C(1 - t/n) \geq \left(\frac{\prod_{j=1}^m j! \binom{2j}{j} \sqrt{2j+1}}{\kappa_m(m+2)^{m/2} [\rho(1-\rho)]^{m/2}} + o(1) \right) \frac{\binom{n}{t}}{n^{m(m+2)/2}}. \quad (15)$$

For the same variances, the leading constant is larger than that from the optimized rectangular Chebyshev bound by $(m+2)2^{m-1}/\kappa_m$. This comparison concerns elementary concentration; it does not assert improvement over conditional counting.

Two moments. Here

$$V_1 = \frac{t(n-t)(n+1)}{12}, \quad V_2 = \frac{t(n-t)(n+1)(n^2-4)}{720},$$

$$\ell_C(1 - t/n) \geq \left\lceil \frac{\binom{n}{t}}{4\pi\sqrt{(V_1 + 1/12)(V_2 + 1/12)}} \right\rceil.$$

The asymptotic leading constant is $3\sqrt{60}/(\pi\rho(1-\rho))$, a factor $8/\pi$ larger than the rectangular constant. At $n = 64, t = 34$, the variances are 5525 and 376805. Rational upper bounds for the denominator certify 2,825,885,263,137 codewords from the smoothed bound. Direct exact summation of the positive part of the quadratic weight $4 - B_1^2/V_1 - B_2^2/V_2$ on the full moment lattice improves this to 2,825,905,919,527. Both exceed the elementary rectangular certificate 1,102,772,374,154; the existing degree-20 conditional certificate 5,074,503,250,115 remains stronger.

A growing number of moments. The gain also holds when $\log n \ll m \ll n$, provided t/n stays in a compact subinterval of $(0, 1)$. Let

$$D_E = \kappa_m(m+2)^{m/2} \prod_{j=1}^m \sqrt{V_j + 1/12}, \quad D_B = \frac{m+2}{2} \prod_{j=1}^m (2\lceil \sqrt{(m+2)V_j} \rceil + 1)$$

be the smoothed ellipsoid and rectangular Gram denominators, respectively. For either $D_\star \in \{D_E, D_B\}$, natural logarithms give the uniform estimate

$$\log D_\star = \frac{m^2}{2} \log(n/m) + (3/4 - \log 2)m^2 + O(m \log n + m^4/n^2).$$

Indeed, Stirling's bounds evaluate $\prod_{j=1}^m \sqrt{2j+1}(2j)!/j!$; the remaining grid correction is $O(m^4/n^2)$. Since $m = o(n)$, eventually $17m^2 \leq n^2 + 4$. The recurrence $V_j/V_{j-1} = (n^2 - j^2)/(4(2j-1)(2j+1))$ then gives $V_j \geq V_1 = \Theta(n^3)$, controlling both the smoothing and rounding errors. The complete-range denominator has the same leading expression with $3/4$ in place of $3/4 - \log 2$ and error $O(m \log n + m^3/n)$. Both errors are $o(m^2)$ in the stated regime. Thus either Gram guarantee improves the unrounded range guarantee by $2^{m^2+o(m^2)}$. At $m \sim c\sqrt{n/\log_2 n}$, this saves $c^2 n / \log_2 n + o(n/\log n)$ bits in the guaranteed logarithmic list size, while preserving the leading condition $c^2 < 4H_2(\rho)$. No central limit theorem in growing dimension is required. For example, exact arithmetic gives the following lower bounds on $\lfloor \log_2 L \rfloor$ from the complete ranges and the *rectangular* Gram denominator D_B , at rate $1/2$ and $t = n/2 + m$:

n	m	Complete ranges	Gram concentration
128	4	42	60
512	8	183	253
4096	20	1915	2333
16384	40	7241	8869

The finite certificates and uniform estimates are recorded in `research/growing_m/`. These comparisons concern interval domains; field-size conditions for a below-Elias radius must still be imposed.

2.6 A finite certificate without conditioning on the first moment

Proposition 2.14 (An exact two-moment interval certificate). *Among the 34-element subsets $S \subseteq \{0, \dots, 63\}$, some class with fixed values of $\sum_{a \in S} a$ and $\sum_{a \in S} \binom{a}{2}$ contains at least*

$$5,133,798,314,667$$

subsets. Consequently, for every prime $p > 64$, $C = \text{RS}(\mathbb{F}_p, \{0, \dots, 63\}, 32)$ satisfies $\ell_C(30/64) \geq 5,133,798,314,667$, with each constructed codeword agreeing in exactly 34 coordinates.

Proof. Taking complements preserves class sizes, so work with 30-subsets. Put

$$X_a = 2a - 63, \quad Y_a = \frac{(2a - 63)^2 - 1365}{4}, \quad x = \frac{1}{2} \sum_{a \in S} X_a, \quad y = \frac{1}{2} \sum_{a \in S} Y_a.$$

The pair $(x, y) \in \mathbb{Z}^2$ is an invertible affine transformation of the two moment sums. Set

$$T = 341x^2 + 5y^2, \quad D = 1,884,025, \quad Q = T/D, \quad A = \frac{851547}{580691}.$$

The ancillary file `radial_certificates/certificate_degree12.json` specifies a rational polynomial h of degree 12. Use the degree-25 radial weight

$$w(Q) = (A - Q)h(Q/A)^2.$$

It is nonpositive outside $Q < A$. If L denotes the largest class, then

$$\sum_{|S|=30} w(Q(S)) \leq L \sum_{(x,y) \in \mathbb{Z}^2} \max\{w(Q(x,y)), 0\}.$$

The numerator is evaluated exactly from the moments through degree 25 of T , obtained by an integer dynamic program on the 32 population pairs $(u, v), (-u, v)$. The denominator is an exact sum over the 210,135 lattice points with $T \leq 2,762,804$. After clearing all coefficient denominators, the ancillary file records the integer numerator and denominator, whose ratio has ceiling 5,133,798,314,667. The standard-library script `radial_certificates/verify_publication.py` recomputes all 676 mixed moments, checks the saved polynomial and ratio, and independently recomputes the denominator by summing expanded monomials over the full signed lattice. Numerical optimization was used only to discover the rational polynomial; it is not an input to the verification. Lemma 2.1 converts the common-moment class into the stated Reed–Solomon list. This improves the degree-20 conditional certificate by approximately 1.17%. The higher-degree conditional certificates above give larger bounds at these parameters. The radial argument offers a separate way to use both moment coordinates at once; no optimality is claimed. $\square$

2.7 A stronger asymptotic constant for fixed dimension

Fix $m \geq 1$, and suppose only that $\min(t, n - t) \rightarrow \infty$. Write V_j for the Gram subset-sum variances above, and let L_n be the largest class of t -subsets sharing their first m binomial moments. Then

$$\liminf_{n \rightarrow \infty} \frac{L_n \prod_{j=1}^m \sqrt{V_j}}{\binom{n}{t}} \geq (2\pi)^{-m/2}. \quad (16)$$

This follows from a finite-population central limit theorem and ordinary pigeonhole counting; a local central limit theorem is unnecessary.

Proof. Let G_j be the mean-zero Gram polynomial of degree j and leading coefficient $1/j!$. For a uniformly chosen t -subset A , put $Z_j = \sum_{a \in A} G_j(a)/\sqrt{V_j}$. Orthogonality gives $\mathbb{E}Z = 0$ and $\text{Cov}(Z) = I_m$. For any fixed nonzero $u \in \mathbb{R}^m$, the population values $a_x = \sum_j u_j G_j(x)/\sqrt{V_j}$ have mean zero and finite-population variance

$$v_n = \frac{1}{n-1} \sum_x a_x^2 = \frac{n\|u\|^2}{t(n-t)}.$$

For fixed m , $\max_x |G_j(x)| = O_m(n^j)$ and the population variance of G_j is $\Theta_m(n^{2j})$. Cauchy–Schwarz therefore gives

$$\frac{\max_x a_x^2}{\min(t, n-t)v_n} = O_{u,m}(1/\min(t, n-t)) \rightarrow 0.$$

Hájek’s finite-population central limit theorem [29], in the sufficient form of [28, Theorem 1], followed by the Cramér–Wold criterion, therefore gives $Z \Rightarrow N(0, I_m)$.

The Gram signatures belong to an affine unit lower triangular image of $\mathbb{Z}^m$. Its centered axis-aligned unit cubes tile $\mathbb{R}^m$, as in the finite bound above. After dividing coordinate j by $\sqrt{V_j}$,

each cell has volume $1/\prod_j \sqrt{V_j}$ and lies within $\epsilon_n = \frac{1}{2}\sqrt{\sum_j V_j^{-1}} = o(1)$ of its center. Consequently a ball of radius R contains at most

$$\kappa_m(R + \epsilon_n)^m \prod_j \sqrt{V_j}$$

possible signatures, where κ_m is unit-ball volume. For fixed $R > 0$, pigeonhole counting and convergence in distribution give

$$\liminf_n \frac{L_n \prod_j \sqrt{V_j}}{\binom{n}{t}} \geq \frac{\Pr(\|N(0, I_m)\| \leq R)}{\kappa_m R^m}.$$

Letting $R \downarrow 0$ after this limit proves (16) by continuity of the Gaussian density. $\square$

If additionally $t/n \rightarrow \rho \in (0, 1)$, then with $k = t - m$ and $p > n$ the moment-class construction gives

$$\ell_{\text{RS}(\mathbb{F}_p, \{0, \dots, n-1\}, k)}(1 - t/n) \geq \left(\frac{\prod_{j=1}^m [\sqrt{2j+1} \binom{2j}{j} j!]}{(2\pi\rho(1-\rho))^{m/2}} - o(1) \right) \frac{\binom{n}{t}}{n^{m(m+2)/2}}.$$

The number of moments is fixed here. This statement supplies neither a finite-length error bound nor the mass of a prescribed individual signature. It is an application of classical limit theory, with no priority claim.

3 From large lists to nearby codewords on a line

List-decoding asks about one received word. Line-decodability asks how nearby codewords can vary along an affine family of received words $f(z) = f_0 + zf_1$. Its selected-witness formulation asks whether many chosen nearby codewords must themselves follow one affine rule in z .

Definition 3.1 (Selected-witness line-decodability). Let $C \subseteq F^n$ be linear. Say that C is line-decodable at radius θ , threshold a , and concurrency b if the following holds. For every affine word line $f(z) = f_0 + zf_1$, every set $S \subset F$ with $|S| > a$, and every choice of codewords $p_z \in C$ within distance θ of $f(z)$, some $P_0, P_1 \in C$ satisfy $p_z = P_0 + zP_1$ for more than b parameters $z \in S$.

The word “selected” matters: the property must hold for every choice of the nearby witnesses. Also, equality uses the *same* parameter z on both sides. We call $\{(z, P_0 + zP_1) : z \in F\}$ a parametrized codeword line. Separately, (f_0, f_1) has *correlated agreement* on t coordinates if two codewords agree with f_0, f_1 simultaneously on one set of t coordinates.

We first turn a large list into a family that has many nearby codewords but no such correlated agreement. Division by $X - \tau$ lowers each polynomial’s degree by one. The value at the chosen pole τ determines which parameter on the word line receives that polynomial.

Lemma 3.2 (A rational word line). Let $D \subset \mathbb{F}_p$ have size n , and let $1 \leq d < t < n$. Suppose N distinct polynomials R_i of degree at most d agree with a common word w on at least t coordinates. Write $Q = p - n > 0$. There is a pole $\tau \in \mathbb{F}_p \setminus D$ for which

$$f_0(x) = \frac{w(x)}{x - \tau}, \quad f_1(x) = \frac{1}{x - \tau}$$

has a set S of distinct parameters, with a selected nearby codeword $P_z \in \text{RS}(\mathbb{F}_p, D, d)$ at each parameter, such that

$$|S| \geq \frac{NQ}{Q + d(N - 1)}. \quad (17)$$

Each selected codeword has at least t agreements. No pair of codewords agrees jointly with (f_0, f_1) on more than d coordinates, and every parametrized codeword line contains at most $\lfloor (n-d)/(t-d) \rfloor < n$ selected pairs. If $N \geq Q$, then $|S| \geq Q/(d+1)$. If $p > n + d\binom{N}{2}$, all N parameters can be made distinct.

Proof. For an admissible pole, assign

$$z_i = -R_i(\tau), \quad P_i(X) = \frac{R_i(X) - R_i(\tau)}{X - \tau}.$$

Then $\deg P_i < d$, and $f_0 + z_i f_1 - P_i = (w - R_i)/(X - \tau)$ on D , preserving the agreement set.

Two parameters collide only at a root of the nonzero polynomial $R_i - R_j$, which has degree at most d . Averaging over the Q admissible poles gives one with at most $dN(N-1)/(2Q)$ colliding pairs. If the multiplicities of its distinct parameters are $m_1, \dots, m_v$, then

$$\sum_j m_j = N, \quad \sum_j m_j^2 \leq N + \frac{dN(N-1)}{Q}.$$

Cauchy–Schwarz gives (17). Select one witness per parameter. If $N \geq Q$, the denominator is at most $(d+1)N$. Under the stronger field condition, the union of D and all pairwise root sets has size less than p , so a collision-free pole exists.

Any agreement of f_1 with a degree- $< d$ polynomial P_1 is a root of $(X - \tau)P_1(X) - 1$. This polynomial has degree at most d and is nonzero, as its value at τ is -1 . Thus even f_1 alone has at most d agreements with any codeword. Finally fix a parametrized codeword line $P_0 + zP_1$. Let $b \leq d$ be the number of coordinates where both coefficient words agree. At every other coordinate the two affine expressions agree for at most one parameter. If v selected witnesses lie on this line, counting agreements gives $v(t-b) \leq n-b$. The ratio $(n-b)/(t-b)$ increases with b , so $v \leq (n-d)/(t-d)$. $\square$

Theorem 3.3 (No uniform finite $n \exp(O_\rho(1/\eta))$ line threshold). *Fix rational $\rho \in (0, 1)$ and finite $c_1, c_2 > 0$. There are arbitrarily large prime-field Reed–Solomon codes of exact rate ρ on consecutive-integer domains that fail selected-witness line-decodability at concurrency n and threshold*

$$a = c_1 2^{c_2 H_2(\rho)/\eta} n, \quad \theta = 1 - \rho - \eta < r_E(\rho).$$

One can take $\eta = \Theta_\rho(1/\sqrt{n \log n})$ and $\log_2 p = \Theta_{\rho, c_2}(\sqrt{n \log n})$.

Proof. Let $d = \rho n$ be the target dimension. Choose $c^2 < 4H_2(\rho)$ and set $s = \lfloor c\sqrt{n/\log_2 n} \rfloor$, $t = d + s + 1$, and $\eta = (s+1)/n$. Apply Lemma 2.1 to the source code of dimension $d+1$. It supplies $N = 2^{\Omega_\rho(n)}$ degree- $\leq d$ polynomials with common agreement size t ; the one-dimensional change does not affect the estimate (3). Choose

$$A > \max\{H_2(\rho)/c, c_2 H_2(\rho)/c\}, \quad \log_2 p = A\sqrt{n \log_2 n} + O(1)$$

using Bertrand’s postulate. Eventually $N > p > n$. Lemma 3.2 gives at least $(p-n)/(d+1)$ selected parameters, whose logarithmic count is $A\sqrt{n \log_2 n} - O(\log n)$. The proposed threshold has logarithm

$$\log_2(c_1 n) + c_2 H_2(\rho) \frac{n}{s+1} = (c_2 H_2(\rho)/c + o(1)) \sqrt{n \log_2 n},$$

which is smaller. Every parametrized codeword line contains fewer than n selected witnesses. Finally $\eta \log_2 p \rightarrow cA > H_2(\rho)$ verifies Elias by the same entropy-continuity argument as above. $\square$

A finite line example. Take $p = 2^{127} - 1$, $n = 64$, target dimension $d = 32$, and agreement size $t = 34$. The first-moment sum 1071 has exactly

$$N = 8,634,334,451,040,106 > 2^{52}$$

34-subsets, as certified in Section 2.3. Their vanishing polynomials have two common leading terms, giving source polynomials of degree at most 32. The exact inequality $p > 64 + 32\binom{N}{2}$ makes all parameters distinct. Here $\eta = 1/32$, the threshold with $c_1 = c_2 = 1$ is 2^{38} , and each parametrized codeword line contains at most 16 selected witnesses. The radius is below Elias. In a smaller exhaustive check, $n = 12$, $d = 6$, $t = 8$, $p = 2^{31} - 1$ yield a moment class of size 33. The pole $\tau = 20$ gives 33 distinct parameters, and the maximum number of selected pairs on one parametrized codeword line is exactly 3, matching the lemma's bound.

3.1 Every large prime admits a short interval-domain counterexample

Holding the number of canceled moments fixed gives a useful alternative to Theorem 3.3. It strengthens the field quantifier and reduces the required block length to logarithmic in the prime.

Theorem 3.4 (Logarithmic-length prime-field lines). *Fix a rational $\rho \in (0, 1)$, write $H = H_2(\rho)$, and fix an integer $r \geq 2$. For every sufficiently large prime p , writing $b = \log_2 p$, there is an interval-domain Reed–Solomon code over $\mathbb{F}_p$ with dimension $k = \rho n$ and*

$$n = H^{-1}b + O_{\rho,r}(\log b), \quad t = k + r, \quad \eta = r/n,$$

and a received line with at least

$$\left(\frac{H}{\rho(1-\rho)} - o(1) \right) \frac{p}{\log_2 p} \quad (18)$$

distinct nearby labels at agreement threshold t . There is no correlated agreement on more than k coordinates, and every parametrized codeword line contains at most $\lfloor (n-k)/r \rfloor$ selected witnesses. The radius is strictly below the characteristic-based Elias radius.

In particular, given any finite $c_1, c_2 > 0$, taking an integer $r > \max\{1, c_2\}$ violates the finite line threshold $c_1 n 2^{c_2 H/\eta}$ over every sufficiently large prime field.

Proof. Put $a = (r-1)(r+2)/2$ and choose

$$n = \frac{b + (a+2)\log_2 b}{H} + O_\rho(1)$$

by rounding upward to a multiple of the denominator of ρ . Apply Lemma 2.1 with source dimension $k+1$ and $s = r-1$. Its moment ranges satisfy $\prod_{j=1}^{r-1} R_j = O_r(n^a)$. Moreover, $\binom{n}{t} \geq 2^{nH_2(t/n)}/(n+1)$ and $nH_2(t/n) = Hn + O_{\rho,r}(1)$. Thus a bank of N distinct polynomials of degree at most k , each with exactly t agreements, has

$$N \geq c_{\rho,r} \frac{2^{Hn}}{n^{a+1}} \geq c'_{\rho,r} p b.$$

In particular $N/p \rightarrow \infty$. Select exactly N members if the class is larger.

We sharpen the collision estimate in Lemma 3.2 by subtracting shared agreement roots. Let I_x be the support incidences, so $\sum_x I_x = Nt$. Across the $Q = p - n$ admissible poles, the total number of pair collisions is at most

$$k \binom{N}{2} - \sum_x \binom{I_x}{2} \leq \frac{N^2(k - t^2/n) + N(t - k)}{2}.$$

Indeed each difference already vanishes on its shared support, and $\sum_x I_x^2 \geq N^2 t^2/n$. Averaging and applying Cauchy–Schwarz to the value multiplicities at one pole gives at least

$$\frac{NQ}{Q + N(k - t^2/n) + (t - k)}$$

distinct labels. Since $k - t^2/n = \rho(1 - \rho)n - 2\rho r - r^2/n$ and $N/p \rightarrow \infty$, this is (18). Lemma 3.2 also gives the correlated-agreement and concurrency assertions.

Finally, $\eta \log_2 p \rightarrow rH > H$, whereas $H_2(1 - \rho - \eta) \rightarrow H$, proving strict Elias admissibility. The logarithm of the proposed threshold is $(c_2/r)b + O_{\rho,r,c_1,c_2}(\log b)$, while the constructed count has logarithm $b - \log_2 b + O_\rho(1)$. Their difference tends to infinity when $r > c_2$. $\square$

The gap here is $\Theta_{\rho,r}(1/\log p)$; the theorem does not assert a superlinear exception count at a fixed positive gap. It uses the same moment classes and rational division as the earlier results, with a different parameter choice.

Finite examples entirely over prime fields. The finite smoothed ellipsoid bound of Proposition 2.13, followed by the same shared-support collision count, gives the following existence certificates. The final column is a strict lower bound on $\log_2(J/(n2^{H_2(k/n)/\eta}))$, with $\eta = (t - k)/n$.

Prime	n	k	t	Nearby labels J	Excess bits
$2^{61} - 1$	136	25	29	$> 8.9248 \cdot 10^{16}$	25.81662
$2^{127} - 1$	257	72	78	$> 2.7727 \cdot 10^{36}$	76.40420

Both examples lie strictly below the characteristic-based Elias radius and have no correlated agreement at their displayed thresholds. All words, witnesses, and challenges are in the stated prime field. The count certifies existence of a suitable moment class and pole; it does not enumerate those large objects. These are ratios of coding counts to the finite prescription with $c_1 = c_2 = 1$.

For exact reproducibility, let $\mathcal{P}(u, v)$ be the minimum number of colliding pairs for u objects in v bins, as in Section C.2. If the concentration bound certifies a source bank of N supports, put

$$B = k \binom{N}{2} - \mathcal{P}(Nt, n), \quad J = \min\{j \geq 1 : \mathcal{P}(N, j) \leq \lfloor B/(p - n) \rfloor\}.$$

For the length-257 example, the certified source and label counts are

$$\begin{aligned} N &= 12326787713559741838360124456674964971, \\ J &= 2772727901311387298989842831775757339. \end{aligned}$$

The standard-library checker in `research/logarithmic_length_lines/` uses rational upper bounds for the concentration denominator, exact integer collision arithmetic, and rational logarithm intervals for the reported decimal comparisons. It also verifies primality by Lucas–Lehmer, checks the strict sufficient Elias inequality $p^{t-k}t^t(n-t)^{n-t} > n^n$, and exhaustively tests 60 small-field support classes over 1,252 admissible poles. The discovery scan is not an optimality proof.

4 A fixed-gap obstruction that survives sublinear remainders

The preceding line theorem uses gaps that shrink with length. It therefore does not settle a proposed bound of the form $c_1 2^{c_2 H_2(\rho)/\eta} n + o(n)$ when the remainder is specified separately for each fixed gap. We now obtain a fixed-gap family.

There are two steps. First, keep one finite list example and replace each evaluation point by the same number of preimages under a power map. Degrees and agreement counts grow by the same factor, preserving both rate and gap. Second, use each nearby codeword on n distinct parameters of a word line. This forces a linear discrepancy from any proposed coefficient smaller than the list size.

Lemma 4.1 (A list example over the integers). *Fix rational $\rho \in (0, 1)$ and finite $c_1, c_2 \geq 1$. There are fixed integers $1 \leq k < t < m$, with $k/m = \rho$, distinct t -subsets $A_1, \dots, A_L \subset \{1, \dots, m\}$, and integer polynomials $W, G_1, \dots, G_L$ satisfying*

$$\deg G_i < k, \quad W(X) - G_i(X) = \prod_{a \in A_i} (X - a), \quad L > c_1 2^{c_2 H_2(\rho)/\eta}, \quad \eta = (t - k)/m.$$

The data is chosen before choosing a reducing prime.

Proof. Apply the moment construction over the integers. Equality of binomial moments implies equality of power sums over $\mathbb{Q}$, and Newton identities imply equality of the leading coefficients as integers. To see that a sufficiently large list exists without fixing a field, write $\rho = a/b$ and take $m = bu^3$, $k = au^3$, $s = u$, $t = k + s$. The logarithm of the lower bound in Lemma 2.1 is $bH_2(\rho)u^3 - O_\rho(u^2 \log u)$, whereas the logarithm of the proposed bound is $\log_2 c_1 + c_2 H_2(\rho)bu^2$. Choose a sufficiently large, now fixed, integer u and a class exceeding the proposed bound. Translating the original domain $\{0, \dots, m - 1\}$ by one, and substituting $X - 1$ in each polynomial, preserves the identities and avoids zero. $\square$

Lemma 4.2 (Replacing each point by a full power-map fiber). *Fix integer data as in Lemma 4.1. For every positive integer B and every bound T_0 , there is a prime $p > T_0$ and a domain $D_B \subset \mathbb{F}_p$ of size mB such that $x \mapsto x^B$ maps D_B onto $\{1, \dots, m\}$ with exactly B preimages per point. The polynomials $G_i(X^B)$ give distinct codewords in $\text{RS}(\mathbb{F}_p, D_B, kB)$, each agreeing with $W(X^B)$ on exactly tB coordinates.*

Proof. Let $K_B/\mathbb{Q}$ be the splitting field of $F_B(X) = \prod_{a=1}^m (X^B - a)$. The completely split case of Chebotarev's density theorem gives infinitely many primes that split completely in this finite Galois extension.¹ Exclude primes at most $\max\{T_0, m, B\}$ and primes dividing the nonzero discriminant of F_B . At any remaining such prime, all roots of F_B reduce to distinct elements of $\mathbb{F}_p$. Let D_B be these roots. The roots of each $X^B - a$ form one fiber of size B , and the fibers are disjoint.

Now $\deg G_i(X^B) \leq (k - 1)B < kB$, and

$$W(X^B) - G_i(X^B) = \prod_{a \in A_i} (X^B - a).$$

The difference vanishes on exactly the t selected fibers. The codewords remain distinct: for $A_i \neq A_j$, a point in a fiber indexed by $A_i \setminus A_j$ makes exactly one of the two displayed products zero. No further coefficient-preservation condition is needed. $\square$

Only the existence of arbitrarily large completely split primes is used; we require no bound on the least such prime. The construction may therefore use fields much larger than those in Theorem 2.2.

Lemma 4.3 (A list forces a selected-witness threshold). *Let $C \subset F^n$ be a nonzero linear code. Suppose a word w has L distinct codewords $q_1, \dots, q_L$ within distance θ , where $L \leq n$ and $|F| \geq Ln$. There is a nonconstant affine word line with Ln distinct parameters and a selected nearby codeword at each, such that no parametrized codeword line contains more than n selected pairs. Thus line-decodability with concurrency n fails at every positive threshold smaller than Ln .*

¹See J. S. Milne, *Algebraic Number Theory*, Theorem 8.31 and Corollary 8.32, [course notes](#). Only infinitude of completely split primes is used.

Proof. Choose nonzero $v \in C$ and disjoint sets $S_1, \dots, S_L \subset F$, each of size n . Put $f(z) = w + zv$ and select $p_z = q_i + zv$ for $z \in S_i$. Translation preserves distance, so every selected codeword is nearby. Fix $P_0, P_1 \in C$. If $P_1 = v$, the equality $P_0 + zP_1 = q_i + zv$ holds only for the branch with $P_0 = q_i$, if any, and that branch has n parameters. If $P_1 \neq v$, the equation $P_0 - q_i = z(v - P_1)$ has at most one solution per branch, giving at most $L \leq n$ selected pairs. The argument compares pairs at the same z ; it does not require the unparametrized images of the lines to be disjoint. $\square$

Theorem 4.4 (Failure at a fixed gap with any sublinear remainder). *For every rational $\rho \in (0, 1)$ and finite $c_1, c_2 \geq 1$, there are a fixed gap $\eta \in (0, 1 - \rho)$, an integer $L > c_1 2^{c_2 H_2(\rho)/\eta}$, and a sequence of prime-field Reed–Solomon codes of lengths $n_B \rightarrow \infty$ and exact rate ρ , all with $\theta = 1 - \rho - \eta$ strictly below their characteristic-based Elias radii, such that the following holds. Each code admits Ln_B selected nearby witnesses on a nonconstant affine word line, but no parametrized codeword line contains more than n_B selected pairs. Consequently, for every sequence $R_B = o(n_B)$, all sufficiently large codes fail line-decodability with concurrency n_B and threshold*

$$a_B = c_1 2^{c_2 H_2(\rho)/\eta} n_B + R_B. \quad (19)$$

The code family is chosen independently of the remainder sequence.

Proof. Fix the integer example from Lemma 4.1; hence m, k, t, L, η are now constants. Let B grow with $n_B = mB \geq L$. For each B , choose the least completely split prime from Lemma 4.2 satisfying

$$p_B > \max\{Ln_B, m, 2^{1/\eta}\}$$

and the separability condition in that lemma. This fixes a prime sequence without reference to R_B . The lifted code has rate $k/m = \rho$ and contains a list of size L at distance exactly $1 - t/m = \theta$. It is nonzero, and the hypotheses of Lemma 4.3 hold. Moreover

$$H_{p_B}(\theta) \leq \theta + 1/\log_2 p_B < \theta + \eta = 1 - \rho,$$

which verifies the Elias condition. Put $A = c_1 2^{c_2 H_2(\rho)/\eta}$. Since $L - A > 0$ is fixed, eventually $R_B < (L - A)n_B$, so $a_B < Ln_B$. Lemma 4.3 contradicts line-decodability at this threshold. $\square$

The gap can depend on the proposed constants, but is fixed along the resulting family. Even a remainder chosen separately for that family cannot absorb its positive linear discrepancy. This last construction concerns the arbitrary selection of nearby witnesses: it has correlated agreement, since (q_i, v) agrees jointly with (w, v) wherever q_i agrees with w . Also, its selected count Ln never exceeds $\ell_C(\theta)n$. Thus the obstruction rules out the proposed quantitative coefficient, not a line threshold expressed using a valid bound on the actual maximum list size.

4.1 Anchored padding gives genuine correlated-agreement failure

The preceding selection argument is not needed if we allow a small padding of the lifted domain. The following construction keeps the rate and gap exactly fixed and makes correlated agreement impossible at the threshold, for every challenge.

Lemma 4.5 (Anchored list-to-line padding). *Let D_0 be a set of m distinct nonzero integers containing 1. Let $1 \leq k < t < m$, and let $A_1, \dots, A_L$ be distinct t -subsets of D_0 , all containing 1, whose monic locator polynomials have the same coefficients in degrees $t, t-1, \dots, k$. Choose an integer $1 \leq r < t - k$. For every positive integer B there is an arbitrarily large prime p and a Reed–Solomon code with*

$$n_B = B(m + r), \quad k_B = Bk, \quad A_B = Bt,$$

admitting a nonconstant received line $f + zg$ with at least $L(Br + 1)$ distinct nearby challenges at agreement threshold A_B . Every nearby word on the line has exactly one nearby codeword, and the parameter-zero word has maximum agreement exactly $A_B - 1$. No degree- $< k_B$ polynomials F, G jointly agree with f, g on A_B coordinates. The radius can be chosen strictly below Elias. If $k \geq 2$, one may instead set $k_B = B(k - 1)$ and allow $1 \leq r \leq t - k$, with the same conclusion.

Proof. Write $F_i(X) = \prod_{a \in A_i} (X - a)$, let W be their common degree- $\geq k$ part, and put $G_i = W - F_i$. Then $\deg G_i < k$ and $G_i(1) = W(1)$ for every i . Use the proof of Lemma 4.2 for the splitting field of $\prod_{a \in D_0} (X^B - a)$, with a prime-size bound specified below, and remove the point 1 from its union of full fibers. On the remaining set D_{old} of size $mB - 1$, put

$$f(x) = F_{\text{old}}(x) := \frac{W(x^B) - W(1)}{x - 1}, \quad g(x) = 0.$$

The candidate polynomials are

$$P_i(X) = \frac{G_i(X^B) - W(1)}{X - 1}.$$

Their numerators vanish at 1, so they are polynomials of degree at most $B(k - 1) - 1 < k_B$, with a zero numerator interpreted in the usual way. Each agrees with F_{old} at exactly $Bt - 1 = A_B - 1$ old coordinates, since their difference is $F_i(X^B)/(X - 1)$. The candidates remain distinct after reduction, since their old agreement supports are distinct. Also $\deg F_{\text{old}} = A_B - 1$.

Let $\mathcal{T}$ contain all degree- $< k_B$ polynomials interpolating F_{old} on some k_B old coordinates. Then $|\mathcal{T}| \leq M := \binom{mB-1}{k_B}$, and every selected P_i belongs to $\mathcal{T}$. Add $q_B = Br + 1$ distinct points x_j outside $D_{\text{old}} \cup \{1\}$ at which the values $\{P(x_j) : P \in \mathcal{T}\}$ are pairwise distinct and avoid $F_{\text{old}}(x_j)$. There are enough such points if

$$p > mB + (k_B - 1) \binom{M}{2} + (A_B - 1)M + q_B,$$

by the root bounds for $P - P'$ and $F_{\text{old}} - P$. Choose nonzero values g_j successively so that all labels

$$\frac{P(x_j) - F_{\text{old}}(x_j)}{g_j}, \quad P \in \mathcal{T},$$

are distinct across all P, j . At step j , at most $(j - 1)M^2$ nonzero values are forbidden, so $p - 1 > (q_B - 1)M^2$ suffices. Set $f = F_{\text{old}}$ on the entire domain and $g(x_j) = g_j$ on the new coordinates. Every selected P_i is nearby at q_B distinct nonzero labels, giving Lq_B such challenges at length n_B .

Any nearby polynomial has at least $A_B - q_B \geq k_B$ old agreements, so belongs to $\mathcal{T}$. It has at most $A_B - 1$ old agreements and, by label injectivity, at most one new agreement. Thus it has exactly $A_B - 1$ old agreements and one new agreement. The same injectivity prevents two polynomials from being nearby at the same challenge. This proves uniqueness for the entire line, not merely for the selected witnesses.

Since $f = F_{\text{old}}$ globally has degree $A_B - 1$, it agrees with any degree- $< k_B$ polynomial on at most $A_B - 1$ coordinates. The selected candidates attain that number at parameter zero, so this word is exactly one coordinate beyond the nearby radius. The same degree bound makes joint agreement on A_B coordinates impossible.

All the prime-size conditions, and $((t - k)/(m + r)) \log_2 p > 1$, can be imposed before applying the infinitude of completely split primes. The last condition verifies the strict Elias inequality. For the sharper dimension $k_B = B(k - 1)$, the candidates still fit, and $A_B - q_B \geq k_B$ still holds when $r \leq t - k$, preserving the exhaustive candidate-pool argument. The larger capacity gap preserves the Elias inequality. $\square$

Theorem 4.6 (The linear coefficient at a fixed positive gap). *Fix rational $\rho \in (0, 1)$ and put $H = H_2(\rho)$. For every sufficiently small rational gap $\eta > 0$, there are prime-field Reed–Solomon codes of lengths $n_B \rightarrow \infty$, exact rate ρ , and exact agreement fraction $\rho + \eta$, and received lines with at least $C_\rho(\eta)n_B$ distinct nearby challenges, where*

$$\log_2 C_\rho(\eta) \geq \frac{H^2/2 - o_\rho(1)}{\eta^2 \log_2(1/\eta)}. \quad (20)$$

No pair of codewords has correlated agreement at that threshold, every nearby word on the line has a unique nearby codeword, and the parameter-zero word has distance $1 - \rho - \eta + 1/n_B$. The nearby radius is strictly below the characteristic-based Elias radius. The $o_\rho(1)$ is as rational η tends to zero; $C_\rho(\eta)$ is fixed as B grows.

Proof. The proof of Lemma 4.5 allows arbitrary integer length n , dimension K , and padding count $q = n - mB + 1$, provided

$$q \geq 1, \quad K \geq B(k - 1), \quad K - 1 + q < Bt.$$

Indeed, the candidates have degree at most $B(k - 1) - 1$, and the global received polynomial has degree $Bt - 1$. The inequality $Bt - q \geq K$ preserves the full candidate-pool argument proving uniqueness at every nearby parameter.

Put $a = \rho + \eta$ and choose

$$m = \left\lfloor \frac{H}{\eta^2 \log_2(1/\eta)} \right\rfloor, \quad t = \left\lceil \frac{am}{1 - \eta/2} \right\rceil, \quad k = \left\lfloor \frac{\rho t}{a} \right\rfloor, \quad s = t - k.$$

For sufficiently small η , one has $1 \leq k < t < m$ and

$$am < t < \frac{am}{1 - \eta}, \quad \frac{t}{m} \rightarrow \rho, \quad s = (1 + o(1))\eta m.$$

The strict upper inequality absorbs the rounding error because $\eta m \rightarrow \infty$. Among the t -subsets containing 1, the same binomial-moment count as in Lemma 2.1 gives a class of size

$$L \geq \left\lceil \frac{\binom{m-1}{t-1}}{\prod_{j=1}^s R_j} \right\rceil,$$

where R_j are the full moment-range sizes for length m . They still bound the ranges after the anchor is fixed. The moment-range estimate used in Theorem 2.2 gives

$$\log_2 \prod_{j=1}^s R_j \leq \frac{s^2}{2} \log_2(m/s) + O(s^2 + s \log m) = (H/2 + o(1))m.$$

Since $\binom{m-1}{t-1} = (t/m)\binom{m}{t}$, it follows that

$$\log_2 L \geq (H/2 - o(1))m.$$

Let B grow through multiples making $n_B = Bt/a$ and $K_B = \rho n_B$ integers, and set $q_B = n_B - mB + 1$. Then $K_B \geq Bk$ and

$$K_B - 1 + q_B = B((1 + \rho)t/a - m) < Bt.$$

The flexible padding construction therefore has exact rate ρ , exact gap η , and more than $C_\rho(\eta)n_B$ nearby labels for $C_\rho(\eta) = (1 - am/t)L$. Since $1 - am/t \sim \eta/2$, this prefactor costs only $O(\log(1/\eta))$ in its logarithm. Substituting the chosen m proves (20). Arbitrarily large splitting primes ensure $n_B = o(p)$ and strict Elias. The rationality of η is used only to keep the displayed rate and agreement fraction exactly fixed at integer lengths. $\square$

The same parameters also give a list lower bound at every sufficiently small rational gap. Keep all mB fiber points and add $n_B - mB$ distinct points outside the fibers. The received polynomial $W(X^B)$ has the L nearby candidates $G_i(X^B)$, each of degree at most $B(k-1) < K_B$. Each difference is the degree- Bt locator $F_i(X^B)$, so each candidate has exactly Bt agreements. Thus, at exact rate ρ and exact gap η , a displayed list has logarithm at least $(H^2/2 - o_\rho(1))/(\eta^2 \log_2(1/\eta))$ on arbitrarily long domains with $n_B = o(p)$ and strict Elias. This is a lower bound on one list; it does not assert growth of that list with B .

In particular, no uniform exceptional-count bound $c_1 2^{c_2/\eta} n + o(n)$ can hold, even when its remainder is specified separately at each fixed gap. Choose a gap for which $C_\rho(\eta) > c_1 2^{c_2/\eta}$; the positive linear discrepancy then absorbs any such remainder. Every exhibited nearby challenge also violates the full common-witness conclusion, because it has no common agreement set of the required size at all. The construction remains linear in n at each fixed gap and does not establish a quadratic lower bound there.

Uniqueness is restricted to the displayed line. Once $q_B \geq L$, the same code has another received word with exactly the L selected codewords nearby: retain the old word, assign each selected codeword its own padding coordinate, and set all remaining padding values outside the corresponding evaluation image of $\mathcal{T}$. Injectivity of those evaluation maps excludes every other candidate. Thus line-wise uniqueness does not imply a small global maximum list.

It also supplies a selected-witness counterexample with concurrency at most n_B . Indeed, on any proposed affine codeword graph let a coordinates agree identically with the received line. Then $a < A_B$, and each other coordinate agrees at at most one selected challenge. If the graph contains v selected nearby pairs, $vA_B \leq va + n_B - a$, giving $v \leq n_B - A_B + 1 \leq n_B$. Exact anchored moment counts and prime-field fixtures of lengths 10, 20, 40 and 11, 22, 44 appear in `research/fixed_gap_padding/`.

Corollary 4.7 (Polynomial-size prime fields). *For each fixed rational $\rho \in (0, 1)$ and every sufficiently small rational $\eta > 0$, the list lower bound and the exceptional-count lower bound $C_\rho(\eta)n$ above also hold along growing lengths with*

$$n = o(p), \quad p = n^{O_{\rho, \eta}(1)}.$$

The rate and gap are exact, the radius is strictly below Elias, and no pair of codewords has correlated agreement at the threshold. The parameter-zero word is exactly one coordinate beyond the nearby radius. This assertion does not include uniqueness of nearby codewords.

Proof. Fix the integer seed with m distinct nodes $a_1, \dots, a_m$ and anchor $a_* = 1$. If $B \mid p-1$, a standard multiplicative-character expansion counts the shifts c for which every $c + a_i$ is a nonzero B th power. The product-character Weil bound [2, Lemma 17], with constant additive phase, gives

$$\#\{c : c + a_i \in (\mathbb{F}_p^*)^B \text{ for all } i\} \geq \frac{p}{B^m} - m\sqrt{p} - m.$$

Indeed, extend all characters by zero at zero. The all-principal term is $p - m$; every other term has absolute value at most $m\sqrt{p} + m$, allowing for the excluded roots. The resulting bound is positive if $p > 16m^2 B^{2m}$.

By Linnik's least-prime theorem [3], there are absolute constants C, L_0 such that the least prime congruent to 1 modulo B^{2m+2} satisfies

$$B^{2m+2} < p \leq CB^{(2m+2)L_0}.$$

For all sufficiently large B , the nodes remain distinct modulo p and the preceding shift exists. Write $W_c(Y) = W(Y - c)$ and $G_{i,c}(Y) = G_i(Y - c)$, and choose $\alpha^B = c + a_*$. Remove α from the union

of the full fibers and replace the anchored quotients by

$$\frac{W_c(X^B) - W(a_*)}{X - \alpha}, \quad \frac{G_{i,c}(X^B) - W(a_*)}{X - \alpha}.$$

Their degrees and exact old-agreement counts are unchanged. For the padding, separate only the L selected candidates, rather than the full interpolation pool $\mathcal{T}$. The sufficient bounds are then

$$p > mB + (K_B - 1) \binom{L}{2} + (Bt - 1)L + q_B, \quad p - 1 > (q_B - 1)L^2.$$

Both are eventually implied by the displayed lower bound on p , since the seed and L are fixed and $K_B, q_B = O(B)$. This gives $Lq_B > C_\rho(\eta)n_B$ distinct nearby labels. The global degree- $Bt - 1$ received word gives the far point and excludes correlated agreement. Restoring the anchor gives the list-only construction as before. Finally $n_B = \Theta(B)$, so $n_B = o(p)$, the field size is polynomial in n_B , and strict Elias holds eventually. The exponent may depend on ρ, η ; the displayed choice is $O_\rho(\eta^{-2}/\log(1/\eta))$ as $\eta \rightarrow 0$. Separating only the selected candidates does not establish whole-line uniqueness. $\square$

Concrete fixed-gap coefficients. The earlier finite integer classes also supply explicit seeds. Among L_0 supports of size t on m points, some point belongs to at least $L = \lceil tL_0/m \rceil$ supports. If that point is a , the integer affine change $x \mapsto 2(x - a) + 1$ makes it the anchor 1 and makes every domain point nonzero. Multiplying transformed locator polynomials by 2^t preserves monicity, integer coefficients, and the common prefix. Using the sharper dimension $B(k - 1)$, the two previously certified classes give

(m, k, t)	r	L	coefficient $C = rL/(m + r)$
$(64, 32, 34)$	2	2949985688957	2949985688957/33
$(157, 63, 68)$	5	12200781436007129430	10167317863339274525/27

The resulting rates and gaps are $(31/66, 1/22)$ and $(31/81, 1/27)$, respectively. In the second family, more than $3.76 \cdot 10^{17}n_B$ distinct challenges are nearby, while correlated agreement at the threshold is impossible. These are growing-prime-field families obtained from the integer classes; they are not statements about retaining the M31 alphabet or a prescribed subgroup domain.

4.2 Padding with collisions improves the finite-field bounds

The preceding constructions required distinct labels. Allowing collisions and averaging their union gives a stronger bound when the field is fixed. The mechanism is simple: many codewords already agree with a fixed core on $A - 1$ coordinates, just one short of the threshold. Each added coordinate lets some of them acquire the missing agreement, at different line parameters. With enough added coordinates, these opportunities cover almost all parameters. Keeping the starting word equal to a polynomial of degree $A - 1$ makes it one coordinate too far from the code and rules out any common agreement explanation. Write $\mathcal{P}(h, N)$ for the minimum number of colliding pairs for h objects in N bins, as in Section C.2.

Lemma 4.8 (Averaged padding). *Let $D_{\text{old}} \subset \mathbb{F}_p$ have N points, and let w be a polynomial of degree $A - 1$. Suppose $P_1, \dots, P_L$ are distinct polynomials of degree less than $K < A$, each agreeing*

with w at exactly $A - 1$ old coordinates, and each pairwise difference has degree at most d . Choose $1 \leq q \leq p - N$ with $K - 1 + q < A$, and put

$$R = p - N, \quad T = d \binom{L}{2} - \mathcal{P}(L(A - 1), N), \quad M = \frac{L^2 R}{LR + 2T}.$$

Then $T \geq 0$. On a domain of size $n = N + q$, there is a received line with at least

$$J = \left\lceil p \left(1 - \left(1 - \frac{M}{p} \right)^q \right) \right\rceil \quad (21)$$

nearby parameters at agreement threshold A , and no pair of codewords has correlated agreement at that threshold.

Proof. For $x \notin D_{\text{old}}$, let s_x be the number of distinct values among the $P_i(x)$, with multiplicities $r_{x,v}$. Each polynomial pair has at most d coincidences. Their old common agreements consume at least $\mathcal{P}(L(A - 1), N)$ such coincidences, so

$$\sum_{x \notin D_{\text{old}}} \sum_v r_{x,v}^2 \leq LR + 2T.$$

Cauchy–Schwarz gives $R^{-1} \sum_x s_x \geq M$. Choose the q points x_j with largest s_x . Set $f = w, g = 0$ on the old coordinates, and $f(x_j) = b_j, g(x_j) = 1$ on the new ones. For independent uniform offsets b_j , the expected size of the union of labels $P_i(x_j) - b_j$ is

$$p \left(1 - \prod_{j=1}^q (1 - s_{x_j}/p) \right) \geq p (1 - (1 - M/p)^q),$$

by arithmetic–geometric mean. Some offsets attain its ceiling. Every label in the union gives a candidate with at least A agreements. For common witnesses F, G , if $G = 0$ the joint agreement is at most $A - 1$ by the degree of $w - F$; otherwise it is at most $K - 1 + q < A$. $\square$

What a growing fixed-gap list would imply. The same averaging mechanism gives a useful transfer without requiring the received word to have a low-degree polynomial representation. The conclusion here concerns the entire agreement support of a nearby candidate; it does not assert the absence of ordinary correlated agreement.

Proposition 4.9 (List amplification with exact parameter scaling). *Let $k + 1 \leq A \leq N$, $k \geq 2$, and let $p \geq 2N + 1$ be prime. Suppose a word on N distinct points of $\mathbb{F}_p$ has L distinct degree- $< k$ candidates, each with at least A agreements. There is a Reed–Solomon code of length $2N$ and dimension k over the same prime field, and a received line $f + zg$, with at least*

$$J \geq \left\lceil \frac{pAL}{p + 3AL} \right\rceil \geq \frac{1}{4} \min\{AL, p\}$$

distinct labels z for which some nearby candidate has at least A agreements and g has no degree- $< k$ interpolant on its entire agreement support. Thus source rate ρ and gap η become exactly $\rho/2$ and $\eta/2$.

Proof. Some coordinate a belongs to at least $\ell \geq AL/N$ candidate agreement supports. Retain those candidates and form

$$Q_P(X) = \frac{P(X) - w(a)}{X - a}, \quad f(x) = \frac{w(x) - w(a)}{x - a}, \quad g(x) = 0 \quad (x \neq a).$$

The Q_P are distinct of degree at most $k - 2$, and have at least $A - 1 \geq k$ agreements on the $N - 1$ retained coordinates. At each of the $R = p - N$ unused coordinates let s_x be the number of distinct values of the Q_P . Counting pairwise coincidences and applying Cauchy–Schwarz gives

$$\frac{1}{R} \sum_x s_x \geq \mu := \frac{\ell R}{R + (\ell - 1)(k - 2)}.$$

Choose the $q = N + 1$ unused coordinates with largest s_x , put $g = 1$ there, and choose their f values independently and uniformly. The expected union of the label sets $\{Q_P(x) - f(x)\}$ is at least $p(1 - (1 - \mu/p)^q)$, exactly as in Lemma 4.8. Every label in this union adds a new agreement to one selected candidate. A degree- $< k$ interpolant for g on that candidate's support would vanish at its at least k old agreements, hence vanish identically, contradicting its value 1 at the new agreement.

For the count, $R \geq p/2$ gives

$$\frac{q\mu}{p} \geq \frac{N\ell}{p + 2(k - 2)\ell} \geq \frac{N\ell}{p + 2N\ell} \geq \frac{AL}{p + 2AL}.$$

Using $1 - (1 - u)^q \geq 1 - e^{-qu} \geq qu/(1 + qu)$ proves the claimed bound. The new length is $N - 1 + q = 2N$, while its dimension and agreement threshold remain k and A , proving exact parameter scaling. $\square$

At fixed positive rate and gap, this gives $J = \Omega(\min\{NL, p\})$. In particular, a list lower bound N^c over superpolynomial prime fields would force $\Omega(N^{c+1})$ full-support exceptions at half the rate and gap. A universal linear full-support bound at the latter parameters would imply constant source list sizes whenever $p/N \rightarrow \infty$. This identifies a sufficient route to exponent lower bounds; it does not supply the growing lists. The full-length lists with $N = p - 1$ fall outside its field-size hypothesis.

Corollary 4.10 (Amplification within an open agreement regime). *Under the source hypotheses of Proposition 4.9, fix rational $0 < \tau \leq 1$ with τN integral. There is a line of length $(1 + \tau)N$, dimension k , and agreement threshold A with at least*

$$\frac{\tau}{5} \min\{AL, p\}$$

full-support MCA-bad labels. Its rate and agreement fraction are $\rho/(1 + \tau)$ and $a/(1 + \tau)$, where $\rho = k/N$ and $a = A/N$. If $a > h(\rho)$ for a continuous sufficient agreement curve h , then every sufficiently small positive τ preserves that strict inequality at the output parameters. If $a^2 < \rho$, the output also remains below Johnson.

Proof. Use the same proof with $q = \tau N + 1$ padding coordinates. Before specializing q , its count gives

$$J \geq \frac{pq\ell}{p + (2(k - 2) + q)\ell} \geq \frac{pqAL}{Np + (2(k - 2) + q)AL}.$$

Here the first inequality follows from $\mu \geq \ell p/(p + 2(k - 2)\ell)$ and $p(1 - (1 - \mu/p)^q) \geq pq\mu/(p + q\mu)$; the second uses monotonicity in ℓ . Since $q \geq \tau N$ and $2(k - 2) + q < 4N$, this is at least $\tau pAL/(p + 4AL) \geq (\tau/5) \min\{AL, p\}$. The old $A - 1 \geq k$ agreements still exclude a full-support direction witness. The parameter identities are exact. The strict inequality $a/(1 + \tau) > h(\rho/(1 + \tau))$ follows by continuity at $\tau = 0$. Finally $a^2/(1 + \tau)^2 < \rho/(1 + \tau)$ whenever $a^2 < \rho$. $\square$

Thus, within any such open regime, source lists of size $\Omega(N^c)$ over sufficiently large primes would force $\Omega(n^{c+1})$ full-support exceptions at nearby parameters in the same regime. This remains a conditional statement: it requires the growing source lists. For arbitrary lengths, taking $q = \lfloor \tau N \rfloor + 1$ preserves the strict margin eventually, with parameters converging to the displayed pair.

The exact finite checks in `research/prime_field_tightness/check_prime_boundary_amplification.py` include parameter halving over $\mathbb{F}_{101}$ and examples where full-support failure coexists with ordinary correlated agreement. The variable-padding checks in `research/prime_field_tightness/check_sparse_padding.py` independently exhaust all direction witnesses for the selected bad supports.

Lemma 4.11 (Padding around a far point). *Keep the core hypotheses and R, T, M of Lemma 4.8, but allow any $1 \leq q \leq p - N$, without requiring $K - 1 + q < A$. There is a nonconstant received line on $n = N + q$ coordinates with at least*

$$J = \left\lceil (p-1) \left(1 - \left(1 - \frac{M}{p-1} \right)^q \right) \right\rceil \quad (22)$$

nearby parameters at agreement threshold A . Its parameter-zero word has distance exactly $(n - A + 1)/n$, one coordinate beyond the nearby radius. No pair of codewords has correlated agreement at threshold A .

Proof. Each $P_i - w$ has degree $A - 1$ and already has $A - 1$ roots on the core. It therefore has no roots outside the core. At an available point x , the difference image $S_x = \{P_i(x) - w(x) : 1 \leq i \leq L\}$ is a subset of $\mathbb{F}_p^*$ of size s_x . The preceding collision bound still gives average image size at least M , so in particular $M \leq p - 1$. Choose the q points with largest images.

Set $f = w$ on the entire domain, let $g = 0$ on the core, and choose $g(x_j)$ independently uniformly in $\mathbb{F}_p^*$ on the padding. The candidate P_i is nearby at the nonzero parameter

$$z = \frac{P_i(x_j) - w(x_j)}{g(x_j)}.$$

For each fixed $z \neq 0$, these independent multiplicative translates hit z with probabilities $s_{x_j}/(p-1)$. The expected union size is

$$(p-1) \left(1 - \prod_j (1 - s_{x_j}/(p-1)) \right) \geq (p-1) (1 - (1 - M/(p-1))^q).$$

Some direction attains the ceiling. Finally $f = w$ agrees with every codeword on at most $A - 1$ coordinates by its degree, and each selected P_i attains that number. This proves the exact distance at zero and also rules out correlated agreement, regardless of the chosen direction. $\square$

4.3 Every point but one, with an almost maximally far exception

Pairing coordinates as $x, -x$ gives a stronger separation for the $c_1 = c_2 = 1$ prescription. The exceptional point can be just one coordinate short of the maximum possible Reed–Solomon distance, while its separation from the nearby radius approaches the entire capacity gap. The construction uses random paired domains and a multiplicative-character estimate, without a matching-moment class.

Lemma 4.12 (Sampling with blocks of paired coordinates). *Let r, m, D, q be positive integers with $D < m$ and $m + rq \leq (p - 1)/2$, where $p > \max\{(4r + 2)^2, 2m^2\}$ is prime. Put*

$$L = \binom{m}{D}, \quad \delta = \frac{p^r}{L} + \frac{(4r + 2)^{2r+2} + 2m^2}{p - 2m^2}.$$

There is an expected $O(n^2)$ -field-operation randomized procedure outputting a length- $n = 2(m + rq)$, dimension- $K = 2D - 1$ Reed-Solomon code and a received line. Every output has maximum agreement exactly $2D = K + 1$ at parameter zero. For every $\gamma > 0$, the probability that some nonzero parameter has fewer than $2D + 2r$ agreements is at most

$$\frac{q\delta}{\gamma} + (p - 1) \left(\frac{\gamma}{1 + \gamma} \right)^q. \quad (23)$$

Whenever all nonzero parameters are nearby, there is no correlated agreement at that threshold.

Proof. Choose $m + rq$ distinct orbits $\{a, -a\}$ uniformly from $\mathbb{F}_p^*$. Use the first m as the core and divide the remaining orbits into q blocks of r orbits each. For every D -subset I of the core, write

$$H_I(X) = \prod_{i \in I} (X^2 - a_i^2).$$

Fix one reference support I_0 , put $w = H_{I_0}$, and consider the candidates $P_I = w - H_I$. Their leading terms cancel and all odd coefficients vanish, so $\deg P_I \leq 2D - 2 < K$. Each candidate has exactly $2D$ core agreements. On each padding pair choose a shared uniform nonzero direction, independently across pairs. Set $f = w$ everywhere and $g = 0$ on the core.

We bound the image of $I \mapsto (-H_I(x_1), \dots, -H_I(x_r))$ for a random outside block. Let $U = p - 1$ and $V = U^r$. Fix representatives $x_1, \dots, x_r$ with distinct nonzero squares. Independently sampling seed variables from

$$T = \mathbb{F}_p \setminus \{0, \pm x_1, \dots, \pm x_r\}$$

induces a probability measure μ on $(\mathbb{F}_p^*)^r$ by $a \mapsto (x_j^2 - a^2)_{j=1}^r$. Every nonprincipal Fourier coefficient of μ has absolute value at most

$$\frac{2r\sqrt{p} + 2r + 1}{p - 2r - 1} \leq \frac{4r + 2}{\sqrt{p}} =: \lambda < 1.$$

Indeed, factor $x_j^2 - a^2$ into the two distinct linear factors at $\pm x_j$ and apply the product-character bound [2, Lemma 17], with constant additive phase. At least one of the $2r$ characters is nonprincipal. The extra $2r + 1$ allows for the deleted arguments. The displayed hypotheses imply the second inequality.

For two supports with u indices on each side of their symmetric difference, independent seeds give vector-collision probability

$$\frac{1}{V} \sum_{\chi} |\hat{\mu}(\chi)|^{2u} \leq \frac{1}{V} + \lambda^{2u}.$$

Conditioning all m seeds to have distinct squares costs at most $1/a_0$, where $a_0 = 1 - 2m^2/p > 0$, by the union bound. After this conditioning, pairs with $1 \leq u \leq r$ cannot collide: cancel common factors and regard the residual products as distinct monic degree- u polynomials in $Y = X^2$. Their difference has degree at most $u - 1 < r$, so cannot vanish at the r distinct values x_j^2 . For $u \geq r + 1$, the conditional collision probability is at most $(V^{-1} + \lambda^{2r+2})/a_0$.

Let E be the collision energy of the block image, counting ordered support pairs, and set $\Delta = VE/L^2 - 1 \geq 0$. Counting the diagonal pairs separately gives

$$\mathbb{E}\Delta \leq \frac{V}{L} + \frac{1 + V\lambda^{2r+2}}{a_0} - 1 \leq \delta.$$

This expectation applies to a uniform core and an outside block. Equivalently one may choose the block first and the admissible core second: each has the same number of compatible choices. Every block in the sampled domain has this marginal law, so Markov's inequality and a union bound give $\Pr(\text{some block has } \Delta > \gamma) \leq q\delta/\gamma$. No independence among these events is assumed.

When $\Delta \leq \gamma$, Cauchy-Schwarz gives an image of size at least $V/(1 + \gamma)$. Conditional on the core and blocks, the random direction vectors are independent and uniform in $(\mathbb{F}_p^*)^r$. For any fixed nonzero label z , multiplication of a direction vector by z preserves this distribution. Thus, if all blocks are good, the expected number of uncovered labels is at most the second term of (23). A block hit supplies $2r$ padding agreements. Combining the two failure events proves the bound.

The polynomial w is monic of degree $2D$, and zero already agrees at its $2D$ core roots. This proves the exact far distance for every output and rules out correlated agreement. Sampling distinct orbits and forming and evaluating w take expected $O(n^2)$ field operations. $\square$

Exact rates and parity. The same construction realizes every fixed rational rate, with at most one additional coordinate. Take n through odd multiples of the reduced denominator of ρ , and set $K = \rho n$. If K is odd, use $D = (K + 1)/2$ and add zero if n is odd, with $f(0) = w(0), g(0) = 0$. No selected candidate agrees there, since $H_I(0) \neq 0$. If K is even, then n is odd. Take $D = K/2$, multiply w and all candidates by X , and add zero with $f(0) = g(0) = 0$. On a padding pair use $xh, -xh$ as its directions. Dimension, threshold and exact far agreement all increase by one. In both cases the nearby threshold is $K + 2r + 1$, and the exact far agreement is $K + 1$. The probability bound is unchanged.

Theorem 4.13 (Fixed separation arbitrarily close to the full gap). *Fix rational $\rho \in (0, 1)$, an integer $r \geq 1$, $c_1 > 0$, and $0 < c_2 < 1 + 1/(2r)$. For every sufficiently large prime p , a code of exact rate ρ and length $n = \Theta(\log p)$ has a received line with every nonzero parameter within radius $\theta = 1 - \rho - \eta$, where $\eta = (2r + 1)/n$, and parameter zero at distance*

$$1 - \rho - 1/n = \theta + \frac{2r}{2r + 1}\eta.$$

The radius is strictly below Elias, there is no correlated agreement, and $c_1 n 2^{c_2 H_2(\rho)/\eta} = o(p)$. The code and line can be sampled in expected polynomial time in $\log p$, with success probability $1 - p^{-\Omega(1)}$.

Proof. Choose $\rho < \beta < 1$ close to ρ , put $\alpha = \rho/\beta$, and choose

$$\frac{2r}{\alpha H_2(\beta)} < C < \frac{2r + 1}{\max\{1, c_2\} H_2(\rho)}.$$

Take $n = C \log_2 p + O(1)$ with the rate parity above, $m = \alpha n/2 + O(r)$, and $D = \rho n/2 + O(1)$, adjusting m so the padding forms complete blocks. Then $L = \binom{m}{D} \geq p^{r+\epsilon}$ for some fixed $\epsilon > 0$, and $q = \Theta(\log p)$. Hence $\delta = p^{-\Omega(1)}$. A sufficiently small fixed γ makes both terms of (23) $p^{-\Omega(1)}$. The degree and threshold identities give the distances. Finally $\eta \log_2 p \rightarrow (2r + 1)/C > \max\{1, c_2\} H_2(\rho)$ proves strict Elias and the numerical comparison. $\square$

4.3.1 Completing the joint image and determining exact distances

A few additional core choices can fill the entire product image. This removes the need for multiple padding blocks and gives an exact distance profile on an affine space. The completion step follows the classical random-translate strategy of Erdős and Rényi [1, Theorem 2]; here a character bound controls nonuniform translates arising from new evaluation points.

Lemma 4.14 (Completion by pairs of alternatives). *Fix r nonzero padding representatives with distinct squares. Let m_0, D_0, t be positive integers with $D_0 < m_0$, put $n = 2(m_0 + 2t + r)$ and $C = 4r + 2$, and assume $p > \max\{(n+1)^2, C^2\}$. Define*

$$L_0 = \binom{m_0}{D_0}, \quad d_0 = \frac{p^r}{L_0} + \frac{C^{2r+2} + 2m_0^2}{p - 2m_0^2}, \quad \Phi(h) = \frac{h^2 + C^2 h/p}{1 - 4/p}.$$

If $(p-1)^r \Phi^{ot}(d_0) < 1$, there is a core of $m_0 + 2t$ sign-orbits and a family of $(D_0 + t)$ -subsets whose locator values on the padding fill $(\mathbb{F}_p^)^r$. More generally, uniform successive sampling of the core succeeds with failure probability at most*

$$\frac{d_0}{h_0} + \sum_{i=1}^t \frac{\Phi(h_{i-1})}{h_i} \quad (24)$$

for any positive thresholds $h_0, \dots, h_t$ with $(p-1)^r h_t < 1$.

Proof. Choose m_0 base orbits uniformly outside the padding. The energy argument in Lemma 4.12 applies with the padding fixed: the mean missing fraction of the product image of all D_0 -subsets is at most d_0 .

At each subsequent step add two new core orbits, represented by a, b . Each candidate chooses exactly one of these two. If S is the current product image and $v(a) = (x_j^2 - a^2)_{j=1}^r$, the new image is $Sv(a) \cup Sv(b)$. For its complement B of density h , the new missing set is the intersection of two translates of B .

First sample a, b independently outside zero, the padding, and all previous core orbits. At most $n+1 < \sqrt{p}$ field elements are excluded. The same character estimate gives every nonprincipal Fourier coefficient of the distribution μ of $v(a)$ absolute value at most $C/\sqrt{p}$. Using normalized Fourier coefficients on $(\mathbb{F}_p^*)^r$, Parseval gives

$$\mathbb{E} \frac{|Bv(a) \cap Bv(b)|}{(p-1)^r} = \sum_{\chi} |\widehat{1_B}(\chi)|^2 |\widehat{\mu}(\chi)|^2 \leq h^2 + \frac{C^2}{p} h.$$

The available set has size at least $p/2$. Conditioning $a \neq \pm b$ therefore costs at most $1/(1-4/p)$, so the conditional mean missing fraction is at most $\Phi(h)$. This bound holds uniformly over every previous choice of the core and its image.

Successive averaging proves the existence assertion: at each step choose a pair attaining at most the conditional mean. Once the missing fraction is less than $1/(p-1)^r$, no vector is missing. Alternatively, Markov's inequality at the base step and conditionally at each later step, followed by a union bound, gives (24). The sampler does not need to compute any of the images. $\square$

Corollary 4.15 (Exact distances and almost the entire capacity gap). *Fix rational $\rho \in (0, 1)$, put $b = \log_2 p$ and $H = H_2(\rho)$, and let $r = r(p) \geq 1$ satisfy $r \log b = o(b)$. Over every sufficiently large prime field, a code of exact rate ρ and length*

$$n = \frac{(2r + 4/5)b}{H} + O_\rho(1)$$

has an affine space with independent directions $e_1, \dots, e_r$, each supported on a disjoint pair of coordinates, such that

$$\text{dist} \left(f + \sum_{j=1}^r z_j e_j, C \right) = 1 - \rho - \frac{1 + 2 \text{wt}(z)}{n} \quad (z \in \mathbb{F}_p^r). \quad (25)$$

Here $\text{wt}(z)$ counts its nonzero entries. At radius $\theta = 1 - \rho - \eta$, $\eta = (2r+1)/n$, the nearby parameters are exactly $(\mathbb{F}_p^*)^r$, and their decoding lists are pairwise disjoint. The radius is strictly below Elias, $n2^{H/\eta} = o(p)$, and there is no correlated agreement. The code and affine space can be sampled in expected polynomial time in $\log p$, with success probability $1 - p^{-\Omega(1)}$.

In particular, every line through f with all direction coefficients nonzero is nearby at all $p-1$ nonzero parameters and has direction weight exactly $2r$. Its exceptional point has distance $1 - \rho - 1/n$, only one coordinate short of maximum distance. If $r \rightarrow \infty$, the separation is $(1 - o(1))\eta$ and $\eta \sim H/b$. Taking $r = \lfloor \log_2 b \rfloor$ gives length $\Theta_\rho(b \log b)$.

Proof. Take $t = 16(r+1)$ and choose n as displayed with the exact-rate parity specified above. Let $a \in \{0, 1\}$ indicate the extra zero coordinate, and let $\epsilon \in \{0, 1\}$ indicate multiplication by X . Set

$$m_0 = (n - a)/2 - 2t - r, \quad D_0 = (\rho n + 1 - \epsilon)/2 - t.$$

These are positive with $D_0 < m_0$ eventually. Since $m_0 = n/2 - O(r)$ and $D_0 = \rho n/2 - O(r)$, the binomial entropy estimate and smoothness of H_2 near the fixed rate give

$$\log_2 L_0 = \frac{nH}{2} + O_\rho(r + \log n) = rb + 2b/5 + o(b).$$

Thus $L_0 \geq p^{r+1/3}$ eventually, while $C^{2r+2} + m_0^2 = p^{o(1)}$. The initial mean missing fraction is at most $2p^{-1/3}$. With probability at least $1 - 2p^{-5/24}$, it is at most $p^{-1/8}$. Whenever $h \leq p^{-1/8}$, for sufficiently large p ,

$$\Phi(h) \leq 4p^{-1/8}h.$$

Consequently the probability that a step fails to reduce the missing fraction by a factor $p^{-1/16}$ is at most $4p^{-1/16}$. A zero missing fraction remains zero. A union bound over t steps gives failure probability at most $2p^{-5/24} + 4tp^{-1/16} = p^{-\Omega(1)}$; on success the final missing fraction is at most $p^{-1/8-t/16} < p^{-r}$, hence is zero.

Fix a reference locator w in the completed family. Its degree is $K+1$ after the parity adjustment, and zero attains its $K+1$ roots, so $\text{dist}(f, C) = (n - K - 1)/n$. The full product image supplies a candidate matching any prescribed nonzero shifts on any subset of padding pairs: extend the unspecified components to arbitrary nonzero values. If z has weight j , this gives $K+1+2j$ agreements. Conversely the received word differs from f on exactly $2j$ positions, so the Hamming-distance inequality gives the matching lower bound. This proves (25).

A codeword within θ at a full-support parameter must agree on all $2r$ padding positions: otherwise changing those positions back to f would give a distance smaller than its established minimum. Distinct full-support parameters have different padding values, so their lists are disjoint. The far point rules out correlated agreement.

Finally $\eta b - H = \Theta_\rho(1/r)$ dominates the $O_\rho(1/b)$ entropy-radius change, proving strict Elias, and

$$\log_2 \left(\frac{n2^{H/\eta}}{p} \right) = \log_2 n - \frac{b}{5(2r+1)} + O_\rho(1/r) \rightarrow -\infty.$$

The construction samples $O(n)$ field elements and forms and evaluates one locator in $O(n^2)$ field operations. It does not recover a nearby witness for an arbitrary supplied parameter. $\square$

Corollary 4.16 (Two far inputs and almost-certain nearby mixtures). *In Corollary 4.15, let $r \geq 2$. When r is even, there are words u, v with*

$$\text{dist}(u, C) = \text{dist}(v, C) = \theta + \frac{r}{2r+1}\eta$$

such that

$$\text{dist}((1-z)u + zv, C) = \theta \quad \text{for every } z \in \mathbb{F}_p \setminus \{0, 1\}.$$

Thus exactly the two endpoints are outside the tested radius, and a uniform affine mixture is nearby with probability $1 - 2/p$. For $r \rightarrow \infty$, both endpoints are almost half a gap outside that radius, although the numerical prescription predicts a nearby fraction tending to zero. All the same rate, domain-length, strict-Elias, and sampling guarantees apply.

More generally, for every divisor $t \geq 2$ of r , the same code has t words $v_1, \dots, v_t$, each at distance

$$\theta + \frac{2r}{2r+1} \left(1 - \frac{1}{t}\right) \eta,$$

whose uniform affine combination is nearby with exact probability

$$\left(1 - \frac{1}{p}\right)^t - \frac{(-1)^t}{p^t} = 1 - O(t/p).$$

Here the coefficients are uniform subject to summing to one. For each fixed batch size t , taking $r \rightarrow \infty$ through multiples of t makes every input almost $(1 - 1/t)\eta$ outside the radius.

Proof. For the first statement, partition the r directions into two equal sets A, B and put $u = f + \sum_{j \in A} e_j$, $v = f + \sum_{j \in B} e_j$. Their affine mixture has parameter coordinates $1 - z$ on A and z on B . At the endpoints exactly $r/2$ coordinates are nonzero; at every other parameter all r are nonzero. The exact distance profile (25) proves every claim about the line. Its $p - 2$ nearby parameters still exceed the prescribed bound by a factor tending to infinity.

For the second statement partition the directions into t groups A_j of size r/t and take $v_j = f + \sum_{i \in A_j} e_i$. Each word has parameter weight r/t , giving the stated distance. An affine combination has coefficient λ_j repeated on A_j , so it is nearby exactly when all t coefficients are nonzero. The number of nonzero-coordinate vectors with sum one is $((p-1)^t - (-1)^t)/p$: this follows by inclusion-exclusion, or by separating zero and nonzero target sums in a recurrence on t . Dividing by the p^{t-1} affine coefficient vectors proves the formula. $\square$

Corollary 4.17 (Linear and multilinear mixtures of far inputs). *Use the t words of Corollary 4.16 and assume $n - K \geq 4r + 1$, as holds throughout its sufficiently large-field asymptotic regime. A uniform unrestricted linear combination is nearby with exact probability*

$$\frac{1}{p} + \left(1 - \frac{1}{p}\right) \left[\left(1 - \frac{1}{p}\right)^t - \frac{(-1)^t}{p^t} \right].$$

If $t = 2^\ell$, index the words by $b \in \{0, 1\}^\ell$ and use the multilinear coefficients

$$\lambda_b(x) = \prod_{j:b_j=0} (1 - x_j) \prod_{j:b_j=1} x_j.$$

For uniform $x \in \mathbb{F}_p^\ell$, the mixture is nearby with exact probability $(1 - 2/p)^\ell$. For every fixed ℓ , all inputs can lie almost $(1 - 2^{-\ell})\eta$ outside the tested radius while this probability tends to one.

Proof. Let s be the sum of the linear coefficients. When $s \neq 0$, scaling by s^{-1} preserves distance to the linear code and produces a uniform affine combination, with the probability already computed. When $s = 0$, the f terms cancel and the mixture is supported on at most $2r$ coordinates. Its distance to the zero codeword is at most $2r/n \leq \theta$. Since s is uniform, the first formula follows. For the multilinear statement the coefficients sum to one. All are nonzero exactly when every x_j avoids zero and one. Apply the exact affine profile and independence of the ℓ parameters. $\square$

Corollary 4.18 (Prescribed exceptional distances on curves). *In the affine space of Corollary 4.15, fix $e \geq 1$, distinct parameters $z_1, \dots, z_t \in \mathbb{F}_p$, and integers $1 \leq a_i \leq r$ with $\sum_i a_i \leq er$. There is a polynomial curve of degree at most e whose exact distance is*

$$\theta + 2a_i/n \quad \text{at } z_i, \quad \theta \quad \text{at every other parameter.}$$

Conversely, every degree-at-most- e curve in this affine space that has any point at distance θ satisfies

$$\sum_{z \in \mathbb{F}_p} (\text{dist}(f_z, C) - \theta) \leq \frac{2er}{n} = e \left(\eta - \frac{1}{n} \right).$$

For lines, two equal endpoint separations cannot exceed half this budget. The two equal separations in Corollary 4.16 attain that bound. This converse is a property of this construction, not a bound for all Reed–Solomon curves.

Proof. Distribute a_i copies of each z_i consecutively around r bins, continuing cyclically between successive parameters. Since $a_i \leq r$, no bin receives the same parameter twice. Since $\sum_i a_i \leq er$, each bin receives at most e parameters. Let its direction coordinate be the product of $z - z_i$ over its assigned parameters, or one if empty. Exactly a_i coordinates vanish at z_i , and none vanish elsewhere. Equation (25) proves the claimed distances. For the converse, each direction coordinate is a polynomial of degree at most e . None is identically zero, since there is a point at distance θ . Each therefore vanishes at at most e field parameters. Sum the zero-coordinate counts and multiply by $2/n$. $\square$

Corollary 4.19 (Interpolation of far inputs). *Let $2 \leq t < p$ divide r , and choose distinct $x_1, \dots, x_t \in \mathbb{F}_p$. The t far words of Corollary 4.16 have a degree-at-most- $(t-1)$ interpolating curve $F(z)$ such that*

$$\text{dist}(F(x_i), C) = \theta + \frac{2r(t-1)}{nt} \quad (1 \leq i \leq t), \quad \text{dist}(F(z), C) = \theta \quad (z \notin \{x_1, \dots, x_t\}).$$

A uniform curve parameter is nearby with probability $1 - t/p$. For fixed t and growing r , all t specified inputs are almost $(1 - 1/t)\eta$ outside the radius. Their equal excess distances attain the degree- $(t-1)$ total budget above.

Proof. Use the Lagrange coefficients $\lambda_j(z) = \prod_{i \neq j} (z - x_i)/(x_j - x_i)$ and set $F(z) = \sum_j \lambda_j(z)v_j$. The coefficients sum to one, are all nonzero away from the interpolation nodes, and form a unit vector at each node. Apply the exact affine distance profile. The total excess distance is $2r(t-1)/n$, as asserted. $\square$

Corollary 4.20 (Witness variation and extension-field challenges). *For a completed line above, a polynomial codeword graph of parameter degree $e \geq 1$ supplies nearby witnesses at at most*

$$\left\lceil \frac{e(n - K - 1)}{2r} \right\rceil$$

parameters. Consequently a polynomial graph selecting a nearby codeword at every nonzero parameter has degree at least $\lceil 2r(p-1)/(n-K-1) \rceil = \Omega_p(p/\log p)$.

Let $E/\mathbb{F}_p$ be any field extension and extend the Reed–Solomon code’s coefficient field to E . In the same affine space, at the same radius, the nearby parameter vectors are exactly $(\mathbb{F}_p^*)^r \subseteq E^r$, and their decoding lists are unchanged. In particular a completed line has exactly the $p-1$ nonzero base-field nearby parameters over E . Every parameter in $E \setminus \mathbb{F}_p$ has the same distance as the exceptional point, namely $1 - \rho - 1/n$.

Proof. Apply Lemma A.8 with $a_0 = K+1$ and $T = K+2r+1$. If a graph supplies a nearby codeword at several parameters, choose those codewords as the selected witnesses in that lemma. The asymptotic degree bound follows from $n = \Theta_\rho(r \log p)$.

The distance of f over E is still $(n-K-1)/n$, by the same root bound. To reach the tested radius after changing at most $2r$ padding positions, every one must change and must match the nearby codeword. That codeword also agrees with f on at least $K+1$ core positions. Interpolation on any K of them forces its coefficients into $\mathbb{F}_p$, since both the evaluation points and the values of f lie in $\mathbb{F}_p$. The padding equalities then force all parameter coordinates into $\mathbb{F}_p^*$. Conversely every such vector already has a base-field nearby codeword, and the same interpolation argument excludes additional extension-field codewords from its list.

Finally fix a line parameter $z \notin \mathbb{F}_p$. There is an $\mathbb{F}_p$ -linear functional $\lambda : E \rightarrow \mathbb{F}_p$ with $\lambda(1) = 1$ and $\lambda(z) = 0$. Applying it coefficientwise to an extension-field codeword shows that every agreement with $f + zg$ is an agreement of a base-field codeword with f . There are at most $K+1$ such agreements. The zero codeword attains $K+1$ on the reference core roots (and the optional common zero coordinate), since g vanishes there. This proves the exact distance. $\square$

The polynomial-degree conclusion concerns algebraic witness selection; it is not a lower bound on decoding running time. The extension-field statement also limits the scope of the example: over a finite extension, the nearby fraction on the line is $(p-1)/|E|$, not $1 - 1/|E|$.

Exact half-rate completion certificates give the following instances. The direction weight is the number of coordinates changed along the line, and the last column bounds $n2^{1/\eta}/p$.

Prime	n	r	Direction weight	Failure <	Prescription <
$2^{61} - 1$	158	1	2	existence	2^{-1}
$2^{127} - 1$	294	1	2	existence	2^{-20}
$2^{521} - 1$	1090	1	2	existence	2^{-147}
$2^{521} - 1$	2142	2	4	existence	2^{-81}
$2^{521} - 1$	2414	2	4	2^{-64}	2^{-26}
$2^{1279} - 1$	7742	3	6	existence	2^{-160}
$2^{1279} - 1$	8014	3	6	2^{-64}	2^{-121}
$2^{1279} - 1$	25970	10	20	2^{-64}	2^{-27}

Rows marked existence use successive averaging; the others have the stated randomized success guarantees. The 61-bit example has distance $78/158$ at zero and exactly $76/158$ at every nonzero parameter. Its line changes only two coordinates. All rows lie strictly below Elias.

The checkers in `research/paired_domain_warp/` bound the recurrence with dyadic or finer fixed-point upper bounds and replay the resulting integer inequalities independently using weaker character and conditioning constants. The search finds sufficient parameters; no optimality is claimed. They separately verify primality, strict Elias, and the numerical comparison. Small-field fixtures verify the entire affine distance profile at all 1369 points of a two-dimensional space over

$\mathbb{F}_{37}$; these fixtures illustrate the geometry and are not claimed to lie below Elias. The stored length-2518 sample has a four-coordinate direction, generator failure below 2^{-88} , and a known nearest codeword, zero, at parameter one. Its coordinates and distances at zero and one are replayed deterministically. The full-profile probability is over the generator's randomness, rather than a deterministic certificate for the individual sample. The following cubic construction separately defeats arbitrary fixed numerical constants c_1, c_2 .

4.3.2 Unambiguous nearby points and an almost maximally far input

Complete nonzero coverage is not needed to refute the numerical bound. A different choice of paired domain makes every nearby point uniquely decodable, while retaining the almost maximally far exceptional point. The following argument uses only polynomial zero bounds.

Lemma 4.21 (A criterion for exact, unambiguous line lists). *Use one padding pair ± 1 and nonzero core representatives $a_1, \dots, a_m$ with distinct squares different from one, and the exact-rate parity construction above, with D selected core orbits. Suppose the only $\varepsilon \in \{-1, 0, 1\}^m$ satisfying*

$$\sum_i \varepsilon_i a_i = 0, \quad \prod_i \left(\frac{1 - a_i}{1 + a_i} \right)^{\varepsilon_i} = 1$$

is the zero vector. Set the direction on the padding to one in the even-locator case, and to x in the case multiplied by X . At agreement threshold $K + 3$, the nearby parameters are exactly

$$-\prod_{i \in I} (1 - a_i^2), \quad |I| = D,$$

and the list at each parameter consists exactly of the candidate codewords from that product's fiber. If these products are distinct, there are exactly $\binom{m}{D}$ nearby parameters and each has a unique nearby codeword. Parameter zero has exact distance $1 - K/n - 1/n$; every nearby parameter has exact distance $1 - K/n - 3/n$.

Proof. The usual locators supply the displayed parameter for every support. Conversely, a nearby codeword must match both padding coordinates and $K + 1$ nonpadding coordinates, because the far-point agreement bound is $K + 1$. Write F for the reference polynomial, including the optional factor X , and let Q be any nearby codeword polynomial. Then $H = F - Q$ is monic of degree $K + 1$, has zero coefficient of X^K , and is the locator of exactly $K + 1$ nonpadding roots. Thus those roots sum to zero.

Remove every sign-orbit whose two elements are roots. Each remaining nonzero root is $\varepsilon_i a_i$ for a nonzero entry of a ternary vector ε . The root-sum condition gives its first equation. The padding equalities give $H(1)/H(-1) = (-1)^{K+1}$. Factoring this ratio into root contributions gives its second equation; a possible zero root contributes one. The hypothesis therefore leaves no single roots. When $K + 1$ is even, zero cannot be a root, and when $K + 1$ is odd, zero must be a root. Consequently H is exactly one of the paired locators in the candidate family, including the optional factor X . This proves the exact list description; distinct products then give uniqueness and the stated count. The root bound at zero and the two-coordinate distance inequality prove the asserted distances. $\square$

Theorem 4.22 (Numerical failure with no local decoding ambiguity). *Fix a rational rate $\rho \in (0, 1)$, $c_1 > 0$, and $0 < c_2 < 3/2$. Over every sufficiently large prime field, there is an exact-rate code of length*

$$n = \frac{1}{2} \log_2 p + O_\rho(1)$$

and a received line with exactly

$$J = 2^{(H_2(\rho)/2 + o(1))n}$$

nearby parameters at radius $\theta = 1 - \rho - 3/n$. Every nearby point has a unique nearby codeword. Parameter zero has distance $1 - \rho - 1/n$, so its separation is $2/3$ of the capacity gap. The radius is strictly below Elias, there is no correlated agreement, and

$$\frac{J}{c_1 n 2^{c_2 H_2(\rho)/(3/n)}} \rightarrow \infty.$$

Uniform sampling of the core constructs the code and line in expected polynomial time in $\log p$, with failure probability at most $p^{-1/2+o(1)}$.

Proof. Choose n with the exact-rate parity above and take $m = (n - a)/2 - 1$, where $a \in \{0, 1\}$ records the optional zero coordinate. Thus $m = \frac{1}{4} \log_2 p + O_\rho(1)$ and $D = \rho m + O_\rho(1)$. First sample the a_i independently and uniformly from $\mathbb{F}_p$. The probability of a zero, a padding collision, or repeated squares is at most $(m^2 + 2m)/p$.

For two distinct supports, equality of their products is a nonzero polynomial equation of degree at most $2m$. A union bound over at most $\binom{m}{D}^2 \leq 4^m$ support pairs bounds the probability of any such collision by $2m4^m/p$.

Fix a nonzero ternary vector with support size u . For $u = 1, 2$, its sum equation is impossible on a valid domain. For $u \geq 3$, change signs of the selected variables and eliminate the last using their sum. The product-ratio equation becomes a nonzero polynomial of degree at most u in the remaining $u - 1$ variables. To see it is nonzero, set all but three original variables to zero and set the remaining three to $X, Y, -X - Y$: the difference of the two products is, up to sign, $2XY(X + Y)$. The probability of the two equations is therefore at most u/p^2 , by the polynomial zero bound. Summing over the ternary vectors gives at most $m3^m/p^2$.

The total bad probability is at most

$$\frac{m^2 + 2m + 2m4^m}{p} + \frac{m3^m}{p^2} = p^{-1/2+o(1)}.$$

Rejection sampling for a valid domain preserves this asymptotic bound. Lemma 4.21 applies on every remaining output. The binomial entropy estimate gives $\log_2 J = nH_2(\rho)/2 + O_\rho(\log n)$, and hence the displayed ratio diverges for $c_2 < 3/2$. Since $(3/n) \log_2 p \rightarrow 6$, the sufficient Elias condition holds with room to spare. The far point rules out correlated agreement. Forming and evaluating the reference locator costs $O(n^2)$ field operations; the sampler need not verify the two combinatorial conditions or recover a witness at a supplied parameter. $\square$

4.3.3 A deterministic family with unique and recoverable witnesses

The unambiguous construction can be made deterministic, with an efficient procedure deciding whether a parameter is nearby and recovering its unique witness. Its core representatives are simply powers of two.

Lemma 4.23 (Decoding products of $4^i - 1$). *For distinct subsets $I \subseteq \{1, \dots, m\}$, the positive integers $T_I = \prod_{i \in I} (4^i - 1)$ are distinct. Given T_I , its support can be recovered using a polynomial number of bit operations in m and $\log T_I$.*

Proof. Write $s = \sum_{i \in I} i$ and $U_I = \prod_{i \in I} (1 - 4^{-i})$, so $T_I = 4^s U_I$. The elementary product inequality gives $2/3 < U_I \leq 1$. For a nonempty support, T_I therefore has exactly $2s$ binary digits, determining s and hence the rational number U_I .

Process indices in increasing order, removing previously selected factors from U_I . At index i , the remaining product is at most $1 - 4^{-i}$ if i is selected. If i is not selected, it is at least

$$\prod_{j=i+1}^m (1 - 4^{-j}) \geq 1 - \sum_{j=i+1}^m 4^{-j} > 1 - \frac{1}{3}4^{-i} > 1 - 4^{-i}.$$

These comparisons determine every membership bit, proving injectivity. They can be implemented with integers: if the remaining integer product is t and the remaining index sum is s , compare $t4^i$ with $(4^i - 1)4^s$. On selecting i , divide t by $4^i - 1$ and subtract i from s . For an arbitrary input, require exact divisibility at every selection and verify the final product and support size. All intermediate integers have polynomial bit length. $\square$

Theorem 4.24 (Deterministic, efficiently decoded counterexamples). *Let $1 \leq D < m$, let p be prime with*

$$p > 2^{D(2m-D+1)},$$

and use core representatives $a_i = 2^i$ for $1 \leq i \leq m$, with padding ± 1 and the exact-rate parity convention above. At threshold $K + 3$, the resulting line has exactly $\binom{m}{D}$ nearby parameters, each with a unique nearby codeword. Parameter zero has distance $1 - K/n - 1/n$, and each nearby parameter has distance $1 - K/n - 3/n$. The code, line, and a procedure deciding membership in the nearby set and recovering the unique witness are deterministic and polynomial in $\log p$.

Consequently, for every fixed rational rate $\rho \in (0, 1)$ and every $c_1 > 0$, $0 < c_2 < 3/2$, all sufficiently large prime fields have such examples at exact rate ρ and length $n = \Theta_\rho(\sqrt{\log p})$, strictly below Elias, with no correlated agreement and

$$\frac{\binom{m}{D}}{c_1 n 2^{c_2 H_2(\rho)/(3/n)}} = 2^{\Omega_{\rho, c_2}(n)}.$$

Proof. Every positive D -subset product is less than

$$\prod_{i=m-D+1}^m 4^i = 2^{D(2m-D+1)} < p.$$

The displayed field bound also gives $p > 2^{m+1}$. No nonzero ternary sum of $2, 4, \dots, 2^m$ can vanish modulo p : its integer absolute value is less than p , and its largest power exceeds the sum of all smaller powers. Thus the signed-relation hypothesis of Lemma 4.21 holds. The nearby labels are

$$z_I = (-1)^{D+1} T_I \pmod{p}.$$

Lemma 4.23 and the no-wrap bound show that these labels are distinct. They also give a decoder: lift $(-1)^{D+1} z$ to $\{0, \dots, p-1\}$, recover and check its support, then output the reference locator minus that support's locator, with the optional factor X . The exact list classification guarantees that rejection means the parameter is outside the tested radius. The degree and distance assertions follow from the same criterion.

For exact rate ρ , choose the admissible n near a sufficiently small constant multiple of $\sqrt{\log_2 p}$. Then $m = n/2 + O(1)$, $D = \rho n/2 + O_\rho(1)$, and the displayed field bound holds. As before, $\log_2 \binom{m}{D} = nH_2(\rho)/2 + O_\rho(\log n)$, which proves the numerical separation for $c_2 < 3/2$. Moreover $(3/n) \log_2 p \rightarrow \infty$, proving strict Elias. The far point rules out correlated agreement. Forming the locators, evaluating them, and performing the integer comparisons use polynomial time in $\log p$. $\square$

For example, exact arithmetic gives the following deterministic rows. Every nearby parameter has a unique witness recoverable by the preceding integer decoder; the last column lower-bounds the ratio to the $c_1 = c_2 = 1$ numerical prescription.

Prime	n	K	D	Nearby count	Ratio $>$
$2^{521} - 1$	74	15	8	$\binom{36}{8}$	1
$2^{1279} - 1$	82	41	21	$\binom{40}{21}$	2^3
$2^{2203} - 1$	106	53	27	$\binom{52}{27}$	2^6
$2^{9689} - 1$	226	113	57	$\binom{112}{57}$	2^{25}

These are proofs for the specified domains, not guarantees over random generation. The checkers in `research/paired_domain_warp/powers_two/` verify primality, the no-wrap bound, strict Elias, and the numerical comparisons with integers, and test support recovery exhaustively on small instances. Uniqueness here concerns every nearby point on this line; it does not assert a global list-size bound for the code.

Proposition 4.25 (An exact two-level distance profile). *In the unadjusted case $n = 2m + 2$, $K = 2D - 1$, assume additionally*

$$p > 2^{m(m+1)/2+2m+2}.$$

Every parameter outside the nearby bank has distance exactly $(n - K - 1)/n$. Thus the line has only two distance levels, and the integer decoder determines which level holds. Both f and $f + g$ are at the far level. The three half-rate rows in the preceding table satisfy this extra bound.

Proof. The zero codeword supplies $K + 1$ reference-core agreements at every parameter, so distance is always at most $(n - K - 1)/n$. Suppose a codeword has at least $K + 2$ agreements. At least one padding position must agree, by the root bound. If exactly one padding position agrees, its $K + 1$ core roots have sum zero; integer dissociation forces them into pairs. Its locator is even, so the other padding position agrees too, placing the parameter in the nearby bank.

It remains to consider two padding agreements and only K known core roots. The difference polynomial is monic of degree $K + 1$, with zero coefficient of X^K . Its last root is therefore minus the sum of those K roots. Cancel complete sign-pairs among the known roots and call the remaining signed powers of two $y_1, \dots, y_u$. Their number is odd. If $u = 1$, the last root completes its pair, again giving a bank locator. For $u \geq 3$, put $S = \sum_i y_i$. Equality of the two padding values would require

$$F = (1 + S) \prod_i (1 - y_i) - (1 - S) \prod_i (1 + y_i) = 0 \pmod{p}.$$

This integer is nonzero. Indeed, writing e_j for the elementary symmetric functions of the y_i gives

$$F/2 = (Se_2 - e_3) + (Se_4 - e_5) + \dots.$$

If the two smallest-magnitude signed powers are $u = \pm 2^a$ and $v = \pm 2^b$, with $a < b$, the monomial with uniquely smallest 2-adic valuation is u^2v in the first parenthesis, with valuation $2a + b$. All other degree-three terms and all higher-degree terms have larger valuation. Thus cancellation is impossible. Finally,

$$|F| \leq 2(1 + |S|) \prod_i (1 + |y_i|) < 2^{m(m+1)/2+2m+2} < p,$$

so F cannot vanish modulo p either. Consequently every improvement over the far distance is already a nearby-bank point. Finally, neither zero nor one is a bank label: each positive product lies strictly between one and $p - 1$. Thus both displayed endpoints are at the far level. $\square$

4.3.4 Unique witnesses with almost the entire gap as separation

Larger prime-order orbits strengthen the separation while preserving uniqueness and efficient witness recovery. The evaluation domain is a union of scaled small orbits, with a bounded number of extra coordinates. It is not a prescribed multiplicative subgroup.

Lemma 4.26 (Rigidity of root sums on scaled prime-order orbits). *Let d be prime, $p \equiv 1 \pmod{d}$, and let $\omega \in \mathbb{F}_p$ have order d . Fix $m \geq 2$ and $0 \leq c < d$, and put*

$$A = 2^{m+1}(d + 2^c).$$

If $p > A^{d-1}$, the following points are distinct:

$$\{2^i \omega^j : 1 \leq i \leq m, 0 \leq j < d\} \cup \{2^{m+t} : 1 \leq t \leq c\} \cup \{\omega^j : 0 \leq j < d\}.$$

Every zero-sum subset of the first two sets is a union of complete d -element orbits from the first set; it contains no extra coordinate.

Proof. Let ζ be a complex primitive d th root of unity. The polynomial $\Phi_d(X) = 1 + X + \dots + X^{d-1}$ is irreducible over $\mathbb{Q}$ by Eisenstein after translating X by one. For an integer polynomial P of degree at most $d - 1$, if $P(\omega) = 0$ but $P(\zeta) \neq 0$, its nonzero integer resultant with Φ_d is divisible by p . On the other hand, if the sum of the absolute values of its coefficients is at most A ,

$$|\text{Res}(\Phi_d, P)| = \prod_{j=1}^{d-1} |P(\zeta^j)| \leq A^{d-1} < p,$$

a contradiction.

A difference between two distinct displayed formal points is nonzero over $\mathbb{C}$ and has coefficient sum less than A , so their reductions cannot collide. Now lift a zero-sum subset of nonpadding points to $P(\zeta)$. Its coefficient sum is less than A , hence $P(\zeta) = 0$. Its degree is at most $d - 1$, so all its coefficients must be equal. Each coefficient records a sum of distinct powers of two. Uniqueness of binary expansion forces the same selected indices in every orbit position. Thus each core orbit is either fully selected or absent, and an extra coordinate, which occurs only in position zero, is absent. $\square$

Theorem 4.27 (Explicit orbit lines with unique witnesses). *With the preceding notation, take $1 \leq D < m$ and assume also*

$$p > 2^{dD(2m-D+1)/2}.$$

There is a length- $n = d(m + 1) + c$, dimension- $K = dD - 1$ Reed–Solomon code and a received line with exactly $\binom{m}{D}$ nearby parameters at radius

$$\theta = 1 - \frac{K}{n} - \frac{d+1}{n}.$$

Every nearby point has a unique codeword, recoverable in polynomial time in $\log p$. The exceptional point has exact distance $1 - K/n - 1/n$; every nearby point has exact distance θ . The direction has weight d , attaining the distance-improvement bound. There is no correlated agreement at the tested threshold. Given ω , construction and decoding are deterministic. An order- d root can be found by a zero-error randomized procedure in expected polynomial time in $\log p$.

Proof. Use the three displayed sets as the core, extra coordinates, and padding, respectively. For D -subsets $I \subseteq \{1, \dots, m\}$ put

$$H_I(X) = \prod_{i \in I} (X^d - 2^{di}).$$

Fix a reference $w = H_{I_0}$, let f be its evaluation word, and let g equal one on the padding and zero elsewhere. The polynomial $w - H_I$ has degree at most $dD - d < K$ and supplies the nearby parameter

$$z_I = -H_I(1) = (-1)^{D+1} \prod_{i \in I} (2^{di} - 1).$$

It matches $dD = K + 1$ core coordinates and all d padding coordinates. The root bound gives exact far agreement $K + 1$ for f ; changing only d coordinates proves the exact nearby distance.

Conversely, any nearby codeword must match all padding coordinates and $K + 1 = dD$ nonpadding coordinates. Its difference from w is monic of degree dD , has zero coefficient of X^{dD-1} , and is the locator of those dD distinct roots. Their sum is zero, so Lemma 4.26 forces them to be exactly D complete core orbits. Thus every nearby codeword belongs to the displayed family.

The positive integer products are less than $2^{dD(2m-D+1)/2} < p$. The proof and greedy decoder of Lemma 4.23 apply with base 2^d in place of four. The normalized product is still greater than $2/3$, and the product of all later factors exceeds the membership threshold. This proves injectivity, the exact count, and unique witness recovery from the signed integer lift of z . Finally, sampling $h \in \mathbb{F}_p^*$ and testing $h^{(p-1)/d} \neq 1$ finds ω in expected $d/(d-1)$ trials. Every accepted root has order d , so this step has no correctness failure event. $\square$

Corollary 4.28 (Unambiguous numerical failure with nearly full separation). *Fix a rational rate $\rho = a/b \in (0, 1)$ in lowest terms and a prime $d \nmid a$. For every sufficiently large prime $p \equiv 1 \pmod{d}$, the construction has exact rate ρ , length $n = \Theta_{\rho,d}(\sqrt{\log p})$, and gap $\eta = (d+1)/n$, with*

$$\text{dist}(f, C) = \theta + \frac{d}{d+1} \eta.$$

Every nearby point has a unique, efficiently recoverable witness. The radius is strictly below Elias and, for every fixed $c_1 > 0$ and $0 < c_2 < 1 + 1/d$,

$$\frac{\binom{m}{D}}{c_1 n 2^{c_2 H_2(\rho)/\eta}} = 2^{\Omega_{\rho,d,c_2}(n)}.$$

In particular, for every fixed $\varepsilon > 0$, infinitely many prime fields admit such counterexamples to the $c_2 = 1$ prescription, with separation at least $(1 - \varepsilon)\eta$ and no decoding ambiguity anywhere on the nearby set.

Proof. Take $n = bt$ with $at \equiv -1 \pmod{d}$, let c be the residue of n in $\{0, \dots, d-1\}$, and put

$$m = (n - c)/d - 1, \quad D = (\rho n + 1)/d.$$

Choosing n near a sufficiently small constant multiple of $\sqrt{\log_2 p}$ ensures both field bounds, with $1 \leq D < m$ eventually. Then

$$\log_2 \binom{m}{D} = \frac{n}{d} H_2(\rho) + O_{\rho,d}(\log n).$$

Comparing this with $c_2 n H_2(\rho)/(d+1)$ proves the ratio. Since $\eta \log_2 p \rightarrow \infty$, strict Elias holds. Choosing a sufficiently large prime d not dividing a gives the final separation claim. There are infinitely many primes congruent to one modulo this d . The gap still tends to zero; no fixed-positive-gap conclusion is asserted. $\square$

Exact half-rate certificates illustrate the tradeoff:

Prime	d	n	K	Separation	Ratio $>$
$2^{9689} - 1$	3	274	137	$3\eta/4$	2^9
$2^{19937} - 1$	5	518	259	$5\eta/6$	2^2
$2^{44497} - 1$	7	908	454	$7\eta/8$	1

Here the ratio compares the exact nearby count $\binom{m}{D}$ with $n2^{1/\eta}$. The respective (m, D, c) are $(90, 46, 1)$, $(102, 52, 3)$, and $(128, 65, 5)$. The accompanying integer certificates verify the field bounds, primality, strict Elias, and these comparisons. Unlike the complete-coverage examples, these lines have a sparse nearby bank. Their witness classification and decoder have no sampling failure.

Smaller finite fields using roots of two. When the field also contains a d th root of two, the same decoder can use the factors $2^i - 1$ instead of $2^{di} - 1$. This reduces the integer product bound substantially. The following statement requires both specified roots in the field; it does not replace the prime-progression hypothesis of Corollary 4.28.

Proposition 4.29 (A root-of-two refinement). *Let d be prime and let $\omega, \alpha \in \mathbb{F}_p$ satisfy $\text{ord}(\omega) = d$ and $\alpha^d = 2$. Take $m \geq 2$, $1 \leq D < m$, $0 \leq c < d$, and put*

$$n = d(m+1) + c, \quad K = dD - 1, \quad A = n2^{\lceil (m+c+1)/d \rceil}.$$

Assume

$$p > A^{d(d-1)}, \quad p > 2^{D(2m-D+3)/2}.$$

Use the core $\{\alpha^i \omega^j : 2 \leq i \leq m+1, 0 \leq j < d\}$, extra coordinates $\alpha^{m+2}, \dots, \alpha^{m+c+1}$, and padding $\{\omega^j : 0 \leq j < d\}$. These points are distinct. The corresponding locator line has exactly $\binom{m}{D}$ nearby parameters at radius $1 - K/n - (d+1)/n$; each has a unique codeword, recoverable in polynomial time in $\log p$. Parameter zero has exact distance $1 - K/n - 1/n$. The construction and decoder are deterministic given the two roots.

Proof. Let $u = 2^{1/d} > 0$ and let ζ be a complex primitive d th root. The field $L = \mathbb{Q}(u, \zeta)$ has degree $d(d-1)$: the degrees d and $d-1$ of its two subfields are relatively prime, so their product both divides and bounds $[L : \mathbb{Q}]$. The ring $\mathbb{Z}[u, \zeta]$ is free on the basis $u^s \zeta^j$ for $0 \leq s < d$, $0 \leq j < d-1$. The given finite-field roots define a ring homomorphism to $\mathbb{F}_p$. Consequently a nonzero element mapping to zero has nonzero integer norm divisible by p , as follows by reducing its multiplication matrix modulo p .

For a lifted subset sum or a difference of two displayed formal coordinates, every conjugate has absolute value at most A . Thus its norm has absolute value at most $A^{d(d-1)} < p$. A modular zero must therefore already be zero in L . Distinct formal coordinates have different positive magnitudes unless their indices agree, in which case their root-of-unity factors are distinct. This proves distinctness after reduction.

For a subset sum of nonpadding coordinates, group exponents of u according to their residue modulo d , using $u^{qd+s} = 2^q u^s$. The powers $1, u, \dots, u^{d-1}$ are linearly independent over $\mathbb{Q}(\zeta)$, so each resulting polynomial in ζ vanishes. Its degree is at most $d-1$, forcing all its coefficients to agree. Within each residue class, each coefficient is a sum of distinct powers of two. Binary uniqueness again forces whole core orbits and excludes all extra coordinates.

The all-codeword classification from Theorem 4.27 now applies, with locators

$$H_I(X) = \prod_{i \in I} (X^d - 2^i), \quad I \subseteq \{2, \dots, m+1\}, \quad |I| = D.$$

Their signed labels lift to positive integers $T_I = \prod_{i \in I} (2^i - 1) < 2^{D(2m-D+3)/2} < p$. Writing $s = \sum_{i \in I} i$, the normalized product satisfies

$$\frac{1}{2} < \prod_{i \in I} (1 - 2^{-i}) < 1,$$

so T_I has exactly s binary digits. At index i , membership is equivalent to the remaining normalized product being at most $1 - 2^{-i}$; the product of all later factors is strictly greater than that threshold. The greedy decoder therefore works in base two. Starting the indices at two is essential to this normalization argument. The remaining degree, count, and distance conclusions are unchanged. $\square$

For a Mersenne prime $p = 2^b - 1$ with $d \neq b$, one can take $\alpha = 2^v$, where $dv \equiv 1 \pmod{b}$. The check $\alpha^d = 2$ is immediate. An order- d root is still required, so $d \mid p - 1$ remains a separate condition.

Exact certificates give the following specific domains. Rates are K/n and need not be one half. Each comparison uses that row's exact binary entropy, evaluated through integer powers rather than floating-point logs.

Prime	d	n	K	Separation	Ratio $>$
$2^{521} - 1$	2	80	29	$2\eta/3$	2^3
$2^{1279} - 1$	3	198	68	$3\eta/4$	2^4
$2^{9689} - 1$	5	905	324	$5\eta/6$	2^{14}
$2^{23209} - 1$	13	3588	1338	$13\eta/14$	2^2
$2^{44497} - 1$	19	7068	2830	$19\eta/20$	1

The ratio is $\binom{m}{D} / (n 2^{H_2(K/n)/\eta})$, with $\eta = (d+1)/n$. The respective (m, D, c) are $(39, 15, 0)$, $(65, 23, 0)$, $(180, 65, 0)$, $(275, 103, 0)$, and $(371, 149, 0)$. All five primes have independent Lucas–Lehmer replays; saved roots, domain distinctness, the two field bounds, strict Elias, and representative witness evaluations are checked independently. These are deterministic specific-domain certificates once the certified roots are supplied. The final row has a uniquely recoverable nearby witness at every accepted parameter, even though the far point is 19/20 of the capacity gap outside the radius and the numerical prescription is too small.

Shorter domains by randomizing the orbit representatives. The preceding decoder uses the integer size of the binary-scaled coordinates. If efficient recovery at an arbitrary supplied parameter is not required, random representatives give shorter domains and a larger nearby bank relative to the field.

Theorem 4.30 (Unique orbit lines of logarithmic length). *Fix a rational rate $\rho = a/b \in (0, 1)$ in lowest terms, a prime $d \nmid a$, and $0 < \alpha < 1$. For every sufficiently large prime $p \equiv 1 \pmod{d}$, there are exact-rate Reed–Solomon codes of length*

$$n = \alpha \log_2 p + O_{\rho,d}(1)$$

and received lines with exactly

$$J = 2^{(H_2(\rho)/d + o(1))n}$$

nearby parameters at radius $\theta = 1 - \rho - (d+1)/n$. Every nearby point has a unique codeword and exact distance θ ; parameter zero has exact distance $1 - \rho - 1/n$. Consequently its separation is

$d/(d+1)$ of the capacity gap. The radius is strictly below Elias, correlated agreement fails, and for every fixed $c_1 > 0$ and $0 < c_2 < 1 + 1/d$,

$$\frac{J}{c_1 n 2^{c_2 H_2(\rho)/((d+1)/n)}} = p^{\Omega_{\rho,d,\alpha,c_2}(1)}.$$

A randomized procedure constructs the code and line in expected polynomial time in $\log p$, with failure probability at most $p^{\alpha-1+o(1)}$. This theorem does not assert efficient witness recovery from an arbitrary nearby parameter.

Proof. Choose n, c, m, D by the exact-rate congruences in Corollary 4.28, now with n near $\alpha \log_2 p$. Fix an order- d root ω . Sample independent uniform $a_1, \dots, a_m, b_1, \dots, b_c \in \mathbb{F}_p$ and use the domain

$$\{a_i \omega^j : 1 \leq i \leq m, 0 \leq j < d\} \cup \{b_1, \dots, b_c\} \cup \{\omega^j : 0 \leq j < d\}.$$

The probability of a collision among its n formal points is at most $\binom{n}{2}/p$.

For $p > d^{d-1}$, no nonempty proper subset of $\{1, \omega, \dots, \omega^{d-1}\}$ sums to zero. Indeed, its lift to $\mathbb{Z}[\zeta_d]$ is nonzero by irreducibility of Φ_d , and its nonzero norm has absolute value at most $d^{d-1} < p$. Thus every subset of the nonpadding domain other than a union of complete core orbits has a sum that is a nonzero linear form in the sampled variables. The probability that any such subset has sum zero is at most

$$\frac{2^{dm+c}}{p} = \frac{2^{n-d}}{p}.$$

For two distinct D -subsets I, J , the polynomial

$$\prod_{i \in I} (1 - a_i^d) - \prod_{j \in J} (1 - a_j^d)$$

is nonzero and has degree at most dD . The probability of any product collision is therefore at most

$$\frac{dD}{p} \binom{\binom{m}{D}}{2}.$$

On every output avoiding these three bad events, the root-locator classification in Theorem 4.27 applies verbatim: every nearby codeword comes from exactly D whole orbits, and the distinct products make all these codewords occur at distinct parameters.

Before conditioning on a distinct domain, the total bad probability is at most

$$B = \frac{\binom{n}{2} + 2^{n-d} + dD \binom{\binom{m}{D}}{2}}{p}. \quad (26)$$

This is $p^{\alpha-1+o(1)}$, since $\binom{m}{D}^2 \leq 2^{2m}$ and $2m \leq n - d$ for $d \geq 2$. Rejecting invalid domains changes the bound by at most the factor $(1 - \binom{n}{2}/p)^{-1} = 1 + o(1)$. The same entropy calculation proves the numerical ratio, and $((d+1)/n) \log_2 p \rightarrow (d+1)/\alpha$ gives strict Elias with room to spare. The sampler checks domain validity and forms the reference polynomial; it does not enumerate subsets or test the two combinatorial good events. $\square$

For example, the exact union bound gives the following half-rate rows. The ratio compares J with $n2^{1/\eta}$; the probability column bounds failure after rejection sampling for distinct coordinates.

Prime	d	n	K	Separation	Ratio $>$	Failure $<$
$2^{521} - 1$	3	466	233	$3\eta/4$	2^{24}	2^{-57}
$2^{521} - 1$	5	468	234	$5\eta/6$	2	2^{-57}
$2^{1279} - 1$	7	1146	573	$7\eta/8$	2^4	2^{-139}

These are guarantees for the sampling procedure. The arithmetic certificates do not deterministically certify the combinatorial good events for a particular sampled domain.

Two padding orbits and a stronger exponent constant. Allowing much larger splitting primes gives a different tradeoff. One can retain unique, efficiently recoverable nearby witnesses and almost full-gap separation even when the numerical prescription's exponent constant is doubled. This construction does not retain logarithmic length in the field size.

Lemma 4.31 (Rigidity for nearby radical radii). *Let d be prime, let q be a power of two with $q - 1 > 3d^{d-1}$, and let $M \geq 1$. Choose an integer $L \geq 1$ with $q^L > 4(dM)^{d-2}$. Write*

$$t = q^{1/d} > 0, \quad R_i = \frac{q^{i+1} - 1}{q^i - 1}, \quad a_i = R_i^{1/d} > 0 \quad (L \leq i < L + M),$$

and let ζ be a complex primitive d th root of unity. Every zero-sum subset of the orbits $a_i\{1, \zeta, \dots, \zeta^{d-1}\}$ is a union of whole orbits. The same holds if some orbits are replaced by single points: no such single point can occur in a zero-sum subset.

Proof. Write $a_i = t + \delta_i$. Since $R_i = q + (q - 1)/(q^i - 1)$, the difference-of-powers identity gives

$$\frac{t}{3d}q^{-i} < \delta_i \leq \frac{t}{d}q^{-i}.$$

Indeed its denominator lies between dt^{d-1} and $d(q + 1)^{(d-1)/d}$; use $q^i - 1 \geq (1 - 1/q)q^i$ for the upper bound and $3(q - 1) \geq q + 1$ for the lower bound.

A selected root sum is $\sum_i a_i C_i$, where C_i is a subset sum of the d th roots of unity. Always $|C_i| \leq d$; if $C_i \neq 0$, its nonzero cyclotomic integer norm gives $|C_i| \geq d^{-(d-2)}$. Similarly, a nonzero $\sum_i C_i$ has absolute value at least $(dM)^{-(d-2)}$, since every other conjugate has absolute value at most dM . In that case

$$\left| \sum_i \delta_i C_i \right| \leq t \sum_{i \geq L} q^{-i} < 2tq^{-L} < \frac{1}{2}t(dM)^{-(d-2)},$$

so it cannot cancel $t \sum_i C_i$. If $\sum_i C_i = 0$, take the least i with $C_i \neq 0$. Its perturbation has absolute value greater than $tq^{-i}/(3d^{d-1})$, while the entire later tail has absolute value at most $tq^{-i}/(q - 1)$, strictly smaller by hypothesis. Thus all C_i must vanish. For prime d , irreducibility of Φ_d then forces each selected subset to be empty or a whole orbit. A singleton cannot qualify. $\square$

Theorem 4.32 (Unique witnesses against a doubled exponent). *Fix a rational rate $\rho = a/b \in (0, 1)$ in lowest terms and a prime $d \nmid a$. For arbitrarily large exact-rate lengths n , and over infinitely many sufficiently large prime fields for each such length, there are Reed–Solomon codes and received lines with gap*

$$\eta = \frac{2d + 1}{n}$$

and exactly $J = 2^{(H_2(\rho)/d + o(1))n}$ nearby parameters. Every nearby point has a unique codeword at exact distance $\theta = 1 - \rho - \eta$, while parameter zero has exact distance

$$1 - \rho - 1/n = \theta + \frac{2d}{2d+1}\eta.$$

The radius is strictly below Elias and correlated agreement fails. For every fixed $c_1 > 0$ and $0 < c_2 < 2 + 1/d$,

$$\frac{J}{c_1 n 2^{c_2 H_2(\rho)/\eta}} = 2^{\Omega_{\rho,d,c_2}(n)}.$$

Given the specified finite-field roots, the code, line, and a decoder for every nearby parameter can be constructed in polynomial time in $n + \log p$. No comparable bound on finding a suitable splitting prime is asserted. In particular, $c_2 = 2$ is compatible with separation at least $(1 - \varepsilon)\eta$ for any fixed $\varepsilon > 0$.

Proof. Choose $n = bu$ with $au \equiv -1 \pmod{d}$, let c be its residue in $\{0, \dots, d-1\}$, and set

$$m = (n - c)/d - 2, \quad D = (\rho n + 1)/d, \quad M = m + c.$$

For large n we have $1 \leq D < m$. Choose q, L as in Lemma 4.31. Use the first m radical orbits as core, the next c radical points as extra coordinates, and the two padding orbits μ_d and $t\mu_d$. These are distinct in characteristic zero, as their magnitudes are distinct except within each orbit.

Reduce without new root relations. Let E be the Galois field generated by ζ, t and all a_i . Its degree is at most $N = (d-1)d^{M+1}$. Put

$$W = \prod_{i=L}^{L+M-1} (q^i - 1), \quad B = (Wn(q+1))^N.$$

All displayed coordinates become algebraic integers after multiplication by W . A nonzero difference of two coordinates or a nonzero subset sum of nonpadding coordinates has, after this multiplication, nonzero integer norm of absolute value at most B : each conjugate has absolute value at most $Wn(q+1)$. Choose a completely split prime $p > B$. The finite-field reductions are therefore distinct and retain exactly the root-sum rigidity just proved. Arbitrarily large such primes exist by the splitting-prime fact used in Lemma 4.2; only infinitude is needed. We may also require

$$p > q^{S_{\max}}, \quad S_{\max} = D(2L + 2m - D - 1)/2.$$

Keep one index-sum class. Among the D -subsets of $\{L, \dots, L + m - 1\}$, choose a sum S of maximum multiplicity. Its class has size

$$J \geq \frac{\binom{m}{D}}{D(m-D) + 1}.$$

For every support I put $H_I(X) = \prod_{i \in I} (X^d - R_i)$. Fix any reference support of size D , set $w = H_{I_0}$, and let f be its evaluation word. Set the line direction to zero on the nonpadding coordinates, to q^S on μ_d , and to one on $t\mu_d$. The identities

$$\frac{H_I(1)}{H_I(t)} = q^{\sum_{i \in I} i}, \quad -H_I(t) = \frac{(-1)^{D+1}(q-1)^D}{\prod_{i \in I} (q^i - 1)}$$

show that $w - H_I$ is a nearby codeword whenever $\sum I = S$. Its degree is at most $dD - d < K = dD - 1$, and it matches $K + 1$ core coordinates and all $2d$ padding coordinates. The root bound and the support of the direction give the exact distances.

Conversely, any nearby codeword must match all padding and $K + 1$ nonpadding coordinates. Its monic difference from w has degree dD and zero penultimate coefficient. Root-sum rigidity forces its roots to be D whole core orbits. The two padding values then imply $q^{\sum I} = q^S$ in $\mathbb{F}_p$. Since both integer powers are less than p , the index sums are equal. This classifies every nearby codeword.

Recover the unique witness. For a nonzero parameter z , lift $(-1)^{D+1}(q-1)^D/z$ to $\{0, \dots, p-1\}$. Every genuine nearby label yields $T_I = \prod_{i \in I} (q^i - 1) < q^{S_{\max}} < p$. The base- q greedy decoder from Lemma 4.23 recovers the support uniquely. Check its cardinality and index sum; rejection proves the parameter is outside the tested radius. Dynamic programming finds a largest index-sum class in polynomial time in m ; the integer products, finite-field locators, and decoder use polynomial time in $n + \log p$.

Finally $m = n/d + O_d(1)$ and $D = \rho n/d + O(1)$, so both the lower bound on J and the trivial upper bound $J \leq \binom{m}{D}$ give $\log_2 J = nH_2(\rho)/d + O_{\rho,d}(\log n)$. Comparison with $c_2 nH_2(\rho)/(2d+1)$ proves the numerical ratio. The splitting primes can be chosen arbitrarily large, ensuring strict Elias. Choosing a large prime d not dividing a proves the final separation statement. The gap tends to zero, and the nearby set is sparse. $\square$

A logarithmic-length version over prescribed prime progressions. The large splitting-prime requirement can be removed if efficient recovery from an arbitrary nearby parameter is not required. Regard q as a finite-field parameter, rather than a fixed small integer. Character sums supply enough parameters for which all required roots exist; polynomial zero bounds exclude unintended root sums and label collisions.

Lemma 4.33 (A finite two-orbit existence certificate). *Let d be prime, $1 \leq D < m$, $0 \leq c < d$, and put*

$$M = m + c, \quad n = d(m + 2) + c, \quad K = dD - 1, \quad U = (M + 1)^2, \quad T = M(M + 1)/2,$$

$$A_0 = \binom{m}{D}, \quad R = D(m - D), \quad S_{\max} = D(2m - D + 1)/2.$$

Let $p \equiv 1 \pmod{d}$ be prime with $p > \max\{n, d^{d-1}\}$. Define

$$Q_0 = \frac{p - U}{d^{M+1}} - U(\sqrt{p} + 1),$$

$$B_0 = (T + 1)((2^d - 1)^M - 1) + S_{\max} \binom{A_0}{2} + \frac{R(R + 1)}{2}.$$

If $Q_0 > B_0$, the two-orbit construction has a valid parameter $q \in \mathbb{F}_p$ for which every nearby point has a unique codeword and

$$J \geq \frac{\binom{m}{D}}{D(m - D) + 1}$$

nearby parameters at radius $1 - K/n - (2d + 1)/n$. The far and nearby distances are exact as in Theorem 4.32. This is an existence certificate; it does not specify a successful parameter q or an arbitrary-parameter witness decoder.

Proof. Let $R_i(Q) = (Q^{i+1} - 1)/(Q^i - 1)$ for $1 \leq i \leq M$. Exclude $Q = 0$ and all zeros of their numerators and denominators. There are at most U excluded elements. On the remaining set, the order of q exceeds $M + 1$, the $R_i(q)$ are distinct and nonzero, and none is 1 or q . Thus the two padding orbits and all core/extra orbits will be disjoint whenever the roots exist.

Count parameters with all roots. Fix a character χ of order d . The indicator that $q, R_1(q), \dots, R_M(q)$ are all d th powers is the product of their d -character averages. Every nonprincipal term is a nontrivial character sum. To check this, take the largest i with a nonzero exponent of R_i . At a primitive $(i+1)$ st root of unity, its valuation is that exponent, nonzero modulo d , while every earlier factor is regular and nonzero. If no such i exists, the nonzero exponent of Q has the same property at zero. Here $p > M+1$ ensures separability.

Replace denominator character powers by positive powers modulo d . The resulting polynomial is not a constant times a d th power and has at most U distinct roots in the algebraic closure. The classical multiplicative-character Weil bound [33, Lemma 2], followed by deletion of at most U arguments, bounds the absolute value of each nonprincipal term by $U(\sqrt{p}+1)$. Character averaging therefore gives at least Q_0 parameters with all required roots.

Exclude unintended zero-sum subsets. Fix an order- d root $\omega \in \mathbb{F}_p$. In the formal series field at infinity choose branches $A_i(Q)^d = R_i(Q)$ with

$$\frac{A_i(Q)}{Q^{1/d}} = 1 + \frac{1}{d}Q^{-i} + O(Q^{-i-1}).$$

For constants C_i , not all zero, the function $\sum_i C_i A_i(Q)$ is nonzero. If $\sum_i C_i \neq 0$ its leading term proves this; otherwise the least index with $C_i \neq 0$ contributes the first nonzero term $C_i Q^{1/d-i}/d$. A proper nonempty subset sum of μ_d is nonzero in $\mathbb{F}_p$: the cyclotomic norm argument bounds its possible nonzero integer norm by $d^{d-1} < p$.

For a support of u nonzero coefficients C_i , take the product of $\sum C_i \epsilon_i A_i(Q)$ over all $\epsilon_i \in \mu_d$. To see explicitly that the product is rational, first form it in independent variables Y_i . Invariance under each substitution $Y_i \mapsto \omega Y_i$ forces every variable exponent in every monomial to be divisible by d . Substitute $Y_i^d = R_i(Q)$. Multiplication by $W_U(Q)^{d^u}$, where $W_U(Q) = \prod_{C_i \neq 0} (Q^i - 1)$, clears all denominators. The resulting polynomial is nonzero because every factor is nonzero in the formal series field. Its degree is at most $d^u(T+1)$: $\deg W_U \leq T$, and each radical has pole order $1/d$ at infinity. Every specialization at which any of these root sums vanishes is a zero of this polynomial.

There are $(2^d - 2)/d$ rotation classes of proper nonempty subsets of μ_d ; primality of d makes every such orbit have size d . Empty and full subsets both contribute coefficient zero. Counting coefficient patterns up to these independent rotations gives the bound

$$(T+1) \sum_{u=1}^M \binom{M}{u} \left(\frac{2^d - 2}{d} \right)^u d^u = (T+1)((2^d - 1)^M - 1)$$

on bad parameters. This also controls any chosen singleton extra coordinates, since their nonzero subset coefficients are included. Outside this set, every nonpadding zero-sum subset is a union of whole core orbits and contains no extra coordinate.

Exclude collisions of the labels and index sums. For distinct D -subsets I, J of $\{1, \dots, m\}$,

$$\prod_{i \in I} (Q^i - 1) - \prod_{j \in J} (Q^j - 1)$$

is a nonzero polynomial of degree at most $S_{\max}$ over any field. If the index sums differ, the leading degrees prove this. If they agree, cancel common factors; in the expansions at infinity, the smallest index in the symmetric difference has coefficient 1 or -1 and cannot cancel. A union bound gives at most $S_{\max} \binom{A_0}{2}$ bad parameters. Finally, exclude roots of $Q^h - 1$ for $1 \leq h \leq R$, costing at most $R(R+1)/2$ parameters. Equal D -support index sums are then the only way their powers of q can agree.

Since $Q_0 > B_0$, at least one root-admitting parameter avoids all bad sets. Choose a largest D -support index-sum class and use the two padding directions q^S and one. The all-codeword locator argument of Theorem 4.32 now applies. Distinct product labels give unique nearby codewords. The integer lift decoder is not available here, since the products may wrap modulo p . $\square$

Corollary 4.34 (Doubling survives at logarithmic length). *Fix rational $\rho = a/b \in (0, 1)$ and a prime $d \nmid a$. Put*

$$C = \max\{\log_2(2^d - 1), 2H_2(\rho)\}, \quad \alpha_* = \frac{d}{\log_2 d + C}.$$

For any fixed $0 < \alpha < \alpha_$ and every sufficiently large prime $p \equiv 1 \pmod{d}$, the conclusions of Theorem 4.32 concerning distances, uniqueness, and numerical violation hold at length*

$$n = \alpha \log_2 p + O_{\rho, d}(1),$$

with no witness-recovery guarantee. In particular they beat every fixed $c_1 > 0$ and $0 < c_2 < 2 + 1/d$ by a factor $p^{\Omega(1)}$. Uniform sampling conditioned on the existence of the required roots has failure probability at most

$$p^{-\gamma+o(1)}, \quad \gamma = 1 - \alpha(\log_2 d + C)/d > 0.$$

Obtaining such a sample by rejection takes expected $d^{M+1}(1 + o(1))$ trials, rather than polynomial time in $\log p$. The simple uniform choice $\alpha = 1/8$ gives $\gamma \geq 3/4$.

Proof. Choose the exact-rate congruences as before with n near $\alpha \log_2 p$. Then $m = \alpha(\log_2 p)/d + O_{\rho, d}(1)$, $M = m + O_d(1)$, and $D = \rho m + O_{\rho, d}(1)$. The principal root-admitting count is $p^{1-\alpha \log_2 d/d+o(1)}$, while the character-sum error is $p^{1/2+o(1)}$. Since $C > \log_2 d$, the hypothesis on α makes the former dominate. The combined bad-parameter count is at most $p^{\alpha C/d+o(1)}$: the root-sum term contributes $p^{\alpha \log_2(2^d-1)/d+o(1)}$ and the product-collision term contributes $p^{2\alpha H_2(\rho)/d+o(1)}$. Thus Lemma 4.33 applies, with the displayed conditional failure ratio. The root-admitting count is $p/d^{M+1}(1 + o(1))$, proving the rejection cost. The same entropy comparison proves the numerical ratio. Finally $\eta \log_2 p \rightarrow (2d + 1)/\alpha > 1$, giving strict Elias. For $\alpha = 1/8$, use $C \leq d$ and $\log_2 d \leq d$. $\square$

Corollary 4.35 (A tradeoff up to exponent constant three). *In either two-orbit construction, one may instead take $K = dD - s$ for any $1 \leq s < d$. The nearby count and uniqueness conclusions persist, with*

$$\eta = \frac{2d + s}{n}, \quad \text{dist}(f, C) = 1 - \rho - s/n = \theta + \frac{2d}{2d + s}\eta.$$

The numerical prescription is violated exponentially for every fixed $c_2 < 2 + s/d$. Thus any fixed $0 < c_2 < 3$ is compatible with unique nearby witnesses and separation greater than two thirds of the gap. Efficient witness recovery remains available in the integer construction; the function-field construction retains its logarithmic length and existence guarantees.

Proof. All constructed differences $w - H_I$ have degree at most $dD - d < K$. For any codeword, the monic difference from w still has degree $dD = K + s$ and zero penultimate coefficient. The same root-sum argument classifies every codeword with $dD + 2d$ agreements. The far word has exactly dD agreements, so both displayed distances follow. For an exact rational rate $\rho = a/b$, choose $n = bu$ with $au \equiv -s \pmod{d}$; this is possible when $d \nmid a$. The support count still has logarithm $nH_2(\rho)/d + O(\log n)$, which gives the threshold $c_2 < (2d + s)/d$. Take $s = d - 1$ and then a sufficiently large prime d for the final claim. In the logarithmic-length version the strict Elias condition still holds; in the integer version choose the splitting prime sufficiently large. $\square$

Remark 4.36 (The dimension cutoff in the integer construction). For $D \geq 2$, lowering K further to $dD-d$ leaves at most one nearby parameter at the same agreement threshold $dD+2d$. Indeed the sums $\sum_{i \in I} R_i$ of distinct D -supports are distinct. To see this over the rationals, write $R_i = q+(q-1)/(q^i-1)$ and take the least index in the symmetric difference. Its nonconstant term exceeds the sum of all later terms, since $(q-1)^2 > q$. Multiplication by W clears denominators, and the nonzero difference has absolute value at most $Wn(q+1) < p$; thus distinctness survives reduction. The coefficient of X^{dD-d} in H_I is $-\sum_{i \in I} R_i$. For $w - H_I$ to have degree below $dD-d$, this sum must equal the reference support's sum, forcing that support alone. The all-codeword classification excludes any other nearby witnesses. This is a limitation of further dimension reduction in this construction, not an upper bound for other constructions.

Lemma 4.37 (A concentrated index-sum class). *Let $A = \binom{m}{D}$ and $V = D(m-D)(m+1)$. The largest index-sum class of D -subsets of $\{1, \dots, m\}$ has size J satisfying*

$$J \geq \left\lceil \frac{A}{\sqrt{V+1}} \right\rceil.$$

The same bound holds for any translated interval of indices. In particular $J = \Omega(A/m^{3/2})$ at fixed density D/m .

Proof. For a uniform D -subset, its index sum S has variance $V/12$. This follows by writing S as the sum of index-weighted selection indicators: their means are D/m and distinct pairs have joint mean $D(D-1)/(m(m-1))$. Let U be independent and uniform on $[-1/2, 1/2]$. The density of $Y = S + U$ is at most $\mu = J/A$, and its variance is $(V+1)/12$. Any real random variable with density at most μ has variance at least $1/(12\mu^2)$: at distance t from its mean, its tail probability is at least $1 - 2\mu t$, and hence

$$\text{Var}(Y) \geq \int_0^{1/(2\mu)} 2t(1 - 2\mu t) dt = \frac{1}{12\mu^2}.$$

Consequently $J^2(V+1) \geq A^2$, proving the claim. Translation changes all index sums by the same amount. $\square$

Finite existence certificates against $c_2 = 2$. Table 1 gives exact half-rate certificates for Lemma 4.33. Every row has positive $Q_0 - B_0$, is strictly below Elias, and beats the prescription even with $c_1 = 1, c_2 = 2$. The last row places the exceptional point $86/87$ of the gap outside the tested radius. These are existence certificates for a suitable parameter and domain; no particular domain is specified by the table.

The certificates use $(m, D, c) = (171, 87, 0), (285, 144, 1), (1325, 664, 3), (2361, 1182, 5), (1895, 949, 17),$ and $(4419, 2211, 41)$, respectively. They replace $\sqrt{p}$ by its integer ceiling and check all bounds using integer arithmetic. Lucas–Lehmer establishes each prime; $(2d+1)(b-1) > n$ gives a strict sufficient Elias certificate. Using $J_0 = \lceil \binom{m}{D} / \sqrt{D(m-D)(m+1)+1} \rceil$ from Lemma 4.37, a displayed exponent w is certified by

$$J_0^{2d+1} > n^{2d+1} 2^{2n+w(2d+1)}.$$

An independent replay verifies these inequalities. Small-field checks separately enumerate every parameter in two fixtures over the field of order 1,000,003, verify the excluded-parameter counts, and classify every potentially nearby codeword on selected lines by interpolation. Integer fixtures also check actual radical domains, the conservative norm bound, and greedy witness recovery. These smaller fixtures test the geometry and proofs; they are not additional violations of the numerical prescription.

$p = 2^b - 1$: b	d	n	K	separation/ η	ratio $>$
521	2	346	173	4/5	2^9
1279	3	862	431	6/7	2^{13}
9689	5	6638	3319	10/11	2^{85}
23209	7	16546	8273	14/15	2^{119}
44497	19	36060	18030	38/39	2^9
216091	43	190144	95072	86/87	2^6

Table 1: Unique-witness two-orbit existence certificates. The ratio is $J/(n2^{2H_2(1/2)/\eta})$, with $\eta = (2d+1)/n$. The table does not claim efficient sampling or decoding.

A length-226 half-rate certificate. An exact index-sum count gives a shorter example over the 338-bit prime

$$p = 204255 \cdot 2^{320} + 1, \quad n = 226, \quad K = 113.$$

Primality has the compact certificate $11^{(p-1)/2} \equiv -1 \pmod{p}$: every prime divisor of p must then be 1 modulo 2^{320} , and hence exceed $\sqrt{p}$. For $m = 111$, $D = 57$, the index-sum class $S = 3192$ contains exactly

$$J = 441734689284929317660654122848$$

supports. This is the coefficient of $y^{57}x^{3192}$ in $\prod_{i=1}^{111}(1+yx^i)$, replayed by two independent recurrences. The finite existence inequality is positive, so some domain over this explicit field has these uniquely nearby parameters, gap $5/226$, and far separation $4\eta/5$. Exact integer comparisons give

$$\frac{J}{n2^{2/\eta}} > \frac{119}{100}, \quad \frac{J}{n2^{1/\eta}} > 2^{45}.$$

The second inequality concerns the original $c_1 = c_2 = 1$ prescription. This certificate specifies the field and the support class, but not a successful domain parameter q .

At rate $19/44$, a separate certificate has $p = 121 \cdot 2^{320} + 1$, $n = 220$, $K = 95$, and $\eta = 1/44$. Here $(m, D, S) = (108, 48, 2616)$ and the exact class contains 31495815243247545087807036124 supports. The ratio exceeds $21/20$ for $c_2 = 2$ and 2^{43} for $c_2 = 1$, with $c_1 = 1$ in both cases. Primality is certified by $3^{(p-1)/2} \equiv -1 \pmod{p}$. The non-half-rate comparison is also entirely integral, using

$$2^{nH_2(K/n)} = \frac{n^n}{K^K(n-K)^{n-K}}.$$

The same existence and domain-specification qualification applies.

4.3.5 Deterministic banks over the 31-bit Mersenne prime

The preceding small-bit full-coverage certificates use averaging. At $p = 2^{31} - 1$, direct enumeration instead certifies the following large banks for concrete domains. These counts do not depend on a sampling success event.

Proposition 4.38 (Explicit M31 banks, verified by enumeration). *Over $\mathbb{F}_{2^{31}-1}$ there are the following Reed–Solomon codes and received lines. Put $\rho = K/n$, $\eta = 3/n$, and $B = n2^{H_2(\rho)/\eta}$.*

n	K	Candidate supports	Certified nearby parameters	Comparison
58	27	40,116,600	39,763,294	$> B$
62	31	145,422,675	140,916,078	$> B$
64	31	300,540,195	281,212,602	$> B$
68	33	1,166,803,110	897,817,238	$> 2B$

Parameter zero has exact distance $(n - K - 1)/n$, and every certified parameter has exact distance $(n - K - 3)/n$. The direction changes only two coordinates. All four radii are strictly below Elias and have no correlated agreement. Thus all four violate the $c_1 = c_2 = 1$ numerical prescription; the last also violates $c_1 = 2, c_2 = 1$.

Construction and computational verification. Let $m = n/2 - 1$ and $D = (K + 1)/2$. Use the first m entries of the following list as core representatives, with ± 1 as the padding pair:

848979899, 738121439, 699266884, 128913239, 806814741, 629503248,
179005809, 134797481, 207154190, 1023774369, 339671428, 287462841,
393054946, 39961481, 1026875673, 850367968, 913861226, 264277071,
48753053, 60664611, 1054291373, 882987304, 134769212, 35446086,
175650241, 263723351, 718238076, 6429846, 428509650, 268381772,
130012564, 236350235, 900863422.

The domain consists of both signs of these representatives and of 1. Set $w(X) = \prod_{i=1}^D (X^2 - a_i^2)$, let f be its evaluation word, and let g equal one on ± 1 and zero elsewhere. A D -subset I supplies the codeword $w - H_I$ at parameter

$$z_I = - \prod_{i \in I} (1 - a_i^2) \neq 0.$$

It has $K + 1$ core and two padding agreements. The root bound at zero and the two-coordinate distance inequality prove both exact distances.

Two independent programs in `research/paired_domain_warp/m31_exact/` enumerate the image of these subset products. One joins product lists from two halves of the core and uses Mersenne reduction; the other recursively enumerates complementary supports, using inverse factors and ordinary modular reduction. Their bitmaps give the same four counts in the table. The verifier also recovers explicit supports and codeword coefficients for selected parameters. The counts are exact for this candidate family and lower-bound the total number of nearby parameters on the line.

The comparison with $c_1 B$ is checked by the integer inequality

$$J^3 K^K (n - K)^{n-K} > c_1^3 n^{n+3},$$

where J is the certified count. In the half-rate row, $B = 62 \cdot 2^{62/3} < 103199661$. Finally $\eta \log_2 p > 90/n > 1$ proves strict Elias in every row. The code's prescribed domain is the displayed paired set; a power-of-two length in the third row does not make it an FFT-domain example. $\square$

Exact local lists in the half-rate row. For $n = 62, K = 31$, the signed-relation condition of Lemma 4.21 is also checked exhaustively. Splitting its 30 variables into two groups gives $3^{15} = 14348907$ assignments per group; the only matching pair is the empty assignment. Two independent implementations agree. The lemma therefore excludes every nearby codeword outside the enumerated locator family. Two further independent counts give its complete list-size distribution:

List size	0	1	2	3	4	5
Parameters	2,006,567,569	136,494,714	4,337,028	83,442	891	3

In particular, even the 136494714 uniquely nearby parameters exceed $B < 103199661$. Every point on this line has at most five nearby codewords at the tested radius. This is an exact statement about the displayed line, not a global list-size upper bound for the code.

4.4 A cubic change of domain covers every nonzero parameter

The outside-collision estimate can be improved by changing the domain before adding the padding. Matching three times as many integer moments allows a two-parameter cubic change of coordinates. The residual collision equations are absolutely irreducible surfaces; averaging their point counts makes the outside evaluation images occupy almost the entire field. The padding can then cover every nonzero parameter.

Lemma 4.39 (Irreducibility of a cubic collision equation). *Let S, T be disjoint sets of distinct field elements with $|S| = |T| = u \geq 1$. Then*

$$F(X, U, V) = \prod_{a \in S} (X - a^3 - Va^2 - Ua) - \prod_{a \in T} (X - a^3 - Va^2 - Ua)$$

is absolutely irreducible, of degree at most u .

Proof. The case $u = 1$ is a nonconstant linear polynomial. Work over the algebraic closure and homogenize using W . The forms $\ell_a = X - aU - a^2V - a^3W$ define planes in $\mathbb{P}^3$ in general position: every four are independent by the Vandermonde determinant. Write the two families as A_i, B_j . Each base line $A_i = B_j = 0$ is generically smooth on $\prod_i A_i - \prod_j B_j = 0$, and hence belongs to a unique irreducible component. The lines indexed by (i, j) and (i', j) meet at $A_i = A_{i'} = B_j = 0$. No other plane passes through this point. The derivative of the second product is nonzero there, so the point is smooth and the two lines belong to the same component. The same argument works for (i, j) and (i, j') . Consequently all base lines lie on one component G .

Restrict G to $A_i = 0$. The restriction is nonzero, since A_i does not divide the difference of products. It vanishes on the u distinct lines cut out by the B_j , so their product divides it. Thus $\deg G \geq u$, the degree of the original homogeneous polynomial. The latter is irreducible; the smooth base points also exclude a repeated factor. Dehomogenizing at $W = 1$ preserves irreducibility, because W is not a factor. The resulting polynomial is nonconstant. $\square$

Lemma 4.40 (Cubic deformation and complete padding). *Let $p > 2m^2$ be prime. On an m -point seed domain in $\mathbb{F}_p$, suppose $L \geq 2$ distinct t -subsets contain a common anchor and have equal power sums of orders $1, \dots, 3s$. Assume $s \geq 0$ and $K = t - s - 1 \geq 1$. Set*

$$N = m - 1, \quad R = p - N, \quad U_0 = p - 1, \quad B = \frac{p + (t - 1)(t - 2)\sqrt{p} + 3t^4}{p - \binom{m}{2}}, \quad M = \frac{LR}{R + (L - 1)B}.$$

If $1 \leq q \leq R$ and

$$U_0(1 - M/U_0)^q < 1, \tag{27}$$

there is a length- $n = N + q$, dimension- K prime-field Reed–Solomon code and a nonconstant received line such that every nonzero parameter has at least t agreements with the code, whereas parameter zero has maximum agreement exactly $t - 1$. There is no correlated agreement at threshold t .

Proof. Translate the seed so the anchor is zero; equal power sums are preserved. For $(a, b) \in \mathbb{F}_p^2$ use the map $\phi_{a,b}(x) = x^3 + bx^2 + ax$. At least $p^2 - \binom{m}{2}p$ maps are injective on the seed: each collision of two distinct nodes specifies one affine line in the parameter plane. The transformed supports have equal power sums through order s , because $\phi_{a,b}(x)^j$ has degree at most $3s$ for $j \leq s$. Newton's identities therefore give equal first s locator coefficients. For transformed locators H_i , choose one H_0 and put $w = H_0/X$, $P_i = (H_0 - H_i)/X$. These are distinct polynomials with $\deg P_i < K$, and each agrees with the degree- $(t-1)$ polynomial w on exactly $t-1$ coordinates of the transformed core.

For a support pair, cancel its common nodes before forming the difference of products. At an outside point, collision of P_i and P_j is exactly the equation in Lemma 4.39. Its degree is at most t . The affine hypersurface estimate of Cafure and Matera [32, Corollary 5.6] bounds its number of triples (X, a, b) by

$$p^2 + (t-1)(t-2)p^{3/2} + 3t^4p.$$

The hypothesis $p > 2m^2$ implies the characteristic condition in that estimate; a linear residual equation has exactly p^2 points. Summing over support pairs and averaging over injective maps gives one map with at most $\binom{L}{2}B$ outside pair collisions.

For this map, the Cauchy-Schwarz image calculation in Lemma 4.8 gives average outside image size at least M . The differences $P_i(x) - w(x)$ are all nonzero outside the core. Choose the q points with largest images and independent uniform nonzero directions there, with direction zero on the core. The expected number of uncovered nonzero parameters is at most the left-hand side of (27). Some choice leaves none. The global degree of w bounds its agreement with every codeword by $t-1$, attained on the core. This proves the exact far-point distance and excludes correlated agreement at threshold t . $\square$

Theorem 4.41 (A full punctured line at fixed rate). *Fix rational $\rho \in (0, 1)$, constants $c_1, c_2 \geq 1$, and an integer $u > c_2$. For every sufficiently large prime p , there are a code of exact rate ρ , length $n = \Theta(\log p)$, and a received line such that all $p-1$ nonzero parameters are within radius $\theta = 1 - \rho - \eta$, while parameter zero has distance exactly*

$$\theta + \eta/u, \quad \eta = u/n.$$

The radius is strictly below Elias, there is no correlated agreement, and $c_1 n 2^{c_2 H_2(\rho)/\eta} = o(p)$.

Proof. Choose rational $\rho < \beta < 1$, close enough to ρ that, with $\alpha = \rho/\beta$, there is a constant C satisfying

$$\frac{1}{\alpha H_2(\beta)} < C < \frac{u}{c_2 H_2(\rho)}.$$

This is possible because $u > c_2$. Put $s = u - 1$, $b = \log_2 p$, and choose $n = Cb + O(1)$ through common denominator multiples for ρ, α . Set $K = \rho n$, $m = \alpha n$, and $t = K + s + 1$. The anchored t -subsets of an integer interval of length m number $2^{mH_2(t/m) - O(\log m)}$. Their first $3s$ integer moments have only $m^{O_s(1)}$ possible vectors. Thus some anchored class has $L \geq p^{1+\epsilon}$ for a fixed $\epsilon > 0$.

In Lemma 4.40, $B = 1 + o(1)$ and $M/(p-1) \rightarrow 1$. Take $q = n - m + 1 = \Theta(\log p)$. Then $(p-1)(1 - M/(p-1))^q \rightarrow 0$, proving complete nonzero coverage. The zero parameter is $1/n = \eta/u$ beyond the radius. Finally $\eta \log_2 p \rightarrow u/C > c_2 H_2(\rho) \geq H_2(\rho)$ gives strict Elias and the stated numerical comparison. $\square$

For every selection of nearby witnesses, an affine codeword graph contains at most $n - K - u + 1 \leq n$ selected pairs, by Lemma A.8. Thus complete coverage also violates the numerical line-decodability threshold at concurrency n .

For $c_1 = c_2 = 1$, one may take $u = 2$: every nonzero point is nearby while the remaining point is $\eta/2$ outside. This is a shrinking-gap construction on cubic images of intervals with added coordinates. It does not prove superlinear nearby counts at a fixed positive gap or transfer the result to a prescribed evaluation subgroup. Its global lists remain large: on the original core, match $3(s+1)$ moments for t -subsets, and then apply the same cubic map. The count still exceeds p , and the $s+1$ preserved moments give degree-below- K candidates at threshold t . Thus it does not replace the separate actual-list-size separation. The displayed line necessarily has local ambiguity as well: at any one padding coordinate, $L > p$ candidates produce only $p-1$ nonzero labels.

Exact finite certificates give the following half-rate instances. Every nonzero parameter is nearby in every row; the last column compares $n2^{1/\eta}/p$ with the $c_1 = c_2 = 1$ prescription.

Prime	n	K	Agreement threshold	Far separation	Prescription $<$
$2^{127} - 1$	214	107	109	$\eta/2$	2^{-12}
$2^{1279} - 1$	3834	1917	1921	$\eta/4$	2^{-308}
$2^{9689} - 1$	18162	9081	9083	$\eta/2$	2^{-593}

The checker `verify_finite_certificate.py` in `research/cubic_domain_warp/` verifies primality, strict Elias, the list count and (27). The independent `audit_finite_independent.py` uses elementary anchored moment boxes and the weaker point-count bound of [32, Theorem 5.2], with no construction-helper imports. Both replays verify the missing-label expectation is less than one. For the two larger fields, $M/(p-1) > 3/4$ and $2q \geq \log_2(p+1)$ suffice; for the 127-bit field, each replay checks the rational power directly. The finite geometry checker verifies general position, smooth base intersections, small-field point counts, and the degree and moment identities on all 5692 injective cubic maps of a 17-node fixture. These are existence certificates for the large examples, not enumerations of their domains, directions or nearby codewords.

4.4.1 Sampling the code and line without enumerating a moment class

The large moment class and the largest evaluation images need not be found explicitly. Random choices produce the code and received line with a quantitative success guarantee.

Proposition 4.42 (A direct randomized construction). *Use the parameters of Lemma 4.40, with seed $\{0, \dots, m-1\}$ and anchor zero. Put*

$$Q = \binom{m-1}{t-1}, \quad \mu = Q/\mathcal{V},$$

where $\mathcal{V}$ is an upper bound on the number of first- $3s$ integer binomial-moment vectors of the anchored supports. Choose an anchored support uniformly, an injective cubic map uniformly, q distinct outside coordinates uniformly, and independent uniform nonzero padding directions. Let w be the locator of the chosen transformed support after removing its anchor, and output $w + zg$. For every $L_* \geq 2$ and $\gamma > 0$, the probability that some nonzero parameter is not nearby is at most

$$\frac{L_*}{\mu} + \frac{\delta}{\gamma} + U_0 \left(\frac{\gamma}{1+\gamma} \right)^q, \quad \delta = \frac{U_0}{L_*} + \frac{U_0 B}{R} - 1. \quad (28)$$

Every output has the exact far-point distance and absence of correlated agreement in Lemma 4.40.

Proof. The chance that the sampled support belongs to a moment class of size less than L_* is at most $\mathcal{V}L_*/Q = L_*/\mu$, by summing the sizes of such classes. Equal binomial moments imply equal

power sums through the same order. Condition on any class of size $L \geq L_*$. For an injective map, let $r_{x,v}$ be the multiplicity of the nonzero difference value $v = P_i(x) - w(x)$, and put

$$E = \sum_{x \text{ outside}} \sum_{v \neq 0} r_{x,v}^2, \quad \Delta = \frac{U_0 E}{RL^2} - 1.$$

Cauchy–Schwarz gives $\Delta \geq 0$. The collision average proved in Lemma 4.40 gives

$$\mathbb{E}E \leq LR + L(L-1)B, \quad \mathbb{E}\Delta \leq \frac{U_0}{L} + \frac{U_0(L-1)B}{RL} - 1 \leq \delta.$$

Hence $\Pr(\Delta > \gamma) \leq \delta/\gamma$. If $\Delta \leq \gamma$, the average outside image size is at least $L^2 R/E \geq U_0/(1+\gamma)$.

For sampling distinct coordinates, the mean product of their nonnegative missing fractions is at most the corresponding power of the population mean. Indeed, replacing two entries by their average increases each elementary symmetric mean: with all other entries fixed, the only changing term is a nonnegative multiple of their product. Iterating this averaging proves the inequality. The expected number of uncovered labels is therefore at most $U_0(\gamma/(1+\gamma))^q$, and the probability of any uncovered label is at most this expectation. Combining the three bad events proves (28). The degree of w and its selected roots give the far-point claim for every output, independently of the coverage event. $\square$

For the parameters in Theorem 4.41, the elementary moment ranges give $\mu \geq p^{1+\epsilon}$ for a fixed $\epsilon > 0$. Taking $L_* = p^{1+\epsilon/2}$ in the analysis gives $\delta = p^{-\Omega(1)}$. Since $q = \Theta(\log p)$, a sufficiently small fixed γ makes the last term in (28) $p^{-\Omega(1)}$ as well. Thus the algorithm succeeds with probability $1 - p^{-\Omega(1)}$. It does not need to compute L_* , the moment class, or any image size. Rejection sampling makes the cubic map uniform among injective maps in expected $O(1)$ trials. Forming and evaluating w takes $O(n^2)$ field operations, so the code and line are output in expected polynomial time in $\log p$. This does not give a procedure for recovering a nearby codeword at a supplied nonzero parameter.

For an exact finite example, take $p = 2^{521} - 1$, $(n, K, m, t) = (990, 495, 900, 497)$, $s = 1$, $L_* = p^{2^{150}}$, and $\gamma = 2^{-8}$. The exact bound in (28) is less than 2^{-137} . An independent integer replay bounds each of its three terms by 2^{-130} , and hence their sum by 2^{-128} . The radius is strictly below Elias, the far point is $\eta/2$ outside, and the numerical prescription fraction is below 2^{-16} . The programs `generate_sample.py` and `verify_sample.py` in `research/cubic_domain_warp/` output and replay a stored sample. Its domain, line values, and exact far-point properties are checked deterministically. Complete coverage has the stated probability guarantee over the generator's randomness; it is not deterministically certified for the individual stored sample.

4.5 Elementary padded-interval refinements

The following bounds use the original interval core directly. They give additional finite examples and tradeoffs without the cubic deformation.

Theorem 4.43 (An arbitrarily large fraction of nearby parameters). *Fix rational $0 < \rho < \beta < 1$, put $H = H_2(\beta)$, and write $b = \log_2 p$. For every sufficiently large prime p , there is a Reed–Solomon code of exact rate ρ , length*

$$n = \left(\frac{2\beta}{\rho H} + o(1) \right) b, \quad \eta = \frac{\rho H / \beta + o(1)}{\sqrt{b \log_2 b}},$$

and a received line with at least

$$\left(1 - \exp\left(-\frac{\beta - \rho}{\rho \beta (1 - \beta)} \right) - o(1) \right) p$$

nearby parameters at gap η . The parameter-zero word has distance $1 - \rho - \eta + 1/n$. No pair of codewords has correlated agreement at the threshold, and the nearby radius is strictly below Elias. Consequently, for any fixed $\delta > 0$, choosing β sufficiently close to 1 gives at least $(1 - \delta)p$ nearby parameters at length $\Theta_{\rho,\delta}(\log p)$ and gap $\Theta_{\rho,\delta}(1/\sqrt{\log p \log \log p})$.

Proof. Put $\alpha = \rho/\beta$. Choose n through common denominator multiples for ρ, α , with $n = 2b/(\alpha H) + O(1)$, and set

$$m = \alpha n, \quad K = \rho n, \quad k = K + 1, \quad s = \left\lfloor 2(1 - \varepsilon_b) \sqrt{b/\log_2 b} \right\rfloor, \quad t = k + s,$$

where $\varepsilon_b \rightarrow 0$ sufficiently slowly, for instance $\varepsilon_b = \sqrt{\log_2 \log_2 b / \log_2 b}$ for large b . The anchored integer moment count gives a class of size L with

$$\log_2 L \geq m H_2(t/m) - \frac{s^2}{2} \log_2(m/s) - O_{\rho,\beta}(s^2 + s \log m + \log m) \geq b + \Omega(\varepsilon_b b).$$

Thus $L/p \rightarrow \infty$. Remove the anchor and divide as above, giving $N = m - 1$ old points and a received polynomial of degree $A - 1$, where $A = t$. Add $q = n - m + 1$ coordinates, and use $d = K - 1$ in the collision bound. Balanced occupancies give

$$\frac{2T}{L^2} = d - \frac{(A - 1)^2}{N} + o(1) = (\rho(1 - \beta) + o(1))n, \quad \frac{M}{p} = \frac{1 + o(1)}{\rho(1 - \beta)n}.$$

Apply Lemma 4.11. Since $q/n \rightarrow 1 - \rho/\beta$ and $(p - 1)/p \rightarrow 1$, substitution into (22) gives the asserted constant fraction and the exact distance at zero. The exact gap is $(s + 1)/n$, and $\eta \log_2 p \rightarrow \infty$ verifies strict Elias. Finally the exponent in the displayed fraction tends to $-\infty$ as $\beta \rightarrow 1$ with ρ fixed. $\square$

Thus almost every parameter can be nearby while no common agreement explanation exists. Here “almost every” means any fixed fraction below one; the length and gap constants depend on that fraction. The gap still shrinks. This does not prove superlinear growth at one fixed positive gap, or uniqueness of nearby codewords. At these parameters, the proposed numerical threshold $c_1 n 2^{c_2 H_2(\rho)/\eta}$ is only $p^{o(1)}$ for fixed c_1, c_2 , whereas the actual count is a constant fraction of p .

Theorem 4.44 (Separation by a fixed fraction of the capacity gap). *Fix rational $\rho \in (0, 1)$, a desired error fraction $\delta > 0$, and finite $c_1, c_2 \geq 1$. There is a constant $\kappa > 0$ and, for every sufficiently large prime p , a code of exact rate ρ and a received line with*

$$n = \Theta(\log p), \quad \eta = \Theta(1/\log p), \quad J \geq (1 - \delta)p,$$

whose parameter-zero word has distance

$$1 - \rho - \eta + \kappa \eta.$$

The nearby radius is strictly below Elias, there is no correlated agreement at the threshold, and $c_1 n 2^{c_2 H_2(\rho)/\eta} = o(p)$. The constants, including κ , may depend on ρ, δ, c_1, c_2 . This does not assert density tending to one with the same positive κ .

Proof. Choose rational $\rho < \beta < 1$ so that

$$\lambda := \frac{\beta - \rho}{\rho\beta(1 - \beta)} > \ln(2/\min\{\delta, 1\}).$$

Put $\alpha = \rho/\beta$, choose a fixed $C > 1/(\alpha H_2(\beta))$, and choose a fixed positive integer s satisfying

$$\frac{s+1}{C} > \max\{H_2(\rho), c_2 H_2(\rho)\}.$$

Write $b = \log_2 p$. Choose $n = Cb + O(1)$ through common denominator multiples for ρ, α , and set $m = \alpha n$, $K = \rho n$, $k = K + 1$, and $t = k + s$. The anchored support count, divided by the number of first- s moment vectors, gives

$$\log_2 L \geq m H_2(t/m) - O_s(\log m) = \alpha C H_2(\beta) b - O_{\rho, \beta, s, C}(\log b) > b + \Omega(b).$$

Thus $L/p \rightarrow \infty$. Apply exactly the anchored core and far-point padding argument of Theorem 4.43, now with this fixed s . Its collision calculation is unchanged to leading order, so $J/p \geq 1 - e^{-\lambda} - o(1) > 1 - \delta$ eventually. The exact gap is $\eta = (s+1)/n$, and the parameter-zero word is $1/n = \eta/(s+1)$ beyond the nearby radius. Set $\kappa = 1/(s+1)$. The choice of s gives strict Elias and

$$c_1 n 2^{c_2 H_2(\rho)/\eta} = p^{c_2 H_2(\rho)C/(s+1)+o(1)} = o(p).$$

The global degree bound excludes correlated agreement as before. $\square$

The dependence of the far-point separation on the desired density can be quantified. For fixed ρ, c_2 , as $\delta \downarrow 0$ the construction permits

$$\kappa = \Omega_{\rho, c_2} \left(\frac{\log \log(1/\delta)}{\log(1/\delta)} \right). \quad (29)$$

Indeed, first take $\delta = 2^{-v}$ with integer $v \geq 2$. Choose a power of two q at least $\max\{2/(1-\rho), 2\rho(v+1)/(1-\rho)\}$, and set $\beta = 1 - 1/q$, $C = 2q/(\rho \log_2 q)$, and $s+1 = \lfloor Cc_2 \rfloor + 2$. Since $H_2(\beta) > (\log_2 q)/q$, the support-count condition holds. Also $(s+1)/C > c_2 \geq c_2 H_2(\rho)$ and $\lambda > (1-\rho)q/(2\rho) \geq v+1$, so $e^{-\lambda} < \delta/2$. Taking the smallest such power of two gives $q = \Theta_\rho(v)$ and $\kappa = 1/(s+1) = \Omega_{\rho, c_2}(\log v/v)$. Rounding an arbitrary δ down to the next dyadic value changes only constant factors. This quantifies the tradeoff across separate families; it still does not give density tending to one at fixed positive κ .

For a concrete interpretation, take $\rho = 1/2$, $\beta = 15/16$, $C = 6$, and $s = 6$. Then $\alpha = 8/15$ and $C\alpha H_2(\beta) > 1$, since $15^{15} < 2^{59}$. This gives

$$n = (6 + o(1)) \log_2 p, \quad \eta = \frac{7/6 + o(1)}{\log_2 p}, \quad \frac{J}{p} \geq 1 - e^{-224/15} - o(1) > 0.999999 \quad (30)$$

for sufficiently large p . Yet the line contains a word at distance $1/2 - \eta + \eta/7$, and the $c_1 = c_2 = 1$ numerical prescription has fraction $p^{-1/7+o(1)} \rightarrow 0$. Thus a point can lie a fixed fraction of the capacity gap outside the radius even when more than 99.9999% of the line is nearby.

There are exact finite certificates of this behavior. Over $p = 2^{4423} - 1$, the parameters

$$n = 26520, \quad K = 13260, \quad A = 13267, \quad \eta = 7/n$$

give $J/p > 0.99999974$, a parameter-zero word exactly $\eta/7$ outside the radius, and numerical prescription fraction below 2^{-619} . A larger separation is also possible: choosing $\beta = 3/4$, $C = 15/8$, and $s = 1$ gives asymptotic nearby fraction at least $1 - e^{-8/3} > 0.93$ with separation $\eta/2$. For $p = 2^{2203} - 1$, the finite parameters $(n, K, A) = (4128, 2064, 2066)$ give $J/p > 0.93155073$ and prescription fraction below 2^{-126} , with exact separation $\eta/2$. The checker `verify_gap_scale_far.py` in `research/fixed_gap_padding/` verifies these statements with exact arithmetic, including primality and strict Elias. The domains are padded intervals; these statements do not transfer automatically to a prescribed evaluation subgroup.

Theorem 4.45 (Nearby density tending to one). *Fix rational $\rho \in (0, 1)$ and a real $c > H_2(\rho)$. Write $b = \log_2 p$ and $\ell = \log_2 b$. For every sufficiently large prime p , there is a Reed–Solomon code of exact rate ρ and a received line with parameters*

$$n = \left(\frac{\sqrt{2}}{c} + o(1) \right) \frac{b^{3/2}}{\sqrt{\ell}}, \quad \eta = \frac{c + o(1)}{b},$$

for which the number J of nearby parameters satisfies

$$\frac{J}{p} \geq 1 - \exp \left[- \left(\frac{1 - \rho}{2\sqrt{2}c} - o(1) \right) \sqrt{b\ell} \right]. \quad (31)$$

The parameter-zero word has distance $1 - \rho - \eta + 1/n$. No pair of codewords has correlated agreement at the threshold, and the nearby radius is strictly below Elias. In particular, $J/p \rightarrow 1$ and $n = o(p)$.

Proof. Choose n a denominator multiple for ρ within $O(1)$ of $\sqrt{2}b^{3/2}/(c\sqrt{\ell})$, put $K = \rho n$, and set

$$s = \lfloor cn/b \rfloor, \quad h = \lceil (4 + \varepsilon_b)b/\ell \rceil, \quad k = K + 1, \quad t = k + s, \quad m = t + h,$$

where $\varepsilon_b = \sqrt{\log_2 \ell / \ell}$. For large p these satisfy $1 \leq k < t < m < n < p$. The anchored t -subsets of an m -point integer interval number $\binom{m-1}{h}$. Since

$$\log_2((m-1)/h) = \ell/2 + (\log_2 \ell)/2 + O_{\rho,c}(1),$$

the bound $\binom{m-1}{h} \geq ((m-1)/h)^h$ gives

$$\log_2 \binom{m-1}{h} \geq (2 + \varepsilon_b/2)b - O_{\rho,c}(b/\ell).$$

The first s binomial moments have a number of possible vectors with logarithm at most

$$\frac{s^2}{2} \log_2(m/s) + O(s^2 + s \log m) = b + O_{\rho,c}(b/\ell + \sqrt{b\ell}).$$

Thus an anchored class has $\log_2 L \geq b + \Omega(\varepsilon_b b)$, so $L/p \rightarrow \infty$.

Remove the anchor and divide, using $N = m - 1 = K + s + h$ old coordinates, threshold $A = t$, and $q = n - N$ padding coordinates. With $d = K - 1$,

$$d - \frac{(A-1)^2}{N} = h - s - 1 - \frac{h^2}{K + s + h} = (1 + o(1))h.$$

The balanced collision count therefore gives $M/p = (1 + o(1))/h$. Apply Lemma 4.11. Since $1 - x \leq e^{-x}$,

$$\frac{J}{p-1} \geq 1 - e^{-qM/(p-1)}, \quad \frac{qM}{p-1} = \left(\frac{1 - \rho}{2\sqrt{2}c} + o(1) \right) \sqrt{b\ell}.$$

The loss from replacing $p-1$ by p is $1/p$ and is absorbed by the $o(1)$ in (31). The lemma also gives the exact far-point distance and excludes correlated agreement. The exact gap is $(s+1)/n$. Finally $\eta b \rightarrow c > H_2(\rho)$ gives strict Elias. $\square$

For any fixed c_1, c_2 , choosing $c > \max\{H_2(\rho), c_2 H_2(\rho)\}$ makes the proposed threshold

$$c_1 n 2^{c_2 H_2(\rho)/\eta} = p^{c_2 H_2(\rho)/c + o(1)} = o(p),$$

whereas the nearby fraction tends to one. This does not assert that all parameters are nearby, and the gap is not fixed. This family also has very large off-line lists. Matching one additional moment on the core loses only $O(\sqrt{b \log b})$ bits from the preceding count and gives more than p codewords near another received word at the same threshold. Thus the comparison here is with the proposed numerical list prescription, not with the code's actual maximum list size; Section 5 addresses the latter distinction.

At rate $1/2$, exact finite certificates give the following contrast. The last column is the finite prescription $n2^{1/\eta}/p$, with $c_1 = c_2 = 1$ and without an unspecified remainder.

Prime	n	K	A	Proven $J/p >$	Prescription $<$
$2^{31} - 1$	92	46	50	0.74303810	2^{-1}
$2^{61} - 1$	216	108	113	0.91025173	2^{-10}
$2^{127} - 1$	468	234	240	0.96160200	2^{-40}
$2^{521} - 1$	2800	1400	1411	0.99485642	2^{-255}

Every row lies strictly below Elias, has a parameter-zero word exactly one coordinate beyond the radius, and has no common agreement pair at the threshold. These are coding-count comparisons on padded interval domains, not protocol-security estimates or prescribed-subgroup results. The exact checker `research/fixed_gap_padding/verify_far_point_padding.py` verifies the density and prescription inequalities. The related checker `verify_near_unit_density.py` in that directory also certifies nearby fractions above 0.99999545, 0.99999976, and 0.99999998 for primes $2^{2203} - 1$, $2^{3217} - 1$, and $2^{4423} - 1$, respectively. Finite list counts use the stronger of the binomial-moment box and Gram bounds; the asymptotic proof needs only the box bound. All certificates use Lucas–Lehmer primality, integer Elias inequalities, and rational lower bounds on the union size.

A separate arithmetic audit, `audit_density_independent.py` in the same directory, rederives all four displayed bounds and the 2203-bit bound without importing the construction checker. The separate small-field replay also verifies the far point and the complete agreement profile.

A far point with a growing coordinate separation. The one-coordinate separation in Theorem 4.45 can be increased, while retaining nearby density tending to one. The convergence estimate becomes weaker. We give the counting argument explicitly because it distinguishes relative distance from the number of coordinates by which a word misses the radius.

Lemma 4.46 (Several padding matches). *Suppose a polynomial w of degree $D \geq K \geq 2$ and L distinct polynomials P_i of degree below K each have exactly D agreements on a core of size N . Put $R = p - N$, $U = p - 1$, $d = K - 1$, and*

$$T = d \binom{L}{2} - \mathcal{P}(LD, N).$$

Here $\mathcal{P}$ is the balanced collision count in Lemma 4.8. Let $1 \leq r \leq q \leq R$, and define

$$v_a = \frac{\binom{r}{a} \binom{q-r}{r-a}}{\binom{q}{r}}, \quad B = v_0 + \sum_{a=1}^r v_a U^a \left(\frac{1}{L} + \frac{2T(d-1)_{a-1}}{L^2(R)_a} \right),$$

where $(x)_a$ denotes a falling factorial and out-of-range binomial coefficients are zero. There is an extension to $n = N + q$ distinct domain points and a line $w + zg$, with $g = 0$ on the core, having at least $\lceil U/B \rceil$ nearby nonzero parameters at agreement threshold $D + r$. At parameter zero the maximum agreement is exactly D , and there is no correlated agreement on $D + r$ coordinates.

Proof. Choose the q new points uniformly without replacement from outside the core, and choose their directions independently and uniformly from $\mathbb{F}_p^*$. For fixed $z \neq 0$, let X count pairs (i, S) , where S is an r -subset of the new positions on which $P_i = w + zg$. Every $P_i - w$ has all its roots in the core. Consequently $\mathbb{E}X = L \binom{q}{r} / U^r$.

Let c_{ij} count outside roots of $P_i - P_j$. The root bound and the balanced core count give $c_{ij} \leq d$ and $\sum_{i < j} c_{ij} \leq T$. Two r -subsets whose intersection has size a contribute probability U^{-2r+a} if $i = j$, and $U^{-2r+a} (c_{ij})_a / (R)_a$ otherwise. Hence

$$\frac{\mathbb{E}X^2}{(\mathbb{E}X)^2} = \sum_{a=0}^r v_a U^a \left(\frac{1}{L} + \frac{1}{L^2} \sum_{i \neq j} \frac{(c_{ij})_a}{(R)_a} \right) \leq B,$$

using $(c_{ij})_a \leq c_{ij}(d-1)_{a-1}$ for $a \geq 1$. Cauchy–Schwarz gives $\Pr(X > 0) \geq 1/B$. Average over the U nonzero parameters to obtain one domain and direction with the claimed count. The global polynomial w has degree D , so its maximum agreement with the code is at most D , attained by the selected polynomials on the core. This also excludes correlated agreement at the larger threshold. $\square$

The common core also gives a uniform bound

$$c_{ij} \leq e := d - \max\{0, 2D - N\} \quad (32)$$

whenever there are at least two selected polynomials. Indeed, their core agreement sets intersect in at least $2D - N$ positions. For $e \geq 1$, one may replace $(d-1)_{a-1}$ by $(e-1)_{a-1}$ in Lemma 4.46. The same proof also permits the simpler bound

$$\frac{\mathbb{E}X^2}{(\mathbb{E}X)^2} \leq \frac{U^r}{L} + \left(\frac{U}{R-r+1} \right)^r \exp\left(\frac{r^2 e}{q-r+1} \right). \quad (33)$$

To see this, let Z be the intersection size of two independent uniform r -subsets of a q -set, so $\Pr(Z = a) = v_a$. Bound $(c_{ij})_a \leq e^a$, and use

$$\mathbb{E}e^Z = \sum_{j=0}^r (e-1)^j \binom{r}{j} \frac{(r)_j}{(q)_j} \leq \left(1 + \frac{r(e-1)}{q-r+1} \right)^r \leq \exp\left(\frac{r^2 e}{q-r+1} \right).$$

Corollary 4.47 (An unbounded number of coordinates outside). *Fix rational $\rho \in (0, 1)$ and $c > H_2(\rho)$. With $b = \log_2 p$ and $\ell = \log_2 b$, there are codes and lines of exact rate ρ satisfying*

$$n = \left(\frac{\sqrt{2}}{c} + o(1) \right) \frac{b^{3/2}}{\sqrt{\ell}}, \quad \eta = \frac{c + o(1)}{b}, \quad \frac{J}{p} \geq 1 - O_{\rho,c}(\ell^{-1/2}).$$

The parameter-zero word is exactly r/n beyond the nearby radius, where $r = \lfloor b^{1/6} \rfloor \rightarrow \infty$. The radius is strictly below Elias and there is no correlated agreement at the threshold. The relative separation r/n still tends to zero. For every choice of nearby witnesses, each affine codeword graph contains at most $\lfloor (n - A + r)/r \rfloor = O_{\rho,c}(n/b^{1/6}) = o(n)$ of the selected pairs, where $A = (\rho + \eta)n$. More generally, a codeword graph of degree at most $e \geq 1$ contains at most $\lfloor e(n - A + r)/r \rfloor$ pairs.

Proof. Take n, K, s as in Theorem 4.45, but now put

$$\varepsilon = \ell^{-1/4}, \quad h = \lceil (3 + \varepsilon)(r + 1)b/\ell \rceil, \quad t = K + 1 + s, \quad m = t + h.$$

Since $r \sim b^{1/6}$, we have $h = o(n)$ and $\log_2((m-1)/h) = \ell/3 + (\log_2 \ell)/2 + O_{\rho,c}(1)$. The anchored support count therefore has logarithm at least $(1+\varepsilon/3)(r+1)b - O_{\rho,c}(rb/\ell)$. The moment-vector count still has logarithm $b + O_{\rho,c}(b/\ell + \sqrt{b\ell})$. Thus $L \geq p^r 2^{\Omega(\varepsilon rb)}$. Remove the anchor as before, setting $D = K + s$, $N = D + h$, and $q = n - N$. The threshold is now $A = D + r$, so $\eta = (s+r)/n \sim c/b$. The outside-pair cap in (32) is $e = h - s - 1 \sim h$, and

$$\frac{r^2 e}{q - r + 1} = O_{\rho,c}(\ell^{-1/2}), \quad \left(\frac{U}{R - r + 1} \right)^r = 1 + o(\ell^{-1/2}).$$

Together with $U^r/L \leq 2^{-\Omega(\varepsilon rb)}$, the bound (33) gives a second-moment ratio at most $1 + O_{\rho,c}(\ell^{-1/2})$. The averaging argument proves the nearby count, and the degree argument gives the exact far-point distance and excludes correlated agreement. Finally $\eta b \rightarrow c > H_2(\rho)$ gives strict Elias. Apply Lemma A.8 with $a_0 = D = A - r$ and $T = A$ to obtain the graph-intersection bounds. $\square$

Thus the far point need not merely miss the radius by one coordinate. This remains a shrinking-gap statement with vanishing relative separation. As before, choosing $c > \max\{H_2(\rho), c_2 H_2(\rho)\}$ makes the numerical prescription $o(p)$ despite nearby density tending to one. The checker `verify_multi_match_far_padding.py` in `research/fixed_gap_padding/` exhausts 55,000 choices of domain and direction over $\mathbb{F}_{11}$ and verifies the exact first and second moments for $r = 1, 2, 3$, as well as the uniform pair cap and 1968 hypergeometric identities. These small fixtures check the mechanism, not Elias.

The relative separation may vanish arbitrarily slowly. The parameter choice in Corollary 4.47 gives a faster approach to density one. A more slowly growing seed surplus permits a larger separation relative to η . The following is another consequence of the same padding lemma, with explicit dependence on a freely chosen schedule.

Corollary 4.48 (Slowly vanishing relative separation). *Fix rational $\rho \in (0, 1)$ and $c > H_2(\rho)$. Write $b = \log_2 p$, and let $s = s(b)$ be any positive integer function satisfying*

$$s \longrightarrow \infty, \quad s^{5/3} \log_2 b = o(b).$$

For every sufficiently large prime, there are a code of exact rate ρ and a received line with

$$n = (1/c + o(1))sb, \quad \eta = (c + o(1))/b, \quad \frac{J}{p} \geq 1 - \frac{3c/(1-\rho) + o(1)}{\log_2 s}.$$

The parameter-zero word is exactly $r/n = \kappa\eta$ outside the radius, where

$$r = \lfloor s^{1/3} \rfloor \rightarrow \infty, \quad \kappa = \frac{r}{s+r} \sim s^{-2/3}.$$

The radius is strictly below Elias, and there is no correlated agreement. Every affine codeword graph contains only $o(n)$ selected witnesses, for every selection of nearby witnesses. If also $c > c_2 H_2(\rho)$, the numerical prescription $c_1 n 2^{c_2 H_2(\rho)/\eta}$ is $o(p)$.

Proof. Choose $n = sb/c + O(1)$ through denominator multiples for ρ , and put

$$K = \rho n, \quad h = \left\lceil \frac{3(r+1)b}{\log_2 s} \right\rceil, \quad D = K + s, \quad t = D + 1, \quad m = t + h, \quad N = m - 1.$$

Here $h = o(n)$ and

$$\log_2(N/h) = \frac{2}{3} \log_2 s + \log_2 \log_2 s + O_{\rho,c}(1).$$

Thus the anchored support count has logarithm $\log_2 \binom{N}{h} \geq h \log_2(N/h) \geq (2 - o(1))rb$. The first s binomial moment ranges have total logarithm $O(s^2 \log m) = O(s^2 \log b) = o(rb)$, by the schedule condition. The anchored class therefore has $\log_2 L \geq (2 - o(1))rb$.

Remove the anchor and use Lemma 4.46 with $q = n - N$ new coordinates. The uniform pair cap is $e = h - s - 1 \sim h$, and

$$\frac{r^2 e}{q - r + 1} = \frac{3c/(1 - \rho) + o(1)}{\log_2 s}.$$

In (33), $U^r/L \leq 2^{-(1-o(1))rb}$ and $(U/(R - r + 1))^r = 1 + o(1/\log s)$, since n, r are polynomial in b . The second-moment ratio is consequently at most $\exp((3c/(1 - \rho) + o(1))/\log_2 s)$. Inverting and averaging gives the stated nearby count. The global degree- D word gives exact far separation r/n ; $\eta = (s + r)/n$ gives the displayed value of κ and strict Elias. The incidence bound in Lemma A.8 gives at most $\lfloor (n - D)/r \rfloor = o(n)$ witnesses on each graph. Finally the numerical prescription is $p^{c_2 H_2(\rho)/c + o(1)} = o(p)$ under the stated condition. $\square$

If one only requires a one-coordinate far point, the nearby density can approach one much faster. Under the stronger schedule condition $s^2 \log_2 b = o(b)$, keep $n = sb/c + O(1)$ but take $r = 1$ and $h = \lceil 2b/\log_2 s \rceil$. Then

$$\kappa = \frac{1}{s + 1}, \quad \frac{J}{p} \geq 1 - \exp\left[-\left(\frac{1 - \rho}{2c} - o(1)\right) s \log_2 s\right].$$

Indeed, the support logarithm is at least $(2 - o(1))b$, the moment cost is $o(b)$, and the averaged image bound satisfies $M/p \sim 1/h$. Lemma 4.11 gives the displayed exponent from $q/h \sim (1 - \rho)s \log_2 s/(2c)$. The $1/p$ loss is negligible because $s \log s = o(b)$. This choice preserves a larger relative separation than the earlier near-unit-density parameter choice, at the cost of a slower approach to density one.

In particular, apply Corollary 4.48 to *any* prescribed function $\varepsilon(p) > 0$ tending to zero. Taking

$$s = \lfloor \min\{\varepsilon(p)^{-3/4}, b^{1/4}\} \rfloor$$

satisfies the schedule condition and gives $\kappa > \varepsilon(p)$ eventually. Hence requiring a point at least $\varepsilon(p)\eta$ outside the radius does not prevent nearby density tending to one. This corollary uses a vanishing separation fraction. The cubic-domain construction in Section 4.4 gives complete nonzero coverage with a fixed positive separation fraction. The qualitative statement can also be obtained by diagonalization from fixed-fraction families; the corollary supplies explicit lengths and growing coordinate deficits. For example, $s = \lfloor \log_2 b \rfloor$ gives $n \sim b \log_2 b/c$, $\kappa \sim (\log_2 b)^{-2/3}$, and $J/p \geq 1 - O(1/\log_2 \log_2 b)$.

These codes still have global lists larger than p . On the core, take $(D + r)$ -subsets and match $s + r$ moments. Their support count has logarithm $(2 - o(1))rb$, while the moment cost is still $O((s + r)^2 \log b) = o(rb)$; their common locator prefix yields degree-below- K candidates at the desired threshold. Extend the received word over the padding. Thus this remains a comparison with the numerical prescription, not an actual-list-size separation.

The checker `verify_slow_separation.py` audits sufficient integer parameter inequalities and an exact prime-field instance; the general schedule is justified by the preceding proof.

For Lemma 4.46, exact finite certificates also give the following half-rate examples. Here r is the number of coordinates by which the parameter-zero word misses the radius; the last column is the same numerical prescription $n2^{1/\eta}/p$, with $c_1 = c_2 = 1$ and no unspecified remainder.

Prime	n	K	A	r	Proven $J/p >$	Prescription $<$
$2^{127} - 1$	694	347	356	2	0.25107817	2^{-40}
$2^{127} - 1$	830	415	426	3	0.02273637	2^{-40}
$2^{521} - 1$	5840	2920	2936	2	0.59499256	2^{-100}
$2^{521} - 1$	8140	4070	4091	3	0.28599616	2^{-100}

All four radii are strictly below Elias. The certificate `verify_multi_match_finite.py` in the same directory verifies primality, the anchored Gram list count, the rational second-moment bound, and the displayed inequalities using exact arithmetic. It also certifies two- and three-coordinate far points over $2^{61} - 1$ and $2^{1279} - 1$. These are existence statements on padded interval domains, not prescribed-domain or protocol-security estimates.

Exact finite comparisons. For a seed with parameters (m, k, t) and $s = t - k$, the same proof uses $N = m - 1$, $K = k - 1$, $A = t$, $q = s + 1$, and $d = k - 2$. The Gram bound supplies L_0 supports, and anchor averaging gives $L = \lceil tL_0/m \rceil$. Exact evaluation of (21) yields the following certificates. The last column is a strict lower bound on $\log_2(J/(n2^{H_2(K/n)/\eta}))$, where $\eta = (A - K)/n$.

Prime	n	K	A	Nearby labels J	Excess bits
$2^{31} - 1$	72	12	15	381222100	6.73559
$2^{61} - 1$	155	35	40	$> 3.2687 \cdot 10^{17}$	27.01590
$2^{127} - 1$	273	70	76	$> 1.6423 \cdot 10^{37}$	78.16602
$2^{521} - 1$	1065	381	393	$> 3.1200 \cdot 10^{155}$	422.98809

All radii are strictly below Elias. These compare coding counts with the prescription $c_1 = c_2 = 1$; they are not protocol-security estimates. The padded domains are not prescribed subgroups. The exact checker `research/fixed_gap_padding/verify_average_padding.py` verifies rational Gram bounds, collision arithmetic, outward logarithm intervals, Lucas–Lehmer primality, and integer Elias inequalities. Two small-field fixtures also exhaust all 5202 offset tuples and 167042 codeword pairs. The finite search is not an optimality proof.

The far-point variant is checked in `research/fixed_gap_padding/verify_far_point_padding.py`: over $\mathbb{F}_{17}$, it exhausts all 65536 nonzero padding directions and then the full agreement profile and all codeword pairs of a selected line. The parameter-zero word on that line has maximum agreement $A - 1$; 15 of its 16 nonzero parameters are nearby.

4.6 An elementary low-rate boundary case

For comparison, if the rate may tend to zero, even the constant code admits a line with exactly one far point. The construction below uses only elementary counting and works on every domain of the specified size, with a field polynomial in the length.

Theorem 4.49 (A punctured nearby line for the constant code). *For every finite $c_1, c_2 \geq 1$, there are a fixed integer $D \geq 2$ and a constant $a > 1$ such that, for every sufficiently large prime p , every dimension-one Reed–Solomon code of length $n = \lfloor p^{1/a} \rfloor$ has the following properties, regardless of its evaluation domain. Put $\rho = 1/n$, $\eta = D/n$, and $\theta = 1 - \rho - \eta$. There is a nonconstant received line $f + zg$ for which*

$$\text{dist}(f + zg, C) \leq \theta \quad \text{for every } z \neq 0, \quad \text{dist}(f, C) = \theta + \frac{\eta}{D}.$$

There is no correlated agreement at the threshold, θ is strictly below Elias, and $c_1 n 2^{c_2 H_2(\rho)/\eta} = o(p)$. Here $n = o(p)$ and $p = \Theta(n^a)$, but $\rho \rightarrow 0$.

Proof. Choose an integer $D > c_2$ and a real $1 + c_2/D < a < 2$. Some monic degree- D polynomial f over $\mathbb{F}_p$ has $\Omega_D(p)$ distinct values with exactly D preimages. Indeed, associate each D -subset of $\mathbb{F}_p$ with its monic locator F , and group the locators by $f = F - F(0)$. There are at most p^{D-1} groups, so one contains at least $\binom{p}{D}/p^{D-1} = \Omega_D(p)$ locators. Within a group, distinct locators give distinct full fibers of f .

Select $L = \lfloor n/(2D) \rfloor$ such fibers as the core, of size $N = DL$, and take their L values as constant codewords. On the core put $g = 0$; add $q = n - N$ distinct outside coordinates. At every outside point x , the L differences between the selected constants and $f(x)$ are distinct and nonzero. Choose $g(x)$ independently and uniformly in $\mathbb{F}_p^*$ on the new coordinates. Each fixed $z \neq 0$ misses all the selected values at every new coordinate with probability

$$\left(1 - \frac{L}{p-1}\right)^q.$$

If it does not miss, one constant has its D core agreements and one new agreement. The expected number of uncovered nonzero parameters is therefore at most

$$(p-1) \left(1 - \frac{L}{p-1}\right)^q \leq p \exp\left(-\frac{qL}{p-1}\right) < 1,$$

since $qL/(p-1) = \Omega_D(n^{2-a})$. Some direction thus makes every nonzero parameter nearby at threshold $A = D + 1$. At zero, the degree of f bounds the agreement with any constant by D , attained on the selected fibers. This gives the exact far-point distance and excludes correlated agreement at $D + 1$.

The sufficient Elias inequality has logarithm

$$\log\left(\frac{p^D A^A (n-A)^{n-A}}{n^n}\right) = (aD - (D+1)) \log n + O(1) > 0.$$

The small-rate entropy expansion gives $c_1 n 2^{c_2 H_2(1/n)/\eta} = n^{1+c_2/D+o(1)} = o(p)$. Finally, the constant code is unchanged by a coordinate permutation. Relabeling the constructed words therefore transfers the statement to any prescribed evaluation domain of the same size. $\square$

For $c_1 = c_2 = 1$, take $D = 2$, $a = 5/3$, and $n = \lfloor p^{3/5} \rfloor$. An explicit core polynomial is

$$f(X) = X^2 - (N+1)X, \quad N = 2\lfloor n/4 \rfloor.$$

The core $\{1, \dots, N\}$ consists of $N/2$ full fibers, namely the pairs $\{x, N+1-x\}$. The unique far point is $\eta/2$ beyond the radius, where $\eta = 2/n$. The exact checker `verify_constant_code_density.py` in `research/fixed_gap_padding/` certifies all $p-1$ nonzero parameters for $p = 2^{61} - 1$, $n = 104159249330$, and for two larger primes. It also verifies the fiber-count identity by exhaustive small-field enumeration. The large-field results are existence certificates, not enumerations of those enormous lines.

The constant code's actual maximum list size at this radius is exactly $\lfloor n/(D+1) \rfloor$: distinct constant values use disjoint agreement sets, and assigning blocks of $D+1$ positions attains the bound. Thus its actual list size times n is $\Theta(n^2) > p$ in this construction. This is not an actual-list counterexample. Its additional feature is that it works on every domain of the given size, whereas the fixed-rate construction in Theorem 4.41 uses chosen domains.

5 Small lists do not bound the number of nearby line points

The previous constructions produced large lists. We now show a different obstruction: even a code with small lists can have an affine line with exponentially many nearby points, each with a unique nearby codeword. The degree argument is elementary. To keep the code's *maximum* list size small, we choose its evaluation domain using a generic-rank theorem for Reed–Solomon agreement matrices and a finite-field union bound. The underlying line is the consecutive-power construction of Ben-Sasson, Carmon, Haböck, Kopparty, and Saraf [18, Theorem 7.1], also used in [17]. Here we choose a domain on which that line coexists with a small global maximum list size at the target radius.

Recall that $\ell_C(\theta)$ is the maximum number of codewords within relative Hamming distance θ of any received word. A selected pair (z, p_z) lies on a parametrized codeword line if $p_z = P_0 + zP_1$ for two fixed codewords $P_0, P_1 \in C$; the scalar z must be the original parameter. Correlated agreement on t coordinates means that $P_0 = f_0$ and $P_1 = f_1$ simultaneously on those coordinates.

Theorem 5.1 (An actual-list-size separation). *Fix any rational rate $0 < \rho < 1$. For every sufficiently large n with $k = \rho n$ integral, there are a prime p , an n -point domain $D \subset \mathbb{F}_p$, and the rate- ρ code $C = \text{RS}(\mathbb{F}_p, D, k)$ with the following properties at*

$$\theta = 1 - \rho - \frac{1}{n}.$$

- (i) *The actual maximum list size satisfies $\ell_C(\theta) = n - k - 1$.*
- (ii) *An affine line $f_0 + zf_1$ has exactly $M = \binom{n}{k+1}$ parameters within distance θ of C . Each has a unique nearby codeword p_z . All remaining $p - M$ points have distance $1 - \rho = \theta + 1/n$, the maximum possible distance from this code.*
- (iii) *Every parametrized codeword line contains at most $n - k$ of these selected pairs (z, p_z) , and this bound is attained. The pair (f_0, f_1) has no correlated agreement on $k + 1$ coordinates.*
- (iv) *The radius θ lies strictly below the characteristic- p Elias radius.*

Consequently, along this family, line-decodability with concurrency n fails at threshold $\ell_C(\theta)n + R_n$ for every $R_n = o(n)$.

Proof. Choose the domain. For sufficiently large n , we have $k \geq 2$ and $a = n - k \geq 2$; take a prime $p \geq n + k2^{10an}$. Sample n distinct domain points uniformly from $\mathbb{F}_p$. We impose three conditions on this domain.

First, put $L = a - 1 \geq 1$. We use the generic-matrix argument underlying [10, Sections 2.2–2.4 and 3.1], with an explicit finite-field union bound. If $L + 1$ distinct codewords lie in a radius- θ ball, their total (unnormalized) distance from its center is at most $(L + 1)(a - 1) = La = L(n - k)$. The non-strict inequality in [10, Lemma 2.3] permits equality and implies that some subcollection of t codewords, $2 \leq t \leq L + 1$, has a k -weakly-partition-connected agreement hypergraph. Its vertices are these codewords, and its i th edge records which agree with the center at coordinate i . Weak partition connectivity means that every partition $\mathcal{P}$ of the vertices satisfies $\sum_i \max\{|\mathcal{P}(e_i)| - 1, 0\} \geq k(|\mathcal{P}| - 1)$, where $\mathcal{P}(e_i)$ denotes the parts meeting e_i .

The associated reduced intersection matrix encodes equalities of polynomial values within each edge, using a spanning star, with the coefficient block for the last polynomial removed. It has $(t - 1)k$ columns and entries of degree at most $k - 1$ in the formal domain points. It has full symbolic column rank by [10, Theorem 2.11]. However, at a domain realizing the proposed subcollection, the

coefficient differences from its last codeword give a nonzero kernel vector. Thus a fixed nonzero full-column minor must vanish there. This minor has total degree at most $(t-1)k(k-1) \leq Lk^2$.

There are at most $\sum_{t=2}^{L+1} 2^{tn} \leq 2^{(L+2)n}$ such labeled hypergraphs. For independent uniform domain points, Schwartz–Zippel and a union bound give failure probability at most $2^{(L+2)n} Lk^2/p$. Since $p > n^2$, the probability that those points are distinct is greater than $1/2$. Conditioning on distinctness therefore gives failure probability at most

$$\frac{2^{(L+2)n+1} Lk^2}{p} < 2^{-2an}.$$

For the last inequality, use $L = a-1$, $p > k2^{10an}$, $2 \leq a < n$, and $1 + \log_2((a-1)k) < (7a-1)n$ for $n \geq 3$. Consequently, except with probability 2^{-2an} , every radius- θ list has size at most $a-1$.

Second, require all $(k+1)$ -subsets of the domain to have distinct sums. For two different index sets T, U , condition on every domain point except one indexed by $T \setminus U$. Equality of the two sums fixes at most one value of this remaining point, among $p-n+1$ possibilities. The probability of any collision is therefore at most $\binom{M}{2}/(p-n+1)$. Our choice of p gives

$$2^{-2an} + \frac{\binom{M}{2}}{p-n+1} < 1.$$

Indeed, $M \leq 2^n$ and $p-n+1 > k2^{10an}$ make the second failure probability less than $2^{2n-1-10an} \leq 2^{-2an}$. Finally, impose the matching-list construction from Proposition 5.2 below, simultaneously at all integer radii strictly below $n-k$. Its proof bounds failure by $(a-1)^2/[2(p-n+1)] < 2^{-2an}$. Together with the preceding two bounds, total failure is less than $3 \cdot 2^{-2an} < 1$. Fix a domain satisfying all three conditions. The matching construction in particular gives $\ell_C(\theta) \geq a-1$, so the bound at our target radius is exact.

Construct the nearby points. For each $(k+1)$ -subset $T \subset D$, put

$$s_T = \sum_{a \in T} a, \quad V_T(X) = \prod_{a \in T} (X - a).$$

Evaluate the affine family

$$f_z(X) = X^{k+1} - zX^k$$

on D . At parameter $z = s_T$, choose

$$p_{s_T}(X) = X^{k+1} - s_T X^k - V_T(X).$$

Its two leading coefficients cancel, so $\deg p_{s_T} < k$. Its difference from f_{s_T} vanishes exactly on T , giving distance $1 - (k+1)/n = \theta$.

Conversely, if a polynomial P of degree less than k agrees with f_z on at least $k+1$ coordinates, then $f_z - P$ is monic of degree $k+1$ and must equal V_T for its root set $T \subset D$. Comparing the coefficient of X^k gives $z = s_T$. Distinct subset sums therefore identify both T and P uniquely. There are exactly M nearby parameters, each with one nearby codeword. Any received word agrees with some degree-below- k polynomial on at least k coordinates, by interpolation. Every remaining parameter therefore has maximum agreement exactly k , hence distance $1 - \rho$. Since $p > M$, such maximally distant points exist on the same line.

Bound concurrency. Suppose $P_0 + zP_1$ contains m selected pairs, and let b count the coordinates where both $X^{k+1} = P_0(X)$ and $-X^k = P_1(X)$. Since $\deg P_1 < k$, the second equality holds on at most k coordinates; hence $b \leq k$, which already rules out the claimed common agreement. At those

b coordinates all m parameters may agree. At every other coordinate two distinct affine functions of z agree at most once. Counting the $k + 1$ agreements of each selected witness yields

$$m(k + 1) \leq bm + n - b, \quad m \leq \frac{n - b}{k + 1 - b} \leq n - k.$$

The bound is attained by fixing a k -subset $A \subset D$ and writing $V_A(X) = \prod_{a \in A} (X - a)$ and $s_A = \sum_{a \in A} a$. The codeword line

$$[X^{k+1} - (X + s_A)V_A(X)] + z[V_A(X) - X^k]$$

has both coefficient polynomials of degree less than k and contains the selected witness for each $T = A \cup \{a\}$ with $a \in D \setminus A$, at its parameter $z = s_A + a$. There are $n - k$ such parameters.

Check the radius and threshold. The entropy estimate $H_p(\theta) \leq \theta + 1/\log_2 p < 1 - \rho$ places θ strictly below the Elias radius. Finally, $M = 2^{H_2(\rho)n - O_\rho(\log n)}$, whereas $\ell_C(\theta)n \leq (n - k - 1)n = O_\rho(n^2)$. Thus $M > \ell_C(\theta)n + R_n$ eventually for every $R_n = o(n)$, without the required concurrency. $\square$

Attribution of the exact profile. The upper bound below follows from [11, Theorem 1.5]. For the lower bound, Roth's Theorem 4 [12] suffices whenever $a - 2 \leq \binom{k+1}{2}$. Indeed, set $L_s = \lfloor (a - 1)/s \rfloor$. If $L_s \geq 2$, put $L = L_s - 1$ and write $a = (L + 1)u + v$ with $1 \leq v \leq L + 1$. Then $u \geq 1$ and $L \leq a - 2 \leq \binom{k+1}{2} \leq \binom{k-1+u+v}{k-1}$. If the maximum list size at radius $(a - s)/n$ were at most L , Roth's bound would imply $a - s \leq La/(L + 1)$, contrary to $L_s s \leq a - 1$. The case $L_s = 1$ is immediate. The sufficient condition holds eventually at every fixed rate in $(0, 1)$. Thus the fixed-rate exact profile is a consequence of published results. The proof below supplies a direct lower construction for every finite $k \geq 2$ and uses [10] for the stated finite-field probability.

Proposition 5.2 (Exact generic list sizes at every integer radius below $n - k$). *Let $k \geq 2$, $a = n - k \geq 2$, and let $p \geq n + k2^{10an}$ be prime. For a uniformly random ordered set of n distinct evaluation points, with probability greater than $1 - 2^{1-2an}$, the Reed–Solomon code C simultaneously satisfies*

$$\ell_C\left(\frac{a - s}{n}\right) = \left\lfloor \frac{a - 1}{s} \right\rfloor \quad (1 \leq s < a).$$

At radius zero the maximum list size is one. The domain in Theorem 5.1 can be chosen to have this entire profile, as well as distinct sums of its $(k + 1)$ -subsets.

Proof. Upper bounds from one rank condition. Use the good-domain event in the proof of Theorem 5.1: every k -weakly-partition-connected agreement hypergraph on at most a vertices has a reduced intersection matrix of full column rank. Its failure probability is less than 2^{-2an} . For each s , put $L_s = \lfloor (a - 1)/s \rfloor$. If $L_s + 1$ distinct codewords are each at distance at most $a - s$ from one word, their total distance is at most

$$(L_s + 1)(a - s) \leq L_s a,$$

since $(L_s + 1)s \geq a$. Lemma 2.3 of [10] then produces one of the excluded hypergraphs on at most $L_s + 1 \leq a$ vertices. Thus the same rank condition supplies all upper bounds simultaneously.

Matching lists from a chain of affine polynomials. Set aside $k - 2$ coordinates H and put $F(X) = \prod_{h \in H} (X - h)$. There remain $a + 2$ coordinates. Fix s and write $L = L_s$. Choose two anchors A, B , another s coordinates, and a seed z_0 . Give the anchors and the other s coordinates received value zero, and give z_0 value one. The zero polynomial already has $s + 2$ agreements. For

each $1 \leq j \leq L-1$, choose a fresh set Z_j of s coordinates and designate one of them z_j . Let $A_j = A$ for odd j and $A_j = B$ for even j , and define

$$c_j = \frac{w(z_{j-1})}{z_{j-1} - A_j}, \quad q_j(X) = c_j(X - A_j), \quad w(x) = q_j(x) \quad (x \in Z_j).$$

This uses $s + 3 + (L-1)s = Ls + 3 \leq a + 2$ coordinates. Give any remaining coordinates value zero. All denominators and slopes are nonzero, and q_j agrees with w at the anchor A_j , the preceding point z_{j-1} , and all s points of Z_j . Each candidate therefore has at least $s + 2$ agreements.

It remains to ensure that the candidates are distinct. Candidates with different anchors have different roots. For $i < j$ with the same anchor, the equality $q_i = q_j$ is equivalent to

$$\prod_{r=i}^{j-1} \frac{z_r - A_r}{z_r - A_{r+1}} = 1.$$

Condition on every domain coordinate except z_i . All factors but one are fixed and nonzero, so the equality takes the form $C(z_i - A_i)/(z_i - A_{i+1}) = 1$ with $C \neq 0$. Because the anchors differ, this equation has at most one solution. Under distinct sampling its probability is therefore at most $1/(p - n + 1)$. There are at most $L^2/4$ pairs with the same anchor, so candidate repetition has probability at most $L^2/[4(p - n + 1)]$. On its complement, multiplication by F gives L distinct polynomials of degree less than k . Set the full received word to zero on H and to $F(x)w(x)$ elsewhere. Each candidate has at least $(k - 2) + (s + 2) = k + s$ agreements, proving the matching lower bound.

Choose one domain for all radii. The sum of the lower-construction failure probabilities is at most

$$\frac{1}{4(p - n + 1)} \sum_{s=1}^{a-1} L_s^2 < \frac{(a-1)^2}{2(p - n + 1)} < 2^{-2an},$$

using $L_s \leq (a-1)/s$ and $\sum_{s \geq 1} s^{-2} < 2$. Together with the single rank-condition failure bound, this proves the probability assertion. Requiring distinct $(k+1)$ -subset sums adds less than 2^{-2an} to the failure probability, as in the preceding proof; the total remains less than one. $\square$

The upper bounds here use the known generic-rank theorem; the lower bounds are explicit interpolation constructions. The domain is generic and the field bound is large. The assertion does not give this list profile on arbitrary prescribed evaluation domains. Dimension $k = 1$ is a real exception: at agreement $1 + s$, the constant code has maximum list size $\lfloor n/(s+1) \rfloor$.

Corollary 5.3 (Extension fields and interleaving). *The domain in Proposition 5.2 has the same exact list profile over every extension of its base field and for every finite interleaving order. Interleaved distance is measured by columns: a coordinate agrees only when all components agree. No further exceptional-domain event is needed.*

Proof. A nonzero full-column minor over the base field remains nonzero over an extension. For interleaved words, agreement of entire columns gives the same agreement hypergraph as in the scalar upper-bound proof. Apply its full-rank matrix to each component of the polynomial coefficient differences. Each component difference vanishes, so two allegedly distinct tuples would coincide. The scalar lower list embeds in one component, with zero in the others, and remains a list over every extension field. This proves equality. The argument does not apply to entrywise distance averaged over all components. $\square$

The companion `checks/astra_rs_all_surplus_check.py` verifies 3,654 large-field parameter fixtures on 378 domains, each reused for all surpluses, including nondivisible surpluses and cases with $k < 2s$. It also checks 450 small-field fixtures, retaining 55 with repeated candidate lines, and enumerates the full lists at eight constructed centers. The 79,079 allocation and probability checks, and 3,003 combined lower-failure checks, pass. Separately, `checks/actual_list_global_small_check.py` exhausts all center classes modulo the code in four small dimension-two examples. Three structured domains have a larger list than the generic profile, confirming the need for the domain qualification.

Sharpness of the global list bound at rates at least one-half. Write $a = n - k \geq 2$. If $k \geq a - 2$, every n -point Reed–Solomon domain has a list of size at least $a - 1$ at radius $(a - 1)/n$. Indeed, set $L = a - 1$ and $h = k - a + 2 \geq 0$, and partition the domain into H , $\{\alpha_1, \dots, \alpha_L\}$, and $\{\beta_1, \dots, \beta_L\}$, where $|H| = h$. Let $F(X) = \prod_{x \in H} (X - x)$ and

$$Q_i(X) = F(X) \left[X^{L-1} - \prod_{j \neq i} (X - \alpha_j) \right] \quad (1 \leq i \leq L).$$

The leading terms cancel, so each has degree at most $h + L - 2 = k - 1$ (for $L = 1$, the sole polynomial is zero). For $i \neq j$, their values at α_i differ by the nonzero quantity $F(\alpha_i) \prod_{r \neq i} (\alpha_i - \alpha_r)$, so they are distinct. Choose a received word equal to zero on H , to $F(\alpha_j) \alpha_j^{L-1}$ at α_j , and to $Q_i(\beta_i)$ at β_i . Each Q_i agrees on H , on every α_j with $j \neq i$, and on β_i , giving $h + L = k + 1$ agreements. This supplies the matching lower bound on every domain in this range. The chain construction in Proposition 5.2 supplies a matching lower bound on a generic domain for every $k \geq 2$.

A sharper collision estimate. For the $s = 1$ chain, the probability of a repeated line is at most

$$\frac{\lfloor \max(a - 5, 0)^2 / 4 \rfloor}{p - n + 1}.$$

For a same-anchor pair $i < j$, condition on all domain points except z_i . Equality of the two slopes has the form $C(z_i - A_i)/(z_i - A_{i+1}) = 1$ with fixed $C \neq 0$. If $C = 1$, equality is impossible; otherwise it fixes at most one of the $p - n + 1$ remaining values of z_i . Pairs with $j - i = 2$ cannot collide: their cleared equation is, up to sign, $(A - B)(z_{i+1} - z_i) = 0$. Among $a - 2$ alternating lines, the number of same-anchor pairs separated by at least four steps is $\lfloor \max(a - 5, 0)^2 / 4 \rfloor$, proving the estimate. In particular, the chain is distinct on every domain for $a \leq 6$. The independent checker `checks/actual_list_chain_conditional_audit.py` exhausts 36,200 allowed values across 1,400 conditional pair cases over $\mathbb{F}_{37}$, verifying the one-solution bound and all 720 gap-two exclusions in those fixtures.

The gap in Theorem 5.1 is $\eta = 1/n$. It therefore rules out a *uniform*, familywise threshold based on the actual list size; it does not rule out an asymptotic statement made separately at each fixed positive gap, with arbitrary dependence of its remainder on that gap. This differs from the separate fixed-gap construction, which defeats a prescribed numerical list coefficient. The present construction instead has provably small actual lists. Its ingredients are the vanishing-polynomial construction and a domain chosen to satisfy the generic-rank guarantee simultaneously with distinct subset sums.

5.1 A general upper bound and a sharp endpoint

The affine-line count in Theorem 5.1 attains a general upper bound at the same radius. The upper bound follows by puncturing to the unique-decoding regime and applying the correlated-agreement

theorem of Ben-Sasson et al. [8, Theorems 1.2 and 1.4]. We include the argument to explain the extremality of the example; no novelty is claimed for this corollary of the known theorem. Stronger bounds are known in parts of the unique-decoding and Johnson regimes [18, Corollary 1.4 and Theorem 1.5]; the elementary bound below is useful for its general range and its sharp endpoint.

Proposition 5.4 (An unconditional puncturing bound). *Let $C = \text{RS}(F, D, k)$, where F is finite, $|D| = n$, and $a = n - k$. Let r be an integer with $a/2 \leq r < a$, and put $s = a - r$ and $d = 2r - a$. Suppose that $f_0, f_1 \in F^D$ have no correlated agreement with degree-below- k polynomials on $n - r$ coordinates. Then the total number of nearby witness pairs satisfies*

$$\sum_{z \in F} |\{P : \deg P < k, \text{dist}(f_0 + zf_1, P|_D) \leq r/n\}| \leq \left\lfloor (n - d) \frac{\binom{n}{d}}{\binom{r}{d}} \right\rfloor = \left\lfloor s \frac{\binom{n}{d+1}}{\binom{r}{d+1}} \right\rfloor.$$

Proof. Delete any d coordinates. The resulting code has length $n - d$, dimension k , and redundancy $a - d = 2s$. Correlated agreement on $(n - d) - s = n - r$ remaining coordinates would also give correlated agreement in the original domain, contrary to the hypothesis. The unique-radius theorem therefore permits at most $n - d$ parameters whose punctured words have at most s errors. At each parameter the nearby polynomial is unique, because the punctured code has minimum distance $2s + 1$. Its relative radius is $s/(n - d) = (1 - k/(n - d))/2$, the included endpoint of the cited unique-radius bound.

For each nearby witness pair (z, P) in the original code, let its error support be E , of size at most r , and extend E to an r -element coordinate set R . Deleting any d -subset of R leaves at most $r - d = s$ errors. Thus this pair is counted by at least $\binom{r}{d}$ punctures. Restriction preserves distinct polynomials because $n - d \geq k$. There are $\binom{n}{d}$ punctures, each counting at most $n - d$ witness pairs. Double counting proves the first bound. The second follows from $r - d = s$ and the adjacent binomial identities. The case $d = 0$ uses the empty puncture. $\square$

At $r = a - 1$, the bound is exactly $\binom{n}{k+1}$. Theorem 5.1 attains it, while its global maximum list size is only $n - k - 1$. Consequently, among affine lines without correlated agreement at this radius, the number of nearby parameters in that theorem is as large as possible. This extremality concerns an algebraic count, not an algorithm for producing its witnesses.

Proposition 5.5 (Pairs with unexplained agreement sets). *Retain the parameters of Proposition 5.4, but allow f_0, f_1 to have correlated agreement. For a nearby pair (z, P) , let $S = \{x \in D : P(x) = f_0(x) + zf_1(x)\}$ be its entire agreement set. The number of such pairs for which no two degree-below- k polynomials jointly explain f_0, f_1 on S is at most*

$$\left\lfloor (n - d) \frac{\binom{n}{d}}{\binom{r}{d}} \right\rfloor.$$

Proof. First consider a length- N puncture of redundancy $2s$. If its coefficient words have no correlated agreement on $N - s$ coordinates, the unique-radius theorem and uniqueness of the nearby polynomial bound all nearby pairs by N . Otherwise, choose a jointly agreeing codeword path $Q_0 + zQ_1$, and let J be its full coefficient agreement set. This path is nearby at every parameter and hence is the unique nearby polynomial. Outside J , an accidental agreement can occur at at most one parameter per coordinate. There are at most $N - |J| \leq s$ such parameters. At every other parameter the entire agreement set is J and is jointly explained. Thus at most N pairs in any puncture have unexplained entire agreement sets.

For an original pair with an unexplained set S , interpolate the two coefficient words on any k positions of S . At least one further position fails one of these interpolants. The resulting $(k + 1)$ -set

$K \subset S$ is itself not jointly explainable. Its complement has size $a - 1 \geq r$, so the error support $E = D \setminus S$ can be extended to an r -set $R \subseteq D \setminus K$. Every d -subset of R leaves at most s errors after puncturing and preserves K . The punctured agreement set therefore remains unexplained. Each original pair is counted in at least $\binom{r}{d}$ punctures, each of which counts at most $n - d$ such pairs. Restriction is injective on the polynomial witnesses, so double counting gives the bound. $\square$

Corollary 5.6 (Polynomial curves). *Retain a, r, s, d from Proposition 5.4, and let $f(z) = \sum_{j=0}^e z^j f_j$ for $f_j \in F^D$ and $e \geq 0$. The number of nearby pairs (z, P) whose entire agreement set cannot be jointly explained by degree-below- k polynomials for all $e + 1$ coefficient words is at most*

$$\left\lceil e(n - d) \frac{\binom{n}{d}}{\binom{r}{d}} \right\rceil.$$

If no such joint agreement exists on $n - r$ coordinates, the same bound holds for all nearby pairs, summed over parameters.

Proof. This is an elementary consequence of the polynomial-curve unique-radius theorem [8, Theorem 6.1]. On a length- N puncture of redundancy $2s$, that theorem bounds nearby parameters by eN unless the coefficient words jointly agree with codewords on $N - s$ positions. Uniqueness makes this a witness bound. In the joint-agreement case, choose the corresponding codeword curve and its full coefficient agreement set J . It is nearby and unique at every parameter. Each coordinate outside J contributes at most e roots of its nonzero discrepancy polynomial. Thus at most $e(N - |J|) \leq es$ parameters have an agreement set larger than J ; all remaining sets are jointly explained.

For each original unexplained support, interpolate every coefficient word on the same k positions and retain one additional position where an interpolant fails. This gives an unexplained $(k + 1)$ -set, independent of e . The protected-support counting argument in Proposition 5.5 now applies with eN in place of N , yielding the displayed bound. For $e = 0$, every nearby support is explained by its nearby polynomial, so the count is zero. The root bound uses formal polynomials and requires no assumption that $e < |F|$. $\square$

The affine bound is also n at $2r = a$. Between the two endpoints it can be much larger than the actual maximum list size times n ; at a fixed gap beyond unique decoding it is generally exponential in n . It therefore leaves the fixed-gap question open. The exact checker `checks/verify_puncturing_bound.py` verifies the parameter and support-count identities and enumerates all 257 parameters on each of five small lines attaining the one-coordinate-gap bound. The independent `checks/astra_syndrome_determinants_check.py` checks the witness and entire-agreement-set bounds on 378 affine lines, including lines with correlated agreement, by exact decoding and 572,226 codeword-distance evaluations. For the curve corollary, `checks/verify_curve_puncturing.py` enumerates all field parameters and codewords on 300 deterministic small curves, including 60 with formal degree at least the field size. All 466,140 distance checks pass; the fixtures include 421 unexplained witness pairs on curves with some joint agreement.

5.2 An incidence bound for all nearby witnesses

Proposition 5.7 (A witness form of the circuit-incidence bound). *Let F be finite, let $1 \leq k < n$, and let $C \subseteq F^n$ be a linear $[n, k]$ MDS code. Let t, e be integers with $k + 1 \leq t \leq n$ and $e \geq 0$, and put $f(z) = \sum_{j=0}^e z^j f_j$ for $f_j \in F^n$. For each pair $(z, c) \in F \times C$, write $S(z, c) = \{i : c_i = f(z)_i\}$. Let $\mathcal{W}$ consist of the pairs with $|S(z, c)| \geq t$ whose entire agreement set does not jointly explain all*

the coefficient words $f_0, \dots, f_e$. Then, for every $k+1 \leq b \leq t$,

$$\sum_{(z,c) \in \mathcal{W}} \binom{|S(z,c)|-1}{b-1} \leq \min\{e, |F|\} \binom{n}{b}.$$

In particular,

$$|\mathcal{W}| \leq \left\lfloor \min\{e, |F|\} \min_{k+1 \leq b \leq t} \frac{\binom{n}{b}}{\binom{t-1}{b-1}} \right\rfloor.$$

If the coefficient words have no joint explanation on any t coordinates, the bound counts every nearby witness pair.

Proof. This extends the incidence proof of Jo [20, Lemma 4.1 and Theorem 4.2]. His local rejection lemma states that a noncodeword on m coordinates of an MDS code rejects at least $\binom{m-1}{b-1}$ tests of size $b \geq k+1$. For each pair in $\mathcal{W}$, apply that lemma to a coefficient word that is outside the code on $S(z, c)$.

Fix a b -set A rejecting at least one coefficient word. The polynomial $f(Z)|_A$ in the quotient space $F^A/(C|_A)$ is formally nonzero and has degree at most e . A linear functional nonzero on one of its coefficient vectors gives a nonzero scalar polynomial of degree at most e . Hence at most $\min\{e, |F|\}$ parameters make $f(z)|_A$ a code restriction. Each such restriction determines at most one global codeword, because $b \geq k$ and the code is MDS. Counting all incidences proves the weighted bound, and then $|S(z, c)| \geq t$ gives the stated count. The argument needs no restriction $e < |F|$. $\square$

For affine lines in the range of Proposition 5.4, put $t = k + s = n - r$. The minimum in Proposition 5.7 is attained at $b = k + 1$, yielding the additional bound

$$|\mathcal{W}| \leq \left\lfloor \frac{\binom{n}{k+1}}{\binom{k+s-1}{k}} \right\rfloor.$$

Indeed, the ratio of the expressions for successive test sizes is $b(n-b)/((b+1)(t-b))$, which exceeds one here because $(k+1)(r+1) > t$. At $(n, k, r) = (10, 2, 6)$ this improves the puncturing bound from 84 to 40. At $s = 1$ both bounds equal $\binom{n}{k+1}$, so the sharp endpoint is unchanged. The puncturing bound can be stronger near the unique radius; we may take the minimum of the two bounds. At fixed positive rate $k/n \rightarrow \rho$ and agreement fraction $t/n \rightarrow \tau$ with $0 < \rho < \tau < 1$, the circuit bound still has exponential rate $H(\rho) - \tau H(\rho/\tau) > 0$, with natural-log entropy H . It therefore does not resolve the fixed-gap question.

Corollary 5.8 (Sharp endpoint for polynomial curves). *Let $1 \leq k \leq n - 2$ and $e \geq 1$. Over a suitable finite field, there are an n -point Reed–Solomon domain and a degree- e curve with exactly $e \binom{n}{k+1}$ nearby parameter–witness pairs at radius $(n - k - 1)/n$. Each nearby word has a unique witness, and the coefficient words have no joint agreement on $k + 1$ coordinates. Every parametrized codeword curve of degree at most e contains at most $e(n - k)$ of these pairs, and this bound is attained.*

Proof. Choose a field of characteristic not dividing e and a domain whose $(k+1)$ -subset sums s_T are distinct. Choose b outside those sums, and pass to a finite extension splitting all polynomials $Z^e + b - s_T$. Each has e distinct roots, and different polynomials have disjoint root sets. Set $h(z) = z^e + b$ and $f(z)(x) = x^{k+1} - h(z)x^k$. As in the affine construction, agreement with a degree-below- k polynomial on $k + 1$ coordinates forces $h(z) = s_T$ and uniquely determines the witness. This gives exactly $e \binom{n}{k+1}$ pairs. The coefficient of z^e is $-X^k$, which precludes joint agreement on $k + 1$ coordinates.

For a parametrized codeword curve $Q(z)$ of degree at most e , let b_0 count coordinates on which $f(z) = Q(z)$ identically in z . Their leading coefficients show $b_0 \leq k$. At each other coordinate the discrepancy has at most e roots. If Q contains m selected pairs, then $m(k+1) \leq b_0 m + e(n-b_0)$, so $m \leq e(n-k)$. The affine codeword line attaining concurrency $n-k$, composed with h , attains equality. If the domain also satisfies the exact generic list profile, that profile persists over the splitting field by the field-extension argument above. $\square$

6 Finite examples on the norm-one circle

The interval construction shows that a uniform list bound fails when the field and block length vary. We now give finite examples over one fixed prime, $p = 2^{31} - 1$. The domains have multiplicative structure, and the construction uses that structure instead of small integer moments. The two methods share a simpler underlying idea: many root polynomials can have the same outer coefficients while differing in their middle coefficients.

6.1 Why the circle helps

Let $i^2 = -1$ in $F_2 = \mathbb{F}_{p^2}$. The norm-one circle is the cyclic group

$$U = \{u \in F_2^\times : u^{p+1} = 1\}.$$

Every element can be written $u = x + iy$ with $x, y \in \mathbb{F}_p$ and $x^2 + y^2 = 1$. On this group $u^p = u^{-1}$. Thus conjugating a polynomial's coefficients and reversing its exponents are closely related operations. For $n \mid p+1$, write μ_n for the subgroup of size n . Our evaluation domains are μ_n and its rotations $g\mu_n$.

For even $K = 2m$, consider the dimension- K circle space

$$C_K = \{A(x) + yB(x) : \deg A < m, \deg B < m\}.$$

Its coefficients may lie in $\mathbb{F}_p$ or in an extension, as specified. In the coordinate u , the space contains all Laurent polynomials in the interior band $-(m-1), \dots, m-1$, with the appropriate coefficient field condition. A Laurent polynomial simply allows negative powers of u ; these are well defined because every evaluation point is nonzero. The interior band leaves room to divide out two common zeros without leaving C_K .

Write $n = MB$. The power map $u \mapsto u^B$ sends μ_n onto μ_M , with B points in each fiber. To choose agreement positions, choose an h -element subset $A \subseteq \mu_M$ containing 1, and form

$$F_A(Y) = \prod_{a \in A} (Y - a) = \sum_{j=0}^h f_{A,j} Y^j.$$

The roots have norm one, so, with $c = f_{A,0}$,

$$f_{A,j} = c f_{A,h-j}^p.$$

Fixing c and r leading coefficients therefore also fixes r trailing coefficients. There are $\binom{M-1}{h-1}$ possible subsets, but at most Mp^{2r} choices of this coefficient data. Some class has at least

$$\left\lceil \frac{\binom{M-1}{h-1}}{Mp^{2r}} \right\rceil$$

members. Subtracting their common outer terms leaves distinct low-degree polynomial parts. Agreement occurs on the selected fibers, so the count of roots in μ_M becomes a count of coordinates in μ_n .

Two additional steps yield the two results below. For a base-field list, one rescales the polynomial parts to make their values lie in $\mathbb{F}_p$. The common anchor $1 \in A$ then permits division at the two circle points $u = 1, -1$. For an affine family, one instead divides by $Y - \tau$, where $\tau \in \mathbb{F}_{p^4} \setminus \mathbb{F}_{p^2}$. As in Lemma 3.2, this produces a scalar label for each polynomial part and a direction word with too little agreement to permit correlated agreement. A collision count selects a pole with many distinct labels. Appendix A gives both arguments in full.

6.2 Two constructions at the same parameters

Theorem 6.1 (Finite circle list and affine-family bounds). *Let $p = 2^{31} - 1$, $E = \mathbb{F}_{p^4}$, and*

$$n = 2^{26}, \quad K = 2^{25}, \quad \rho = \frac{1}{2}, \quad \eta = \frac{33}{1024} - \frac{2}{n}, \quad T = (\rho + \eta)n, \quad \theta = 1 - \rho - \eta.$$

On $D = \mu_n$, and on any rotated domain $g\mu_n \subseteq U$, the following statements hold.

(i) *There is an $\mathbb{F}_p$ -valued received word with more than*

$$2^{40}$$

distinct codewords from the base-field circle space C_K agreeing with it in at least T positions.

(ii) *There are words $f_0, f_1 \in E^D$ with no correlated agreement at threshold T with the extension-field circle code C_K , but more than*

$$2^{74}$$

distinct scalars $z \in E$ for which $f_0 + zf_1$ has a codeword agreeing in at least T positions. Witnesses can be selected so that every parametrized affine codeword graph contains at most fifteen selected pairs (z, P_z) .

The radius θ is strictly below the characteristic- p Elias radius at rate $1/2$.

Proof. For (i), apply Theorem A.2 with $M = 2048$, $d = 1024$, $r = 32$, $h = 1090$, and $B = 2^{15}$. For (ii), apply Theorem A.4 with $M = 1024$, $r = 15$, $h = 545$, and $B = 2^{16}$. Both choices give $K = n/2$ and the same certified agreement threshold $T = hB - 2$. The exact coefficient counts and collision bound are evaluated in (42) and Section A.1.2. Corollary A.9 gives the selected-graph bound. The strict Elias comparison has the rational certificate in Section A.1.3. $\square$

These are two different received-word constructions with the same length, rate, and radius. Neither the theorem nor its proof requires enumerating the very large coefficient classes. The formulas and integer lower bounds are explicit; the choice of a sufficiently large class, and of a suitable pole in (ii), is guaranteed by counting.

Transfer to ordinary Reed–Solomon codes. The second construction also gives an ordinary Reed–Solomon example. Remove the two anchor coordinates and use the stereographic coordinate $t = y/(1 + x) \in \mathbb{F}_p$. Multiplication by the nonzero weight $(1 + t^2)^{m-1}$ identifies the interior Laurent band with ordinary polynomials of degree at most $K - 2$. The resulting code has

$$n' = n - 2, \quad k' = K - 1, \quad \rho' = \frac{1}{2}, \quad \eta' = \frac{T}{n - 2} - \frac{1}{2} = \frac{33 \cdot 2^{16} - 1}{n - 2}.$$

It has the same scalar labels and at least T agreements per selected witness, with no correlated agreement at threshold T . Theorem A.10 proves that the coordinate change and scaling preserve these equalities. The new gap is η' , not η : puncturing preserves the number of certified agreements but changes the block length. Comparisons for this Reed–Solomon code must use n' and η' .

6.3 Stronger finite examples at other rates

The preceding examples use rate $1/2$ to make the two constructions directly comparable. Neither construction is restricted to that rate. In the list construction, choose any even packet dimension d and put $h = d + 2r + 2$. The resulting rate is d/M ; the gap remains $(2r + 2)/M - 2/n$. In particular, no coefficient reserve is needed for the following larger list.

Corollary 6.2 (A quarter-rate base-field list). *Let $p = 2^{31} - 1$, $n = 2^{26}$, $K = 2^{24}$, and*

$$\rho = \frac{1}{4}, \quad \eta = \frac{1}{32} - \frac{2}{n}.$$

On μ_n and its rotations, some $\mathbb{F}_p$ -valued word has more than 2^{43} codewords in C_K agreeing on at least $(\rho + \eta)n$ coordinates. The radius $1 - \rho - \eta$ is strictly below the characteristic- p Elias radius. If L denotes the certified list size, then

$$\frac{L}{2^{H_2(\rho)/\eta}} > 200,000.$$

Proof. Use Theorem A.2 with $M = 64$, $B = 2^{20}$, $d = 16$, $r = 0$, and $h = 18$. The count is $\lceil \binom{63}{17}/64 \rceil$. Appendix D.2 gives exact certificates for the count, radius, and displayed comparison. $\square$

Corollary 6.3 (An eighth-rate line without correlated agreement). *Let $p = 2^{31} - 1$, $E = \mathbb{F}_{p^4}$, $n = 2^{26}$, $K = 2^{23}$, and*

$$\rho = \frac{1}{8}, \quad \eta = \frac{5}{256} - \frac{2}{n}, \quad T = (\rho + \eta)n.$$

On μ_n and its rotations, there are words $f_0, f_1 \in E^D$ with no correlated agreement at threshold T with C_K , but more than 2^{75} distinct scalars z for which $f_0 + zf_1$ has a codeword agreeing on at least T coordinates. If J is the certified scalar count, then

$$\frac{J}{n2^{H_2(\rho)/\eta}} > 2^{22}.$$

The radius is strictly below the characteristic- p Elias radius. Witnesses can be selected so that every parametrized affine codeword graph contains at most 55 selected pairs.

Proof. Apply Theorem A.4 with $M = 256$, $B = 2^{18}$, $r = 1$, and $h = 37$. Its polynomial degree is $d = 33$, so its collision factor is $d - 1 = 32$. The second word agrees with every outer-band codeword at at most $a_0 = K + B = 33B < T = 37B - 2$ coordinates. Lemma A.8 bounds the number of selected pairs on one graph by

$$\left\lfloor \frac{223B}{4B - 2} \right\rfloor = 55.$$

The exact scalar count, Elias comparison, and ratio certificate are in Appendix D.2. $\square$

The small packet group explains the improvement in the list construction: only 64 possible constant coefficients divide the $\binom{63}{17}$ anchored root sets. Reciprocity fixes the opposite end of each root polynomial for free. The degree allowance at rate $1/4$ is already enough to contain the remaining

middle terms after removing the anchors. The line construction uses a different rate and one pair of reserved coefficients. Its scalar count exceeds the numerical line prescription by more than four million. Both examples transfer to ordinary Reed–Solomon codes by Theorem A.10, with length $n - 2$ and dimension $K - 1$. At these rates the exact transferred rate is $(K - 1)/(n - 2)$, rather than precisely $1/4$ or $1/8$.

6.4 The quadratic field already suffices

The larger scalar count above uses a pole in $\mathbb{F}_{p^4} \setminus \mathbb{F}_{p^2}$. An outside pole is not essential to the counterexample. It is enough to avoid the packet labels μ_M . Choosing the pole inside $\mathbb{F}_{p^2}$ gives fewer available poles, but also keeps the entire construction over the quadratic field.

For example, take $p = 2^{31} - 1$, $n = 2^{26}$, $K = 2^{23}$, and

$$\rho = \frac{1}{8}, \quad \eta = \frac{3}{128} - \frac{2}{n}.$$

Corollary A.5, with $M = 128$, $B = 2^{19}$, $r = 0$, and $h = 19$, gives an $\mathbb{F}_{p^2}$ -valued pair with no correlated agreement at threshold $(\rho + \eta)n$, and more than 2^{57} nearby scalar combinations. Its certified scalar count exceeds $n2^{H_2(\rho)/\eta}$ by more than 400. The radius remains strictly below Elias, and the same example transfers to an ordinary Reed–Solomon code with a prime-field evaluation domain and quadratic alphabet. Appendix D.2 records the exact count and inequalities.

6.5 Shorter codes give larger separations

The number of selected scalar labels depends on the packet group and the reserved coefficients, but not on the number B of coordinates in each packet. Reducing B therefore preserves that number while reducing the length factor in the proposed line threshold. The gap also changes, so the comparison must be recomputed.

Corollary 6.4 (Compact eighth-rate examples). *Let $p = 2^{31} - 1$ and $\rho = 1/8$. Each row below gives a circle code on μ_n , or any rotation, and a line with no correlated agreement at threshold $T = (\rho + \eta)n$. Write J for its certified number of distinct nearby parameters.*

Alphabet	n	K	η	$J/(n2^{H_2(\rho)/\eta})$
$\mathbb{F}_{p^2}$	2048	256	23/1024	$> 2^{22}$
$\mathbb{F}_{p^4}$	8192	1024	79/4096	$> 2^{34}$

Both radii are strictly below the characteristic- p Elias radius. In the first row, the words, witnesses, and labels lie in $\mathbb{F}_{p^2}$; the absence of correlated agreement holds even against the larger circle code over $\mathbb{F}_{p^4}$.

Proof. Use $(M, B, r, h) = (128, 16, 0, 19)$ in Corollary A.5 and $(256, 32, 1, 37)$ in Theorem A.4. The respective agreement counts are 302 and 1182, exceeding the direction-word root bounds $K + B = 272$ and 1056. The label counts are unchanged from the preceding eighth-rate examples. Appendix D.3 gives integer certificates for the two new radii and comparisons. $\square$

7 Stronger finite certificates

The following refinements strengthen the finite examples without changing our asymptotic claims. All comparisons in this section use the literal constants $c_1 = c_2 = 1$: the list prescription is $2^{H_2(\rho)/\eta}$ and the line prescription is $n2^{H_2(\rho)/\eta}$. The evaluation domain and alphabet are specified separately in each example.

7.1 Exact enumeration at length 64

An exact enumeration strengthens the finite interval example further. Among the 34-subsets $A \subseteq \{0, \dots, 63\}$, exactly

$$5,552,914,238,035$$

satisfy $\sum_{a \in A} a = 1071$ and $\sum_{a \in A} \binom{a}{2} = 22139$. Their monic root polynomials share the coefficients of degrees 34, 33, and 32, the last being $\binom{1071}{2} - 22139$. Subtracting their common prefix therefore gives distinct codewords of degree at most 31, each agreeing with the prefix word at exactly 34 coordinates. This proves the displayed list lower bound over every prime field with $p > 64$. The count uses an exact take-or-omit recurrence in subset size, first moment, and binomial second moment, with reachability pruning. An independent ascending-label recurrence on complementary 30-subsets verifies every histogram entry; all 129 previously computed moments also agree.

This is a 5.05% improvement on the degree-40 polynomial-weight certificate 5285900426578. At $p = 2^{127} - 1$, the original strict below-Elias comparison still applies. The enumeration counts one integer moment class exactly; it does not claim that this is the largest possible Reed–Solomon decoding list.

7.2 Conditional radial weights and a displaced boundary

Proposition 7.1 (Conditional signed radial weights). *Let $\mathcal{A}$ be a family of t -subsets of $\{0, \dots, n-1\}$ sharing their first binomial moment, and put $C = |\mathcal{A}|$. For $j = 2, 3, 4, 5$, choose*

$$b_j(x) = \binom{x}{j} + \sum_{i < j} c_{ji} \binom{x}{i}, \quad c_{ji} \in \mathbb{Q}.$$

For a uniform $A \in \mathcal{A}$, write $B_j = \sum_{a \in A} b_j(a)$. Let $U_2, \dots, U_5$ be independent uniform variables on $[-1/2, 1/2]$, independent of A , and choose positive rational scales s_j . Set

$$X_j = \frac{B_j - \mathbb{E}B_j + U_j}{s_j}, \quad S = \sum_{j=2}^5 X_j^2, \quad \mu_r = \mathbb{E}S^r.$$

For $w \in \mathbb{Q}[T]$ with integrable positive part on $[0, \infty)$, suppose $E = \mathbb{E}[w(S)] > 0$. Then some class sharing all five binomial moments has size at least

$$\left\lceil \frac{CE}{\pi^2 (\prod_{j=2}^5 s_j) \int_0^\infty w(u)_+ u \, du} \right\rceil.$$

Consequently, for every prime $p > n$, the ordinary Reed–Solomon code on $\{0, \dots, n-1\}$ with dimension $t-5$ has a list of that size at agreement threshold t .

Proof. Because the size and first moment are fixed, the remaining moment vectors lie in an affine lower-triangular lattice with diagonal entries one. Its translates of $[-1/2, 1/2]^4$ tile $\mathbb{R}^4$: successively rounding coordinates determines the lattice point uniquely. If L is the largest fiber, the density of $B+U$ is at most L/C , and that of X is at most $(L/C) \prod_j s_j$.

Radial integration in four dimensions gives

$$\int_{\mathbb{R}^4} w(\|x\|^2)_+ \, dx = \pi^2 \int_0^\infty w(u)_+ u \, du.$$

Replacing the signed weight by its positive part and applying the density bound proves the inequality. Triangular inversion fixes the first five integer binomial moments. Newton identities then fix the

leading six terms of the monic support locators. Subtracting those locators from their common prefix gives distinct polynomials of degree less than $t - 5$, agreeing exactly on their supports. Distinctness after reduction follows from uniqueness of the monic locator of a subset of distinct field elements. $\square$

Choosing the weight. The subclass $w(u) = (R - u)P(u)^2$ is convenient. For $P(u) = \sum_{i=0}^d c_i u^i$, its expectation and radial integral are quadratic forms with matrices

$$A_{ij}(R) = R\mu_{i+j} - \mu_{i+j+1}, \quad B_{ij}(R) = \frac{R^{i+j+3}}{(i+j+2)(i+j+3)}.$$

The matrix $B(R)$ is positive definite. At fixed R , its generalized Rayleigh quotient gives the largest ratio in this polynomial class. Numerical eigenvectors propose a weight; rational rounding and exact evaluation certify the selected weight independently of the optimization.

A verified M31 instance. Take $p = 2^{31} - 1$, $n = 157$, $t = 68$, and condition on $\sum_{a \in A} a = 5304$. The conditional population is

$$C = 4037124019196117766480536278705909334612974.$$

Using normalized population Gram polynomials of degrees 2 through 5, the scales

$$(s_2, s_3, s_4, s_5) = (5751, 76272, 754066, 5946077),$$

and a degree-seven weight

$$w(u) = - \prod_{i=1}^7 (u - r_i),$$

with positive ordered rational roots approximately

$$(r_1, \dots, r_7) \approx (2.78693, 5.20044, 5.74234, 11.14408, 11.19106, 20.21394, 20.21451),$$

gives a list of at least

$$28165503856884755099.$$

Exact rational logarithm intervals verify that the radius is strictly below Elias and that the list exceeds $2^{H_2(63/157)/(5/157)}$ by more than $2^{34.09946}$. The positive intervals are $[0, r_1]$, $[r_2, r_3]$, $[r_4, r_5]$, and $[r_6, r_7]$, so their integrals are exact polynomial endpoint differences. The certificate uses 280 scalar directions through degree fourteen and exact cube-noise moments. Of 560 subset-mode outputs, 351 were counted directly and 209 obtained through independently checked reflections. All complement, prior-moment, and projection-polynomial identities pass. The exact rational roots, rather than their displayed decimal approximations, are in `research/overnight_2026-09-16/certificates/radial_moment_optimum7_verification.json`.

Limit of the truncated radial calculation. An exact rational contraction certificate proves existence of ordered roots and a constant $\lambda > 0$ satisfying

$$\mu_j = \frac{\lambda}{j+2} \sum_{i=1}^7 (-1)^{i+1} r_i^{j+2}, \quad 0 \leq j \leq 7.$$

The density equal to λ/π^2 on these four radial bands has the prescribed moments. It therefore supplies an upper limit on every degree-at-most-seven radial weight certificate; the corresponding signed root polynomial attains it. The validated interval identifies the same rounded integer bound displayed above. This is an optimum of the truncated radial moment method, not an upper bound on the actual RS list.

Proposition 7.2 (Displacing a radial boundary). *Let $X \in \mathbb{R}^d$ have density at most M , where $d \in \{3, 4\}$. Let $0 < r_1 < \dots < r_7$, put $l(s) = \prod_{j=2}^7 (s - r_j)$, and define*

$$w_0(s) = l(s)(r_1 - s), \quad w_v(x) = l(|x|^2)(r_1 - |x|^2 + v \cdot x).$$

Suppose $|v|^2 r_2 < (r_2 - r_1)^2$. Write $a = v/2$ and $R = r_1 + |a|^2$. Then

$$\begin{aligned} \int_{\mathbb{R}^d} (w_v)_+ &= \int_{\mathbb{R}^d} (w_0(|x|^2))_+ - \int_{|x|^2 < r_1} w_0(|x|^2) dx \\ &\quad + \int_{|y|^2 < R} (R - |y|^2) l(|y + a|^2) dy. \end{aligned}$$

In particular, if $\mathbb{E}w_v(X) > 0$, then $M \geq \mathbb{E}w_v(X) / \int (w_v)_+$.

Proof. Completing the square gives $r_1 - |x|^2 + v \cdot x = R - |x - a|^2$. The assumed inequality places this displaced ball strictly inside $\{|x|^2 < r_2\}$. The other positive annuli remain unchanged. On each origin-centered annulus the added term $(v \cdot x)l(|x|^2)$ integrates to zero. Substituting $x = y + a$ on the displaced ball proves the integral identity. Finally, $\mathbb{E}w_v \leq \mathbb{E}(w_v)_+ \leq M \int (w_v)_+$. $\square$

For the normalized smoothed moment lattice, $M \leq (L/C) \prod_j s_j$. The required expectations use no additional subset counts: differentiating an identity $|x|^{2j+2} = \sum_u c_u (u \cdot x)^{2j+2}$ gives $x_i |x|^{2j} = \sum_u c_u u_i (u \cdot x)^{2j+1}$. The displaced-ball integral is exact by angular moments or Cartesian monomial integration. At $n = 157$, $k = 63$, $t = 68$, the verified four-dimensional certificate gives $L \geq 28169451256663519418$ over $\mathbb{F}_{2^{31}-1}$, with finite-prescription excess greater than 34.09967 bits. The three-dimensional certificate gives source class 508108952862977950462384886 and quartic-extension line bank 508108952659354448051620618, with excess greater than 50.90955 bits. The selected rational vectors, exact integrals, and separate raw-moment checks are recorded in the ancillary displaced-boundary certificates.

For the three-dimensional version, retain Gram coordinates of degrees 2, 3, 4 after conditioning on the first moment. The radial integration factor is 2π , and its radial density is $s^{1/2}$. The same triangular lattice argument certifies a class with four fixed integer moments. Its locator prefix is fixed through degree 64, so its residuals have degree at most 63. The rational-line lemma lowers that degree to 62. The quartic-extension instance therefore has length 157, dimension 63, and 68 agreements, with common agreement exactly 63 and selected-graph concurrency at most 18. Its evaluation points remain in $\mathbb{F}_p$; its received words and labels use $\mathbb{F}_{p^4}$. The Elias comparison uses the characteristic $p = 2^{31} - 1$.

7.3 Exact incidences and prime-alphabet lines

Proposition 7.3 (Coordinate and next-coefficient collision savings). *Let N distinct support locators share a prefix leaving residual degree at most k over $\mathbb{F}_p$, on an evaluation domain of size $n < p$. Let I_x count the supports containing x . If their next coefficients take at most K values, then some pole outside the domain has at most*

$$\left\lfloor \frac{k \binom{N}{2} - \sum_x \binom{I_x}{2} - P(N, K)}{p - n} \right\rfloor$$

colliding unordered pairs, where $P(N, K) = K \binom{a}{2} + ab$ for $N = aK + b$, $0 \leq b < K$.

Proof. At least $P(N, K)$ pairs share the next coefficient and have difference degree at most $k - 1$; the others have degree at most k . Shared support roots account for exactly $\sum_x \binom{I_x}{2}$ roots of these differences on the evaluation domain. Subtracting them and averaging over the remaining $p - n$ poles proves the assertion. $\square$

The least J for which $P(N, J)$ does not exceed this pair budget is a guaranteed number of distinct line labels. For the interval $\{0, \dots, 81\}$, the selected twelve-subset class has first sum 486, second binomial moment 12792, and $N = 221190005$. Its exact coordinate incidences and a rationally certified next-moment range give $J = 138752510$ over $\mathbb{F}_{2^{31}-1}$, with $k = 9$, $t = 12$, common agreement 9, and concurrency at most 24. This exceeds the stated finite line prescription by more than $2^{7.04613}$. The largest two-moment class has more supports, but yields a weaker line certificate.

Here the source locators share the prefix $X^{12} - 486X^{11} + 105063X^{10}$. The exact shared-root count is $\sum_x \binom{l_x}{2} = 43128963788215318$. Write $Z = \sum_{x \in A} \binom{x}{3}$. Newton identities give

$$e_3 = \binom{486}{3} + (2 - 486)12792 + 2Z.$$

Exact rational dual inequalities place Z between 222145 and 276503. Explicitly, for $\sigma \in \{1, -1\}$ and rational multipliers $\lambda = (\lambda_0, \lambda_1, \lambda_2)$, every selected support satisfies

$$\sigma Z \geq 12\lambda_0 + 486\lambda_1 + 12792\lambda_2 + \sum_{x=0}^{81} \min \left\{ 0, \sigma \binom{x}{3} - \lambda_0 - \lambda_1 x - \lambda_2 \binom{x}{2} \right\}.$$

This follows termwise from $z_x r_x \geq \min\{0, r_x\}$ for $z_x \in \{0, 1\}$. The choices $(\sigma, \lambda) = (1, (1421, -509, 107/3))$ and $(-1, (-9568, 2447/3, -130/3))$ give the exact lower bounds $666433/3$ and $-829511/3$, respectively. Integer rounding gives the stated interval for Z . Thus at least 449907051360 source pairs share the next coefficient, and the averaged allowed-pole collision budget is 82437495. Inverting the balanced-pair function gives the stated 138752510 labels. The coordinate counts have a separate formal-division check, using the same underlying histogram engine in a different order. The next-coefficient bounds use rational termwise inequalities, independently of the exploratory minimum/maximum recurrence.

7.4 Stronger native circle examples

Proposition 7.4 (A refined circle-pencil bank). *Let N distinct residual polynomials R_A be obtained from the anchored h -subsets of $G = \mu_M$ in one reciprocal coefficient class, and put $d = h - 2r - 2$. For integers $v \geq 0$ and $b \geq 1$, define*

$$\Pi(v, b) = (b - e) \binom{a}{2} + e \binom{a+1}{2}, \quad v = ab + e, \quad 0 \leq e < b.$$

Set

$$C = d \binom{N}{2} - \binom{N}{2} - \Pi(N(h-1), M-1) - (p+1-M)\Pi(N, p).$$

For poles in $\mathbb{F}_{p^2} \setminus \mu_{p+1}$ take $Q = p^2 - p - 1$; for poles in $\mathbb{F}_{p^4} \setminus \mathbb{F}_{p^2}$ take $Q = p^4 - p^2$. There is an admissible pole with at least

$$\min\{j \in \{1, \dots, N\} : \Pi(N, j) \leq \lfloor C/Q \rfloor\}$$

distinct labels. The agreement and absence-of-correlated-agreement conclusions of the original construction are unchanged.

Proof. The function $\Pi(v, b)$ is the minimum number of colliding unordered pairs in a partition into at most b classes. For each $y \in G$, let s_y count supports containing y . Every shared support point is a root of $R_A - R_{A'}$, so double counting gives

$$\sum_{A < A'} |A \cap A'| = \sum_{y \in G} \binom{s_y}{2} \geq \binom{N}{2} + \Pi(N(h-1), M-1).$$

At every norm-one point, reciprocity gives $R_A(y)^p = c^{-1}y^{-d}R_A(y)$ with the same c for the entire class. Thus at most p evaluation values occur there. Each of the other $p+1-M$ norm-one points contributes at least $\Pi(N, p)$ pair roots. These points and G are disjoint. Subtracting their contributions from the total degree budget $d \binom{N}{2}$ bounds the sum of pair collisions over admissible poles by C . Some pole has at most $\lfloor C/Q \rfloor$ collisions. Balancing its label multiplicities proves the assertion. $\square$

For $p = 2^{31} - 1$, $M = 128$, $h = 19$, and $r = 0$, this gives

$$J \geq 365153906390146587$$

with quadratic-field poles, improving the earlier 285003988493147037. At length 2^{26} and rate $1/8$, the bank exceeds the compared finite prescription by more than 2^9 . At length 2048, dimension 256, and 302 agreements, it exceeds that prescription by more than 2^{23} . Both radii are strictly below the characteristic-based Elias bound. These numerical statements have exact integer and rational certificates in `research/overnight_2026-09-16/certificates/circle_threshold_verification.json`.

Counting omitted-support collisions as well. There is a further exact saving at each $y \in G$. Exactly s_y residuals take the value $v(y)$, whereas the others occupy at most $p-1$ values in the same one-dimensional prime-field subspace. Their collision count is therefore at least

$$f_N(s_y) = \binom{s_y}{2} + \Pi(N - s_y, p-1).$$

This counts disjoint pairs and can replace the support-root term above. Its forward difference is $s - \lfloor (N - s - 1)/(p-1) \rfloor$, an increasing function, so balancing unknown incidences gives a valid lower bound. At reserve $r = 0$, an exact cyclic subset-product count supplies the class population and every s_y . For M dyadic, if $C_k(c)$ counts k -subsets of $\{1, \dots, M-1\}$ with exponent sum $c \pmod{M}$, then

$$C_k(c) = \frac{1}{M} \sum_{a|M} \mathcal{R}_a(c) (-1)^{k+\lfloor k/a \rfloor} \binom{M/a-1}{\lfloor k/a \rfloor},$$

where $\mathcal{R}_a$ is the Ramanujan sum. This follows by extracting the coefficient of z^k in $(1 - (-z)^a)^{M/a} / (1 + z)$ for each character. For a nonanchor exponent b , removing its generating factor gives

$$I_b = \sum_{j=0}^{h-2} (-1)^j C_{h-2-j}(c - (j+1)b).$$

Applying these exact populations and collisions gives the following banks. All rows have $p = 2^{31} - 1$, native circle domains, and radii strictly below the characteristic-based Elias bound.

Alphabet	n	K	T	Guaranteed labels	Excess bits
$\mathbb{F}_{p^2}$	2048	256	302	365153907657996934	> 23
$\mathbb{F}_{p^2}$	4096	64	86	334185312123996677	> 24.5949
$\mathbb{F}_{p^4}$	8192	1024	1182	73497635817388704528460	> 34.77733

The first row improves the same-parameter quadratic bank in Appendix D.3 by about 28.1%. The last two rows maximize the checked guarantee within the stated one-anchor dyadic family, using complete product classes at $r = 0$ and generic prefix pigeonholing at $r > 0$. This is an optimization of that formula, not of arbitrary circle lists or actual coefficient classes. The checked range is dyadic $M \geq 4$, $B \geq 2$, $MB \leq 2^{30}$, dyadic packet dimensions $K/B < M$, and every reserve satisfying the strict Elias condition and the construction’s support-size requirements. A separate coefficient-sum implementation checks all 703 computed product classes and 1406 collision certificates.

Scope and reproducibility. The interval-domain results and the native circle results above are different constructions. None lowers the pinned better.codes display of 116.13, and none is a revised security bound for the complete deployed Stwo protocol. Their contribution is a stronger set of counterexamples to the specified finite coding prescriptions. The proof statements, portable inputs, and verification commands are in `research/overnight_2026-09-16/README.md`.

8 Implications for hash-based SNARK security

The preceding results concern properties of codes. Their relevance to SNARKs comes from the role those properties play in a soundness proof. This section explains that connection before considering a particular protocol.

8.1 The coding statement used by a reduction

An interactive oracle proof can ask a prover to supply evaluations of polynomials and then test those evaluations through a small number of queries. In a hash-based argument, Merkle commitments bind the prover to the supplied words. The algebraic analysis must still explain what passing the tests says about those words. List decoding and proximity gaps are two ingredients in such analyses [7, 8].

For example, consider a code $C \subseteq E^n$, two fixed words $f_0, f_1 \in E^n$, and a uniformly random challenge $z \in E$. Define

$$B_\theta(f_0, f_1) = \{z \in E : \text{dist}(f_0 + zf_1, C) \leq \theta\}.$$

Suppose a coding theorem guarantees that, unless f_0 and f_1 have correlated agreement with two codewords, the set B_θ has size at most T . Then, for words without that correlated agreement,

$$\Pr_z[\text{dist}(f_0 + zf_1, C) \leq \theta] \leq \min\{1, T/|E|\}.$$

This is a bound on a specified algebraic event. Using it in a soundness proof also requires a reduction from verifier acceptance to that event, with the words fixed before the challenge is sampled. The evaluation field, coefficient field, and challenge field must be the fields covered by the coding theorem.

The size of a decoding list enters a different counting step. If an analysis associates a received word with at most L candidate codewords and proves a failure bound for each candidate, it can apply a union bound over those candidates. A numerical estimate for L is therefore part of the quantitative argument. Replacing the actual list size by a conjectured smaller value can invalidate that argument. It does not determine the probability of the verifier’s acceptance event.

8.2 What the counterexamples rule out

Three distinctions are needed to apply the results correctly.

Minimum distance and list size. The root bound proves the usual minimum-distance guarantee for Reed–Solomon codes. It does not by itself establish the proposed quantitative list bound beyond the unique-decoding radius. Theorem 2.2 rules out a uniform $\exp(O_\rho(1/\eta))$ bound over prime fields and evaluation domains, even below the characteristic-based Elias radius. An analysis that needs such a bound must impose further restrictions on the code or use a different quantitative estimate.

A list and the nearby points on a line. A maximum-list bound concerns one received word at a time. It does not say that the nearby codewords for different points on an affine line belong to a common small family. Theorem 5.1 makes this distinction explicit: the global maximum list size is $n - k - 1$, while one affine line has $\binom{n}{k+1}$ nearby points, each with a unique witness. Its gap is $1/n$. This theorem does not settle the analogous question with a separately fixed positive gap.

Uniform estimates and fixed-parameter estimates. A security calculation uses a particular field, domain, rate, and radius. A theorem quantified over all prime-field codes would apply to that code, but a counterexample to the universal theorem need not use that particular code. Conversely, a bound with constants depending on the field or the gap cannot be used uniformly without controlling that dependence. The fixed-gap lifts in Theorems 4.4 and 4.6 rule out prescribed numerical coefficients with remainders sublinear along their increasing-length families. The anchored version also prevents correlated agreement altogether at its threshold. Their fields grow, and their chosen seed lists remain fixed. These quantifiers are part of the result.

8.3 From coding counterexamples to protocol conclusions

A coding counterexample is enough to refute a coding premise under the same quantifiers. A conclusion about a complete SNARK requires more. Its oracle words may be constrained by an arithmetization, previous commitments, claimed evaluations, and later consistency tests. A soundness analysis must account for those constraints and the distribution and order of all challenges. Passing to a noninteractive protocol through Fiat–Shamir introduces a further cryptographic analysis.

The circle constructions give finite examples in polynomial spaces used by Circle STARKs. The quotient calculations in Appendices B and C examine additional algebraic conditions on derived words. Their statements retain their own sampling, radius, and conditioning hypotheses. In particular, the numerical raw-quotient example lies above the Elias radius, whereas the principal list and affine-line examples lie below it. No full-protocol security level follows from the quotient counts in this paper.

The application below uses this distinction to identify which quantitative premises of a published analysis are contradicted and which questions about the instantiated protocol remain open.

9 Application to S-two’s quantitative coding assumptions

The S-two whitepaper uses list- and line-decodability assumptions in its analysis of polynomial proximity tests [23]. This provides one application of the preceding coding results. We separate three questions: the asserted uniform list bound, the coefficient in a line-decodability threshold, and finite numerical instantiations. They require different counterexamples and different quantifiers.

9.1 The uniform list prescription

Conjecture 1 of the whitepaper takes the form

$$\ell_C(\theta) \leq c_1 2^{c_2 H_2(\rho)/\eta}, \quad \theta = 1 - \rho - \eta, \quad (34)$$

with constants depending only on the rate and a stated range below the characteristic-based Elias radius. Under its universal prime-field and evaluation-domain quantifiers, this prescription is false for every fixed pair of finite positive constants. Theorem 2.2 gives a single sequence with

$$\log_2 \ell_C(\theta) = \Omega_\rho(n), \quad 1/\eta = \Theta_\rho(\sqrt{n \log n}).$$

The logarithm of the right-hand side of (34) is sublinear in n , whereas the constructed list has a logarithm linear in n . Changing fixed constants cannot close that gap.

Even a polynomial dependence of the prefactor on the prime cannot repair a constant exponent coefficient. More precisely, suppose the prescription is required for every eligible code over each sufficiently large $\mathbb{F}_p$, allowing $c_1 = c_1(p) \leq p^K$ for a fixed K and $c_2 = c_2(p)$. Corollary 2.3, with $b = \log_2 p$, forces

$$c_2(p) \geq (1/2 - o(1)) \frac{\log_2 p}{\log_2 \log_2 p}.$$

Indeed its list logarithm is at least $(1/2 - o(1))b^2/\log_2 b$, whereas the proposed logarithm is at most $Kb + (1 + o(1))c_2(p)b$.

This conclusion uses growing prime fields and a decreasing gap. It does not assert exponential lists at a fixed positive gap, and it does not rule out stronger guarantees for a restricted class of evaluation domains or a restricted relation between field size and length.

Proposition 2.4 gives a separate obstruction at rate $1/4$ and gap $1/8$: when $n = p - 1$, even a finite list bound depending only on rate and gap is impossible. The short-domain theorem above shows that restricting to $n = o(p)$ does not rescue the proposed reciprocal-gap exponent.

9.2 The quantitative line coefficient and its remainder

Conjecture 2 in Appendix A.5 of the whitepaper has a threshold of the form $a = \ell(\theta)n + o(n)$, with concurrency number n . Substituting the proposed quantitative list bound gives

$$a = c_1 2^{c_2 H_2(\rho)/\eta} n + o(n). \quad (35)$$

Here the asymptotic remainder matters. A finite example, or a sequence whose gap decreases, need not refute a remainder statement made only for each separately fixed gap.

Theorem 4.4 addresses that issue directly. For any fixed constants, it preserves the prescribed rate and chooses a gap fixed along an increasing-length family. A preserved list has size $L > A$, where $A = c_1 2^{c_2 H_2(\rho)/\eta}$. The family has Ln selected nearby witnesses, but no parametrized affine codeword graph contains more than n selected pairs. Its discrepancy from the proposed coefficient is

$$(L - A)n.$$

Since $L - A > 0$ is fixed, no remainder that is $o(n)$ along this family can absorb the discrepancy. The family is selected independently of the remainder. Here $o(n)$ means sublinear along admissible increasing-length families, allowing the prime field to grow; the whitepaper does not explicitly state its field-uniformity convention. The theorem is stated for constants at least one; any smaller positive constants are covered by enlarging them to one.

Theorem 4.6 strengthens this conclusion: its lines have no correlated agreement at their exactly fixed rate and gap, yet have at least $C_\rho(\eta)n$ nearby challenges with $\log_2 C_\rho(\eta) = \Omega_\rho(\eta^{-2}/\log(1/\eta))$ at every sufficiently small rational gap. Their selected witnesses also have concurrency at most n . Thus the fixed-gap coefficient obstruction does not depend on selecting among persistent good witness lines.

The shrinking-gap constructions show how large the numerical failure can be. Corollary 4.15 makes every nonzero parameter nearby while the exception is $(1 - o(1))\eta$ outside the radius, only one coordinate short of maximum distance. The proposed numerical fraction tends to zero. Separately, Theorem 4.41 has a length-214, half-rate certificate over $2^{127} - 1$ that compares nearby fraction $1 - 1/p$ with a prescription below 2^{-12} ; the larger certificate over $2^{9689} - 1$ gives a prescription below 2^{-593} . These comparisons concern the prescribed numerical coefficient; they do not replace the fixed-gap argument needed to address a remainder controlled separately at each gap.

These arguments do not establish failure of a fixed-gap threshold expressed using the *actual* maximum list size $\ell_C(\theta)$. In the preliminary construction, for example, the selected count is $Ln \leq \ell_C(\theta)n$. Replacing a false numerical estimate for list size and retaining a theorem that uses actual list size are different operations.

Theorem 5.1 addresses the actual-list question separately. At every fixed rate it has $\ell_C(\theta) = n - k - 1$ but exponentially many uniquely nearby line points and maximum concurrency $n - k$. At rate one-half these bounds are $n/2 - 1$ and $n/2$, respectively. Thus even the threshold $\ell_C(\theta)n + o(n)$ fails along this shrinking-gap family. This rules out a familywise interpretation of that threshold; it does not settle a version whose remainder is controlled only after a positive gap has been fixed. The original conjecture does not specify this uniformity convention.

9.3 Finite comparisons on the circle code

The whitepaper’s concrete examples set $c_1 = c_2 = 1$ and omit the asymptotic remainder. At the common parameters of Theorem 6.1, the resulting comparison values are $2^{1/\eta}$ for a list and $n2^{1/\eta}$ for the line threshold. Let L_0 and B_0 denote the list-size and scalar-count lower bounds certified in Appendix A. They give

$$\frac{L_0}{2^{1/\eta}} > 500, \quad \frac{B_0}{n2^{1/\eta}} > 200,000.$$

These are ratios of coding-theoretic counts. They are not multipliers for a complete attack’s success probability and are not security-bit losses. In particular, the logarithm of a list-size ratio is not, by itself, a reduction in a proof system’s security level.

The two bounds also concern the displayed radius. A parameter-table row evaluated at a different gap must be checked at that gap. For example, the value 2^{16} at rate $1/2$ corresponds to $\eta = 1/16$, which differs from $33/1024 - 2/n$. A counterexample to a universal prescription does not automatically supply a counterexample to every individual numerical row.

The circle tensor space and canonical full-coset domains described above are the algebraic spaces appearing in the examined public proving source [24]; Proposition A.12 and Proposition A.13 explain their mathematical identification. This connection makes the fixed-prime examples pertinent to the coding assumptions used for that family of tests.

Concrete parameters and the challenge schedule. The inspected production configuration uses a nominal 96-bit target: 26 bits of query grinding, 70 queries, and blowup factor two. The source’s parameter counter computes $26 + 70 \cdot 1 = 96$, and the project documentation calls its default soundness conjectured [25]. This arithmetic records the parameter convention; a security

theorem must justify the contributions attributed to each stage. The whitepaper’s illustrated 100-bit configuration is a different choice [23, Section 5.5].

There is a further distinction between that illustrated calculation and the cited implementation. The former allocates grinding to intermediate challenge stages, while the latter performs query grinding after those challenges and omits the corresponding intermediate checks [26]. Grinding after a challenge cannot be credited to an earlier stage without a supporting argument. A quantitative analysis of the implementation must therefore use its actual challenge schedule and accepted parameters. This issue is separate from the counterexamples to the coding assumptions, and the comparison concerns only the pinned source revisions.

The remaining protocol question. A proximity test inside a proof system does not necessarily receive an arbitrary pair of words. Its inputs can be derived quotient polynomials constrained by earlier commitments, algebraic identities, and sampled evaluation points. Appendix C resolves one of these algebraic conditions: fixed tables and fixed, generally false claimed openings produce raw quotient pencils with a certified joint proximity event. The transfer also handles specified nonlinear batching schedules (Theorem C.4), with explicit quotient-layer realizations. Its numerical raw-quotient example lies above the Elias radius (Section C.4); it is distinct from the below-Elias counterexamples. It does not prove that these inputs satisfy the complete set of commitment, AIR, and FRI conditions. Nor does a count of exceptional scalars determine the cost of finding favorable claims and witnesses or arranging a preceding transcript. Thus the coding results invalidate the stated quantitative justifications under their stated quantifiers; turning them into an attack on a complete protocol is a separate problem.

10 Further coding-theoretic questions

The main open questions concern the restrictions needed for useful coding bounds and their role in concrete soundness analyses.

1. **Restrictions on field size and domain.** What quantitative bounds hold under a prescribed restriction $p \leq n^D$, with D fixed independently of the gap, or on a specified multiplicative subgroup or circle coset? Corollary 4.7 already realizes the fixed-gap coefficient obstruction with polynomial-size fields, but its field exponent depends on the gap. The domain-specific questions remain distinct.
2. **The correct dependence on the gap.** For a prescribed relation between p and n , what dependence on η is necessary? The exponential-list family has shrinking gap; the fixed-gap lifts retain fixed seed lists rather than exponentially growing ones. The anchored lift forces the logarithm of any universal linear proximity-gap coefficient to grow at least as $\eta^{-2}/\log(1/\eta)$ at every sufficiently small rational gap. Whether a fixed positive gap permits superlinear growth in length remains open here. Theorem G.1 excludes all candidate families with boundedly many distinct roots as a route to such growth in sufficiently large characteristic, even allowing varying root positions and multiplicities. A third-order equation within this family still has an actual solution variety of quadratic degree.
3. **Bounds for a prescribed code.** Which list and proximity-gap bounds hold for a fixed field and a fixed FFT domain at the radii used in a proof system? A generic-domain existence theorem and a counterexample on an integer interval leave this question open. Such a theorem should expose its finite constants and its dependence on the challenge field.

4. **Complete coverage with unique nearby codewords.** Can the full punctured-line behavior of Corollary 4.15 coexist with a unique nearby codeword at every nonzero parameter? The cubic construction necessarily has local ambiguity: at a single padding coordinate, more than p candidates produce only $p-1$ labels. It also has very large global lists. The paired constructions in Theorems 4.22, 4.24, and 4.30 have whole-line uniqueness and exponentially many nearby parameters, but do not give complete nonzero coverage. The fixed-gap construction gives whole-line uniqueness with a linear lower bound on the nearby count.
5. **Recovering nearby witnesses.** Can one efficiently recover nearby codewords for the full-coverage constructions, or enumerate enough of the equal-moment classes to match the counting lower bounds? The deterministic powers-of-two family in Theorem 4.24 and the orbit family in Theorem 4.27 have efficient witness-recovery algorithms. Lemma 4.12 and Proposition 4.42 output the code and line in expected polynomial time in $\log p$, but do not recover a witness for a supplied nonzero parameter. Theorem F.1 limits one possible description: an affine binary family with a fixed locator prefix has dimension at most $\lfloor \min\{t, n-t\}/(s+1) \rfloor$. At a fixed positive prefix fraction this permits only a constant-size family.
6. **Lines and actual list size.** What line-decodability statements follow from the actual list size at a fixed positive gap? Theorem 5.1 rules out the proposed linear threshold along a shrinking-gap family, even with unique nearby witnesses on the selected line.

A Circle-code constructions in detail

A.1 Base-field lists for circle evaluation codes

We first construct many codewords close to a single received word. The construction uses the roots of polynomials to choose agreement positions, then fixes a few coefficients so that the remaining polynomial parts have low degree. On the norm-one circle, the leading and trailing coefficients are linked. This lets one impose both sets of conditions at the cost of fixing only one set.

A.1.1 The norm-one circle and Laurent polynomials

Let $p \equiv 3 \pmod{4}$ be prime. Since -1 is not a square in $\mathbb{F}_p$, adjoining a root i of $i^2 = -1$ gives the quadratic extension $F_2 = \mathbb{F}_{p^2}$. Its nontrivial field automorphism is $a \mapsto a^p$, and $i^p = -i$. Thus, for $x, y \in \mathbb{F}_p$,

$$(x + iy)^p = x - iy, \quad (x + iy)^{p+1} = x^2 + y^2.$$

The solutions of $x^2 + y^2 = 1$ can therefore be written as the group

$$U = \{u \in F_2^\times : u^{p+1} = 1\}.$$

This group is cyclic of order $p+1$: it is the kernel of the norm map from $F_2^\times$ to $\mathbb{F}_p^\times$. For $n \mid p+1$, let $\mu_n \subseteq U$ be its subgroup of order n . Multiplying all points by $g \in U$ gives a rotated domain $g\mu_n$.

The coordinate $u = x + iy$ simplifies computations on the circle because $u^{-1} = x - iy$. In particular,

$$x = \frac{u + u^{-1}}{2}, \quad y = \frac{u - u^{-1}}{2i}.$$

A *Laurent polynomial* is a finite sum $\sum_j a_j u^j$ in which negative exponents are allowed. They cause no poles on U , since $u \neq 0$. Its support is the set of exponents with nonzero coefficients. We call a

Laurent polynomial with coefficients in F_2 *base-field-real* when

$$a_{-j} = a_j^p \quad \text{for every } j.$$

This coefficient condition makes its values lie in $\mathbb{F}_p$ on U : taking the p th power both conjugates the coefficients and replaces u by u^{-1} . The word “real” here denotes this finite-field condition, not a condition on real numbers.

For even $K = 2s$, the circle polynomial space used below is

$$\mathcal{C}_K = \{A(x) + yB(x) : A, B \in \mathbb{F}_p[x], \quad \deg A < s, \deg B < s\}.$$

It has dimension K as a polynomial-function space on the algebraic circle. Whenever the stated evaluation domain is large enough to make evaluation injective, evaluating these functions gives a dimension- K linear code. We use the term *circle FFT space* for this space; its tensor basis is explained in Proposition A.12. The constructions below lie in its slightly smaller interior Laurent subspace. Working in that subspace makes rotation straightforward.

For a length- n , dimension- K evaluation code, write $\rho = K/n$ for its rate. A word agreeing with a received word in at least $(\rho + \eta)n$ positions lies at relative Hamming distance at most $\theta = 1 - \rho - \eta$. Thus η measures the agreement beyond the rate. Throughout, H_2 denotes binary entropy, and

$$H_p(t) = t \log_p(p-1) - t \log_p t - (1-t) \log_p(1-t).$$

The characteristic- p Elias radius at rate ρ is the solution $r_E(\rho)$ of $H_p(r_E) = 1 - \rho$ on the increasing branch of H_p .

Lemma A.1 (Interior band of the circle FFT space). *Let $p \equiv 3 \pmod{4}$ be prime, let $i^2 = -1$ in $\mathbb{F}_{p^2}$, and write $u = x + iy$ on the norm-one circle. For even $K = 2s$, the dimension- K circle FFT space*

$$\mathcal{C}_K = \mathbb{F}_p[x]_{<s} + y\mathbb{F}_p[x]_{<s}$$

contains every base-field-real Laurent polynomial supported on $[-s+1, s-1]$. This interior subspace is preserved by rotations of the circle.

Proof. For $0 \leq j < s$, the functions $u^j + u^{-j}$ are polynomials in x of degree j , and $(u^j - u^{-j})/i$ are y times polynomials in x of degree $j-1$. The real combinations of these functions span the indicated interior band. Replacing u by a norm-one scalar times u preserves the band and reality. $\square$

Theorem A.2 (Reciprocal-reserve real circle lists). *Let $p \equiv 3 \pmod{4}$ be prime, let $n = MB$ divide $p+1$, and suppose $M \geq 4$ and $B \geq 2$ are powers of two. Choose an even integer d with $0 < d < M$, and an integer $r \geq 0$ such that*

$$h = d + 2r + 2 < M.$$

Set $K = dB$ and $s = K/2$. On $D = \mu_n$, and on any rotation of this domain, there exists an $\mathbb{F}_p$ -valued received word with at least

$$L = \left\lceil \frac{\binom{M-1}{h-1}}{Mp^{2r}} \right\rceil$$

distinct codewords in the exact space $\mathcal{C}_K$ of Lemma A.1, each agreeing with the received word at at least $hB - 2$ positions. The rate and gap are exactly

$$\rho = \frac{d}{M}, \quad \eta = \frac{2r+2}{M} - \frac{2}{n}, \quad \theta = 1 - \rho - \eta.$$

Every witness lies in the strict interior Laurent band, so no boundary-mode identification is needed to transport the construction to a coset.

Proof. For each h -subset $A \subset \mu_M$ containing 1, put

$$F_A(Y) = \prod_{a \in A} (Y - a) = \sum_{j=0}^h f_j Y^j, \quad c = f_0.$$

The integer h is even. Consequently $c \in \mu_M$, and the norm-one roots give the reciprocal coefficient identity

$$f_{h-j} = c f_j^p. \quad (\text{R1})$$

Partition all $\binom{M-1}{h-1}$ anchored subsets by c and the r leading coefficients $f_{h-1}, \dots, f_{h-r}$. There are at most $M p^{2r}$ classes. In a largest class, containing at least L supports, the bottom r coefficients are fixed as well, by (R1).

Write $T_A(Y) = F_A(Y)/Y^{r+1}$. Let $v(Y)$ consist of its common top $r+1$ and bottom $r+1$ Laurent terms. Thus these are precisely the terms coming from $f_0, \dots, f_r$ and $f_{h-r}, \dots, f_h$. The remainder

$$R_A(Y) = v(Y) - T_A(Y)$$

is an ordinary polynomial of degree at most $d = h - 2r - 2$. Distinct supports give distinct R_A , and the anchor gives $R_A(1) = v(1)$.

For norm-one Y , (R1) implies

$$T_A(Y)^p = c^{-1} Y^{-d} T_A(Y). \quad (\text{R2})$$

The involution in (R2) pairs the retained top and bottom blocks, so v and R_A satisfy the same identity. Choose $\alpha \neq 0$ with $\alpha^p/\alpha = c$, possible because c has norm one. Explicitly, $\alpha = 1 + c^{-1}$ works if $c \neq -1$, and $\alpha = i$ works otherwise. Define

$$V(u) = \alpha u^{-s} v(u^B), \quad S_A(u) = \alpha u^{-s} R_A(u^B).$$

Since $dB = 2s$, (R2) proves that V and S_A are fixed by base-field Frobenius and hence are $\mathbb{F}_p$ -valued on the circle. Both B and s are even, so every S_A has the same value $\beta = \alpha v(1) \in \mathbb{F}_p$ at $u = 1$ and $u = -1$.

Let $y(u) = (u - u^{-1})/(2i)$. Away from these anchors define

$$W(u) = \frac{V(u) - \beta}{y(u)}, \quad P_A(u) = \frac{S_A(u) - \beta}{y(u)}.$$

Give W arbitrary base-field values at the two anchors. For the witness, multiplication of its numerator by u^s gives the polynomial

$$\alpha(R_A(u^B) - v(1)u^s),$$

of degree at most K , divisible by $u^2 - 1$. Therefore P_A is a Laurent polynomial supported on $[-s+1, s-1]$, with removable values at the anchors. Its reality, including at the removable points, follows from the Frobenius identity of the rational expression. Lemma A.1 places it in the exact dimension- K circle space. When K is a power of two, this is the corresponding FFT space.

Off the anchors,

$$W(u) - P_A(u) = \frac{\alpha u^{-s} F_A(u^B)}{u^{B(r+1)} y(u)}.$$

Among the nonanchor coordinates, agreement holds exactly on the selected h fibers. The map $u \mapsto u^B$ has fibers of size B on D , so there are exactly $hB - 2$ nonanchor agreements, and at least this many agreements on the full domain. Distinct F_A give distinct P_A ; a nonzero difference in the interior band, after clearing its Laurent monomial, has degree at most $K - 2 < n$, so the codewords are distinct on D . Finally, rotation preserves the real interior band and all agreement counts. $\square$

A.1.2 A numerical example over a Mersenne prime

Take $p = 2^{31} - 1$ and $n = 2^{26}$. For $M = 2048$, $r = 32$, and $h = 1090$, Theorem A.2 gives the exact integer lower bound

$$L = \left\lceil \frac{\binom{2047}{1089}}{2048(2^{31} - 1)^{64}} \right\rceil = 1,199,728,024,606.$$

For comparison, the same theorem with no reserved leading coefficients has $M = 32$, $r = 0$, $h = 18$, and $L = 8,286,954$. Consider the candidate finite list-size bound $2^{H_2(\rho)/\eta}$. At rate $1/2$, its base-two logarithm is $1/\eta$. The following table compares that bound with the two lower bounds.

M	r	h	$\log_2 L$	$1/\eta$	$\log_2 \text{ratio}$
32	0	18	22.98241048	16.00000763	6.98240285
2048	32	1090	40.12584453	31.03033173	9.09551280

The last column is the base-two logarithm of the ratio between the proved list-size lower bound and the candidate upper bound. The pigeonhole argument establishes the existence of the large coefficient class; it does not require enumerating that class.

A.1.3 Strict below-Elias certificate

For the second row, the gap is

$$\eta = \frac{33}{1024} - \frac{2}{2^{26}} = 0.0322265326976776123046875.$$

It is slightly smaller than $1/\log_2 p$, so that convenient sufficient condition must not be used. Direct high-precision evaluation instead gives

$$H_p(1/2 - \eta) = 0.49993479960319703654 \dots < 1/2.$$

The conclusion also has an exact integer/rational certificate. Direct integer arithmetic verifies $p^{100} > 2^{3099}$, hence $\log_2 p > 3099/100$. The binary entropy satisfies $H_2(1/2 - \eta) \leq 1 - 2\eta^2$: its second derivative is at most -4 , and its derivative vanishes at $1/2$. Direct rational arithmetic gives

$$\eta \frac{3099}{100} > 1 - 2\eta^2.$$

Consequently

$$H_p(1/2 - \eta) < (1/2 - \eta) + \frac{H_2(1/2 - \eta)}{\log_2 p} < 1/2.$$

This radius is strictly below characteristic- p Elias. Independently of rounded logarithms, the exact checks $L^8 > 2^{321}$ and $1/\eta < 31031/1000$ certify a list excess greater than 9.094 bits.

A.1.4 A small exact-arithmetic example

An independent implementation over $\mathbb{F}_{31}[i]/(i^2 + 1)$ uses $n = 32$, $M = 16$, $B = 2$, $r = 1$, and $h = 12$. It enumerates all 1,365 anchored supports, finding 1,336 coefficient classes and a largest class of size three. For every member of that class it verifies all reciprocal coefficient identities, real-valued centering, both removable anchors, exact division by $u^2 - 1$, the strict Laurent band $[-7, 7]$, distinctness, and agreement at exactly 22 non-anchor domain positions. This small example checks the algebraic construction; its parameters are not claimed to violate the numerical candidate bound.

The finite-field checks use exact field arithmetic and integers; the displayed logarithms use 100-digit decimal arithmetic. Negative controls with $(M, r) = (512, 7)$ and $(1024, 15)$ have larger counts but are above Elias for this list construction and therefore do not give examples below Elias for this list construction.

A.2 Affine families from simultaneous coefficient constraints

We now seek a different phenomenon: many scalar combinations of two received words have nearby codewords, even though the two original words have no nearby codewords on a sufficiently large common set of positions. Such a scalar-parameterized affine family is also called a *pencil*.

Throughout this section, $p \equiv 3 \pmod{4}$ is prime, $F_2 = \mathbb{F}_{p^2} = \mathbb{F}_p(i)$ with $i^2 = -1$, and $E = \mathbb{F}_{p^4}$. A bar denotes the nontrivial automorphism of $F_2/\mathbb{F}_p$. The norm-one group

$$U = \{u \in F_2^\times : u\bar{u} = 1\}$$

has order $p + 1$. For each $n \mid p + 1$, write μ_n for its unique subgroup of order n . We use the split coordinate $u = x + iy$ on the circle; then $x = (u + u^{-1})/2$ and $y = (u - u^{-1})/(2i)$.

For $m \geq 1$, define the Laurent spaces

$$\mathcal{L}_m^- = \text{span}_E\{u^j : -(m-1) \leq j \leq m-1\}, \quad \mathcal{L}_m^+ = \text{span}_E\{u^j : -m \leq j \leq m\}.$$

If $D \subseteq U$, evaluation of either space on D will be denoted by the same symbol when the domain is unambiguous. We consider any code $C \subseteq E^D$ satisfying

$$\mathcal{L}_m^- \subseteq C \subseteq \mathcal{L}_m^+. \tag{36}$$

The dimension- $2m$ circle FFT space satisfies this condition, as shown in Section A.4. The constructions below place every witness in the smaller space, and rule out correlated agreement even in the larger one.

For words $f, g \in E^D$, let $\text{agr}_D(f, g)$ be the number of coordinates on which they agree. A pair (f_0, f_1) has correlated agreement at threshold T with C if there exist $P_0, P_1 \in C$ agreeing with the two words simultaneously on at least T coordinates. When no such pair exists, a scalar $z \in E$ is *globally bad at threshold T* if

$$\text{agr}_D(f_0 + zf_1, P_z) \geq T$$

for some $P_z \in C$. A *bank* is a set of distinct such scalars, together with one witness for each scalar. Here the bank records only which scalars have nearby codewords; it does not require those codewords to depend affinely on the scalar.

Lemma A.3 (Global failure and line-decodability). *Suppose (f_0, f_1) has no correlated agreement at threshold T on a domain of length n . Select a witness P_z with at least T agreements for each scalar in a bank. No affine line of codewords contains more than $n - T + 1$ of the selected pairs (z, P_z) . In particular, it contains at most n such pairs.*

Proof. Suppose $P_z = A + zB$ for q distinct selected scalars. Let a be the number of coordinates at which both $f_0 = A$ and $f_1 = B$. At every other coordinate, the affine equality $f_0 + zf_1 = A + zB$ can hold for at most one scalar. Hence

$$qT \leq qa + n - a.$$

Since $a \leq T - 1$, this gives $q(T - a) \leq n - a$, and therefore $q \leq n - T + 1$. $\square$

Consequently, a bank larger than a proposed line-decodability threshold contradicts the definition with concurrency number n . This observation will connect the explicit pencils to the finite threshold discussed below.

Theorem A.4 (Simultaneous coefficient reserve pencil). *Let $n = MB \mid p + 1$, where $B \geq 2$ is even. Let $r \geq 0$ and $2r + 4 \leq h \leq M - 1$, and put*

$$a = 2r + 3, \quad K = (h - a)B = 2m, \quad s = m - 1, \quad T = hB - 2.$$

Assume $K + B < T \leq n$, and let C be any code on $D = \mu_n$ satisfying (36). Define

$$N = \left\lceil \frac{\binom{M-1}{h-1}}{Mp^{2r}} \right\rceil, \quad Q = p^4 - p^2, \quad d = h - 2r - 2. \quad (37)$$

There exist words $\gamma_0, \gamma_1 \in E^D$ which have no correlated agreement at threshold T with C , and which have a bank of size at least

$$\left\lceil \frac{NQ}{Q + (d - 1)(N - 1)} \right\rceil. \quad (38)$$

Each selected witness lies in $\mathcal{L}_m^-$, and agrees on at least T coordinates outside $\{1, -1\}$. The same conclusion holds on any rotated domain $g\mu_n \subseteq U$.

Proof. Put $G = \mu_M$. For every h -element subset $A \subseteq G$ containing 1, write

$$F_A(Y) = \prod_{a \in A} (Y - a) = \sum_{j=0}^h f_{A,j} Y^j, \quad f_{A,h} = 1.$$

Partition these subsets by $c = f_{A,0}$ and the r coefficients $f_{A,h-1}, \dots, f_{A,h-r}$. The constant coefficient has at most M possible values, and each other coefficient lies in F_2 , so there are at most Mp^{2r} classes. Fix a class of size at least N , and retain exactly N distinct subsets in it.

The norm-one condition on the roots gives the identity

$$F_A(Y) = cY^h \overline{F_A(1/Y)}, \quad f_{A,j} = c \overline{f_{A,h-j}}. \quad (39)$$

Thus the bottom r coefficients are common to the class as well. In $F_A(Y)/Y^{r+1}$, let $v(Y)$ be the sum of the common terms with indices $0 \leq j \leq r$ and $h - r \leq j \leq h$. Then

$$R_A(Y) := v(Y) - F_A(Y)/Y^{r+1}$$

is an ordinary polynomial of degree at most $d = h - 2r - 2$. The polynomials R_A are distinct, and all satisfy $R_A(1) = v(1)$.

Choose a pole $\tau \in E \setminus F_2$, to be specified below. Define

$$f_0(u) = \frac{v(u^B)}{u^B - \tau}, \quad f_1(u) = \frac{1}{u^B - \tau}, \quad \gamma_j(u) = u^{-s} \frac{f_j(u) - f_j(1)}{u^2 - 1}.$$

At $u = \pm 1$ use the removable values. There are no other denominator zeros on D : every u^B lies in F_2 , while $\tau \notin F_2$. For each A , put

$$z_A = -R_A(\tau), \quad Q_A(Y) = \frac{R_A(Y) - R_A(\tau)}{Y - \tau}, \quad P_A(u) = u^{-s} \frac{Q_A(u^B) - Q_A(1)}{u^2 - 1}.$$

The polynomial Q_A has degree at most $d - 1 = h - a = K/B$. Since B is even, its composed numerator vanishes at $u = \pm 1$. Dividing by $u^2 - 1$ leaves degree at most $K - 2$, and multiplying by u^{-s} places P_A in the Laurent band $[-s, s]$. Thus $P_A \in \mathcal{L}_m^- \subseteq C$. At every nonanchor point with $u^B \in A$, the identity $v(u^B) = R_A(u^B)$ holds; it also holds at the subtracted anchor value. Therefore $\gamma_0(u) + z_A \gamma_1(u) = P_A(u)$ at at least $hB - 2 = T$ coordinates outside $\{1, -1\}$.

To rule out correlated agreement, it suffices to rule out sufficient agreement with γ_1 alone. Put

$$S_B(u) = 1 + u^2 + \dots + u^{B-2}.$$

Cancellation yields

$$\gamma_1(u) = -\frac{u^{-s} S_B(u)}{(1 - \tau)(u^B - \tau)}. \quad (40)$$

For $P \in \mathcal{L}_m^+$, the function $H(u) = u^m P(u)$ is an ordinary polynomial of degree at most K . Every agreement with γ_1 is a root of

$$(1 - \tau)(u^B - \tau)H(u) + uS_B(u). \quad (41)$$

This polynomial is nonzero: the second summand has degree $B - 1$, whereas the first has degree at least B when $H \neq 0$. Its degree is at most $K + B$. Hence no word in the entire outer-band code agrees with γ_1 on more than $K + B < T$ coordinates. Formula (40) also holds at the removable anchor values.

It remains to select a pole with sufficiently many distinct labels. For $A \neq A'$, the nonzero polynomial $R_A - R_{A'}$ has degree at most d and has the root $1 \in F_2$. It therefore has at most $d - 1$ roots outside F_2 . Averaging over the Q possible poles, some pole has at most $(d - 1)N(N - 1)/(2Q)$ colliding unordered pairs. If the multiplicities of its distinct labels are $b_1, \dots, b_t$, then

$$\sum b_i = N, \quad \sum b_i^2 \leq N + (d - 1)N(N - 1)/Q.$$

Cauchy–Schwarz gives $t \geq NQ/[Q + (d - 1)(N - 1)]$, proving (38). Finally, substituting u/g transports all words to $g\mu_n$. Rotation preserves both Laurent bands and all agreement counts. $\square$

Corollary A.5 (Poles in the quadratic coefficient field). *Under the hypotheses of Theorem A.4, the defining words, selected witnesses, and scalar labels can all lie in $F_2 = \mathbb{F}_{p^2}$. With*

$$Q_2 = p^2 - M,$$

the guaranteed number of distinct labels is

$$\left\lceil \frac{NQ_2}{Q_2 + (d - 1)(N - 1)} \right\rceil.$$

The pair has no correlated agreement at threshold T , even against outer-band codewords over an extension of F_2 .

Proof. Choose the pole from $F_2 \setminus \mu_M$. The denominators only require $u^B \neq \tau$ and $1 \neq \tau$, so every pole in this set is allowed, including zero. All coefficients and values in the pencil formulas then belong to F_2 . A difference $R_A - R_{A'}$ still has the root 1, which is excluded from the pole set, and therefore has at most $d - 1$ roots among its Q_2 elements. The same collision average proves the count with Q_2 in place of Q . The nonzero polynomial in (41) has degree at most $K + B < T$ over any ambient extension. Its root bound, and hence the absence of correlated agreement, are unchanged. $\square$

Remark A.6 (Existence and explicitness). The formulas specify a concrete family of pencils. The large coefficient class and a sufficiently good pole are guaranteed by counting; a large coefficient class or a suitable pole need not have been enumerated to prove the lower bound. We distinguish this existence-certified bank from a bank whose labels and witnesses have all been listed explicitly.

A.2.1 A numerical affine-family example over a Mersenne prime

Let $p = 2^{31} - 1$ and take

$$M = 1024, \quad r = 15, \quad h = 545, \quad n = 2^{26}, \quad B = 2^{16}.$$

Then $a = 33$, $K = n/2$, $d = 513$, and

$$T = 545B - 2, \quad \theta = 1 - T/n = \frac{479}{1024} + \frac{2}{n}, \quad \eta = \frac{33}{1024} - \frac{2}{n}.$$

The theorem gives the integer bank lower bound

$$30,608,283,697,878,843,892,998, \tag{42}$$

whose base-two logarithm exceeds 74.6963402382.

At exact rate $1/2$, the finite prescription $n2^{H_2(1/2)/\eta}$ has logarithm less than 57.0303317264. The excess in the logarithm is greater than 17.6660085118 bits.

For clarity, define the characteristic- p Elias radius $r_E(\rho)$ by $H_p(r_E) = 1 - \rho$, with

$$H_p(t) = t \log_p(p-1) - t \log_p t - (1-t) \log_p(1-t).$$

The above distance is strictly below $r_E(1/2)$. Equivalently, the quantity

$$(1/2 - \theta) \log_2 p - H_2(\theta) - \theta \log_2(1 - 1/p)$$

is positive; its value exceeds 0.0020212123. The exact count and high-precision entropy comparison are reproduced in the accompanying `counterexample_parameters.py` and its machine-readable output. The probability that a uniform scalar in E belongs to the certified bank is at least the count in (42) divided by p^4 . This probability describes the scalar-combination event for the two words in the theorem.

Remark A.7 (The comparison radius matters). This example contradicts a universal finite threshold prescription at the displayed radius. It does not, by itself, contradict each numerical row of a parameter table that evaluates the prescription at a different radius. For example, substituting $\eta = 1/16$ into $2^{1/\eta}$ gives 2^{16} ; that evaluates the candidate at a different radius from the one used here.

A.2.2 A sharper bound on selected-witness concurrency

The single-word agreement bound in the reserve construction gives a much stronger concurrency conclusion than the generic bound in Lemma A.3.

Lemma A.8 (Incidence bound from a quantitative agreement deficit). *Let $C \subseteq E^D$ be a linear code on n coordinates, and let $0 \leq a_0 < T \leq n$ be integers. Suppose either the direction f_1 , or some point $f_0 + z_0 f_1$, agrees with every codeword in at most a_0 coordinates. Select codewords p_z at distinct scalars with $\text{agr}_D(f_0 + z f_1, p_z) \geq T$. Every parametrized codeword graph $Q(z) = \sum_{j=0}^e z^j P_j$, where $P_j \in C$ and $e \geq 1$, contains at most*

$$\left\lfloor e \frac{n - a_0}{T - a_0} \right\rfloor$$

of the selected pairs (z, p_z) . In particular, take $e = 1$ for affine codeword graphs.

Proof. Let a count coordinates where $f_0 + zf_1 = Q(z)$ identically as a polynomial in z . If the hypothesis concerns f_1 , comparison of linear coefficients gives $a \leq a_0$. If it concerns a point at z_0 , evaluation there gives the same bound because $Q(z_0) \in C$. At every other coordinate the equality has at most e roots. If the graph contains q selected pairs, counting agreements gives $qT \leq qa + e(n - a)$, so

$$q \leq e \frac{n - a}{T - a} = e \left(1 + \frac{n - T}{T - a} \right) \leq e \frac{n - a_0}{T - a_0}.$$

Taking the integer part proves the claim. $\square$

Corollary A.9 (At most fifteen selected witnesses on one graph). *Every parametrized affine codeword graph contains at most 15 selected pairs from the M31 bank in (42). The same upper bound holds after the two-anchor puncturing and stereographic scaling of Theorem A.10.*

Proof. The root certificate (41) bounds agreement of γ_1 with every outer-band codeword by $a_0 = K + B$. In the displayed M31 instance, $n = 1024B$, $K = 512B$, $T = 545B - 2$, and $B = 65536$. Lemma A.8 therefore gives

$$q \leq \left\lfloor \frac{511B}{32B - 2} \right\rfloor = 15.$$

Indeed, subtracting $15(32B - 2)$ from the numerator gives $31B + 30 > 0$, whereas $16(32B - 2) - 511B = B - 32 > 0$.

Puncturing removes only coordinates outside the certified T agreements, and cannot increase the maximum agreement with γ_1 beyond a_0 . The nonzero stereographic scaling preserves equalities; every transformed codeword lifts to the interior band, which is contained in the outer band. Thus the transformed code of length $n - 2$ satisfies

$$q \leq \left\lfloor \frac{511B - 2}{32B - 2} \right\rfloor = 15$$

by the same lemma. This is an upper bound on selected-witness concurrency, not a claim that a graph attaining fifteen witnesses has been exhibited. $\square$

A.3 Transfer to prime-domain Reed–Solomon codes

Theorem A.10 (Two-anchor puncturing and stereographic scaling). *Let $p \equiv 3 \pmod{4}$, let E contain $\mathbb{F}_{p^2}$, and let $D = \mu_n$, where $n \mid p + 1$ is even and $K = 2m < n$. Suppose words γ_0, γ_1 and a bank of distinct scalars have witnesses in $\mathcal{L}_m^-$, each with at least T agreements outside $\{1, -1\}$. Suppose the pair has no correlated agreement at threshold T with $\mathcal{L}_m^-$. Then there is a set $D' \subseteq \mathbb{F}_p$ of size $n - 2$ and a pair of words over E with the same bank and the same agreement threshold for the ordinary Reed–Solomon code*

$$\text{RS}_{<K-1}(E, D').$$

This code has exact rate $(K - 1)/(n - 2)$, and the transformed pair has no correlated agreement at threshold T .

Proof. Delete the two anchors. Writing $u = x + iy$, map each retained point to

$$t = \frac{y}{1 + x} \in \mathbb{F}_p.$$

Stereographic parametrization is injective away from $(-1, 0)$, which was deleted, and has inverse

$$x = \frac{1 - t^2}{1 + t^2}, \quad y = \frac{2t}{1 + t^2}, \quad u = \frac{1 + it}{1 - it}.$$

The resulting set D' has $n - 2$ distinct finite points. Since $p \equiv 3 \pmod{4}$, the function $w(t) = (1 + t^2)^{m-1}$ is nonzero on $\mathbb{F}_p$. For every $j \in [-(m-1), m-1]$,

$$w(t)u^j = (1 + it)^{m-1+j}(1 - it)^{m-1-j}. \quad (43)$$

These are polynomials of degree at most $2m - 2 = K - 2$, and they form a basis of that polynomial space. Indeed they are the images of the homogeneous monomial basis of binary forms of degree $2m - 2$ under the invertible change of variables $(X, Y) \mapsto (X + iY, X - iY)$. This proof is valid in positive characteristic and does not require any binomial coefficient to be nonzero.

Multiply each retained coordinate of each word and witness by $w(t)$. Equation (43) shows that the witness code becomes exactly $\text{RS}_{<K-1}(E, D')$. All equality tests and scalar labels are preserved by nonzero coordinate scaling, so every witness keeps its T guaranteed agreements. Any correlated agreement of the transformed pair would lift to two elements of $\mathcal{L}_m^-$ agreeing on the same original coordinates, a contradiction. Finally,

$$\frac{\dim \text{RS}_{<K-1}(E, D')}{|D'|} = \frac{K - 1}{n - 2}.$$

This equals $1/2$ when $K = n/2$. □

Corollary A.11 (Transfer of the reserve bank). *In Theorem A.4, suppose $h = M/2 + a$, where $a = 2r + 3$, so that $K = n/2$. The same bank exists for an ordinary Reed–Solomon code with a prime-field domain, extension alphabet E , exact rate $1/2$, length $n' = n - 2$, and gap*

$$\eta' = \frac{T}{n - 2} - \frac{1}{2} = \frac{aB - 1}{n - 2}.$$

Together with Lemma A.3, a bank larger than $(n - 2)2^{1/\eta'}$ contradicts the corresponding finite line-decodability threshold with concurrency number $n - 2$.

The M31 instance above satisfies this inequality as well. It is therefore an example under the literal prime-field-domain and extension-alphabet quantifiers of the finite prescription, rather than only a circle-code analogue. The transfer uses a change of coordinates and nonzero coordinate scaling. It therefore preserves equality sets and scalar labels exactly.

The accompanying exact-arithmetic fixture checks all 189 archived circle witnesses over $\mathbb{F}_{81914}$ after this transformation. They become degree-at-most-14 polynomials on 30 distinct prime-field coordinates, retain 189 distinct labels, and each has at least 20 agreements. The no-correlated-agreement argument uses the algebraic bound $18 < 20$; it is not represented as an exhaustive enumeration of all codewords.

A.4 Circle polynomial spaces and coset domains

For completeness, we verify two algebraic facts used by the constructions: the relation between the circle tensor basis and Laurent bands, and the relation between a full coset and its two half-cosets. Neither fact depends on an ordering of evaluation coordinates.

Proposition A.12 (The circle tensor basis). *For $K = 2m$ a power of two, the circle FFT tensor basis spans*

$$C_K = \{A(x) + yB(x) : \deg A < m, \deg B < m\}.$$

After scalar extension to E it satisfies $\mathcal{L}_m^- \subseteq C_K \subseteq \mathcal{L}_m^+$.

Proof. The tensor factors are $y, x, \pi(x), \pi^2(x), \dots$, with $\pi(x) = 2x^2 - 1$. Products not using y have distinct leading degrees $0, \dots, m-1$ and nonzero leading coefficients, so they form a basis of the polynomials in x of degree less than m . Multiplication by y gives the second summand. Substitution of $x = (u + u^{-1})/2$ and $y = (u - u^{-1})/(2i)$ proves the outer-band containment. The functions $1, x, \dots, x^{m-1}, y, xy, \dots, x^{m-2}y$ span the inner band. The same basis remains a basis after extending coefficients from $\mathbb{F}_p$ to E . $\square$

Proposition A.13 (Canonical domains). *Suppose n is even and $2n \mid p+1$. If $g \in U$ has order $2n$, the canonical coset $g\mu_n$ is the disjoint union*

$$\{g^{1+4j} : 0 \leq j < n/2\} \cup \{g^{-(1+4j)} : 0 \leq j < n/2\}.$$

Thus describing a canonical domain as two conjugate half-cosets changes only the ordering of its points.

Proof. In additive point-index notation modulo $2n$, the full odd coset is $\{1 + 2j : 0 \leq j < n\}$. Its half-coset representation is $\{1 + 4j : 0 \leq j < n/2\}$ and its negatives. These two half-cosets are disjoint and their union is exactly the full odd coset. Passing to the split multiplicative coordinate gives $g\mu_n$. $\square$

Consequently, a construction on a full multiplicative coset also applies when its coordinates are listed using the two half-cosets. Rotation preserves both Laurent bands.

Remark A.14 (Sizes and target spaces). The M31 circle has order $p+1 = 2^{31}$; a canonical coset uses a subgroup of twice its own size and has maximum length 2^{30} . Here n denotes evaluation length. For example, the rate-1/2 code at length 2^{26} has dimension 2^{25} . The full outer-band circle code has dimension $K+1$, rather than K ; the proof excluding correlated agreement still covers it, but its rate is $1/2 + 1/n$ and any comparison naming that code must use that rate.

A.5 What the finite counterexamples establish

The list construction and the affine-family construction address two different finite bounds. At rate $1/2$, the first gives a list larger than $2^{1/\eta}$. The second gives more than $n2^{1/\eta}$ exceptional scalars, with no correlated agreement at the prescribed threshold. The Reed–Solomon transfer shows that the latter phenomenon also occurs with a prime-field evaluation domain and an extension-field alphabet.

One must distinguish such finite statements from a proposed asymptotic threshold of the form

$$a = \ell(\theta)n + o(n), \quad \ell(\theta) \leq c_1 2^{c_2 H_2(\rho)/\eta}, \quad \theta = 1 - \rho - \eta.$$

The displayed numerical examples contradict the remainder-free choice $c_1 = c_2 = 1$ at their stated fields, lengths, and radii. A single finite example does not determine the unspecified quantifiers in $o(n)$ and does not refute every choice of constants c_1, c_2 .

For an asymptotic claim, the field, domain length, and gap must be allowed to vary in a stated way. In particular, an ordinary Reed–Solomon code with a prime-field evaluation domain has length at most p . Holding p fixed while allowing an arbitrary prime-dependent remainder leaves no unbounded sequence of such lengths. The separate fixed-gap lifting construction in this manuscript gives an increasing-length family for the quantitative coefficient: its discrepancy is linear and cannot be absorbed by a sublinear remainder for that family. That selected-graph construction has correlated agreement and is distinct from the globally non-correlated affine families constructed here. Neither argument refutes a line bound whose coefficient validly bounds the actual list size.

A.6 The common-core form of the circle pencil

The anchored construction in Theorem A.4 can be written in the fixed-core form familiar from the Proximity Prize's orbit pencil [22]. This is a change of presentation, with the same witnesses and agreement counts.

Use the theorem's notation, and fix one support A_0 . For each support $A = \{1\} \cup U$, write

$$F_A(Y) = (Y - 1)G_U(Y), \quad D_U(Y) = \frac{G_{U_0}(Y) - G_U(Y)}{Y^{r+1}}.$$

The common leading and trailing coefficients imply $\deg D_U \leq h - 2r - 3 = K/B$. Since $R_A - R_{A_0} = (Y - 1)D_U$, the old scalar labels and the new labels $t_U = D_U(\tau)$ satisfy

$$z_A - z_{A_0} = (1 - \tau)t_U.$$

Set $Q_U = (D_U - t_U)/(Y - \tau)$ and $S_B(u) = (u^B - 1)/(u^2 - 1)$. Then the new pencil and witnesses are

$$g_0 = \frac{u^{-s}S_B(u)G_{U_0}(u^B)}{u^{B(r+1)}(u^B - \tau)}, \quad g_1 = -\frac{u^{-s}S_B(u)}{u^B - \tau}, \quad W_U = u^{-s}S_B(u)Q_U(u^B).$$

Their difference is

$$g_0 + t_U g_1 - W_U = \frac{u^{-s}S_B(u)G_U(u^B)}{u^{B(r+1)}(u^B - \tau)}.$$

The zeros consist of the $h - 1$ varying fibers and the fixed core of $B - 2$ points with $u^B = 1$ and $u \neq \pm 1$. The agreement count is $(h - 1)B + B - 2 = hB - 2$. Also $\deg(S_B Q_U(u^B)) \leq K - 2$, so the centered witnesses lie in the required interior Laurent band.

Direct subtraction in the original formulas gives

$$g_0 = \gamma_0 + z_{A_0}\gamma_1 - P_{A_0}, \quad g_1 = (1 - \tau)\gamma_1, \quad W_U = P_A - P_{A_0}.$$

Thus an invertible affine change of scalar and a codeword translation identify the two presentations. The number of labels and the failure of correlated agreement are unchanged. In particular, introducing this common-core notation does not itself improve the existing numerical bound.

B A two-fiber pencil in raw quotient form

This section addresses one algebraic compatibility condition with the conjugate-line quotient, using the unweighted formula in² the source. It does not construct an accepted Stwo transcript. In particular, the claimed openings below are zero and generally false.

Write $p = 2^{31} - 1$, $H = \mathbb{F}_p[i]$, $E = H[w]$, where $i^2 = -1$ and $w^2 = \kappa = 2 + i$. Conjugation over H sends w to $-w$. Let $n = 2M \mid p + 1$, $K = 2m \mid p + 1$, with m even and $m + 3 \leq M - 1$. The circle code C_K contains every Laurent polynomial with exponents between $1 - m$ and $m - 1$, and is contained in the band $[-m, m]$.

Proposition B.1 (Fixed tables realizing a two-fiber pencil). *There are two $\mathbb{F}_p$ -valued tables on μ_n , fixed independently of the sample, and an out-of-domain circle sample over E , such that their normalized raw quotients r_0, r_1 with claimed values zero have at least*

$$J = \left\lceil \frac{NQ}{Q + m(N - 1)} \right\rceil, \quad N = \left\lceil \frac{\binom{M-1}{m+2}}{M} \right\rceil, \quad Q = \frac{p^2 - 1}{2},$$

²Source checked at Stwo revision 826591c6c371376810ca8213b5812d5daf6d5092, `core/constraints.rs` and `core/pcs/quotients.rs`.

distinct parameters α for which $r_0 + \alpha r_1$ agrees with C_K on at least $K + 4$ coordinates. The direction r_1 agrees with any codeword on at most $K + 2$ coordinates, so the pencil has no common agreement on $K + 4$ coordinates.

Proof. Choose a class of $(m + 2)$ -subsets $U \subset \mu_M \setminus \{1\}$ with common constant coefficient c in $G_U(Y) = \prod_{a \in U} (Y - a)$. There are at least N members. Fix U_0 , and set

$$D_U = (G_{U_0} - G_U)/Y, \quad \deg D_U \leq m.$$

These polynomials are distinct. Average collisions of their evaluations over the Q nonsquares $\tau \in H^*$. Every difference has at most m roots, so Cauchy–Schwarz gives a nonsquare τ with at least J distinct labels $t_U = D_U(\tau)$. Every element of μ_M is a square in H , so these poles avoid the domain.

Put $Y = u^2$ and

$$g_0 = \frac{u^{1-m} G_{U_0}(u^2)}{u^2(u^2 - \tau)}, \quad g_1 = -\frac{u^{1-m}}{u^2 - \tau}, \quad W_U = u^{1-m} \frac{D_U(u^2) - t_U}{u^2 - \tau}.$$

The last expression lies in the strict Laurent band. The difference $g_0 + t_U g_1 - W_U$ is $u^{1-m} G_U(u^2)/[u^2(u^2 - \tau)]$, giving $2(m + 2)$ agreements. For any codeword C , multiplying $g_1 - C$ by $u^m(u^2 - \tau)$ gives a nonzero polynomial of degree at most $K + 2$. Nonzeroness follows by evaluating at a root of $u^2 - \tau$, where the remaining term is $-u$. This proves the direction bound.

Choose $a_0 \neq 0$ with $a_0^p/a_0 = c$: take $a_0 = 1 + c^{-1}$ if $c \neq -1$, and $a_0 = i$ otherwise. The two fixed tables are

$$f_0(u) = a_0 u^{-(m+2)} G_{U_0}(u^2), \quad f_1(u) = u^m + u^{-m}.$$

Conjugating coefficients and using $u^p = u^{-1}$ proves that both are $\mathbb{F}_p$ -valued. Neither depends on τ .

Because κ and τ are nonsquares in H , choose $b \in H^*$ with $\kappa b^2 = \tau$. Set $a = wb$ and

$$z_x = (a + a^{-1})/2, \quad z_y = (a - a^{-1})/(2i).$$

These coordinates satisfy the circle equation and are anti-invariant under conjugation over H . Writing $z_x = wX_1, z_y = wY_1$, we have $Y_1 \neq 0$ and

$$U_z = (x - (X_1/Y_1)y)/2 = L_0 u^{-1}(u^2 - \tau), \quad L_0 = \frac{1}{2(1 - \tau)}.$$

For claimed opening zero, the raw conjugate-line formula is $q = (\bar{z}_y - z_y)f/[(-x)Y_1 - (-y)X_1]$, hence $q/w = f/U_z$. Thus, with $r_j = q_j/w$, direct multiplication gives

$$r_0 = \lambda_0 g_0, \quad r_1 = \lambda_1 g_1 + P_\tau,$$

where

$$\lambda_0 = a_0/L_0, \quad \lambda_1 = -(\tau^m + 1)/L_0, \quad P_\tau = \frac{u^{1-m}}{L_0} \sum_{j=0}^{m-1} \tau^{m-1-j} u^{2j} \in C_K.$$

Here $\lambda_1 \neq 0$: otherwise $\tau^{2m} = 1$, which would make τ a square in H , since $2m \mid p + 1$. For $\alpha_U = \lambda_0 t_U/\lambda_1$, the witness is $\lambda_0 W_U + \alpha_U P_\tau$. The scalar changes and codeword shift preserve distinct labels, agreement counts, and the direction bound. Multiplication by w also preserves membership in the E -linear code. $\square$

Finite instance and remaining protocol gap. Taking $(n, K, M, m) = (128, 32, 64, 16)$ gives agreement 36/128 and $N = 40,448,653,414,754$, yielding $J = 40,437,303,920,543$. The expression above is an exact lower bound; no enumeration of this bank is claimed. A separate deterministic fixture over $p = 127$, with the same extension polynomial, checks the raw quotient identities and agreement counts for multiple nonsquare poles, holding the two base-field tables fixed throughout. An additional fixture over the actual M31 field checks three supports at one nonsquare pole: all 1,536 identities pass, with 36 agreements for each support. This result concerns a specially chosen sample family. It does not establish its probability under Stwo’s sampler, map the actual challenge power schedule or its precise evaluation cosets, or address interpolation and extension of the raw quotient subdomain. Nor does it show that the tables and false openings survive the remaining proof checks. The bounds for larger fibers do not automatically apply to this two-fiber case.

There is a concrete reason to keep the sampler issue separate. Under an idealized uniform sample from the E -rational circle and an independent uniform batching parameter in E , this particular family is rare. The map $a \mapsto ((a + a^{-1})/2, (a - a^{-1})/(2i))$ bijects E^* with that circle. Samples used here have $a \in wH^*$, a fraction $1/(p^2 + 1)$ of the circle. All constructed batching labels lie in H , a fraction $1/p^2$ of E . Thus the event consisting of this sample family and these labels has probability at most $1/[p^2(p^2 + 1)]$, approximately 2^{-124} , even before limiting the label bank. This is an upper bound for the constructed event in the stated idealized model, not a soundness bound for Stwo. In particular, raw quotient compatibility by itself does not establish a practical attack or a reduction below a 96-bit security target.

C From large real lists to raw quotient pencils at typical samples

Sampling outside the evaluation domain to constrain candidate codewords is the established DEEP technique [9]. Here we study large explicit families under the conjugate-line quotient, fixed claims, and prescribed scalar schedules.

The special sample restriction in Appendix B can be removed. We first prove a bound with sample-dependent claimed openings; Theorem C.2 then fixes the claims before the sample and improves the joint probability bound. Sample-dependent claims are consistent with the inspected source order: column commitments precede the sample, while claimed values are mixed before the batching challenge. Both results concern raw quotient agreement; neither proves full proof acceptance.

Theorem C.1 (A list-to-quotient transfer). *Let $p = 2^{31} - 1$, $H = \mathbb{F}_{p^2}$ and $E = \mathbb{F}_{p^4}$, with the tower and normalization of Appendix B. Let $\mathcal{D}$ be a set of n base-field circle points and $K = 2m$. Suppose a fixed base-field word f has N distinct real Laurent polynomials F_j , supported in $[-m, m]$, each agreeing with f on at least $T > K + 2$ points of $\mathcal{D}$. Set*

$$S = (p^2 - 1)^2, \quad D_* = \left\lceil \frac{NS}{S + 2K(N - 1)} \right\rceil, \quad J = \left\lceil \frac{D_*}{p^2} \right\rceil.$$

For at least half of the S admissible circle samples described below, there is a claimed value v such that the two fixed tables $(f, 0)$, with claimed values $(v, 1)$, have normalized raw quotients r_0, r_1 for which at least J distinct $\alpha \in H$ satisfy

$$r_0 + \alpha r_1 \text{ agrees with } C_K(H) \text{ on at least } T \text{ coordinates.}$$

The direction r_1 has at most $K + 2$ agreements with any codeword, so these pencils have no common agreement on T coordinates. The claims are chosen after the sample and before α .

Proof. Parameterize the E -circle by $z_x = (a + a^{-1})/2$, $z_y = (a - a^{-1})/(2i)$, $a \in E^*$. Call a admissible when $a \notin H$ and $z_y \notin H$. For $a \notin H$, the second condition fails exactly when $a\bar{a} = -1$. The sets H^* and this norm fiber have sizes $p^2 - 1$ and $p^2 + 1$, and intersect at $\pm i$. Their complement therefore has size S .

For $j \neq k$, multiplying $F_j - F_k$ by u^m gives a nonzero polynomial of degree at most K . Thus at most K admissible values of a cause $F_j(z) = F_k(z)$. The expected number of unordered collisions is at most $\binom{N}{2}K/S$. By Markov's inequality, at least half the admissible samples have at most $N(N-1)K/S$ collisions. Cauchy-Schwarz then gives at least D_* distinct values among $F_j(z)$. One of the p^2 additive H -cosets in E contains at least J of them. Write this coset as $v + H$, and retain one index for each distinct value. The labels $\alpha_j = F_j(z) - v$ are distinct and belong to H .

Use the source functions U_z and ℓ_b defined in Appendix B and the preceding note. Explicitly, for $z_x = X_0 + wX_1$, $z_y = Y_0 + wY_1$, and $b = B_0 + wB_1$,

$$U_z = (x - X_0)/2 - (X_1/Y_1)(y - Y_0)/2, \quad \ell_b = B_0 + (B_1/Y_1)(y - Y_0).$$

The polynomial uU_z has degree two and distinct nonzero roots $a, \bar{a}$. The numerator $F_j - \ell_{F_j(z)}$ vanishes at both roots. Indeed $\ell_b(z) = b$, and conjugating over H supplies the other equality. Division therefore yields

$$P_j = \frac{F_j - \ell_{F_j(z)}}{U_z} \in \text{span}_H\{u^{1-m}, \dots, u^{m-1}\} \subset C_K(H).$$

Since $b \mapsto \ell_b$ is H -linear and $\ell_1 = 1$, we obtain

$$r_0 + \alpha_j r_1 - P_j = \frac{f - \ell_v - \alpha_j}{U_z} - P_j = \frac{f - F_j}{U_z}.$$

No denominator vanishes on the base-field circle, so the claimed agreements follow. Finally $r_1 = -1/U_z$. For any codeword P in the outer Laurent band, clearing $r_1 - P$ by $u^m(uU_z)$ gives a polynomial of degree at most $K + 2$. It is nonzero at $u = a$, where its value is $-a^{m+1}$. This proves the direction bound. $\square$

An explicit family and finite count. The construction in Appendix B supplies such a family: put $f = R_{U_0}$ and

$$R_U = a_0 u^{-(m+2)} G_U(u^2), \quad F_U = f - R_U.$$

The common leading and constant coefficients give support $F_U \subseteq [-m, m]$, while R_U has $K + 4$ roots on the domain. For $(n, K) = (512, 128)$, the exact counts give

$$N = \left\lceil \binom{255}{66} / 256 \right\rceil, \quad J \geq 18,014,398,492,704,768.$$

On every good sample the certified raw proximity event therefore has probability at least $J/p^4 \approx 2^{-70}$ under a uniform linear batching challenge. The good-sample fraction costs approximately one further bit. This is not a protocol security estimate.

Domains and the source sampler. The transfer theorem applies to arbitrary base-field circle domains. The displayed family transports to a coset $u_0\mu_n$ by replacing u with u/u_0 . Reality and Laurent support are preserved, and its quotient witnesses lie in the rotation-invariant interior band. Stwo's canonical domain is precisely such a coset, with u_0 of order $2n$. The source samples using stereographic coordinates $((1 - t^2)/(1 + t^2), 2t/(1 + t^2))$. For uniform $t \in E$, each admissible circle point has exactly one preimage; the two singular parameters and the omitted point are outside this admissible set. Thus, under independent uniform secure-field draws, the joint sample-and-batching proximity event has probability at least $SJ/(2p^8)$.

Scope and checks. Exact generic-sample fixtures pass over $\mathbb{F}_{127}$ and M31; a separate $\mathbb{F}_{23}$ fixture checks the fixed second claim 1 and distinct labels on additive H -cosets. Section C.6 gives a compact description of the large support family and constructs its fixed table efficiently. An efficient algorithm for finding a favorable claim and its witnesses is still not established. Additional columns, periodicity terms, claim constraints from the AIR, commitment consistency, FRI queries and grinding remain to be analyzed. In particular, neither the proximity probability nor the existence of a nearby codeword establishes an accepted false proof.

C.1 A fixed claim and a stronger joint probability bound

The preceding argument selects a favorable claimed value separately for each sample and discards up to half the admissible samples. Both losses can be avoided when the objective is a joint sample-and-batching bound. The following averaging argument fixes a single claim before either challenge. Its guarantee concerns the total number of successful pairs; it does not promise the same number of labels at every sample.

Theorem C.2 (Fixed-claim list-to-quotient transfer). *Under the hypotheses of the list-to-quotient transfer above, let*

$$A_* = \left\lceil \frac{NS^2}{p^2(S + K(N-1))} \right\rceil, \quad S = (p^2 - 1)^2.$$

There is one value $v_ \in wH$, independent of the sample and batching challenge, such that the fixed tables $(f, 0)$ with fixed claims $(v_*, 1)$ have at least A_* distinct pairs (a, α) with a admissible and $\alpha \in H$, for which*

$$r_{0,a} + \alpha r_{1,a} \text{ agrees with } C_K(H) \text{ in at least } T \text{ coordinates.}$$

For every admissible sample, the direction has at most $K + 2 < T$ agreements with any codeword. Under independent uniform draws $t, \alpha \in E$, with t mapped to the circle by the stereographic sampler, the probability of the certified raw proximity event is at least

$$\frac{A_*}{p^8} \geq \frac{NS^2}{p^{10}(S + K(N-1))}. \quad (44)$$

The two singular sampler parameters are counted as failures in this unconditioned bound.

Proof. Let $\mathcal{A}$ be the set of admissible parameters. For $a \in \mathcal{A}$ and $b \in E$, let

$$m_{a,b} = \#\{j : F_j(z(a)) = b\}, \quad \mathcal{I} = \{(a, b) : m_{a,b} > 0\}.$$

Then $\sum_{a,b} m_{a,b} = NS$. Each unordered pair of distinct Laurent polynomials collides at at most K parameters, by the same degree- K root bound used above. Consequently

$$\sum_{a,b} m_{a,b}^2 = NS + 2 \sum_a \sum_b \binom{m_{a,b}}{2} \leq NS + KN(N-1).$$

Cauchy–Schwarz, applied once to all incidences, gives

$$|\mathcal{I}| \geq \frac{(NS)^2}{NS + KN(N-1)} = \frac{NS^2}{S + K(N-1)}. \quad (45)$$

Fix the p^2 representatives wH for the additive cosets of H in E . Partition $\mathcal{I}$ according to the coset containing b . One fixed coset $v_* + H$ contains at least A_* incidences. For each of these

incidences choose one witness F_j and put $\alpha = b - v_* \in H$. This is an injective map from the retained incidences to pairs (a, α) , even when several polynomials have the same value at one sample. Both v_* and the tables are fixed throughout this count.

For the sample a , write $U_a = U_{z(a)}$ and use the interpolation operator $\ell_{a,b}$ from the preceding proof. The quotient witness is

$$P_{a,j} = \frac{F_j - \ell_{a,b}}{U_a} \in C_K(H), \quad r_{0,a} + \alpha r_{1,a} - P_{a,j} = \frac{f - F_j}{U_a}.$$

Here $r_{0,a} = (f - \ell_{a,v_*})/U_a$ and $r_{1,a} = -1/U_a$. Thus all retained incidences give the required agreements. The degree- $K + 2$ direction argument applies to every admissible sample, regardless of the chosen coset. Finally, every admissible a has exactly one nonsingular stereographic preimage $t \in E$. The A_* distinct pairs therefore occupy A_* of the p^8 equally likely pairs (t, α) , proving the probability claim. $\square$

The same finite instance with fixed claims. For the preceding $(n, K, T) = (512, 128, 132)$ instance,

$$N = \left\lceil \binom{255}{66} / 256 \right\rceil, \quad S = (p^2 - 1)^2, \quad p = 2^{31} - 1.$$

Exact integer arithmetic gives

$$A_* = 766247768292072891947921916124690454684786490150411277, \quad 2^{69} A_* > p^8.$$

The certified joint raw proximity probability is therefore strictly greater than 2^{-69} . The earlier half-sample bound is $SJ/(2p^8)$, approximately 2^{-71} at these same parameters; the improvement is approximately a factor of four. The earlier 2^{-70} figure was conditional on a good sample, so it should not be compared directly with this joint probability.

Quantifiers and limits. This strengthens the timing statement to *there exists one fixed claim, followed by random sample and batching draws*. It does not identify an efficiently computable v_* . Section C.6 constructs the table and describes the support bank implicitly; efficient witness recovery remains open. Individual samples may have fewer labels than the previous per-good-sample bound, or no certified label. The false claim 1 for the zero table remains part of the construction. The result concerns raw quotient agreement under the stated independent uniform challenge model. It does not establish a Fiat–Shamir attack, an accepted false proof, or a 69-bit security level for Stwo.

Reproducible checks. The companion script `checks/verify_fixed_claim.py` checks the displayed large-field counts using exact integers and rational numbers. An independent toy instance over $\mathbb{F}_{74}$ exhausts all 2,304 admissible samples. It checks the sampler bijection, counts distinct evaluation incidences and additive cosets, fixes one claim globally, and verifies every retained quotient identity and agreement count. A separate small family with actual evaluation collisions tests the multiplicity term in (45). These finite checks support the formulas; the proof above establishes the general statement.

C.2 Subtracting roots forced by shared agreements

The degree bound in Theorem C.2 counts all roots of a difference as possible sample collisions. Roots on the base-field evaluation domain cannot be admissible samples. Agreements with the same received word force many such roots and give a sharper bound.

Proposition C.3 (Agreement-aware collision bound). *Suppose that, for each of the N polynomials in the transfer theorem, we select T agreement coordinates from a common pool of n_0 base-field circle points. Write*

$$NT = n_0q + r, \quad 0 \leq r < n_0, \quad R = (n_0 - r) \binom{q}{2} + r \binom{q+1}{2}, \quad C_* = K \binom{N}{2} - R.$$

Then the total number of unordered polynomial-pair collisions over admissible samples is at most C_ . In particular, the assumed family implies $C_* \geq 0$. The total number I of distinct sample/value incidences satisfies*

$$I \geq \frac{N^2 S^2}{NS + 2C_*}. \quad (46)$$

Consequently Theorem C.2 holds with the stronger pair count

$$A_{\text{agr}} = \left\lceil \frac{N^2 S^2}{p^2(NS + 2C_*)} \right\rceil.$$

Proof. Let d_x count selected agreement sets containing coordinate x . Then $\sum_x d_x = NT$, and double counting gives

$$\sum_{j < k} |A_j \cap A_k| = \sum_x \binom{d_x}{2} \geq R.$$

The last inequality follows by balancing the integer counts: moving one unit from a count at least two larger than another strictly decreases the sum of binomial coefficients. At a coordinate of $A_j \cap A_k$, both polynomials equal the received word. These are distinct roots of $u^m(F_j - F_k)$, in addition to any admissible collision roots, and are disjoint from the admissible sample set. Hence pair (j, k) has at most $K - |A_j \cap A_k|$ admissible collisions. Summing proves the bound. The same incidence Cauchy–Schwarz argument as before yields (46); partition into the p^2 additive H -cosets gives the final statement. $\square$

In the displayed two-fiber family, all 132 agreement coordinates avoid the two points above $1 \in \mu_{256}$. Thus $n_0 = 510$ is valid, with $N = \lceil \binom{255}{66} / 256 \rceil$. The exact bound becomes

$$A_{\text{agr}} = 1045232662865816582797566371469227027334159065108360767.$$

The negative base-two logarithm of A_{agr}/p^8 is approximately 68.5521. This is a further factor of approximately 1.3641 over Theorem C.2 at the same parameters.

C.3 Fixed claims under nonuniform batching schedules

The coefficient multiplying the second quotient need not itself be a uniform field element. A second averaging argument handles a prescribed function of the original uniform challenge. Here the claimed value is allowed to range over all of E , rather than just the representatives wH .

Theorem C.4 (Transfer for a prescribed scalar schedule). *Use the same list, admissible samples, and raw quotients as above. Let $g_0, g_1 : E \rightarrow E$ be fixed functions, and set*

$$\mathcal{B} = \{\alpha \in E : g_0(\alpha) \neq 0, g_1(\alpha)/g_0(\alpha) \in H\}, \quad G = |\mathcal{B}|.$$

Let I_0 be any real lower bound for the number of distinct sample/value incidences, for example the right-hand side of (46). There is a single claim $v \in E$, fixed before both challenges, for which at least

$$A_{\text{sched}} = \left\lceil \frac{GI_0}{p^4} \right\rceil$$

distinct admissible pairs (a, α) satisfy

$$g_0(\alpha)r_{0,a} + g_1(\alpha)r_{1,a} \text{ agrees with } C_K(E) \text{ in at least } T \text{ coordinates,}$$

with fixed tables $(f, 0)$ and fixed claims $(v, 1)$. Under the same independent uniform challenge model, the joint probability is at least A_{sched}/p^8 .

Proof. For every distinct incidence (a, b) and every $\alpha \in \mathcal{B}$, assign the triple to

$$v = b - \phi(\alpha), \quad \phi(\alpha) = g_1(\alpha)/g_0(\alpha) \in H.$$

There are GI triples and p^4 possible values of v , so one fixed value receives at least A_{sched} triples. For this fixed v , the map to (a, α) is injective: these two coordinates uniquely determine $b = v + \phi(\alpha)$. Choose a polynomial witnessing each retained incidence. The raw quotient identity gives

$$g_0(\alpha)r_{0,a} + g_1(\alpha)r_{1,a} - g_0(\alpha)P_{a,j} = g_0(\alpha)\frac{f - F_j}{U_a}.$$

The witness $g_0(\alpha)P_{a,j}$ lies in the scalar-extended code $C_K(E)$. Nonzero multiplication preserves agreement coordinates. Counting the distinct sampler preimages proves the probability bound. No injectivity or uniform-output assumption about ϕ is used. $\square$

Additional terms that are codewords. The conclusion is preserved after adding any $B_{a,\alpha} \in C_K(E)$ to the batched word: add the same codeword to its witness. Thus other columns need not have zero quotients if their complete contribution is proved to lie in this code. A common preceding factor α^e , $e > 0$, removes at most the challenge $\alpha = 0$ from the certified set. These statements require code membership of the additional contribution; they do not discard arbitrary additional quotient or AIR terms.

Polynomial schedules over the subfield. If $g_0, g_1 \in H[X]$ and g_0 is a nonzero polynomial of degree $d_0 < p^2$, then $G \geq p^2 - d_0$: every $\alpha \in H$ outside the roots of g_0 qualifies. Thus even a nonlinear relative coefficient incurs no additional loss from the degree of g_1 . More precisely, for $g_0 = 1$, $g_1 = X^d$, $d \geq 1$, cyclicity of E^* gives

$$G = 1 + (p^2 - 1) \gcd(d, p^2 + 1). \quad (47)$$

Indeed a nonzero α qualifies exactly when $\alpha^{d(p^2-1)} = 1$. The assertion is about the number of original challenges, not a claim that α^d is uniformly distributed.

A finite dependence on the gap between powers. For $p = 2^{31} - 1$, each of $d = 1, 2, 5, 10$ divides $p^2 + 1$. The same list with the agreement-aware incidence bound gives:

Power gap d	One-sample columns	Qualifying challenges G	$-\log_2(A_{\text{sched}}/p^8)$
1	2	p^2	68.5521
2	3	$2p^2 - 1$	67.5521
5	6	$5p^2 - 4$	66.2301
10	11	$10p^2 - 9$	65.2301

The final column is approximate; exact integer certificates prove, respectively, probabilities greater than 2^{-69} , 2^{-68} , 2^{-67} , and 2^{-66} . All rows use the same $(n, K, T) = (512, 128, 132)$; the favorable fixed claim may differ by row. At the quotient layer the schedule $1, \alpha^d$ is realized by putting $d - 1$ zero columns with truthful zero claims between the two designated columns, with one sample per column at the same point. Every inserted numerator is identically zero, while its sample consumes one challenge power. This establishes the displayed quotient expression and column count; it does not show that any particular AIR allows this arrangement.

Corollary C.5 (The periodicity coefficient $\alpha + \alpha^3$). *For $g_0 = 1$ and $g_1 = X + X^3$, one has $G = 2p^2 - 1$. Using the agreement-aware bound at $(n, K, T) = (512, 128, 132)$ gives*

$$A_{\text{cubic}} = \left\lceil \frac{(2p^2 - 1)N^2S^2}{p^4(NS + 2C_*)} \right\rceil, \quad 2^{68}A_{\text{cubic}} > p^8.$$

Its joint raw-proximity lower bound is therefore strictly greater than 2^{-68} ; its negative base-two logarithm is approximately 67.5521.

Proof. Put $q = p^2$ and write $\alpha = x + wy$, with $x, y \in H$ and $w^2 = \kappa$. For $y = 0$ there are q qualifying challenges. For $y \neq 0$, equality of $\alpha + \alpha^3$ with its conjugate is equivalent to

$$3x^2 + \kappa y^2 = -1.$$

The element $3 \in \mathbb{F}_p^*$ is a square in H (here $p \neq 3$). Choose $s \in H$ with $s^2 = 3$. Since $i^2 = -1$,

$$\text{Norm}_{E/H}(sx + wiy) = 3x^2 + \kappa y^2.$$

The nonzero norm fiber at -1 has $q + 1$ elements. Exactly two have $y = 0$, since $-1/3$ is a square in H . Thus $q - 1$ further challenges qualify, giving $G = 2q - 1$. The numerical inequality is certified with exact integers by the companion script. $\square$

The source schedule that gives this coefficient. At the pinned proving revision [24], `build_samples_with_randomness_and_periodicity` assigns powers across trees, columns, and samples in that order. Two consecutive one-sample columns at the same point receive $1, \alpha$ if they are the first nonempty columns. No periodicity sample is added to either.

There is also an exact three-power realization of the cubic coefficient. Place the (f, v) column first with its sole sample at z . Place the zero column next, with claims 0 at a previous point z_- and 1 at z . Give the zero column the maximal lifted log size. Its periodicity point then coincides with z , and the source assigns the three powers $\alpha, \alpha^2, \alpha^3$ to its periodicity, previous, and current samples. Its previous-sample numerator is zero. The normalized sum is therefore exactly

$$r_{0,z} + (\alpha + \alpha^3)r_{1,z}.$$

If the previous point makes the source's conjugate-line formula singular, that sample must additionally be excluded. For a fixed base-field rotation $z_- = hz$, at most $p^2 - 1$ initially admissible samples are lost: the excluded parameters satisfy $a\bar{a} = -h^{-2}$, a norm fiber of size $p^2 + 1$, of which two parameters lie in H . Replacing S by the conservative $S' = S - (p^2 - 1)$ in the incidence bound gives a version covering both source denominators. The strict 2^{-68} comparison still holds.

This realizes the scalar schedule at the quotient layer for the stated column arrangement. It does not establish that an AIR admits these claims, that a complete transcript accepts, or that the favorable claim and corresponding witnesses can be found efficiently.

C.4 The radius of the raw-quotient example

The (512, 128, 132) example has rate $\rho = 1/4$, gap $\eta = 1/128$, and radius $\theta = 95/128$. It is *above* the characteristic-based Elias radius: $H_p(95/128) \approx 0.7687496752 > 3/4$. This comparison follows without floating-point assumptions from

$$p^{96}95^{95}33^{33} < (p-1)^{95}128^{128}.$$

It follows that this raw-quotient example is not itself a counterexample within the below-Elias range of the whitepaper’s list prescription. The below-Elias counterexamples elsewhere in the paper remain separate constructions. Neither the 2^{-69} bound nor its stronger variants should be interpreted as a protocol security level.

C.5 What a nearby codeword supplies at the FRI query stage

The source permits one further conditional statement. Its first FRI layer combines the quotient values computed at sampled positions with complementary values supplied by the FRI proof. Agreement is required at the sampled positions; the surrounding folding block need not consist entirely of agreement coordinates.

Lemma C.6 (Conditional query survival). *Let $R \in E^D$ be the quotient word computed from fixed committed tables and claimed openings, and suppose $P \in C_K(E)$ agrees with R on $A \subseteq D$. Assume that an honest FRI proof is formed for P on this same domain and degree bound, and that the original table decommitments are valid. If every sampled quotient-query position lies in A , the first-layer value substitution leaves the honest FRI verification unchanged. This holds for the source’s supported folding steps and leaf packing.*

Under q_{qry} independent uniform draws from D , with replacement before sorting and deduplication, the probability of this sufficient event is exactly

$$\left(\frac{|A|}{|D|}\right)^{q_{\text{qry}}}.$$

If the preceding raw-proximity event has probability at least π and supplies $|A| \geq T$ on a length- n domain, then this particular two-stage sufficient event has probability at least $\pi(T/n)^{q_{\text{qry}}}$, under independent future query draws.

Proof. The source function `compute_decommitment_positions_and_rebuild_evals` partitions the sorted query positions into folding blocks. At an actual query position it uses the provided quotient value; at any other position in that block it takes a value from the FRI witness. Form the latter values and all authentication paths using the honest proof for P . When every queried position is in A , the supplied quotient value is also P at that position. Thus every reconstructed block, including any packed leaf, equals its honest counterpart. Its Merkle check, subsequent fold, and all later FRI checks are unchanged.

The raw query draws all land in A with the displayed probability. Sorting and removing duplicates does not change that event. The final statement follows by conditioning on the earlier sample and batching challenges, including the selected witness, and then averaging. $\square$

Numerical consequence and scope. For the (512, 128, 132) example, $T/n = 33/128$. At 70 raw query draws the additional sufficient-event factor is

$$(33/128)^{70} \approx 2^{-136.8924}.$$

For illustration, multiplying this by the certified raw bound for the eleven-column, power-gap-ten schedule gives a two-stage lower bound of approximately $2^{-202.1225}$. This uses 70 as a stated query count; it does not identify this rate-1/4 example with a production parameter set. These are lower bounds for a particular sufficient event, not upper bounds on all possible strategies or proof-system soundness guarantees.

The lemma is conditional on constructing the codeword and honest FRI proof, providing valid table decommitments, and meeting the prescribed domain and degree metadata. It does not establish the AIR conditions, efficient recovery of witnesses or the favorable fixed claim, or a successful native verifier execution. Proof-of-work search and Fiat–Shamir retry costs are not included in this probability. In particular, an inverse one-trial probability is not automatically an attack work factor: reusing a successful prefix and searching later challenges requires its own analysis.

The companion `checks/verify_query_survival.py` enumerates all $8^3 = 512$ ordered query triples for an eight-coordinate word with five agreements. For each of folding steps 1, 2, 3, precisely 125 triples give unchanged reconstructed blocks, including cases where unqueried positions in the block are disagreement coordinates. This checks the reconstruction argument against an explicit vector commitment model; it does not replace a native Merkle or FRI test.

C.6 An explicit table with an exponentially large implicit list

The support family need not be enumerated to construct its received word. On μ_M , equality of the constant coefficients of the root polynomials is a subset-sum condition modulo M . This yields an efficient exact construction of the fixed table used above.

Proposition C.7 (Constructing the table without enumerating its list). *Let λ generate μ_M and consider the h -element subsets of $\{\lambda, \lambda^2, \dots, \lambda^{M-1}\}$. The sizes of all classes with equal constant coefficient, a largest class, and one support in that class can be computed with $O(M^2h)$ additions of integers of $O(M)$ bits, storing $O(Mh)$ such integers and support masks. For the displayed $M = 256$, $h = 66$ instance, the largest count is*

$$N_{\text{exact}} = 471959451942648482921857753174390587612228702156784439968875.$$

The residue 1 attains this count. It gives an explicitly computed 512-entry $M31$ -valued table on a canonical coset and an implicit family of N_{exact} distinct real Laurent polynomials with support in $[-64, 64]$, each agreeing with that table in exactly 132 coordinates.

Proof. Let $D_{j,k,r}$ be the number of k -subsets of $\{1, \dots, j\}$ whose sum is r modulo M . The initial condition is $D_{0,0,0} = 1$, with all other counts zero, and

$$D_{j,k,r} = D_{j-1,k,r} + D_{j-1,k-1,r-j \bmod M}.$$

Updating k in decreasing order permits an in-place computation using $O(Mh)$ counts. Whenever a count first becomes positive, retain one witness support mask from the responsible predecessor, adding j if the second term supplied it. These masks have at most M bits. Every count is at most 2^{M-1} . The stated operation and storage bounds follow. Also $\sum_r D_{M-1,h,r} = \binom{M-1}{h}$, which checks the total count.

The constant coefficient of $G_U(Y) = \prod_{j \in U} (Y - \lambda^j)$ is $(-1)^h \lambda^{\sum_{j \in U} j}$. Thus the residue classes counted by the recurrence are exactly the constant-coefficient classes. Choose a largest class and its retained support U_0 . The earlier construction sets

$$f(u) = a_0 u^{-(m+2)} G_{U_0}(u^2), \quad F_U(u) = a_0 u^{-(m+2)} (G_{U_0}(u^2) - G_U(u^2)).$$

Here $m = 64$, and a_0 is the explicit phase factor from Appendix B. The common leading and constant coefficients put every F_U in $[-64, 64]$. Its difference from f vanishes at precisely the two-point fibers of U , giving 132 agreements. Distinct supports give distinct polynomials. Rotation to the canonical coset preserves these statements. The table requires only G_{U_0} , not the other polynomials in the family. $\square$

A closed form and a fixed defining support. In this instance, every odd residue has exactly

$$N_{\text{exact}} = \frac{\binom{255}{66} + \binom{127}{33}}{256}$$

supports. This also follows from the roots-of-unity filter. For a character of order $d = 2^s > 1$, the support generating function is $(1 - t^d)^{256/d} / (1 + t)$; its degree-66 coefficient is $A_d = (-1)^{\lfloor 66/d \rfloor} \binom{256/d-1}{\lfloor 66/d \rfloor}$. Thus the count at residue r is

$$\frac{1}{256} \left(\binom{255}{66} + \sum_{d=2,4,\dots,256} c_d(r) A_d \right),$$

where $c_d(r)$ is $d/2$ if $d \mid r$, $-d/2$ if $d/2 \mid r$ but $d \nmid r$, and zero otherwise. For odd r only $d = 2$ contributes, giving the displayed closed form. For even r the $d = 2$ contribution has the opposite sign; the exact inequality $\sum_{d \geq 4} (d/2) |A_d| < 2 \binom{127}{33}$ shows its remaining terms cannot reach the odd-residue count. The script independently checks this character formula against all 256 dynamic-programming counts.

One may fix the defining support outright as $U_0 = \{2, 3, \dots, 68\} \setminus \{40\}$: it has 66 elements and sum $2305 \equiv 1 \pmod{256}$. Hence even the search for a defining support is unnecessary for this particular table.

Concrete artifact. The companion `checks/construct_m31_bank.py` performs 3,759,360 integer updates; the recorded run took approximately 0.6 seconds for this counting stage. It writes all 256 exact residue counts, a defining support, the canonical domain parameters, and the 512 base-field table entries to `checks/explicit_m31_bank.json`. A checked generic-sample fixture also supplies a normalized quotient witness in the interior Laurent band $[-63, 63]$ and verifies all 512 coordinate identities, with exactly 132 agreements.

The fixture's claimed value is chosen to realize one specified sample/challenge pair. It is not asserted to be the favorable fixed claim promised by Theorem C.4. The family is now explicit as a compact membership rule and exact count, and its fixed table is fully constructed. Finding the favorable claim and recovering a witness efficiently for a fresh challenge remain open. The numerical probability bounds above may continue to use the slightly smaller $N = \lceil \binom{255}{66} / 256 \rceil$, by retaining that many members of the class.

C.7 Witness recovery as a constrained subset product

The implicit description of the family permits an exact search for a witness after sampling. The following reduction isolates the remaining algorithmic problem and gives a complete, although expensive, baseline. The claim v is held fixed throughout the search.

Proposition C.8 (Exact recovery criterion and a complete search). *Use the two-fiber family of Section C.6, with $h = m + 2$, defining support U_0 , residue r , and relative sample parameter $a \notin H$.*

Assume $\mu_{2M} \subset H$. For any desired value $\beta \in E$, a support U in this family satisfies $F_U(a) = \beta$ if and only if

$$|U| = h, \quad \sum_{j \in U} j \equiv r \pmod{M}, \quad \prod_{j \in U} (a^2 - \lambda^j) = G_{U_0}(a^2) - \frac{a^h}{a_0} \beta. \quad (48)$$

Each factor on the left is nonzero. A zero target therefore has no solution. For a nonzero target, partition $\{1, \dots, M-1\}$ into parts of sizes s and $M-1-s$. There is a complete search using a table of at most

$$R = \sum_{k=0}^{\min(h, M-1-s)} \binom{M-1-s}{k}$$

right-hand subset records and at most

$$L = \sum_{k=0}^{\min(h, s)} \binom{s}{k}$$

dictionary lookups per target, in addition to constructing the subset products. Keeping all records with equal keys recovers every matching support; keeping one per key suffices to find a witness when one exists.

Proof. Substitute $F_U(a) = a_0 a^{-h} (G_{U_0}(a^2) - G_U(a^2))$ and rearrange. If $a^2 = \lambda^j$, then a is one of its two square roots in $\mu_{2M} \subset H$, contradicting $a \notin H$.

For a right-hand subset V , store its support under the key

$$\left(|V|, \sum_{j \in V} j \pmod{M}, \prod_{j \in V} (a^2 - \lambda^j) \right).$$

For each left-hand subset W , with cardinality k , residue t , and nonzero product P_W , look up $(h-k, r-t \pmod{M}, \tau/P_W)$, where τ is the target in (48). The parts are disjoint, so every returned union has the required cardinality, residue, and product. Conversely, each solution has exactly one such decomposition, which the search visits. The subset counts give the stated bounds. $\square$

For the scheduled transfer, set $\beta = v + g_1(\alpha)/g_0(\alpha)$ when $g_0(\alpha) \neq 0$ and this ratio belongs to H . A recovered support supplies F_U , and exact polynomial division then supplies the interior-band quotient witness from Theorem C.4. Preparing the subset-product table depends on a and must generally be repeated when the sample changes. It can be reused for multiple targets at the same sample. This reduction does not identify the favorable fixed claim from the averaging theorem.

Fresh-query check on a smaller valid circle code. The companion `checks/verify_witness_search.py` fixes $v = 0$ before generating samples. It uses $p = 23$, $n = 24$, $M = 12$, $h = 4$, $m = 2$, and the full base-field circle, with defining support $U_0 = \{1, 2, 3, 7\}$ and residue 1. The resulting family has 27 members of outer Laurent degree at most 2, each with 8 agreements. The tower $H = \mathbb{F}_{23}[i]$, $E = H[w]$, $w^2 = 2 + i$, is a field: the script checks $(2 + i)^{(23^2-1)/2} = -1$. This is a circle-code experiment; its length is not a Stwo FFT size.

The deterministic experiment chooses 128 distinct admissible samples using a recorded pseudorandom seed. At each sample it tests every one of the 529 challenges in H , together with eight specified outside- H controls. On all 68,736 queries, the returned support set agrees exactly with an independent enumeration of all 27 family members. There are 135 successful queries: 128 have

$\alpha = 0$, which always recovers $F_{U_0} = 0$, and seven have $\alpha \neq 0$. No outside- H control succeeds in this fixture. For all 135 recovered witnesses, exact division yields coefficients in H with Laurent support $[-1, 1]$; all 3,240 coordinate identities pass, with exactly eight agreements per witness. The split uses 31 left-hand and 57 right-hand records per sample. These challenges are deliberately exhaustive in H ; the observed success fraction is not an estimate for uniform challenges in E .

Cost at the displayed M31 parameters. For the direct implementation with $s = 127$, $M = 256$, and $h = 66$, the complete right-hand table would have

$$R = \sum_{k=0}^{66} \binom{128}{k} = 228210150822262702568674774219878072803 \approx 2^{127.4236}$$

records, before accounting for bytes per record. Thus this baseline does not give a practical M31 decoder. This is the cost of the specified complete enumeration, not a lower bound on witness-search complexity or a claim that more selective algorithms cannot improve it. The useful conclusion is an exact algebraic target and a checked recovery procedure for fresh queries with a claim fixed in advance.

D Checks and provenance

D.1 Exact interval certificates

The comparisons in the main text follow from the moment-class bounds and the exact certificates below. All examples have $k = n/2$, agreement $t = k + s$, and gap $\eta = s/n$. The last column gives a strict lower bound on $\log_2 L$.

p	n	s	Certified strict lower bound on $\log_2 L$
$2^{31} - 1$	154	5	61.74656
$2^{61} - 1$	548	9	242.16447
$2^{127} - 1$	64	2	42
$2^{127} - 1$	2158	17	995.98234
$2^{521} - 1$	4096	8	3634
$2^{521} - 1$	29176	56	14233.01899

For the length-64 row the original denominator product is 32,285,041, giving 50,186,958,429 code-words. The conditional dispersion certificate (8) strengthens this to 2,674,048,230,075; the exact enumeration in Section 7.1 further gives 5,552,914,238,035 $> 2^{42}$. The length-4096 row uses Lemma 2.1. The four decimal rows use Proposition 2.13, with rational upper bounds for its denominator and outward rational intervals for logarithms. Their checker is in `research/prime_exponent_coefficients/`. It verifies primality by Lucas–Lehmer and checks the strict sufficient Elias inequality

$$p^s t^t (n - t)^{n-t} > n^n$$

using integers. The two remaining rows satisfy the stronger sufficient condition $p^s > 2^n$. For the last row, $\eta = 1/521$, so a bound $2^{c_2/\eta}$ requires

$$c_2 > \frac{14233.01899}{521} > 27.31865.$$

The corresponding forced coefficients for the three smaller primes are 2.00475, 3.97715, and 7.84601. These comparisons set $c_1 = 1$; for a different prefactor, subtract $\eta \log_2 c_1$ from each necessary lower bound on c_2 . They concern interval domains, not the prescribed circle domain. The finite rational-line example uses the conditional first-moment count

$$N = 8,634,334,451,040,106$$

from Section 2.3. The exact inequality $2^{127} - 1 > 64 + 32 \binom{N}{2}$ certifies a pole with all N scalar labels distinct. These integers are retained here for reproducibility; the comparisons in the main text only need their rounded powers of two.

D.2 Certificates at additional rates

For Corollary 6.2, the exact guaranteed list size is

$$L = \left\lceil \binom{63}{17} / 64 \right\rceil = 15,827,733,944,904, \quad \eta = \frac{1,048,575}{33,554,432}.$$

Integer arithmetic verifies $p^{100} > 2^{3099}$, $3^{100000} > 2^{158496}$, and $L^{2000} > 2^{87695}$. Since $H_2(1/4) = 2 - \frac{3}{4} \log_2 3 < 0.81128$, these give

$$\log_2 L - \frac{H_2(1/4)}{\eta} > \frac{87695}{2000} - \frac{81128}{100000\eta} = \frac{937767635977}{52428750000} > 17.8865.$$

This implies the rounded ratio in the corollary. For the Elias check, write $\theta = 3/4 - \eta$. The exact rational inequality

$$\eta \frac{3099}{100} > 1 - 2(\theta - 1/2)^2 \geq H_2(\theta)$$

and $\log_2 p > 3099/100$ imply $H_p(\theta) < 3/4$. The same construction at rates $1/8$, $1/16$, and $3/4$ gives list ratios exceeding 2^{13} , 2^{15} , and 2^{13} , respectively. The rate-3/4 space has even dimension but is not a native power-of-two FFT size. No such restriction is needed for the mathematical code.

For Corollary 6.3, put $Q = p^4 - p^2$. The exact counts are

$$N = \left\lceil \frac{\binom{255}{36}}{256p^2} \right\rceil = 73,497,635,817,391,996,020,834,$$

$$J = \left\lceil \frac{NQ}{Q + 32(N-1)} \right\rceil = 73,497,635,817,383,868,141,347.$$

Here $\eta = 655359/33554432$. Integer arithmetic verifies $7^{100000} > 2^{280735}$, $7^{1000} < 2^{2808}$, and $J^{1000} > 2^{75960}$. Set

$$H^+ = 3 - \frac{7 \cdot 280735}{8 \cdot 100000} > H_2(1/8).$$

Concavity gives $H_2(7/8 - \eta) < H^+ + \eta(2808/1000)$, whereas exact rational arithmetic gives $\eta(3099/100 - 2808/1000) > H^+$. Together with $\log_2 p > 3099/100$, this certifies the strict Elias bound. Finally,

$$\log_2 J - 26 - \frac{H_2(1/8)}{\eta} > \frac{75960}{1000} - 26 - \frac{H^+}{\eta} = \frac{9064121863}{409599375} > 22.129.$$

After puncturing, the exact parameters are $\rho' = (2^{23} - 1)/(2^{26} - 2)$ and $\eta' = (5 \cdot 2^{18} - 1)/(2^{26} - 2)$. Exact rational logarithm intervals separately certify the Elias bound and a ratio greater than 2^{22} at these changed parameters.

For the quadratic-field example, put $Q_2 = p^2 - 128$. Then

$$N_2 = \left\lceil \left(\frac{\binom{127}{18}}{128} \right) \right\rceil = 25,460,924,804,916,891,029,$$

$$J_2 = \left\lceil \frac{N_2 Q_2}{Q_2 + 16(N_2 - 1)} \right\rceil = 285,003,988,493,147,037.$$

Here $\eta = 786431/33554432$. The same H^+ above satisfies $\eta(3099/100 - 2808/1000) > H^+$, proving the strict Elias bound. The integer inequality $J_2^{10000} > 2^{579837}$ gives

$$\log_2 J_2 - 26 - \frac{H_2(1/8)}{\eta} > \frac{579837}{10000} - 26 - \frac{H^+}{\eta} = \frac{13827665031}{1572862000} > 8.791.$$

Thus the ratio exceeds 400. An independent fixture over $\mathbb{F}_{312}$ checks all 945 allowable poles for an anchored coefficient class of 87 supports. It verifies the collision bound and all witness identities for a best pole and for the allowed pole zero; these yield 87 and 84 distinct labels, respectively.

D.3 Certificates for the compact examples

Corollary 6.4 uses the same integers J_2 and J as the two eighth-rate constructions above. The two new Elias comparisons follow from

$$p^{896} 873^{873} 151^{151} > (p-1)^{873} 1024^{1024},$$

$$p^{3584} 3505^{3505} 591^{591} > (p-1)^{3505} 4096^{4096}.$$

Taking logarithms gives $H_p(873/1024) < 7/8$ and $H_p(3505/4096) < 7/8$. Also,

$$J_2^2 > 2^{115}, \quad J^3 > 2^{227}, \quad 7^5 > 2^{14}.$$

The last inequality gives $H_2(1/8) < 11/20$, so the logarithms of the two ratios exceed

$$\frac{115}{2} - 11 - \frac{11}{20} \frac{1024}{23} > 22, \quad \frac{227}{3} - 13 - \frac{11}{20} \frac{4096}{79} > 34.$$

All these comparisons use exact integer or rational arithmetic.

D.4 Verification and source notes

The accompanying files give exact integer certificates for the finite bounds and exhaustive small-field checks of the constructions. They also include a small explicit test of the power-fiber lift. The proofs establish the asymptotic claims; the experiments check formulas and boundary cases. Large circle lists and suitable poles are obtained by counting, rather than enumerated.

The numerical certificates and construction checks are included with this draft. They use exact integer or finite-field arithmetic. The protocol-related models do not constitute native verifier executions. The source comparisons concern the cited repository revisions; they are not claims about later implementations. This is a working draft with machine-assisted internal checks, and it has not undergone external peer review.

E Nearly complete rational fibers and the Frobenius index

We prove Theorem 2.11. The unit-equation and Wronskian method follows the approach of Corvaja–Zannier [5, Proposition 2]; the linear-independence argument below uses the special relation $p = \ell n + \epsilon$. We give the needed height calculation explicitly.

Proof of Theorem 2.11. Work over $k = \overline{F}$. The hypotheses imply $B < p$, $2\ell B < p$, and $p > n$. In particular, R is separable.

The collision components and their boundary. The reduced curve $R(x) = R(y)$ in $\mathbb{P}^1 \times \mathbb{P}^1$ has bidegree (B, B) and contains the diagonal. Suppose it has a component other than a torus graph $x/y = \text{constant}$ or $xy = \text{constant}$. On its smooth projective normalization C , let $a = \deg x$ and $b = \deg y$. Neither projection is constant, and $1 \leq a, b \leq B - 1$ because the diagonal is a different component. Put $K = k(C) = k(x, y)$. Let S consist of the zeros and poles of x, y , together with the points above $x \in \mu_n \setminus T$ or $y \in \mu_n \setminus T$. The genus and boundary bounds give

$$\begin{aligned} g_C &\leq (a-1)(b-1), & |S| &\leq 2a + 2b + c(a+b), \\ \chi &:= 2g_C - 2 + |S| &\leq 2ab + c(a+b). \end{aligned} \tag{49}$$

The four units. Put $u = x^n$, $v = y^n$, and $w = (u-1)/(v-1)$. Because $\deg x, \deg y < p$, both coordinates are separating; so are u and v , since $p \nmid n$. In particular, $[K^p(v) : K^p] = p$. Outside S , $u-1$ and $v-1$ have exactly the same zeros, all simple. Indeed, outside S , $x \in \mu_n$ means $x \in T$; its complete geometric fiber consists of B distinct points in T , so $y \in T$, and conversely. Such fibers are unramified, and the collision curve is smooth with both coordinate projections unramified at their pairs. Consequently w is an S -unit, meaning that all its zeros and poles lie in S . The four S -units

$$-1, \quad u, \quad w, \quad -wv$$

sum to zero.

Independence over K^p . We claim that $w, 1, u$ are linearly independent over K^p . Otherwise there are $A_0, A_1, A_2 \in K^p$, not all zero, such that

$$P(U, V) = A_0(U-1) + (A_1 + A_2U)(V-1)$$

vanishes at (u, v) . This polynomial is nonzero and has bidegree at most $(1, 1)$. Remove the common factor of its two coefficients as a polynomial in U ; that factor has degree at most one in V and cannot vanish at v . The resulting primitive polynomial is linear in U , since an equation in v alone of degree at most one would contradict $[K^p(v) : K^p] = p$. It defines a graph $U = T(V)$. The function T is nonconstant, since $u \notin K^p$, and is therefore a Möbius transformation.

Set $c = x^p, d = y^p \in K^p$. The congruence defining ℓ gives

$$x = (c/u^\ell)^\epsilon, \quad y = (d/v^\ell)^\epsilon.$$

Hence (u, v) also annihilates the numerator $G(U, V)$ of

$$R((c/U^\ell)^\epsilon) - R((d/V^\ell)^\epsilon),$$

whose bidegree is at most $(\ell B, \ell B)$ over K^p . If the primitive P and G were coprime, their resultant in U would be a nonzero polynomial in V of degree at most $2\ell B < p$ vanishing at v . This is impossible. Thus P divides G , and

$$R((c/T(V)^\ell)^\epsilon) = R((d/V^\ell)^\epsilon) \quad (50)$$

as rational functions of V .

Over an algebraic closure of K^p , choose $\alpha^\ell = c$, $\beta^\ell = d$, and set

$$F_0(Z) = R(Z^{\ell\epsilon}), \quad M(Z) = \alpha/T(\beta/Z).$$

Equation (50) says $F_0 \circ M = F_0$. The deck group of F_0 has order at most $\ell B < p$ and contains the scalings by ℓ th roots of unity. The tame finite-subgroup classification for PGL_2 gives cyclic, dihedral, A_4 , S_4 , or A_5 [6]. The last three have no element of order at least six. In a cyclic or dihedral group, an element of order at least three has the rotation subgroup's pair of fixed points. Thus every deck map preserves the pair $\{0, \infty\}$ fixed by the scalings, and

$$M(Z) = \zeta Z \quad \text{or} \quad M(Z) = \zeta/Z.$$

Here $\zeta \in k$: since F_0 is defined over the algebraically closed field k , any deck map takes every k -point into a finite fiber consisting of k -points, and the images of three points determine its coefficients.

In the first case $u/v = \alpha/(\zeta\beta)$. Taking ℓ th powers and using $p = \ell n + \epsilon$ gives $(x/y)^\epsilon = \zeta^\ell$, so x/y is constant. In the second case $uv = \alpha\beta/\zeta$, and the same argument gives $(xy)^\epsilon = \zeta^\ell$, so xy is constant. Either case makes the chosen component a torus graph, contrary to its selection. This proves the claimed independence.

The height contradiction. Denote the four units by f_0, f_1, f_2, f_3 . Any three are independent over K^p , and their ordinary three-row Wronskians are nonzero and agree up to sign. For an explicit check, $u' \neq 0$ and

$$W(1, u, w) = u'w'' - u''w' = (u')^2(w'/u')'.$$

If this were zero, $w'/u' \in K^p$ and then $w - (w'/u')u \in K^p$, contradicting independence. Here K^p is the constant field of a nonzero derivation on K . Multiplying a Wronskian with derivative orders 0, 1, 2 by $(dt)^3$ gives a well-defined nonzero rational cubic differential Ω on C .

At a point $P \in S$, choose the three functions that omit a function of smallest valuation. Differentiation in a local parameter lowers valuation by at most one, so

$$\mathrm{ord}_P \Omega \geq \sum_{i=0}^3 \mathrm{ord}_P f_i - \min_i \mathrm{ord}_P f_i - 3.$$

Outside S , all local derivatives are regular, giving $\mathrm{ord}_P \Omega \geq 0$. Since each f_i has a degree-zero divisor supported on S , summing yields

$$H := - \sum_P \min_i \mathrm{ord}_P f_i \leq 3(2g_C - 2 + |S|) = 3\chi.$$

The tuple contains a nonzero constant and u , while v is a ratio of two coordinates. The degree of any such ratio is at most the projective height, so $H \geq n \max(a, b)$. By (49),

$$n \max(a, b) \leq 6ab + 3c(a + b) \leq 6(\min(a, b) + c) \max(a, b).$$

This contradicts $n > 6(B - 1 + c)$. Every component is therefore a torus graph. Each has bidegree $(1, 1)$, so the degree- B map has B geometric deck transformations and is Galois. Each deck map preserves T , which is nonempty. Evaluating it at a point of T shows that its scaling constant belongs to μ_n . Thus the entire deck group preserves μ_n .

Small defect forces free action. The rotation subgroup has order $d \mid n$. If there are inversions, the whole group has order $B = 2d$; otherwise $B = d$. Thus $B \mid 2n$. Since T is nonempty, $B \leq n$, so dyadic n forces $B \mid n$ even without a small-defect assumption. A nonzero finite point is fixed by no nonidentity rotation and by at most one inversion, so every nonfree orbit has $B/2$ points. Fibers of a Galois quotient are orbits; thus any nonfree orbit in μ_n lies entirely outside T . The inequality $c < B/2$ rules this out. If the group has only rotations, its action is already free. In either case all domain orbits have B points, proving full balance and $B \mid n$. Corollary 2.10 now applies. $\square$

The cutoff $c < B/2$ is strict. On $\mu_{15} \subset \mathbb{F}_{181}$, $R = X + X^{-1}$ has seven two-point fibers and one fixed point. Here $B = 2$, $c = 1 = B/2$, and $15 > 6(B - 1 + c)$, but $B \nmid 15$.

The exact checks in `research/structured_domains/frobenius_index/` verify the pinned arithmetic, nonzero Wronskians on sample non-diagonal collision conics in both congruence classes, and small finite-field deck groups. A separate exhaustive enumeration checks 27,273,130 disjoint pairs of possible complete fibers on five small domains, including non-Galois examples outside the theorem's hypotheses. These computations supplement the proof; they do not establish the general classification. The additional near-balance checker tests 26,880 exact arithmetic inequalities and 1,823,695 rational pencils. Its qualifying positive-defect cases all have torus deck groups; planted ramified examples also check the strict free-action cutoff.

F A limit on binary-affine support transfer

The large equal-prefix classes in this paper need not have a simple binary-linear description. The following bound gives one precise obstruction to such a description. It concerns affine families of *indicator vectors*, with addition over $\mathbb{F}_2$, even though the evaluation points lie in a field of different characteristic.

Theorem F.1 (Binary-affine locator classes). *Let $D = \{\alpha_1, \dots, \alpha_n\}$ be distinct elements of a field of characteristic different from two. Let $\mathcal{A} \subseteq \{0, 1\}^n$ be an affine subspace over $\mathbb{F}_2$ of dimension r . Suppose every word has integer weight t , and the corresponding monic degree- t locator polynomials have the same coefficients in degrees $t - 1, \dots, t - s$, where $0 \leq s < t$. Then*

$$r \leq \left\lfloor \frac{\min\{t, n - t\}}{s + 1} \right\rfloor, \quad |\mathcal{A}| \leq 2^{\lfloor \min\{t, n - t\} / (s + 1) \rfloor}.$$

The dimension bound is attained over suitable odd prime fields.

Proof. The case $r = 0$ is immediate. Parameterize the family injectively by $u \in \mathbb{F}_2^r$, writing the j th indicator as

$$x_j(u) = b_j + \langle a_j, u \rangle \pmod{2}, \quad a_j \in \mathbb{F}_2^r, \quad \epsilon_j = (-1)^{b_j}.$$

The nonzero column labels a_j span $\mathbb{F}_2^r$. For a nonzero label a , let C_a^+ and C_a^- be its coordinate classes with $b_j = 0$ and $b_j = 1$, respectively. The constant integer weight condition, expanded in the real-valued characters $\chi_a(u) = (-1)^{\langle a, u \rangle}$, gives $|C_a^+| = |C_a^-| =: h_a$. In every word exactly h_a coordinates of their union are selected. If c nonzero labels occur, then

$$r \leq c, \quad \sum_{a \neq 0} h_a \leq \min\{t, n - t\}.$$

We show that each occurring label has $h_a \geq s + 1$.

Let $L_u(X)$ denote the locator at u . Equality of the specified coefficients means, in the Laurent series field at infinity, that

$$L_u/L_0 = 1 + O(X^{-s-1}).$$

For a fixed nonzero a , take the product $\prod_u L_u(X)^{\chi_a(u)}$. Since $\sum_u \chi_a(u) = 0$, this product is also $1 + O(X^{-s-1})$. Character orthogonality over the integers gives

$$\sum_u \chi_a(u) x_j(u) = \begin{cases} -\epsilon_j 2^{r-1}, & a_j = a, \\ 0, & a_j \neq a. \end{cases}$$

Writing $F_a^\pm$ for the locators of $C_a^\pm$, we obtain

$$\left(\frac{F_a^-}{F_a^+} \right)^{2^{r-1}} = 1 + O(X^{-s-1}).$$

The ratio has leading term one, since both polynomials have degree h_a . Raising such a Laurent series to an integer power invertible in the field preserves its first nonzero nonconstant term. Thus $F_a^-/F_a^+ = 1 + O(X^{-s-1})$. The two disjoint nonempty root sets give different monic polynomials, so their nonzero difference has degree at most $h_a - s - 1$. Hence $h_a \geq s + 1$. Summing and using $r \leq c$ proves the bound.

For sharpness put $B = s + 1$ and use r disjoint pairs of full B -element fibers of X^B . Select one fiber in each pair independently, then add fixed selected and unselected coordinates as needed. Each fiber locator is $X^B - a$, so the varying locator product has the same first $B - 1$ coefficients. The common fixed factor preserves that prefix. The fiber-lifting argument of Lemma 4.2 supplies arbitrarily large odd primes for the required finite collection of fibers. With $r = \lfloor \min\{t, n - t\}/B \rfloor$, the remaining fixed-coordinate counts $t - rB$ and $n - t - rB$ are nonnegative, giving equality. $\square$

Finite consequences. For the $(n, t, s) = (64, 34, 2)$ integer class, every affine binary subfamily has at most $2^{10} = 1024$ members. For $(157, 68, 5)$ the bound is $2^{11} = 2048$. The much larger certified classes therefore require nonlinear families of support indicators. Likewise, choosing 272 of 511 packet labels with fourteen fixed leading coefficients permits at most $2^{15} = 32768$ members in an affine binary subfamily. This does not exclude nonlinear support families or other transfers from binary-field constructions.

The characteristic-two exception. On $\mathbb{F}_{16}$, let a range over an affine binary hyperplane not containing zero and take the supports

$$\{x \in \mathbb{F}_{16} : \text{Tr}_{\mathbb{F}_{16}/\mathbb{F}_2}(ax) = 1\}.$$

Their indicators form an affine binary family of dimension three, and every support has size eight. The monic locator is a normalized trace polynomial plus a constant, with possible degrees only 8, 4, 2, 1, 0. Its first three coefficients below the leading term are zero. Thus $(n, t, s, r) = (16, 8, 3, 3)$, violating the bound $r \leq 2$ that would follow in odd characteristic. Here taking a power of two can raise a Laurent vanishing order, exactly the step excluded above.

The checker in `research/binary_affine_locator/` enumerates 103,416 small affine binary spaces and verifies all 8,407 constant-weight families, including an odd-characteristic example with characteristic no greater than the prefix length. It also checks twenty sharpness examples and the $\mathbb{F}_{16}$ exception. These finite checks supplement the proof.

Corollary F.2 (Subquadratic transfer of punctured binary subspaces). *Let $V = \mathbb{F}_2^m$, $N = 2^m$, and $m \geq 4$. Injectively label $V \setminus \{0\}$ by elements of a field of characteristic different from two. Let $\mathcal{C}$ be a collection of codimension-two linear subspaces $W \leq V$ whose labeled punctured supports $W \setminus \{0\}$ have monic locators with the same first $N/8 - 1$ coefficients below the leading term. Put*

$$M_N = \left\lfloor \frac{1 + \sqrt{4N - 7}}{2} \right\rfloor.$$

Then

$$|\mathcal{C}| \leq \left\lfloor \frac{(N-1)M_N}{3} \right\rfloor = O(N^{3/2}).$$

Proof. Fix a nonzero functional $a \in V^*$. The codimension-two subspaces with $a \in W^\perp$ are $W_b = \ker a \cap \ker b$, indexed by nonzero $b \in Q := V^*/\langle a \rangle$, a binary group of size $N/2$. Let $S_a \subseteq Q \setminus \{0\}$ index the members of $\mathcal{C}$ of this form. On $\ker a$, the binary indicator of W_b is $1 + b(x)$; outside $\ker a$ it is zero. This dependence on b is affine and injective, and puncturing the origin preserves affine dimensions.

If S_a contained an affine binary plane, its four punctured supports would form an affine family of dimension two and constant weight $t = N/4 - 1$. Theorem F.1, with $s = N/8 - 1$, would give

$$2 \leq \left\lfloor \frac{t}{s+1} \right\rfloor = 1,$$

a contradiction. Thus S_a contains no affine plane. In a binary group, equality of the sums of two different unordered pairs of distinct elements gives four distinct elements forming an affine plane. Hence all $\binom{|S_a|}{2}$ pair sums are distinct and nonzero, so

$$\binom{|S_a|}{2} \leq N/2 - 1, \quad |S_a| \leq M_N.$$

Every $W \in \mathcal{C}$ has three nonzero elements in $W^\perp$. Double-counting gives $3|\mathcal{C}| = \sum_{a \neq 0} |S_a| \leq (N-1)M_N$. $\square$

In characteristic two, every binary additive domain of size N has all $(N-1)(N-2)/6$ codimension-two subspaces in this prefix class. Indeed, their degree- $N/4$ subspace locators are linearized: the next possible degree is $N/8$. Dividing by X gives exactly the punctured prefix above. The linearized form follows inductively from $L_{U \oplus \langle b \rangle} = L_U^2 + L_U(b)L_U$. The corollary therefore rules out preserving a positive asymptotic fraction of this quadratic family under any odd-characteristic relabeling. It does not exclude different locator identities or other transfers from binary-field constructions.

The additional checker tests 24,825 prefix classes across 390 injective relabelings, including characteristic three with prefix length three. It verifies the dual-incidence counts and absence of repeated pair sums, and checks 14,364 affine-plane parameterizations. The full binary families over $\mathbb{F}_{16}$ and $\mathbb{F}_{32}$ give negative controls of sizes 35 and 155, exceeding the respective odd-characteristic bounds 20 and 62.

A sublinear bound within every fixed multiplicative orbit

The preceding bound still permits a linear-sized shared-prefix family, which could be enough for the padding reduction. A stronger obstruction applies within a fixed multiplicative orbit of binary subspaces.

Theorem F.3 (Transfer within a multiplicative orbit). *Let $K = \mathbb{F}_{2^m}$, $N = 2^m$, and let $U \subset K$ be an r -dimensional binary subspace, with $1 \leq r < m$. For $a \in K^*$ put*

$$H_a = \{v \in K : \text{Tr}_{K/\mathbb{F}_2}(auv) = 0 \text{ for all } u \in U\}.$$

Injectively label $K \setminus \{0\}$ by elements of a field of characteristic different from two. Let $\mathcal{C}$ be a collection of distinct H_a whose punctured locators share their first s coefficients below the leading term, where $0 \leq s \leq N/2^r - 2$. Define

$$b = \left\lceil \log_2 \frac{(2^r - 1)N}{s + 1} \right\rceil, \quad h = |\{c \in K^* : cU = U\}|,$$

and assume $b \leq m$. Then

$$|\mathcal{C}| \leq \frac{4}{h} N^{1-2^{1-b}}.$$

For fixed r and a fixed positive prefix fraction $(s + 1)/N$, this is $o(N)$. In particular, for $r = 2$, $s = N/8 - 1$, and $m \geq 6$,

$$|\mathcal{C}| \leq \frac{4}{h} N^{15/16}, \quad h \in \{1, 3\}.$$

The case $h = 3$ is the family of $\mathbb{F}_4$ -linear hyperplanes.

Proof. Let $A \subset K^*$ consist of all normals for members of $\mathcal{C}$. Nondegeneracy of the trace pairing shows that $H_a = H_{a'}$ exactly when $aU = a'U$, so $|A| = h|\mathcal{C}|$. Newton identities express each power sum in the corresponding elementary symmetric functions without division. Thus the moments

$$M_j(a) = \sum_{v \in H_a \setminus \{0\}} x_v^j$$

are independent of $a \in A$ for $1 \leq j \leq s$; the zeroth moment is the common size $N/2^r - 1$. This step does not require the label-field characteristic to exceed s .

Suppose A contains an affine binary b -flat $a_\epsilon = a_0 + \sum_{i=1}^b \epsilon_i w_i$ with independent w_i . For $\beta \in \mathbb{F}_2^b \setminus \{0\}$ define

$$c_\beta(v) = 2^{-b} \sum_{\epsilon \in \mathbb{F}_2^b} (-1)^{\beta \cdot \epsilon} \mathbf{1}_{v \in H_{a_\epsilon}}.$$

All its moments vanish: $\sum_{v \neq 0} c_\beta(v) x_v^j = 0$ for $0 \leq j \leq s$. The character expansion of the annihilator of U gives, with $\text{Tr} = \text{Tr}_{K/\mathbb{F}_2}$,

$$c_\beta(v) = 2^{-r} \sum_{\substack{u \in U \setminus \{0\} \\ \text{Tr}(uw_i v) = \beta_i \ (1 \leq i \leq b)}} (-1)^{\text{Tr}(ua_0 v)}.$$

For fixed $u \neq 0$, multiplication by u preserves independence of the w_i , so the displayed fiber has $N/2^b$ points. Therefore c_β is supported on at most $(2^r - 1)N/2^b \leq s + 1$ points.

At least one such vector is nonzero over the label field. Otherwise invertibility of the binary Walsh transform would make all 2^b incidence rows identical. A fixed H_a can have at most $2^r - 1$ normals: for any fixed nonzero $u_0 \in U$, their distinct products au_0 all belong to its same annihilator. Our choice of b has $2^b > 2^r - 1$, a contradiction. On the support of a nonzero c_β , the square Vandermonde matrix of moments from zero to one less than the support size is invertible, because the labels are distinct. Its vanishing moments give the final contradiction. Thus A has no affine binary b -flat.

An elementary induction bounds an affine- j -flat-free set B in a binary group of size q by $4q^{1-2^{1-j}}$. For $j = 1$ it has at most one point. For $j \geq 2$, pair its elements along each nonzero direction v . The occupied pairs form an affine- $(j-1)$ -flat-free set in the quotient of size $q/2$. With $\alpha = 2^{2-j}$,

$$|B|(|B| - 1) = \sum_{v \neq 0} |B \cap (B + v)| \leq 8(q-1)(q/2)^{1-\alpha}.$$

Consequently $|B| \leq 1 + 2 \cdot 2^{\alpha/2} q^{1-\alpha/2} \leq 4q^{1-\alpha/2}$. Apply this with $j = b$ and $q = N$ to A .

Finally, $\{c \in K : cU \subseteq U\}$ is a finite subfield: it is closed under addition and multiplication, and every nonzero element acts bijectively on U . Its binary degree divides r . For $r = 2$ it has size two or four, so $h = 1$ or 3 ; the latter case makes U a scalar copy of $\mathbb{F}_4$. $\square$

This rules out a linear-sized transfer from any bounded number of fixed-codimension multiplicative orbits. At codimension two, a cN -sized shared-prefix family would have to occupy at least $(c/4)N^{1/16}$ orbits. Arbitrary codimension-two families can span linearly many orbits, so the general linear-sized-transfer question remains open.

The exact checker verifies the character support formula on all 63 affine binary five-flats avoiding zero at $N = 64$, and 20 such flats at $N = 256$. It separately checks multiplicative orbits with stabilizers one and three, and a codimension-three orbit with seven-dimensional affine sections. Odd-field prefix tests, a characteristic-two negative control, and a small complete odd-field transfer complement these checks. They supplement the universal proof.

Short prefixes determine subspaces under multiplicative relabeling

For the natural multiplicative relabeling, a stronger statement applies to all binary subspaces simultaneously, without restricting their orbits.

Theorem F.4 (Multiplicative relabeling rigidity). *Let $K = \mathbb{F}_{2^m}$, $N = 2^m$, and let $\chi : K^* \rightarrow E^*$ be an injective multiplicative character into a field of characteristic different from two. Label $v \neq 0$ by $\chi(v)$, or by $a\chi(v) + b$ with $a \neq 0$. For each r -dimensional binary subspace $U \subset K$, with $1 \leq r < m$, let H_U be its annihilator under the absolute trace pairing, and put $t = 2^r - 1$. The first $2t - 1$ coefficients of the punctured locator of H_U determine U uniquely whenever that prefix fits in the locator. In characteristic zero or characteristic greater than t , the first t coefficients suffice.*

The latter cutoff is sharp: if $r \mid m$, the $(N-1)/t$ subspaces $U = c\mathbb{F}_{2^r}$ have punctured locators with their first $t-1$ coefficients all zero under the unshifted labeling.

Proof. Write $\psi(v) = (-1)^{\text{Tr}_{K/\mathbb{F}_2}(v)}$ and

$$G_j = \sum_{v \in K^*} \psi(v) \chi(v)^j, \quad 1 \leq j < N-1.$$

Every G_j is invertible. Indeed, substituting $x = uy$ in $G_j G_{-j}$ gives

$$G_j G_{-j} = \sum_{u \neq 0} \chi(u)^j \sum_{y \neq 0} \psi((u+1)y) = N.$$

The inner sum is $N-1$ at $u = 1$ and -1 otherwise, and the nontrivial multiplicative character χ^j has total sum zero. The integer N is nonzero in E .

Expand the indicator of H_U with additive characters and substitute $y = uv$ for each nonzero $u \in U$. The zero- u term vanishes, giving

$$\sum_{v \in H_U \setminus \{0\}} \chi(v)^j = \frac{G_j}{2^r} \sum_{u \in U \setminus \{0\}} \chi(u)^{-j}. \quad (51)$$

Thus equal locator prefixes, and hence equal power sums, give equal moments of the two t -element dual label sets $\{\chi(u)^{-1} : u \in U \setminus \{0\}\}$. When the characteristic is zero or exceeds t , reverse Newton identities recover the dual set from its first t moments.

In every other allowed characteristic, subtract the indicator measures of the two dual sets. Their zeroth moments agree, and the difference is supported on at most $2t$ distinct points. Moments zero through $2t - 1$ and an invertible square Vandermonde minor force the difference to vanish. Injectivity of χ then recovers U . All moments used have index less than $N - 1$, since $r < m$. For affine relabeling, the binomial expansion recovers the original χ -moments inductively; its diagonal coefficient is $a^j \neq 0$.

For sharpness, if $U = c\mathbb{F}_{2^r}$, its annihilator is $\mathbb{F}_{2^r}$ -linear. Its nonzero elements are a union of multiplicative cosets of $\mathbb{F}_{2^r}^*$. Their labels are full cosets of the t th roots of unity, so the locator is a polynomial in X^t . Its first $t - 1$ coefficients vanish. The scalar stabilizer has size t , giving $(N - 1)/t$ distinct subspaces. $\square$

At codimension two, three coefficients suffice in characteristic greater than three, and five suffice in every odd characteristic. Thus a fixed positive shared-prefix fraction cannot survive this multiplicative relabeling, even for two supports, once N is large enough. The sharpness example has only a constant prefix length; its agreement gap shrinks with N and does not provide the desired fixed-gap list.

The exact checker exhausts 26,114 subspaces in six prime-field fixtures, checking both multiplicative and affine-multiplicative prefixes. A separate $\mathbb{F}_{330}$ fixture checks all 155 codimension-two subspaces of $\mathbb{F}_{32}$, without the large-characteristic Newton assumption. It checks 126,361 primal-dual moment identities and 27 Gauss products. A codimension-three example at $N = 512$ has 73 locators with six common zero coefficients and pairwise distinct seventh coefficients. Arbitrary nonmultiplicative relabelings across many subspace orbits remain outside these results.

G Boundedly many roots imply linear full-support MCA

A polynomial solution variety of quadratic degree need not produce quadratically many exceptional labels on a received line. We prove an $O(n)$ bound for every candidate family with a bounded number of distinct roots, allowing both the roots and their multiplicities to vary.

Let $\mathcal{D}$ be a set of n distinct field elements and let f, g be arbitrary words on it. For a candidate P of degree at most D , set

$$S(P, z) = \{x \in \mathcal{D} : f(x) + zg(x) = P(x)\}.$$

Call (P, z) bad at threshold $A > D$ if $|S(P, z)| \geq A$ and no polynomials F_0, G_0 of degree at most D agree with f, g , respectively, on all of $S(P, z)$. Equivalently, g has no degree- D interpolant on this full support, since otherwise $F_0 = P - zG_0$ works. A label is bad if it has at least one bad candidate in the family.

Theorem G.1 (Bounded root count). *Fix $r \geq 1$ and $0 < \rho_- \leq \rho_+ < a \leq 1$. Suppose $\rho_- \leq D/n \leq \rho_+$ and $A \geq an$. There are constants $C_{r,a}$, $c_{r,a}$, and $n_0(r, a, \rho_-)$ such that, for $n \geq n_0$, every received line has at most $C_{r,a}n$ bad labels witnessed by polynomials with at most r distinct roots over an algebraic closure, if the characteristic is zero or $p > c_{r,a}n$. Nonzero scalar coefficients, root positions, and root multiplicities may all vary. Including the zero polynomial costs at most n additional labels.*

The constants below are explicit but very large. No general first-order MCA theorem or improved better.codes certificate follows. We may work over an algebraic closure when discussing roots: a degree- D interpolant on more than D base-field points descends to the base field.

G.1 Pencils and clusters with small residual degree

An affine polynomial pencil $P_z = U + zV$, with $\deg U, \deg V \leq D$, contains at most n bad labels. Indeed a bad full support contains a coordinate where $(f, g) \neq (U, V)$; agreement there forces $g \neq V$ and determines the unique label $z = (U - f)/(g - V)$.

Suppose a selected family of distinct bad labels has candidates HQ , where H is fixed with at most s distinct roots, and $\deg Q \leq K$. We can take $K \leq D - \deg H$. If three graph points (z, Q) are not collinear, their line-incidence determinant is a nonzero polynomial of degree at most K . Their common agreement support has size at most $K + s$, after dividing by H away from its roots. Assume $K + s \leq a^3 n/16$. For $L \geq 4/a$, counting ordered triples through coordinates gives a lower bound $a^3 n L^3/8$: if b_x counts agreeing candidates, use $(b_x)_3 \geq \max(b_x - 2, 0)^3$ and convexity. Noncollinear triples contribute at most $(a^3 n/16)L^3$. Each ordered pair determines a unique graph line, which contains at most n selected bad labels by the pencil observation. Collinear triples therefore contribute at most $n^2 L^2$. Consequently

$$L \leq 16n/a^3. \quad (52)$$

The same bound covers $L < 4/a$. This argument needs no characteristic hypothesis.

Set $\delta = a^3/(128r)$. Fix any set S of at most $3r$ algebraic roots and consider candidates whose total root multiplicity outside S is at most $r\delta D$. Partition their exponents on S into boxes of side $w = \lfloor \delta D \rfloor + 1$. Within each box factor the product H given by its lower-corner exponents. The residual degree is at most

$$|S|(w - 1) + r\delta D \leq 4r\delta D \leq a^3 n/32.$$

If $3r \leq a^3 n/32$, equation (52) applies to each box. There are at most

$$B_0 = (1 + \lceil 1/\delta \rceil)^{3r}$$

boxes, so this cluster contains at most Cn selected bad labels, where

$$C = 16B_0/a^3. \quad (53)$$

The bound is uniform in S . Factors may be considered over the algebraic closure; a pencil through two original base-field candidates has base-field coefficients.

G.2 Four candidates with private high-multiplicity roots

Lemma G.2. *Let $P_1, \dots, P_4$ be nonzero polynomials of degree at most D , each with at most r distinct roots. Suppose each P_i has a root of multiplicity at least δD which is not a root of any other P_j . For an integer $m \geq 1$, put*

$$N = \binom{m+3}{3}, \quad J = m+1, \quad M = N - J.$$

Assume

$$\delta D > \binom{N-1}{2}(4r-1), \quad \text{char } \mathbb{F} = 0 \quad \text{or} \quad p > mD.$$

For two independent homogeneous linear forms F, G in four variables, the gcd of $F(P_1, \dots, P_4)$ and $G(P_1, \dots, P_4)$, after removing factors supported on the union of the roots of the P_i , has degree at most

$$\frac{24D}{m+5} + \frac{(4r-1)(M-1)}{2}. \quad (54)$$

Proof. The N homogeneous degree- m monomials in the P_i are linearly independent. For otherwise take a minimal relation with q terms $f_1, \dots, f_q$. A ratio of two distinct monomials has an order of absolute value at least δD at some private root, so $q = 2$ is impossible. Any $q - 1$ terms have a nonzero ordinary Wronskian: their degrees are less than p , and row reduction to distinct leading degrees gives a nonzero Vandermonde leading coefficient. The same argument holds in characteristic zero.

Let S be the union of the roots of the P_i and A_0 the common gcd of the f_i . All Wronskians obtained by omitting one term are constant multiples. Omit a term of largest degree for the global degree bound, and a term of smallest order separately at each root for the local bounds. Comparing them yields

$$\max_i \deg f_i - \deg A_0 \leq \binom{q-1}{2} (|S| - 1).$$

The left side is at least the rational degree of any ratio f_i/f_j , and hence at least δD . Since $q \leq N$ and $|S| \leq 4r$, this contradicts the assumed threshold.

The degree- m part of the homogeneous ideal (F, G) has dimension M : its quotient is a polynomial ring in two variables, of degree- m dimension J . Evaluation in the P_i is injective by the preceding paragraph. Write d_α for the degrees of the evaluated monomials. A basis of the evaluated ideal has sum of degrees at most the sum of the largest M of the d_α . Indeed, order the formal monomials by degree and row-reduce from highest degree; each basis element has degree at most the degree of its distinct pivot. Thus its nonzero Wronskian W satisfies

$$\deg W \leq \sum_\alpha d_\alpha - \binom{M}{2}.$$

At a root $s \in S$, let $a_\alpha(s)$ be the monomial orders. Intersecting a codimension- J subspace with the span of monomials of order at least each integer k gives an adapted basis whose sum of orders is at least $\sum_\alpha a_\alpha(s) - J \max_\alpha a_\alpha(s)$. The dimension of each such intersection loses at most J ; summing these filtration dimensions proves the bound. Cancellations can only increase orders. A constant basis change preserves the order of W , so, using

$$\sum_{s \in S} \sum_\alpha a_\alpha(s) = \sum_\alpha d_\alpha, \quad \sum_{s \in S} \max_\alpha a_\alpha(s) \leq m \sum_{i=1}^4 \deg P_i,$$

we obtain

$$\sum_{s \in S} \text{ord}_s W \geq \sum_\alpha d_\alpha - Jm \sum_i \deg P_i - |S| \binom{M}{2}.$$

Let T be the gcd in the statement. It divides every evaluated ideal element. The identity $\text{Wr}(Th_1, \dots, Th_M) = T^M \text{Wr}(h_1, \dots, h_M)$ contributes at least $M \deg T$ additional zeros outside S . Combining the degree and order bounds gives

$$\deg T \leq \frac{4JmD}{M} + \frac{(4r-1)(M-1)}{2}.$$

Since $M = m(m+1)(m+5)/6$, this is (54). □

This is an elementary specialization of the high-power gcd phenomenon. A broader characteristic-zero result is Guo–Sun–Wang [31, Theorem 11]; the finite-characteristic statement above follows directly from the Wronskian proof.

G.3 Counting the remaining quadruples

Select one nonzero bad candidate at each bad label. Call a root heavy if its multiplicity is at least δD . A quadruple without a private heavy root for every member has some member P_i whose heavy roots all lie in the union of the other three root sets. Fix the other three candidates. Their root union has size at most $3r$, and every root of P_i outside it has multiplicity less than δD . The cluster bound (53) gives at most Cn choices for this selected bad label. Summing over the four possible members, there are at most $4CnL^3$ such ordered quadruples. This includes candidates with no heavy roots.

For every other quadruple at distinct labels, common agreement coordinates satisfy the two independent linear forms

$$\begin{aligned}(z_2 - z_3)P_1 + (z_3 - z_1)P_2 + (z_1 - z_2)P_3 &= 0, \\ (z_2 - z_4)P_1 + (z_4 - z_1)P_2 + (z_1 - z_2)P_4 &= 0.\end{aligned}$$

Lemma G.2, together with the at most $4r$ roots in the union, bounds their common support. Choose

$$m \geq 1536/a^4 - 5, \quad \frac{(4r-1)(M-1)}{2} + 4r \leq a^4 n/64,$$

as well as the lemma's hypotheses and $3r \leq a^3 n/32$. Their common support then has size at most $a^4 n/32$. For $L \geq 6/a$, fourth incidence gives

$$\frac{a^4 n L^4}{16} \leq \frac{a^4 n L^4}{32} + 4Cn^2 L^3.$$

Therefore

$$L \leq \frac{128C}{a^4} n = \frac{2048}{a^7} (1 + \lceil 128r/a^3 \rceil)^{3r} n. \quad (55)$$

The small- L case also satisfies this bound. At fixed positive rate, all degree and length thresholds eventually hold, and a sufficiently large fixed lower bound on p/n ensures $p > mD$. The zero polynomial contributes at most n more labels by the same single-coordinate argument as for a fixed pencil. This proves Theorem G.1.

G.4 A quadratic-degree solution surface

Fix $e \geq 1$ in characteristic zero or $p > 2e$. The nonzero polynomial solutions of degree at most $2e$ to

$$e^2 P^2 P''' + 3e(1-e)PP'P'' + (1-e)(1-2e)(P')^3 = 0 \quad (56)$$

are exactly $P = cH^e$ with $\deg H \leq 2$. At a root of multiplicity m , the coefficient of the lowest possible term, after removing a nonzero local scalar, is $m(m-e)(m-2e)$. Thus every root has multiplicity e or $2e$. Conversely substitution verifies every cH^e ; equivalently this is the denominator-cleared equation $(P^{1/e})''' = 0$. The monic base descends by uniqueness.

The projective map $[H] \mapsto [H^e]$ from the plane of quadratics is injective and base-point free, with injective tangent maps since $e \neq 0$ in the field. Its image has degree e^2 . Explicitly, cut by $P(u) = P(v)$ and $P(w) = P(v)$ for three distinct points u, v, w . There is no nonzero solution with $H(v) = 0$, so normalize $H(v) = 1$. The values $H(u), H(w)$ independently range over the e -th roots of unity, giving e^2 distinct reduced points by quadratic interpolation. This fixed third-order, jet-degree-three equation has an actual solution variety of degree $\Theta(n^2)$ at fixed positive rate, but Theorem G.1 gives only $O(n)$ bad labels in its large-characteristic regime.

The checkers in `research/bounded_root_mca/` verify unequal-degree Wronskian bookkeeping and cluster counting, including every full support of 1,025 bad labels in a length-4,096 example with at most two roots per candidate. The earlier `research/power_family_mca/` package verifies the quadratic-power classification and degree sections. These checks supplement the proofs. Families with a growing number of distinct roots and the fixed-gap quadratic lower-bound target remain open here.

H Actual first-order components have linear positive-dimensional degree

The degree of a Taylor-reconstruction cover can exceed the degree of the actual polynomial-solution locus. The following bound separates its curve and surface components. We then exhibit quadratically many actual isolated regular solutions, even at derivative degree one, and prove that this example still has only constantly many nearby labels at a fixed positive agreement gap. The full first-order MCA conjecture remains unresolved.

Theorem H.1 (Positive-dimensional actual components). *Let E be algebraically closed of characteristic zero or $p > D$, with $D \geq 1$. Let $Q(X, z, u, v) \neq 0$ have total degree at most $B \geq 1$ in (u, v) and degree at most H in z , with no restriction on its X -degree. In affine (z, P) coefficient space, let $\mathcal{V}$ be the Zariski closure of the actual regular solutions*

$$\deg P \leq D, \quad Q(X, z, P, P') = 0, \quad Q_v(X, z, P, P') \neq 0.$$

Put

$$\tau = \max\{0, 2D - 3\}, \quad b = 1 + \tau(B - 1), \quad T = b + \tau H, \quad K = H + BT, \quad q = B + H.$$

Every component of $\mathcal{V}$ has dimension at most two. If J_i is the sum of the projective degrees of its i -dimensional irreducible components, then

$$J_1 \leq qK, \quad J_2 \leq qT.$$

In particular $J_1 + J_2 = O_{B,H}(D)$. The reduced generic challenge fiber of the surface components has total degree at most $\deg_v Q$. The bounds concern actual components defined over E before choosing a generic Taylor center; they do not apply to arbitrary Taylor images.

Proof. If the regular locus is empty, the claims are immediate. Otherwise adjoin a Taylor center t transcendental over E . Put $S = Q_v$, $A_0 = Q_X + vQ_u$, and

$$\mathcal{D} = \partial_X + v\partial_u - (A_0/S)\partial_v,$$

holding z constant. On $S \neq 0$, implicit differentiation reconstructs every derivative through order D . For $j \geq 2$, write this derivative as N_j/S^{2j-3} , with $N_2 = -A_0$. If $e = 2j - 3$, the recurrence is

$$\begin{aligned} N_{j+1} &= S^2(\partial_X N_j + v\partial_u N_j) - A_0 S \partial_v N_j \\ &\quad - e N_j (S(\partial_X S + v\partial_u S) - A_0 \partial_v S). \end{aligned}$$

Induction gives jet degree at most $1 + (2j - 3)(B - 1)$ and challenge degree at most $(2j - 3)H$. The common-denominator Taylor polynomial through degree D is $F(X)/S(t, z, u, v)^\tau$, where every coefficient of F has jet degree at most b and challenge degree at most τH . Division by the Taylor factorials is valid since $p > D$.

The actual locus in initial data. Substitute this polynomial and its derivative into Q and clear $S^{\tau B}$. In a term of jet degree $e \leq B$, the numerator has jet degree at most

$$eb + \tau(B - e)(B - 1) \leq Bb$$

and challenge degree at most $H + \tau BH$. Thus every X -coefficient of the residual has total (z, u, v) degree at most K . Include the initial equation $Q(t, z, u, v) = 0$, of total degree at most q . On $S(t, z, u, v) \neq 0$, these equations describe exactly the initial data of actual degree $\leq D$ solutions. Reconstruction and the initial-jet map are inverse regular maps on these open loci. Consequently all actual regular components have dimension at most two. Their closures acquire no irreducible component supported entirely on the boundary of this open locus.

Curve components. Choose λ, c independently generic after adjoining t , and cut the actual locus by

$$P(t) + \lambda z = c.$$

Each projective curve component has finitely many points at infinity, defined over E . At any such point the coefficient direction and the z direction are not all zero, so $\sum_i a_i t^i + \lambda z \neq 0$. The chosen hyperplane avoids these points. Its generic intersection also avoids the finite separant boundary and the finite intersections of each curve with other actual components, including surfaces.

The function $P(t) + \lambda z$ on each curve is separable. The coefficient functions and z generate the curve's function field, so their differentials cannot all vanish; independence of t, λ prevents cancellation. Its generic fiber is therefore reduced. The section selects exactly J_1 distinct affine points in the regular reconstruction locus, away from the surface components.

In initial data the section substitutes $u = c - \lambda z$ into $Q(t, z, u, v)$, a nonzero plane equation of degree at most q . Each residual restricts to degree at most K . A plane component on which all residuals vanish, unless contained in $S = 0$, maps to a positive-dimensional part of this generic hyperplane section of the actual locus, and hence to a section of a surface component. None of the selected J_1 points lies on it. Nor do they lie on components contained in $S = 0$. On every remaining plane component at least one residual is nonzero. A generic linear combination of the residuals vanishes identically on none of them. Bézout's bound for its intersections with their union gives $J_1 \leq qK$.

Surface components. Use the same generic section. It contains no curve at infinity of an actual surface: a generic form $P(t) + \lambda z$ is nonzero on each such curve. Thus the affine section closures retain the full surface degrees. On the regular initial-data locus the projection to (z, u) is generically separable, because $Q_v \neq 0$. A generic line $u + \lambda z = c$ therefore gives generically reduced sections.

Their initial-data curves are distinct components of the plane equation $Q(t, z, c - \lambda z, v) = 0$ and have summed degree at most q . The projective reconstruction coordinates can be written as

$$S^\tau, \quad zS^\tau, \quad \text{the coefficient numerators of } F.$$

After the substitution, all have degree at most T : the first two have degrees at most $T - 1, T$, and the remaining numerators have degree at most $b + \tau H = T$. Homogenizing to degree T , the degree of the rational image of each source curve is at most T times its degree, as seen by pulling back a generic hyperplane. Summing gives $J_2 \leq qT$.

Finally fix the generic challenge and work over its algebraic closure. At an independent generic Taylor center, regular reconstruction identifies every positive-dimensional actual solution component with a component of the plane equation in (u, v) . Projection to u has total degree at most $\deg_v Q$.

The hyperplane $P(t) = u_0$ avoids all curve points at infinity by the same transcendence argument, and its generic fiber is reduced. It therefore counts the full coefficient-space degree of these curves, proving the generic-fiber assertion. $\square$

Linear degree cannot be replaced by a constant. Let $D \geq 2$, in characteristic zero or $p > D + 1$, and put

$$R(X, z) = X^{D+1} + zX - 1, \quad Q = RY_1 - R_X Y_0 + Y_0^2.$$

Here $B = 2$, $H = 1$. Every nonzero polynomial solution satisfies $(R/P)' = 1$. The rational function $R/P - X$ has degree at most $D + 1$ and derivative zero, so it is constant. Indeed, in positive characteristic its degree is less than p , whereas a nonconstant rational p th power has degree at least p . Hence all nonzero solutions are

$$z = a^{-1} - a^D, \quad P_a(X) = \sum_{i=0}^{D-1} a^i X^{D-i} + a^{-1}, \quad a \neq 0.$$

Conversely these are solutions, since $R(a, z) = 0$ and $P_a = R/(X - a)$. The coefficient of X^{D-1} recovers a . After multiplying the projective coordinates by a , they span all monomials $1, a, \dots, a^{D+1}$, so the curve has degree $D + 1$. The additional zero-solution line gives $J_1 = D + 2$.

Linear surface growth also occurs: $(X - z)P' - DP = 0$ has the actual family $P = c(X - z)^D$. Two generic affine hyperplanes restrict to $A(z) + cB(z) = 0$ and $C(z) + cE(z) = 0$, with A, C of degree at most one and B, E of degree at most D . Their elimination polynomial $AE - CB$ generically has degree $D + 1$ and no extraneous roots. Thus this surface has degree $D + 1$, although its generic challenge fiber is a line.

Corollary H.2 (Nonaffine actual curves contribute linearly). *Fix a received line $f + zg$ on n distinct evaluation points and an agreement threshold $D < A \leq n$. Among the actual curve components in Theorem H.1, omit those that are affine codeword graphs $P = F + zG$, with $\deg F, \deg G \leq D$. The remaining components contain at most*

$$\frac{n - D}{A - D} qK$$

polynomial-challenge pairs with at least A agreements. In particular, their exceptional-label contribution is $O_{B,H,\eta}(n)$ when $A - D \geq \eta n$ and $D \leq n$.

Proof. On an irreducible curve C , call a coordinate persistent if its agreement hyperplane contains C . If more than D coordinates were persistent, interpolation at $D + 1$ of them would force $P = F + zG$ identically on C . A vertical curve would then be a point; otherwise C would be that affine codeword graph. Thus every retained curve has at most $h \leq D$ persistent coordinates.

Each of its other $n - h$ agreement hyperplanes meets C in at most $\deg C$ distinct affine points. Every selected nearby pair lies on at least $A - h$ of these hyperplanes. Its number is therefore at most

$$(\deg C) \frac{n - h}{A - h} \leq (\deg C) \frac{n - D}{A - D}.$$

Sum over the retained curves and use $J_1 \leq qK$. $\square$

Scope and verification of the component bounds. The theorem gives no linear bound on the number of isolated joint solutions; the next construction shows that such a bound is false. Nor does it bound by a constant the number of persistent affine codeword graphs that can each contribute accidental agreements. These obstacles prevent an inference of general $O(n/|F|)$ first-order MCA from the component-degree bounds alone. The checks in `research/first_order_actual_components/` verify the reconstruction recurrence on five equations, 576 finite Taylor fixtures, and 40 sharp-order linear-equation curves. A second checker exhausts 33,849 polynomial–challenge pairs for the nonlinear curve family, verifies 39 split reduced hyperplane sections, and detects extra solutions outside that example’s stricter characteristic hypothesis $p > D + 1$. These checks supplement the geometric proof; they do not certify the generic-section argument by themselves.

H.1 Quadratically many isolated regular first-order solutions

Theorem H.3 (Quadratic isolated count at derivative degree one). *Let $D \geq 1$, in characteristic zero or $p > D + 2$, and let R be a monic squarefree polynomial of degree $N = D + 1$ with nonzero roots. The first-order equation*

$$(z - X^2)(RP' - R'P + P^2) + 2XRP - 2R^2 = 0 \quad (57)$$

has exactly $\binom{D+2}{2}$ polynomial–challenge solutions of degree at most D , all regular. They are indexed by unordered multisets $\{a, b\}$ of roots of R , including $a = b$, and are

$$z = ab, \quad P_{a,b} = \frac{R}{X - a} + \frac{R}{X - b}.$$

Thus its reduced actual joint solution locus consists of exactly that many isolated points. Its jet degree is two, derivative degree one, and challenge degree one. For every D , these points can all have distinct labels over a prime field of size $O(D^3)$.

Proof. Put $U = (X^2 + z)P - 2XR$ and $V = R - XP$. Direct differentiation gives

$$U'V - UV' = (z - X^2)(RP' - R'P + P^2) + 2XRP - 2R^2.$$

Since $R(0) \neq 0$, V is not identically zero. The rational function U/V has degree at most $D + 2$ and derivative zero, so it is constant. In positive characteristic this uses the strict bound $D + 2 < p$: a nonconstant rational p th power has degree at least p . Writing the constant as $-u$ gives

$$(X^2 - uX + z)P = (2X - u)R.$$

For $H = X^2 - uX + z$, this says $P = RH'/H$. Every root of H must belong to R : a root of multiplicity one or two gives a nonzero simple-pole residue in H'/H , since the characteristic is not two. Conversely every such multiset gives the displayed solution. Its partial-fraction residues distinguish the multisets. The separant is $(z - X^2)R$, nonzero as a polynomial in X at every solution, so all solutions are regular.

For distinct prime-field labels, choose the roots to form a multiplicative Sidon set including diagonal products. If k roots have already been chosen and S is their $k(k + 1)/2$ products, a new root x is excluded by at most

$$(k + 2)|S| + k + 1$$

field elements: those with $xa \in S$ for an old root a , those with $x^2 \in S$, the old roots, and zero. Products among the new pairs cannot collide otherwise. For $k \leq N - 1$ this bound is at most $(N^3 + N)/2$. A prime larger than this bound and $D + 2$ therefore suffices, and a prime of size $O(N^3)$ exists. This constructs all the solutions and their pairwise distinct labels in the prime field. $\square$

The solution count by itself therefore cannot prove linear first-order MCA. The following agreement bound also shows why the example is not a quadratic proximity-gap lower bound.

Proposition H.4 (Constant fixed-gap count for the isolated family). *Over a finite field satisfying Theorem H.3, fix a received line $f + zg$ on n distinct coordinates and an agreement threshold $D < A \leq n$. Let L be the number of distinct labels with an A -agreeing candidate from the displayed family. Then*

$$L \leq \frac{(D+2)n}{A-D}.$$

More strongly, let r_0 be the number of domain points that are roots of R , put $m = n - r_0$ and $t = A - r_0$, and suppose $t^3 > 4m^2$. Then

$$L \leq \frac{2m}{t - (4m^2)^{1/3}}.$$

In particular $L = O_\eta(1)$ for $A - D \geq \eta n$ and sufficiently large n . Neither bound requires that all family labels be distinct.

Proof. Every candidate has degree D , hence at least $A - D$ of its agreements have nonzero value. At a root c of R , only the N multisets containing c have nonzero value. At a point x outside the roots, write $\alpha = f(x)/R(x)$ and $\beta = g(x)/R(x)$. For fixed a , agreement is

$$\frac{1}{x-a} + \frac{1}{x-b} = \alpha + \beta ab.$$

Clearing $x - b$ gives a nonzero polynomial of degree at most two in b . If it vanished identically, its quadratic coefficient would give $\beta = 0$ since $a \neq 0$, its linear coefficient would give $\alpha = 1/(x - a)$, and its constant coefficient would still be nonzero. Thus there are at most $2N$ ordered matches. Diagonal matches satisfy $2 = (\alpha + \beta a^2)(x - a)$, a nonzero cubic equation, so there are at most three of them. The number of unordered matches is consequently at most $\lfloor (2N + 3)/2 \rfloor = N + 1$. Counting nonzero agreement incidences proves the first bound, even for the number of nearby pairs.

For the stronger bound, choose one nearby candidate at each of the L distinct labels and put

$$W_{a,b} = P_{a,b}/R = \frac{1}{X-a} + \frac{1}{X-b}.$$

No three of the labeled functions $(ab, W_{a,b})$ at distinct labels are affine-collinear. To prove this, suppose their affine relation has coefficients c_1, c_2, c_3 , all nonzero, with $\sum c_i = \sum c_i z_i = 0$. A root belonging to only one support cannot have its nonzero residue canceled. Each root in the union therefore belongs to at least two supports, so the union has at most three roots. If it has three, the supports must be the three edges of a triangle. Their residue matrix has determinant 2, so they are linearly independent. If it has two roots a, b , the three distinct multisets must be $\{a, a\}, \{a, b\}, \{b, b\}$. Their only affine relation is $W_{a,a} - 2W_{a,b} + W_{b,b} = 0$, but the same coefficients give $a^2 - 2ab + b^2 = (a - b)^2 \neq 0$ on the labels. One root cannot support three distinct candidates. This proves the claim.

For any three distinct labels, take $(c_1, c_2, c_3) = (z_2 - z_3, z_3 - z_1, z_1 - z_2)$. The nonzero rational function $\sum c_i W_i$ has a denominator of degree at most six. Using the product of the three monic quadratic denominators, its numerator has degree at most four: each summand has leading term $2X^5$, which cancels because $\sum c_i = 0$. Thus three candidates can agree with the received line at at most four common coordinates outside the roots of R .

Each selected candidate has at least t agreements on these m coordinates. If h_x counts the agreeing candidates at x , then $\sum h_x \geq tL$ and

$$\sum_x \binom{h_x}{3} \leq 4 \binom{L}{3} \leq \frac{4L^3}{6}.$$

For every nonnegative integer h , $6\binom{h}{3} \geq (h-2)_+^3$. Convexity therefore gives

$$m \left(\frac{tL}{m} - 2 \right)_+^3 \leq 4L^3.$$

Taking cube roots and rearranging proves the bound (the case $tL \leq 2m$ also satisfies it). Finally $r_0 \leq D+1$, so $t \geq \eta n - 1$ and $m \leq n$ at a fixed gap. The denominator is positive and of order n for sufficiently large n . $\square$

Exact checks. The isolated-family checker exhausts 17,410 polynomials, covering 180,384 polynomial–challenge pairs, and checks every pair of local received values at each coordinate of four small-field fixtures. It verifies the rational derivative identity and constructs prime-field Sidon examples through $D = 64$ with 2,145 distinct regular isolated labels. It also checks the nonzero degree-four triple numerators and the integer convexity inequality. A characteristic three negative control has an extra solution outside the theorem’s strict characteristic hypothesis. These finite checks supplement the classification and incidence proofs above.

H.2 A positive density of nonsingular agreements gives linear MCA

The preceding component counts do not by themselves bound all bad labels linearly. Conditioning Taylor reconstruction at an agreeing coordinate gives a different sufficient condition that also covers surface components. It isolates singular agreement coordinates as the remaining obstruction to this argument.

Theorem H.5 (Nonsingular-agreement bound). *Let F have characteristic zero or $p > D$, with $D \geq 1$, and let $Q(X, z, u, v) \neq 0$ have jet degree at most $B \geq 1$ and challenge degree at most H . Its X -degree is unrestricted. Put*

$$\tau = \max\{0, 2D - 3\}, \quad b = 1 + \tau(B - 1), \quad T_0 = b + \tau H, \quad K = H + BT_0, \quad q = B + H.$$

Fix a received line $f + zg$ on n distinct coordinates, an agreement threshold $D < A \leq n$, and an integer $L \geq 1$. Consider labels admitting a degree- $\leq D$ polynomial solution $Q(X, z, P, P') = 0$ whose full agreement support S satisfies $|S| \geq A$, has no degree- $\leq D$ polynomial agreeing with g on all of S , and contains at least L coordinates x with

$$Q_v(x, z, P(x), P'(x)) \neq 0.$$

The number of these labels is at most

$$\frac{n}{L} \left(qK + qT_0 \frac{n-D}{A-D} + qn \right). \tag{58}$$

In particular it is $O_{B,H,\eta,\lambda}(n)$ when $A - D \geq \eta n$ and $L \geq \lambda n$, with $\eta, \lambda > 0$ fixed.

Proof. Fix a domain point x and substitute $u = f_x + zg_x$ into the initial equation. The resulting polynomial in (z, v) has degree at most q . If it vanishes identically, so does its v derivative, and there are no nonsingular initial data to count at this coordinate. Otherwise work on the open locus $S_0 = Q_v(x, z, f_x + zg_x, v) \neq 0$ of this plane curve.

The Taylor reconstruction in Theorem H.1 applies at the fixed center x on this open locus. Its common denominator is S_0^τ , of degree at most $T_0 - 1$; its coefficient numerators and zS_0^τ have degree at most T_0 . Thus its projective coordinate degree is at most T_0 . Exact solutions are selected by residual equations of degree at most K . Reconstruction and the initial-jet map are inverse regular maps on this open exact-solution locus.

The isolated points of the residual locus number at most qK . Indeed, on every component of the initial plane curve not wholly satisfying the residuals, some residual is nonzero. A generic linear combination over an algebraic closure is nonzero on each such component; Bézout gives the bound. Components contained in $S_0 = 0$ are omitted, and points on a retained curve are accounted for with that curve.

The retained source curves are components of the initial plane equation, so there are at most q and their degrees sum to at most q . Their reconstructed image curves have total degree at most qT_0 , by pulling back a generic hyperplane under the rational map of degree at most T_0 . Initial-jet recovery ensures none of these images collapses to a point.

For every nonaffine image curve, the incidence argument of Corollary H.2 gives at most its degree times $(n - D)/(A - D)$ nearby pairs. That argument applies to these curves whether or not they are components of the full joint locus. The total contribution is at most $qT_0(n - D)/(A - D)$.

Each remaining image curve is a graph $P = F_0 + zG_0$ and has at most n bad labels. A support consisting only of persistent agreements has the common witnesses F_0, G_0 ; a bad support therefore contains an accidental agreement at a nonpersistent coordinate, which forces one label. If the graph does not descend to F , it has at most one F -rational point at distinct challenge labels: two such points would recover F_0, G_0 over F . This also satisfies the bound n . There are at most q graphs.

Hence this fixed coordinate occurs nonsingularly in a bad nearby witness for at most $qK + qT_0(n - D)/(A - D) + qn$ labels. Select one qualifying witness per label. Each supplies at least L such coordinate-label incidences. Summing over all n coordinates and dividing by L proves (58). $\square$

Corollary H.6 (Value-independent separant). *Suppose*

$$Q(X, z, u, v) = R(X, z)v - \mathcal{A}(X, z, u), \quad R \neq 0,$$

under the same degree and characteristic hypotheses, and put $r = \deg_X R$. If $A > \max\{D, r\}$, all its full-support bad labels number at most

$$H + \frac{n}{A - r} \left(qK + qT_0 \frac{n - D}{A - D} + qn \right). \quad (59)$$

This is $O(n)$ for fixed B, H and fixed positive lower bounds on $(A - D)/n$ and $(A - r)/n$. The value degree of $\mathcal{A}$ is arbitrary. At each fixed label with $R(X, z) \not\equiv 0$, its ordinary list size at threshold A is at most $\lfloor (n - r)/(A - r) \rfloor$.

Proof. There are at most H labels with $R(X, z) \equiv 0$, using any one nonzero X -coefficient of R . At all others, at most r domain points are zeros of R , so every nearby candidate has at least $A - r$ nonsingular agreements. Apply the theorem with $L = A - r$.

For the ordinary list bound, two distinct solutions at the same label have a nonzero difference Δ satisfying $R\Delta' = C(X)\Delta$. At a zero of Δ where R is nonzero, the multiplicity $1 \leq m \leq D < p$ makes the left side have order $m - 1$ and the right side order at least m , a contradiction. Thus a

value at a nonsingular coordinate determines at most one solution. If $r_z \leq r$ domain points are singular, nearby candidates have disjoint regular agreement supports of size at least $A - r_z$. Their number is at most $(n - r_z)/(A - r_z) \leq (n - r)/(A - r)$. $\square$

Sharp linear order in a nonlinear subclass. For a nonzero polynomial W , the equation

$$WP' - W'P - P(P - W) = 0$$

has exactly the polynomial solutions $0, W$. To see this, write $Z = P/W$, giving $Z' = Z(Z - 1)$. If $Z \neq 0$, then $U = 1 - 1/Z$ satisfies $U' = U$. In characteristic zero, a nonzero rational logarithmic derivative U'/U vanishes at infinity and cannot equal one. In characteristic p , the p -fold ordinary derivative is zero on $F(X)$, whereas $U' = U$ would make it equal U . Hence $U = 0$.

Let $n = 12s$, $D = 3s$, $A = 6s$, and choose a prime $p > 12s$. On domain $\{0, \dots, n - 1\}$ let W be the locator of the first $3s$ points. Put $f = g = 0$ on these roots and on a second block of $3s$ points; put $f = W, g = 0$ on a third block of $3s$. At the final $3s$ coordinates, indexed by $j = 1, \dots, 3s$, put $f = 3jW, g = W$. Each candidate cW , $c \in \{0, 1\}$, has $6s$ persistent agreements and acquires one more at each label $z = c - 3j$. These $6s = n/2$ labels are distinct modulo p . On each resulting support, g has $6s > D$ zeros and one nonzero value, so it has no degree- $\leq D$ explanation. Every other label has only persistent supports and is good. Thus exactly $n/2$ labels are bad, with jet degree two, challenge degree zero, code rate tending to $1/4$, and agreement gap $1/4$.

Scope and checks. The theorem does not assert that interpolation equations have many nonsingular agreement coordinates. At fixed gap its bound is $O(n^2/L)$; a superlinear example must concentrate on witnesses with a vanishing fraction of such coordinates. The general first-order conjecture remains open. The checks in `research/quasilinear_first_order/` enumerate small-field polynomial solutions, verify regular-coordinate uniqueness and list incidences, include implicit equations with actual solution surfaces, and check the exact $n/2$ lower construction at five lengths. These finite checks supplement the reconstruction and incidence proof; no independent review or literature-priority claim is made.

I Sharp isolated-solution growth at fixed derivative order

The quadratic isolated family of Appendix H extends to every fixed derivative order. Here we give a matching general upper bound and an explicit family attaining its exponent. These are bounds on the actual polynomial-solution locus, before any agreement filter. The attaining family still has a constant nearby-label count at a fixed positive gap, so this sharpness does not establish sharp MCA error.

Theorem I.1 (Isolated regular solution upper bound). *Let $d \geq 1$, $D \geq d$, and let E be algebraically closed of characteristic zero or $p > D$. Let $Q(X, z, u_0, \dots, u_d)$ have total jet degree at most $B \geq 1$ and challenge degree at most H , with unrestricted X -degree. Let J_0 count the isolated points of the reduced closure of actual regular pairs*

$$Q(X, z, P, P', \dots, P^{(d)}) = 0, \quad \deg P \leq D, \quad Q_{u_d}(X, z, P, \dots, P^{(d)}) \neq 0.$$

Put

$$\tau = \max\{0, 2(D - d) - 1\}, \quad b = 1 + \tau(B - 1), \quad K = H + B(b + \tau H), \quad q = B + H.$$

Then $J_0 \leq qK^{d+1} = O_{d,B,H}(D^{d+1})$.

Proof. If $J_0 = 0$ there is nothing to prove. Every isolated point of this closure belongs to the actual regular locus, since that locus is dense. Choose a Taylor center t at which the separant is nonzero at every isolated regular solution; there are finitely many such points and only finitely many excluded centers for each. Write

$$S = Q_{u_d}, \quad A_0 = Q_X + \sum_{i < d} u_{i+1} Q_{u_i}, \quad \mathcal{D} = \partial_X + \sum_{i < d} u_{i+1} \partial_{u_i} - (A_0/S) \partial_{u_d}.$$

The quotient-rule recurrence in the proof of Theorem H.1, with this derivation, gives for the j th derivative, $j > d$, denominator $S^{2(j-d)-1}$, jet numerator degree at most $1 + (2(j-d) - 1)(B-1)$, and challenge degree at most $(2(j-d) - 1)H$. The initial derivatives $u_0, \dots, u_d$ need no denominator. Thus the degree- D Taylor reconstruction, with common denominator S^τ , has coefficient numerators of jet degree at most b and challenge degree at most τH . Taylor factorials are invertible.

Substitute this polynomial and its first d derivatives into Q , clear $S^{\tau B}$, and take coefficients in X . Every residual has total initial-data degree at most K : a term of jet degree $e \leq B$ has numerator jet degree at most $eb + \tau(B-e)(B-1) \leq Bb$ and challenge degree at most $H + \tau BH$. Include the initial equation $Q(t, z, u_0, \dots, u_d) = 0$, of degree at most q . On $S(t, z, u_0, \dots, u_d) \neq 0$, these equations describe precisely the initial jets of actual degree- $\leq D$ solutions. The initial-jet and reconstruction maps are inverse regular maps there, so they preserve the isolated points being counted.

The initial hypersurface has dimension at most $d+1$ in $d+2$ variables. Cut successively by $d+1$ generic linear combinations of the residuals, each of degree at most K . At each stage, discard positive-dimensional components on which every residual vanishes; none can contain a target isolated point in the regular open locus. On each remaining component through a target point, some residual is nonzero, and a generic combination cuts its dimension. Choose the combinations simultaneously for the finitely many components at each step. This process retains every target isolated point. Bézout bounds the sum of the degrees of the retained intersections by qK^{d+1} . This argument does not require that the whole unpruned intersection be zero-dimensional: other positive-dimensional components may remain away from the target points. $\square$

Theorem I.2 (Attaining every isolated-count exponent). *Fix $r = d+1 \geq 2$ and $D \geq 1$, in characteristic zero or $p > D+r$. Let R be monic and squarefree of degree $N = D+1$, with nonzero roots. Put*

$$\phi_j = X^j \ (1 \leq j < r), \quad \phi_r = X^r + z, \quad F_j = R\phi_j' - P\phi_j,$$

and let Q_r be the Wronskian of $F_1, \dots, F_r$, with derivative rows $0, \dots, r-1$ and z held constant. Then Q_r is a nonzero differential polynomial of order d , jet degree r , challenge degree one, and degree one in its highest derivative. Its degree- $\leq D$ polynomial solutions are exactly

$$P = R \frac{H'}{H}, \quad z = H(0),$$

where H ranges over monic degree- r polynomials whose root multisets lie among the roots of R . All are regular, and there are exactly

$$\binom{D+r}{r} = \Theta_d(D^{d+1})$$

isolated polynomial-challenge pairs, with no positive-dimensional solution component. For fixed d , all labels can be distinct over prime fields of size $O_d(D^{2d+1})$.

Proof. The separant has the explicit value

$$\frac{\partial Q_r}{\partial P^{(r-1)}} = (-1)^r C_r R^{r-1} (X^r + (-1)^{r-1} z), \quad C_r = \prod_{j=1}^{r-1} j!. \quad (60)$$

Indeed, the highest jet occurs only in the last derivative row, with coefficient row $-(\phi_1, \dots, \phi_r)$. Move that row to the top. In the other rows the successive highest derivatives of the ϕ_j have coefficient R , giving $(-1)^r R^{r-1} \text{Wr}(\phi_1, \dots, \phi_r)$. The monomial Vandermonde formula gives

$$\text{Wr}(\phi_1, \dots, \phi_r) = C_r (X^r + (-1)^{r-1} z).$$

This is nonzero for every constant z . The pure P^r coefficient is $(-1)^r \text{Wr}(\phi_1, \dots, \phi_r)$, so the jet degree is exactly r . All claimed degree bounds follow; the X -degree is also $O_r(D)$, although the upper bound does not need it.

Each specialized F_j has degree at most $D+r < p$. In this range a polynomial Wronskian vanishes exactly when its columns are linearly dependent over constants. To see the nontrivial direction, echelon any independent polynomial basis to distinct leading degrees. The leading Wronskian coefficient is their nonzero leading-coefficient product times the Vandermonde of these distinct degrees, all less than p . The same argument works in characteristic zero.

The first $r-1$ columns are independent for every polynomial P . Otherwise $RH' - PH = 0$ for a nonzero $H \in \text{span}(X, \dots, X^{r-1})$. Its root at zero has multiplicity between one and $r-1$, so H'/H has a nonzero simple pole there, contradicting $R(0) \neq 0$ and polynomial P . Consequently $Q_r = 0$ if and only if a unique monic polynomial

$$H = X^r + \sum_{j=1}^{r-1} h_j X^j + z$$

satisfies $RH' - PH = 0$. Its root multiplicities are at most $r < p$; polynomiality forces every root to belong to R , and conversely every such root multiset works. Partial-fraction residues distinguish all the multisets. The leading coefficient of P is r , so its degree is exactly D . Equation (60) proves regularity, and the classification proves finiteness and the count.

To separate the labels, greedily choose N nonzero roots whose products of r factors, with repetitions allowed, are all different. Suppose $k \geq 1$ roots have been chosen. Products of any fixed smaller number of factors are also distinct, by padding with one old root. A new product has form $x^j c$, with $1 \leq j \leq r$ and c an old product of $r-j$ factors. Its collision with an old r -fold product excludes at most j choices of x . Collisions between $x^i c$ and $x^j c'$ for $i < j$ exclude at most $j-i$ nonzero choices. Equal-power collisions are already excluded. Writing $M_s(k) = \binom{k+s-1}{s}$, including $M_0(k) = 1$, a bound on all forbidden choices, including old roots and zero, is

$$k+1 + \sum_{j=1}^r j M_{r-j}(k) M_r(k) + \sum_{1 \leq i < j \leq r} (j-i) M_{r-i}(k) M_{r-j}(k) = O_r(k^{2r-1}).$$

Start from one nonzero root and choose a prime larger than this bound for every $k < N$ and larger than $D+r$. A prime of size $O_r(N^{2r-1})$ suffices. All resulting H , P , and distinct labels $H(0) = (-1)^r \prod a$ lie in that prime field. $\square$

Proposition I.3 (The attaining family still has constant nearby count). *For the family in Theorem I.2, let a received line have L distinct labels with candidates agreeing on at least A of n*

coordinates, where $1 \leq A \leq n$. Let r_0 count domain roots of R , and put $m = n - r_0$, $t = A - r_0$, and $C = 3r - 2$. If $t^3 > 8Cm^2$, then

$$L \leq \max \left\{ \frac{4m}{t}, \frac{8m^3(r-1)}{t^3 - 8Cm^2} \right\}.$$

In particular $L = O_{r,\eta}(1)$ at fixed $A - D \geq \eta n$ and sufficiently large n . This conclusion remains valid for any assignment of one label to each candidate, counting distinct labels.

Proof. Choose one candidate at each selected label, and normalize it to $W = P/R = H'/H$. Its residues at the roots of R belong to $\{0, \dots, r\}$, and their sum is r . Distinct candidates have different residue vectors. On an affine line of labeled functions (z, W) through two of them at distinct labels, choose a residue coordinate where their vectors differ. That residue is an injective affine function of z along the line and has only $r + 1$ allowed values. Thus every such line contains at most $r + 1$ candidates. There are at most $(r - 1)\binom{L}{2}/3$ collinear triples.

A noncollinear triple gives a nonzero rational relation $\sum_i c_i W_i$, with $\sum_i c_i = \sum_i c_i z_i = 0$. On the product of its three degree- r denominators, the numerator has degree at most $3r - 2$: each W_i starts with r/X at infinity, so the highest coefficient cancels. Such a triple has at most $C = 3r - 2$ common agreeing coordinates outside the roots of R . A collinear triple has at most m .

Let h_x count agreements at each outside coordinate. Each selected candidate contributes at least t agreements. The same convexity estimate as in Proposition H.4 gives

$$m \left(\frac{tL}{m} - 2 \right)_+^3 \leq 6 \sum_x \binom{h_x}{3} \leq CL^3 + m(r - 1)L^2.$$

If $L \geq 4m/t$, the left side is at least $t^3 L^3 / (8m^2)$; rearranging proves the second displayed alternative. Otherwise the first alternative holds. Finally $r_0 \leq D + 1$ gives $t \geq \eta n - 1$ and $m \leq n$, proving the constant fixed-gap bound. $\square$

Collinear triples cannot simply be omitted. For $r = 3$, take two roots $a = 1$, $b = -2$ and $H_j = (X - a)^{3-j}(X - b)^j$ for $j = 0, 1, 3$. Their logarithmic derivatives are affine in j , and their labels $-1, 2, 8$ are also affine at these three indices. Over sufficiently large characteristic this is a distinct-label collinear triple. The extra L^2 term in the proof handles such configurations.

Verification and limits. The package `research/isolated_solution_sharpness/` checks the separant through order five, exhausts 760 polynomials covering 5,270 polynomial-challenge pairs at orders one through three, and includes characteristic-three and characteristic-five negative controls. Prime-field examples have 969 distinct isolated labels at order two, 495 at order three, and 462 at order four. A separate checker verifies the reconstruction recurrence on four equations of orders two through five and 360 finite Taylor fixtures. The checks support the written proofs; they do not replace the generic intersection argument. No fixed-gap superlinear MCA lower bound follows: the polynomially many isolated solutions in the sharpness family fail the required nearby-label condition.

I.1 A rational envelope has only linearly many bad labels

The agreement obstruction above extends beyond a fixed residue alphabet. The following elementary incidence bound allows arbitrarily many candidates and arbitrary characteristics. It gives another reason why a large algebraic solution count need not make the proximity bound sharp.

Proposition I.4 (Rational-envelope bound). *Let $\mathcal{D}$ consist of n distinct points of a field F , and fix polynomials $S, R \in F[X]$ with $R \neq 0$. Consider any family of polynomials P of degree at most D satisfying*

$$\frac{P - S}{R} = \frac{a_P}{b_P}, \quad \deg a_P, \deg b_P \leq r, \quad b_P \neq 0.$$

Let h be the number of zeros of R on $\mathcal{D}$. For an integer agreement threshold $1 \leq A \leq n$, suppose

$$A - h \geq \beta n > 0, \quad n \geq 48r/\beta^3.$$

On every received line $f + zg$, the number of full-support MCA-bad labels witnessed by this family is at most

$$\frac{16(n - A + 1)}{\beta^3}.$$

Here badness means that g has no degree-at-most- D interpolant on the candidate's entire agreement set. There is no characteristic hypothesis.

Proof. First, any polynomial pencil $U + zV$, with $\deg U, \deg V \leq D$, has at most $M = n - A + 1$ bad labels. Let c count its persistent agreements, where $(f, g) = (U, V)$. Each other coordinate agrees at at most one label. If $c < A$, the number of nearby labels is at most $\lfloor (n - c)/(A - c) \rfloor \leq n - A + 1$. If $c \geq A$, a bad label must have a nonpersistent agreement, giving at most $n - c \leq n - A$ labels.

Choose one bad candidate P_z at each of L distinct labels and put $W_z = (P_z - S)/R$. Delete the h zeros of R and normalize the received line by subtracting S and dividing by R . Each candidate retains at least βn agreements. Its reduced denominator is nonzero at every retained coordinate, since $W_z = (P_z - S)/R$ is regular there.

A noncollinear triple of graph points (z, W_z) over $F(X)$ has a nonzero affine-incidence determinant. Clearing its three denominators gives a polynomial of degree at most $3r$, so the triple has at most $3r$ common retained agreements. A graph line through two selected points corresponds, on multiplying by R and adding S , to a polynomial pencil of degree at most D : interpolate the two actual polynomials at their distinct scalar labels. It therefore contains at most M selected bad labels.

Let b_x count retained agreements at coordinate x , and set $b_x = 0$ at deleted coordinates. If $L \geq 4/\beta$, convexity gives

$$\sum_x b_x(b_x - 1)(b_x - 2) \geq n(\beta L - 2)^3 \geq \beta^3 n L^3 / 8.$$

Noncollinear ordered triples contribute at most $3rL^3$. Each ordered pair determines one graph line with at most M choices of a third selected label; these triples contribute at most nML^2 . Consequently

$$\beta^3 n L^3 / 8 \leq 3rL^3 + nML^2.$$

Since $3r \leq \beta^3 n / 16$, this implies $L \leq 16M/\beta^3$. The case $L < 4/\beta$ satisfies the same bound. Over an infinite field apply the argument to any finite selection of bad labels. $\square$

For example, if $\deg R \leq D + c$ and $A - D \geq \eta n$, one can take $\beta = \eta - c/n$ when positive. For fixed $c, r, \eta > 0$ the bound is therefore $O_\eta(n)$. It applies to $P = RH'/H$ with $\deg H \leq r$ without a restriction on the residues, and even permits $r \leq \beta^3 n / 48$. A bounded union of such families still has a linear bound. The hypothesis does not hold for arbitrary positive-rate candidates merely by setting $R = 1$, since then the necessary residual degree r is generally too large.

The checker `research/prime_field_tightness/check_rational_envelope.py` verifies 220 rational triple determinants over $\mathbb{F}_{101}$ and all full agreement supports in a pencil attaining $n - A + 1 = 13$ bad labels. These checks supplement the incidence proof.

I.2 Many nearby labels force a rational path to be affine

There is a stronger restriction when the candidate itself is a rational function of the challenge of bounded degree. No bound on the coefficient heights in X is needed.

Theorem I.5 (Rational-path rigidity). *Let $D < A \leq n$ be integers, and let $f + Zg$ be a received line on n distinct points of a field F . Suppose $T(X, Z) \in F(X)(Z)$ has numerator and denominator degrees in Z at most $h \geq 1$. Either*

$$T(X, Z) = U(X) + ZV(X), \quad \deg U, \deg V \leq D,$$

or the number of scalar labels z at which $T(X, z)$ is defined, is a polynomial of degree at most D , and has at least A agreements with $f + zg$ is at most

$$\frac{(8h + 10)n}{A - D}.$$

In the affine case there are at most $n - A + 1$ full-support MCA-bad labels. These statements hold in every characteristic.

Proof. Write $T = a/b$ with coprime $a, b \in F[X, Z]$, $b \neq 0$, and both Z -degrees at most h . For every polynomial specialization P_z under consideration, $a(X, z) = b(X, z)P_z(X)$ as polynomials. Put

$$F_x(Z) = a(x, Z) - b(x, Z)(f(x) + Zg(x)).$$

Call a coordinate persistent if $F_x \equiv 0$, and let H count them. At a persistent coordinate $b(x, Z) \not\equiv 0$, since otherwise $X - x$ would divide both a and b . Thus all but at most h selected labels agree there. At any other coordinate agreement requires a root of the nonzero polynomial F_x , of degree at most $h + 1$. This argument includes points where specialization makes both $a(x, z)$ and $b(x, z)$ zero.

Put $\delta = A - D$. If $H \leq D + \delta/2$, each nearby label needs at least $\delta/2$ nonpersistent agreements. Their total incidence is at most $(h + 1)n$, giving at most $2(h + 1)n/\delta$ nearby labels.

Now suppose $H > D + \delta/2$, and choose L nearby labels. If T is affine in Z over $F(X)$, two polynomial values give the stated polynomial affine form; the case $L \leq 1$ already satisfies the numerical bound. Otherwise any graph line over $F(X)$ meets the graph of T in at most $h + 1$ labels: the numerator of $T - U - ZV$ is nonzero and has Z -degree at most $h + 1$.

The selected graph points (z, P_z) therefore have at most $(h - 1)L(L - 1)$ ordered collinear triples. A noncollinear triple has a nonzero affine-incidence determinant, a polynomial in X of degree at most D , so it shares at most D persistent agreeing coordinates. Every persistent coordinate has at least $L - h$ agreements. For $L \geq h + 2$, counting ordered triples gives

$$H(L - h - 2)^3 \leq DL^3 + H(h - 1)L^2.$$

Using $(L - h - 2)^3 \geq L^3 - 3(h + 2)L^2$ yields

$$(H - D)L \leq (4h + 5)H, \quad L \leq \frac{(4h + 5)H}{H - D} < \frac{(8h + 10)n}{\delta}.$$

The smaller- L case satisfies the same bound. In the affine alternative, the sharp polynomial-pencil bound from the proof of Proposition I.4 gives $n - A + 1$ bad labels. $\square$

Corollary I.6 (Degree required by a rational witness formula). *Suppose a rational function $T(X, Z)$ selects a degree-at-most- D nearby witness at each of $L > n - A + 1$ distinct full-support MCA-bad labels, with agreement threshold $A > D$. If its numerator and denominator degrees in Z are at most h , then*

$$h \geq \frac{(A - D)L}{8n} - \frac{5}{4}.$$

Proof. The affine alternative can cover at most $n - A + 1$ of the selected bad labels. Rearrange the other alternative of Theorem I.5. A degree-zero formula is a constant polynomial pencil whenever it has a polynomial value, so it cannot cover the hypothesized set either. $\square$

For the complete-coverage examples, $L = p - 1$ and $A - D = \eta n + 1$. Any rational formula selecting witnesses throughout the nearby set consequently has challenge degree $\Omega(\eta p)$; when $\eta = \Theta(1/\log p)$ this is $\Omega(p/\log p)$. This quantifies a failure of low-degree witness coherence. It is a degree bound, not a computational lower bound: efficient witness recovery may use branching rather than one rational formula.

The reciprocal-gap dependence in the rigidity theorem is necessary up to constants for fixed h . The construction and proof in `research/prime_field_tightness/RATIONAL_PATH_RIGIDITY.md` give a non-affine projective path with order $1/\eta$ nearby bad labels at fixed rate. The exact checker considers lengths 160, 320, 640, all at rate $1/4$ and capacity gap $1/8$: the path has respectively 41, 81, 161 polynomial members, but exactly ten nearby labels, all full-support bad. This count is for the specified path, not for all codewords on the line.

J Positive-dimensional actual components save one degree power

Appendix I proves that isolated actual solutions can require a D^{d+1} count. Positive-dimensional actual components satisfy a smaller bound, extending Appendix H to every fixed derivative order. We retain the projective degree in coefficient space throughout.

Theorem J.1 (Degrees by actual component dimension). *Use the hypotheses and notation of Theorem I.1, and additionally put $T = b + \tau H$. Let J_j be the sum of the projective degrees of the j -dimensional irreducible components of the reduced closure of the actual regular joint solution locus. All components have dimension at most $d + 1$, and*

$$J_0 \leq qK^{d+1}, \quad J_j \leq qK^{d+1-j}T^{j-1} \quad (1 \leq j \leq d+1).$$

Thus every positive-dimensional stratum has total degree $O_{d,B,H}(D^d)$. For an ordinary equation with the challenge fixed, let I_j count the degrees of its actual regular components of dimension j . Then

$$I_0 \leq B(Bb)^d, \quad I_j \leq B(Bb)^{d-j}b^{j-1} \quad (1 \leq j \leq d).$$

In particular the positive-dimensional part of a reduced generic challenge fiber has degree $O_{d,B}(D^{d-1})$. This last assertion excludes the isolated points of that fiber.

Proof. The initial-jet description and Taylor reconstruction are as in Theorem I.1. They give dimension at most $d + 1$ and the stated J_0 bound. We explain why a special family of linear sections counts the full actual coefficient-space degree.

Evaluation projections count actual degrees. Let C be an actual j -dimensional component, defined before any Taylor center is adjoined. Choose independent generic parameters t_i, λ_i for $0 \leq i < j$, and use the affine linear forms

$$\ell_i(P, z) = P(t_i) + \lambda_i z.$$

At a projective point at infinity, the coefficient direction of P and the z direction cannot all be zero. Consequently $P(t) + \lambda z$ is not the zero polynomial in (t, λ) at that point. The incidence of a point of C at infinity and j vanishing forms has dimension at most $(j - 1) + j = 2j - 1$, less than the

2j-dimensional parameter space. Thus the projective projection center misses the closure of C . The resulting map to $\mathbb{P}^j$ is finite: its hyperplane pullback is the ample hyperplane bundle on C . Its degree is $\deg C$.

This map is generically separable. The coefficient functions of P and z generate the function field of C , so their differentials span its j -dimensional cotangent space. The evaluation forms span these linear functions; hence the differentials of j independent generic evaluation forms are independent. Their generic affine fiber is therefore reduced and consists of exactly $\deg C$ points. The same choice works simultaneously for the finitely many actual j -components. Generic target values $\ell_i = c_i$ avoid their singular loci, intersections with other actual components, and the separant boundary. The latter is proper after a generic t_0 is chosen, since each actual component has a dense regular locus. These sections therefore select exactly J_j distinct regular points.

Count the section in initial data. Use t_0 as Taylor center. The first section is $u_0 = c_0 - \lambda_0 z$, leaving $d + 1$ initial-data variables. The initial equation has degree at most q . The coefficient residuals have degree at most K . Each of the other $j - 1$ sections, pulled back through reconstruction and cleared by S^τ , has degree at most T : its coordinates are coefficient numerators of degree at most T , zS^τ of degree at most T , and S^τ of degree at most $T - 1$.

These further sections cut the initial hypersurface properly on $S \neq 0$, even before the residuals are imposed. Indeed the Taylor map there has a linear inverse given by its derivatives through order d and z , so it is an immersion. Evaluation hyperplanes span all its coefficient coordinates and cut each positive-dimensional image properly for generic parameters and target values. After the $j - 1$ further sections, the local source dimension at each selected point is at most $d + 1 - j$.

Now impose $d + 1 - j$ generic linear combinations of the residuals. Their full common zero set already isolates every selected point: those points lie on no higher-dimensional actual component and the actual j -components have just been cut to points. The pruning argument of Theorem I.1 therefore retains and isolates all selected points. Bézout bounds their number by $qT^{j-1}K^{d+1-j}$, as claimed. Components on $S = 0$ contribute no selected point and need not be interpreted as actual solutions.

Fixed challenge. Work over an algebraic closure of the coefficient field with that challenge held constant. There are $d + 1$ initial variables, initial equation degree at most B , coefficient-numerator degree at most b , and residual degree at most Bb . Use the evaluation forms $P(t_i)$ without a z coordinate. The same projection argument applies: a nonzero coefficient direction cannot vanish at every t . The first section is $u_0 = c_0$. The identical count gives $B(Bb)^{d-j}b^{j-1}$ for $j \geq 1$ and $B(Bb)^d$ for isolated points. Apply this in particular over the algebraic closure of the generic challenge field. $\square$

Corollary J.2 (Nonaffine actual curves at every order). *For a received line on n distinct coordinates and $D < A \leq n$, the actual curve components in Theorem J.1, other than affine codeword graphs $P = F + zG$, contain at most*

$$\frac{n - D}{A - D} qK^d$$

pairs with at least A agreements. This is $O(n^d)$ at fixed jet and challenge degrees, derivative order, and positive gap.

Proof. The agreement-hyperplane count in Corollary H.2 uses only actual curve degrees. Each retained curve has at most D persistent agreement coordinates, and its nearby-pair count is at most its degree times $(n - D)/(A - D)$. Sum and apply $J_1 \leq qK^d$. $\square$

Theorem J.3 (Sharp positive-dimensional and fiber exponents). *For every $r \geq 1$ and $e \geq 1$, put $D = re$ and assume characteristic zero or $p > D + r - 1$. There is an equation of derivative order r , jet degree r , challenge degree one, and X -degree bounded in terms of r alone, whose reduced actual regular solution closure is irreducible of dimension $r + 1$ and degree at least e^r . Its reduced challenge fibers have degree e^{r-1} . Thus the exponents D^d and D^{d-1} in Theorem J.1 cannot be lowered uniformly, even with a single actual component.*

Proof. Set $\phi_j = X^j$ for $1 \leq j < r$ and $\phi_r = X^r + z$, and define

$$Q = \text{Wr}(eP\phi'_1 - P'\phi_1, \dots, eP\phi'_r - P'\phi_r).$$

This is homogeneous of jet degree r , and its separant is

$$Q_{P(r)} = (-1)^r C_r (eP)^{r-1} (X^r + (-1)^{r-1} z), \quad C_r = \prod_{j=1}^{r-1} j!.$$

The proof is the same row calculation as Equation (60). Its coefficient of P^r is $e^r \text{Wr}(\phi'_1, \dots, \phi'_r)$, which is nonzero; hence the jet degree is exactly r . The separant proves exact order r , linear highest-derivative degree, and challenge degree one. Every nonzero polynomial solution is regular.

Each specialized Wronskian column has degree at most $D + r - 1 < p$. The polynomial Wronskian criterion from Theorem I.2 therefore says that $Q = 0$ precisely when some nonzero

$$H \in \text{span}(X, \dots, X^{r-1}, X^r + z)$$

satisfies $ePH' - P'H = 0$. For nonzero P , this makes $(P/H^e)' = 0$. The rational degree of this quotient is at most $D < p$, so $P = cH^e$ for a nonzero constant c . Conversely every such power satisfies the equation. The zero polynomial also satisfies it; for $r > 1$ it is singular and belongs to the closure of the nonzero regular locus.

Write $H = \sum_{i=0}^r h_i X^i$ projectively. The condition on H is exactly $h_0 = zh_r$. The power map

$$\mathbb{P}^r \longrightarrow \mathbb{P}^D, \quad [H] \longmapsto [H^e]$$

is base-point-free and generically one-to-one, with hyperplane pullback $\mathcal{O}(e)$. It is birational onto its image: on the monic degree- r chart its coefficients are recovered successively from the coefficients of H^e , dividing by e , which is invertible. Its image consequently has degree e^r .

The projective subfamily $h_0 = zh_r$ is closed over the affine z -line. Its power-map image is closed by properness, and its relative affine cone is exactly the actual solution closure just classified. The family of hyperplanes $h_0 = zh_r$ is a projective bundle, so that cone is irreducible of dimension $r + 1$. On the dense monic chart, z is recovered from the constant coefficient of H . Projection forgetting z is thus birational onto the affine cone of the whole power-map image, of degree e^r . Linear projection cannot increase degree, giving the claimed joint-degree lower bound. For each fixed z , the hyperplane of H is a $\mathbb{P}^{r-1}$; its power-map image, and hence its affine cone, has degree e^{r-1} . $\square$

Proposition J.4 (Sharp isolated fibers and affine-graph counts). *Fix $d \geq 1$ and $D \geq d$, in characteristic zero or $p > D + d$. There is a challenge-independent equation of order d and jet degree $d + 1$ with exactly*

$$\binom{D + d + 1}{d} = \Theta_d(D^d)$$

polynomial solutions of degree at most D , all regular and isolated. Adjoining a free challenge turns these into that many actual affine codeword-graph curve components of degree one. Thus both the I_0 and J_1 exponents of Theorem J.1 are sharp as algebraic counts.

Proof. Let R be monic and squarefree of degree $N = D + 1$, and take the Wronskian of

$$R(X^j)' - PX^j, \quad j = 0, \dots, d.$$

Its separant in $P^{(d)}$ is $(-1)^{d+1} R^d \prod_{j=1}^d j!$, a nonzero polynomial independent of P . Each column has degree at most $D + d < p$, so the Wronskian vanishes exactly when $RH' - PH = 0$ for some nonzero polynomial H of degree at most d . Hence $P = RH'/H$. For nonconstant H , its roots must belong to R ; conversely every multiset of at most d roots gives a solution. Constant H gives $P = 0$. The residues are integers between zero and $d < p$ and distinguish all the multisets. The number is

$$\sum_{s=0}^d \binom{N+s-1}{s} = \binom{N+d}{d}.$$

Every solution is regular by the separant formula. The finite classification proves the isolated count; a free challenge gives one constant-polynomial graph per solution. $\square$

Proposition J.5 (Linear MCA for the logarithmic-derivative family). *For fixed d , let R be monic and squarefree of degree $D + 1$, in characteristic zero or $p > d$. Consider the family*

$$\mathcal{P} = \{RH'/H : H \text{ monic, } \deg H \leq d, \text{ all roots of } H \text{ belong to } R\}.$$

This includes the zero polynomial. For $D < A \leq n$ with $A - D \geq \eta n$, every received line has $O_{d,\eta}(n)$ full-support bad labels witnessed by $\mathcal{P}$, for sufficiently large n . A candidate can be reused at many challenges; no prior assignment of labels is assumed.

Proof. Select one bad candidate at each of L distinct bad labels, and write $W = P/R$. At each root of R its residue belongs to $\{0, \dots, d\}$. On any affine line of graph points (z, W) where W varies, one residue is an injective affine function of z , so the line has at most $d + 1$ selected points. If W is constant, the candidate P is fixed. It has at most n bad labels: a bad full support must contain an agreeing coordinate with $g(x) \neq 0$, which determines z ; otherwise $(P, 0)$ explains that support. Thus every graph line has at most $M = \max\{n, d + 1\}$ selected points.

A noncollinear triple gives a nonzero rational relation of numerator degree at most $C = 3d - 1$, after clearing its three denominators of degree at most d . It has at most C common agreeing coordinates outside roots of R . There are at most $(M - 2)\binom{L}{2}/3$ collinear triples, by counting pairs. Let r_0 count domain roots of R , and put $m = n - r_0$ and $t = A - r_0$. Triple incidence and convexity give

$$m \left(\frac{tL}{m} - 2 \right)_+^3 \leq CL^3 + m(M - 2)L^2.$$

For sufficiently large n , $t \geq \eta n - 1$ and $m \leq n$ imply $t^3 > 8Cm^2$. If $L < 4m/t$ the claim is immediate. Otherwise the left side is at least $t^3 L^3 / (8m^2)$, giving

$$L \leq \frac{8m^3(M - 2)}{t^3 - 8Cm^2} = O_{d,\eta}(n).$$

This counts genuinely bad labels rather than all nearby challenges, which may fill the field on a constant candidate graph. $\square$

Verification and MCA scope. The exact checker in `research/actual_higher_order_components/` checks the complete solution classification on four small-field fixtures, the separant through order five, and 40 reduced sections of the projected power images and fixed-challenge fibers. These sections have exactly e^r and e^{r-1} points. It includes a small-characteristic failure and explicitly checks lower-degree boundary powers, which must not be lost by using only monic degree- r charts. The geometric upper bound is a separate written argument. A second checker exhausts 7,203 polynomials for the fixed-equation family and verifies its separant through order four.

The positive-dimensional degree saving alone does not prove an $O(n^d)$ full-MCA bound. Isolated actual joint points can number $\Theta(D^{d+1})$, and actual affine codeword-graph curve components can each contribute accidental labels. Their generic challenge fibers are isolated points, outside the positive-dimensional fiber bound. The sharpness family here has at most r distinct roots per candidate and is already covered by the linear-MCA result of Appendix G in its sufficiently large characteristic regime. The fixed-gap quadratic MCA lower bound remains open.

K A nonlinear first-order family near the characteristic cutoff

The bounded-root result of Appendix G requires characteristic larger than a suitable constant times the length. For the inverse Bernoulli family below, a direct classification also handles characteristic just above the message degree. The extra Frobenius constants available in that range do not produce large lists.

Theorem K.1 (Inverse Bernoulli solution count). *Let $s \geq 2$, $D \geq 1$, and let E be algebraically closed of characteristic zero or $p > \max\{D, s+1\}$. For every nonzero polynomial $A(X)$, the equation*

$$P^{s-1}P' = A, \quad \deg P \leq D,$$

has at most $2s$ polynomial solutions, all regular. In characteristic zero, or when $p > sD$, it has at most s solutions. The bound $2s$ is attained for suitable degrees and primes with $p \equiv 1 \pmod{s}$.

Proof. Every solution has a positive degree $d < p$ (with the obvious characteristic-zero interpretation), and $\deg A = sd - 1$. If its leading coefficient is a , the leading coefficient of A is da^s . Thus all solutions have the same degree and their leading coefficients differ by s th roots of unity. Normalize one leading coefficient to one. It suffices to count monic solutions of the resulting equation and multiply by s . For two such solutions P, Q ,

$$(P^s - Q^s)' = 0.$$

In characteristic zero their difference of powers is constant. Factoring it as $\prod_{\zeta^s=1} (P - \zeta Q)$ shows that it cannot be a nonzero constant: a factor with $\zeta \neq 1$ has degree d . If it is zero, monicity gives $P = Q$. The same argument applies when $p > sD$. Otherwise, in positive characteristic, a distinct pair has

$$P^s - Q^s = C(X)^p, \quad 1 \leq \deg C \leq s-1.$$

At least three factors. Suppose $s \geq 3$. Cancel the monic gcd G of P, Q and put $U = P/G$, $V = Q/G$, both monic of the same degree $h \geq 1$. The rational map U/V has degree $h < p$ and is separable. The preimages of the s distinct s th roots of unity all lie among the roots of C and infinity. These are at most s points; the s fibers are disjoint and nonempty. Consequently each fiber has exactly one point, of ramification index h .

For completeness, an elementary degree count forces $h = 1$. Make a generic fractional linear change of the X coordinate so that all those fiber points are finite and the new infinity maps to

none of the s values, nor to zero or infinity. The new numerator and denominator have degree h . At each of the s points their derivative numerator $U'V - UV'$ vanishes to order $h - 1$, since $h < p$. Its degree is at most $2h - 2$. Hence $s(h - 1) \leq 2h - 2$, which for $s \geq 3$ gives $h = 1$.

Return to the original coordinate, and write $U = X - u$, $V = X - v$, with $u \neq v$. The polynomial $S = U^s - V^s$ has $s - 1$ distinct roots, none equal to u or v . These exhaust the roots of C , which therefore has degree $s - 1$ and is squarefree. Comparing valuations in $G^s S = C^p$ proves

$$p = sg + 1, \quad G = S_0^g, \quad d = (s - 1)g + 1,$$

where S_0 is the monic normalization of S and $g = (p - 1)/s$. There are no other roots of G . The strict assumption $p > s + 1$ ensures $g \geq 2$.

Direct differentiation, using $sg = -1$ in the field, gives

$$A = \text{nonzero constant} \cdot (UV)^{s-1} S_0^{p-2}.$$

At a root of multiplicity m in any other solution T , the multiplicity in $T^{s-1}T'$ is $sm - 1$, since $m < p$. Thus T can choose roots only from S_0 , with multiplicity g , and from u, v , with multiplicity one. If it chooses t of the former and k of the latter, its degree requires

$$tg + k = (s - 1)g + 1, \quad 0 \leq t \leq s - 1, \quad 0 \leq k \leq 2.$$

For $g \geq 2$ this forces $t = s - 1$ and $k = 1$, so $T = P$ or Q . There are at most two monic solutions. Conversely the displayed U, V, S_0 construction supplies two, and their s scalar multiples attain the bound whenever $p = sg + 1$ with $g \geq 2$.

Two factors. For $s = 2$, a distinct monic pair satisfies $(P - Q)(P + Q) = c(X - a)^p$. Since $P + Q$ has degree d and leading coefficient two, its two factors give

$$P = (X - a)^d + v(X - a)^{p-d}, \quad Q = (X - a)^d - v(X - a)^{p-d}, \quad v \neq 0.$$

In particular $d > p/2$. If $p - d \leq d - 2$, the coefficient of X^{d-1} in P fixes a , and then v is fixed. If $p - d = d - 1$, then $p = 2d - 1 \geq 5$ and a is a root of P of multiplicity $d - 1 > d/2$, so it is again unique. Every monic solution has at most one distinct partner. The displayed pair and its negatives attain four solutions. Finally, the separant P^{s-1} is nonzero for every solution of a nonzero- A equation, proving regularity. $\square$

The strict characteristic condition has a real exception. For $s = p - 1$ and $D = p - 1$, the polynomials

$$P_a = \frac{X^p - X}{X - a}, \quad a \in \mathbb{F}_p, \quad P_a^{p-2} P'_a = -(X^p - X)^{p-2},$$

and all their nonzero $\mathbb{F}_p$ scalar multiples give $p(p - 1)$ solutions. Here the jet degree s grows with p . This does not contradict Theorem K.1, which assumes $p > s + 1$.

Corollary K.2 (Linear full MCA for inverse Bernoulli families). *Under the theorem's hypotheses, for nonzero A the solution family witnesses at most $2sn$ full-support bad labels on any received line. For $A = 0$, the corresponding count at agreement threshold $a \geq 2$ is at most $\binom{n}{2}/(a - 1)$. In particular both counts are $O(n)$ at fixed s and fixed positive agreement fraction.*

Proof. A fixed candidate has at most n bad labels: each bad full support contains an agreeing coordinate with nonzero received direction, which determines the label; otherwise $(P, 0)$ explains that support. Apply Theorem K.1 when $A \neq 0$.

When $A = 0$, the degree guard makes every solution constant. Each coordinate gives an affine line $c = f(x) + zg(x)$ in the (z, c) plane. A bad support of size at least a contains at least two distinct line equations, since otherwise constant polynomials F, G explain the whole support. Among its coordinates there are at least $a - 1$ pairs with distinct line equations. Each such pair intersects at at most one point. Select one candidate per bad label and count pairs to obtain the stated bound. $\square$

Checks and scope. The exact standard-library checker in `research/inverse_bernoulli/` exhausts 164,803 monic polynomials across thirteen fixtures, checks six sharpness examples, and verifies the exceptional family at $p = 3, 5, 7, 11$. The argument is specific to inverse Bernoulli equations. It does not settle the first-order conjecture or give a quadratic MCA construction at any higher order.

L An exact cubic complete-fiber bound on the pinned subgroup

The near-complete-fiber classification leaves maps with many uncovered points outside its range. The following finite, computer-assisted result closes one such proposed packet construction.

Theorem L.1 (Exact cubic maximum). *Let $p = 2130706433$ and $S = \mu_{256} \subset \mathbb{F}_p^*$. Every rational map R of degree three, with coefficients in any extension of $\mathbb{F}_p$, has at most three fibers consisting of three distinct points of S . This bound is attained. Consequently a fixed map $R(X^{1024})$ has at most three complete degree-3072 fibers on μ_{262144} and supplies at most eight unions of such fibers.*

Proof. Write the monic locator of a complete fiber as $X^3 - e_1X^2 + e_2X - e_3$. Its coefficients belong to $\mathbb{F}_p$, with $e_3 \in \mu_{256}$. All these coefficient triples lie on one affine line: they are the monic members of the numerator–denominator pencil of R . Two full fibers determine that pencil over $\mathbb{F}_p$, regardless of the original coefficient field. Two locators on a degree-three pencil have no common root, since a common factor would reduce the degree of the map.

First suppose e_3 is constant. Scaling all roots in μ_{256} normalizes $e_3 = 1$, because cubing is bijective on this group. There are exactly

$$M = \frac{(256 - 1)(256 - 2)}{6} = 10795$$

three-element subsets with product one. Enumerating all $\binom{M}{2} = 58,260,615$ pairs of their (e_1, e_2) points and counting collinear points, with common-factor pencils excluded, gives maximum three and an explicit attaining line.

Otherwise e_3 parametrizes the line injectively. Normalize one fiber’s product to one and write

$$e_1 = a_1 + s_1(e_3 - 1), \quad e_2 = a_2 + s_2(e_3 - 1).$$

The associated map is

$$\frac{X^3 + (s_1 - a_1)X^2 + (a_2 - s_2)X}{s_1X^2 - s_2X + 1}.$$

It is degenerate exactly when the denominator vanishes at one of the three roots of the anchor locator $X^3 - a_1X^2 + a_2X - 1$. Every line with at least three complete fibers is found by taking one of the M product-one anchors, enumerating all $255M$ triples at the remaining product levels,

and identifying repeated pairs of slopes (s_1, s_2) . For every nondegenerate repeated pair, lookup among the normalized triples counts its complete fibers at all 256 product levels. The exhaustive enumeration of

$$255M^2 = 29,715,666,375$$

anchor/triple pairs gives maximum three, also attained. Together the two cases prove the first assertion. The map $X \mapsto X^{1024}$ is 1024-to-one from μ_{262144} onto S , so a complete degree-3072 fiber of the composition corresponds exactly to a complete three-point fiber of R . Three disjoint packets have at most 2^3 unions. $\square$

Reproducibility and scope. The source, mathematical reduction, all 85 disjoint anchor-range checkpoints, witnesses, and resource reports are in `research/structured_domains/general_cubic_packets/`. The general enumeration found 22,692 nondegenerate candidate occurrences and took 524.46 seconds, with sampled peak process-group RSS 176,192 KiB under a 384 MiB limit. A separate Python checker verified every checkpoint witness on the entire domain, validated the source hash and complete disjoint coverage, independently exhausted all pencils on $\mu_8/\mathbb{F}_{97}$ and $\mu_{16}/\mathbb{F}_{193}$, and reran the full pinned constant-product enumeration. The optional `-full` flag repeats the general enumeration too.

Eight unions, even with one fixed core appended, are far short of the required bank size 274980728111395088. This excludes the fixed composed-cubic packet route; maps varying with the candidate and constructions using partial fibers remain outside the conclusion. No better.codes score improvement follows from this exclusion.

M Sharp list bounds for polynomial Riccati solutions

The cross-ratio identities of Riccati equations, combined with intersection multiplicities in positive characteristic, give an ordinary-list bound even when all agreement coordinates are singular. For a challenge-independent nonlinear equation it also gives a linear full-support exceptional-label bound. The parametrization and its gcd form are classical; see Gasull, Torregrosa, and Zhang [4, Lemmas 6–7]. We prove the finite-characteristic statements and the agreement consequences used here.

Theorem M.1 (Riccati lists and solution counts). *Let $D \geq 1$, and let E have characteristic zero or $p > D$. Consider one fixed equation*

$$a(X)P' = b_0(X) + b_1(X)P + b_2(X)P^2, \quad a \neq 0, \quad (61)$$

with coefficients in $E[X]$, with no restriction on their degrees. On any n distinct evaluation points, the number of distinct solutions of degree at most D agreeing with an arbitrary received word in at least $A > D$ coordinates is at most

$$\left\lfloor \frac{n}{A - D} \right\rfloor. \quad (62)$$

This bound is attained over prime fields at unbounded lengths and positive limiting rate and agreement gap. If $b_2 \neq 0$, the total number of degree-at-most- D polynomial solutions is at most

$$C_D = \begin{cases} D + 2, & \text{in characteristic zero or } p > D + 1, \\ 2D + 2, & \text{if } p = D + 1. \end{cases} \quad (63)$$

The bound $D + 2$ is sharp. The boundary bound $2D + 2$ is attained at $p = 2, D = 1$.

Proof. Total candidate count. Assume $b_2 \neq 0$. Work at a transcendental Taylor center t , where $a(t)b_2(t) \neq 0$, and write the equation as $P' = F(X, P)$, with $F = \alpha + \beta u + \gamma u^2$ and $\gamma \neq 0$. Put $\Delta = \partial_X + F\partial_u$ and $F_j = \Delta^j u$. For $j < p$, or in characteristic zero, F_j has u -degree $j+1$ and leading coefficient $j!\gamma^j$. This follows inductively: $F\partial_u$ gives the highest degree and ∂_X gives a lower degree. In characteristic zero or $p > D+1$, every solution satisfies

$$F_{D+1}(t, P(t)) = P^{(D+1)}(t) = 0.$$

This is a nonzero polynomial of degree $D+2$ in $P(t)$. Distinct polynomial solutions have distinct values at the transcendental t , proving the first count.

If $p = D+1$, use instead the truncated reconstruction

$$T(U, X) = \sum_{j=0}^D F_j(t, U) \frac{(X-t)^j}{j!}.$$

Every candidate is $T(P(t), X)$. Its U -degree is $D+1$, with leading coefficient $\gamma(t)^D(X-t)^D$. The residual $a\partial_X T - b_0 - b_1 T - b_2 T^2$ therefore has U -degree exactly $2D+2$, with nonzero leading coefficient $-b_2(X)\gamma(t)^{2D}(X-t)^{2D}$. Any nonzero X -coefficient of this residual is a nonzero polynomial in U of degree at most $2D+2$ vanishing at every candidate value $P(t)$. This proves the boundary count. In particular, in positive characteristic the total count is at most $2D+2 \leq 2p$.

Lists by intersection multiplicities. For $M \geq 2$ distinct candidates and a domain coordinate x , put

$$C_x = \sum_{i < j} \text{ord}_x(P_i - P_j).$$

The root bound gives $\sum_x C_x \leq D \binom{M}{2}$. If h candidates match a given received value at x , we claim

$$(h-1)(M-1) \leq 2C_x. \quad (64)$$

If $b_2 = 0$, differences solve one homogeneous linear equation. The ratio of two nonzero differences has derivative zero and rational degree at most $D < p$, and hence is constant. The same is true in characteristic zero. Thus the candidates have form $P_0 + cW$, and their values at x are either all distinct or all equal; (64) follows immediately.

Now suppose $b_2 \neq 0$. Differences satisfy

$$a(P_i - P_j)' = (b_1 + b_2(P_i + P_j))(P_i - P_j),$$

so every cross ratio has derivative zero. In characteristic zero it is constant and different from $0, 1, \infty$. A colliding pair and two candidates outside its value class would force that cross ratio to be one. Therefore either all values are distinct or one value occurs at least $M-1$ times. For $h \in \{0, 1, M-1, M\}$, the inequality follows from $C_x \geq \binom{h}{2}$.

In positive characteristic a cross ratio need not be constant. However, any zero of a nonzero rational function with derivative zero has order divisible by p . Choose a colliding pair and two candidates outside its value class. The denominator of their cross-ratio identity is nonzero at x , and the numerator of the cross ratio minus one, up to sign, is the product of the two within-pair differences. If the two outside values were distinct, this product would have positive order at most $D < p$, impossible. Consequently every collision pattern other than those already handled has exactly two value classes, both of size at least two.

Let the matching class have size h and the other class have size $b = M - h$. Let μ, ν be the minimum positive vanishing orders of differences within these classes. The same cross-ratio

argument for every choice of two within-class pairs gives $\mu + \nu \geq p$. Write $A_0 = \binom{h}{2}$, $B_0 = \binom{b}{2}$, so $C_x \geq A_0\mu + B_0\nu$. If $h \leq b$, already

$$2(A_0 + B_0) - (M - 1)(h - 1) = (b - 1)(b - h + 1) \geq 0.$$

If $h \geq b$, minimizing with $\mu, \nu \geq 1$ and $\mu + \nu \geq p$ gives $C_x \geq A_0 + (p - 1)B_0$, and hence

$$2C_x - (M - 1)(h - 1) \geq (b - 1)(pb - M + 1) \geq 0,$$

because $b \geq 2$ and the total-count bound gives $M \leq 2p$. An unmatched received value has $h = 0$. This proves (64) in all cases. Summing it gives $AM \leq n + DM$, proving (62). The case $M = 1$ is immediate, and the same proof allows real agreement thresholds.

For sharpness of $D + 2$, choose a split and squarefree of degree $D + 1$. The equation $aP' = a'P - P^2$ has the $D + 2$ solutions 0 and $a/(X - r)$, one for each root r of a . At the boundary $p = 2, D = 1$, $(X^2 + X)P' = P^2 + P$ has exactly 0, 1, $X, X + 1$ as its four solutions. $\square$

Proposition M.2 (Sharp prime-field lists). *Fix $M \geq 3$ and a prime $p \equiv 1 \pmod{2M}$. Put*

$$s = \frac{p - 1}{2M}, \quad n = p - 1, \quad D = (M - 2)s, \quad A = Ms.$$

There is a nonlinear equation (61) over $\mathbb{F}_p$ and a word on $\mathbb{F}_p^$ whose list of degree-at-most- D solutions at agreement threshold A has exactly $M = n/(A - D)$ elements.*

Proof. Partition the image of X^s on $\mathbb{F}_p^*$ into $a_1, \dots, a_M, b_1, \dots, b_M$ and set

$$R(Y) = \prod_{i=1}^M (Y - a_i), \quad T(Y) = Y^{M-1}, \quad C_i(Y) = \frac{R(Y)}{Y - a_i} - T(Y), \quad P_i(X) = C_i(X^s).$$

Leading terms cancel, so $\deg P_i \leq D$. On the fiber over a_j choose received value $-T(a_j)$, and on the fiber over b_j choose $C_j(b_j)$. Exactly $M - 1$ candidates match a root fiber, since the remaining difference is $R'(a_j) \neq 0$. The values $C_i(b_j)$ are all distinct, since $R(b_j) \neq 0$ and the denominators $b_j - a_i$ are distinct. Each P_i therefore agrees in exactly $(M - 1)s + s = A$ coordinates, and the polynomials are distinct.

The polynomials $U_i = R/(Y - a_i)$ satisfy $RU'_i = R'U_i - U_i^2$. Translating $U = C + T$ gives

$$RC' = (R'T - T^2 - RT') + (R' - 2T)C - C^2.$$

Substitute $Y = X^s$ and multiply the right side by sX^{s-1} . This is one polynomial Riccati equation for all the P_i , with nonzero quadratic coefficient. Since $p > 2D$, the preceding theorem shows that no additional solution can enter this list. Primes in the progression $1 \pmod{2M}$ give unbounded lengths for fixed M . The code rate $(D + 1)/n$ tends to $(M - 2)/(2M)$ and the capacity gap $(A - D - 1)/n$ tends to $1/M$. $\square$

Corollary M.3 (Fixed-equation full-support exceptional labels). *Under Theorem M.1, assume $b_2 \neq 0$. For a received line $f + zg$ on n distinct points and $A > D$, the number of labels possessing a solution of the fixed equation with at least A agreements and bad full agreement support is at most*

$$\frac{2n(n + C_D)}{A - D}. \tag{65}$$

Here a support is bad if no polynomial of degree at most D interpolates g on the entire support. Thus the bound is $O(n)$ when $A - D$ is a fixed positive fraction of n .

Proof. Put $\delta = A - D$. A candidate's persistent coordinates have $g_i = 0$ and $P(x_i) = f_i$. Call a candidate heavy if it has at least $D + \delta/2$ persistent coordinates. These candidates form a list for f at that threshold, so there are at most $2n/\delta$ by (62). Each has at most n bad labels: a bad support must contain an agreement at a coordinate with $g_i \neq 0$, and this coordinate determines a unique label. Otherwise $g = 0$ on the support, contradicting badness.

A light candidate needs more than $\delta/2$ nonpersistent agreements at every nearby label. It has at most n such coordinate-label incidences altogether, so it has at most $2n/\delta$ nearby labels. There are at most C_D candidates in total, with C_D as in (63). Adding the heavy and light bounds proves the claim. $\square$

Characteristic guard and scope. The constant-cross-ratio argument alone cannot justify the weaker guard $p > D$; intersection multiplicities are essential. In characteristic three, put $D = 2$, $U = 1 + X + X^2$, $V = X + X^2$, $W = 1 + X$, and $N = UV$. The equation with $a = N$, $b_0 = 0$, $b_1 = N'$, and $b_2 = U'V - UV'$ has the solutions $0, U, V, W$. Their values at zero are $0, 1, 0, 1$, contradicting the collision property. Their cross ratio is a nonconstant rational function with zero derivative. Exact enumeration gives five degree-at-most-two solutions, also exceeding $D + 2$. This is consistent with the boundary value $C_D = 2D + 2$ in (63).

The failure of the $D + 2$ count at the boundary is not confined to small primes. For every prime p , the equation

$$(X^p - X)P' = -P - P^2, \quad \deg P \leq p - 1,$$

has exactly $p + 2 = D + 3$ solutions over every extension of $\mathbb{F}_p$: 0 , -1 , and $(X^p - X)/(X - r)$ for $r \in \mathbb{F}_p$. Indeed, evaluating at $\mathbb{F}_p$ forces each value of P to be 0 or -1 . Let S be the set of coordinates with value -1 and $f = \prod_{r \in S} (X - r)$. Lagrange interpolation gives $P = (X^p - X)f'/f$. Substitution reduces the equation to $f'' = 0$. Since $\deg f \leq p$, a nonconstant proper divisor f of $X^p - X$ with $f'' = 0$ must be linear: its only possible nonzero coefficients have degrees $0, 1, p$. Thus S is empty, a singleton, or all of $\mathbb{F}_p$, proving the assertion. This family does not prove that $D + 3$ is a universal upper bound at the boundary.

The ordinary-list bound applies separately at each label of a challenge-dependent Riccati equation. The full-support bound (65) requires a single fixed equation and does not extend by this argument to challenge-dependent families. The checks in `research/riccati_cross_ratio/` enumerate 183,581 candidate polynomials, test 616 received lines, and verify nine prime-field equality fixtures for Proposition M.2, together with the characteristic-three negative control. A further boundary-characteristic checker enumerates 1,084,807 polynomials for 549 equations, checks 2,790 coordinate-family instances of the multiplicity inequality, and includes nonconstant cross ratios and the sharp characteristic-two boundary count. An independent extension-field implementation exhausts all 729 degree-at-most-two polynomials over $\mathbb{F}_9$ and checks 85 received lines, including bad full-support labels at $p = D + 1$. These restricted results do not settle the general first-order conjecture.

N Superlinear ordinary correlated-agreement failures in quadratic extensions

The quadratic lower bound of Corollary 2.6 concerns full agreement-set MCA. A separate construction excludes even ordinary correlated agreement, at the cost of a smaller exponent and a gap that is bounded below rather than fixed exactly. We make no novelty claim for the standard sieve input or the combination below.

Theorem N.1. *For every $0 < \varepsilon < 1/8$, there are unbounded lengths n and primes $p = 12n/13 + 5$ for which a Reed–Solomon code over $\mathbb{F}_{p^2}$, of exact rate $3/13$, has a received affine line with $\Omega_\varepsilon(n^{5/4-\varepsilon})$ nearby labels at an agreement threshold M , but no ordinary correlated agreement of size M . The capacity gap satisfies $M/n - 3/13 > 3/26$, and the radius is strictly below the characteristic-based Elias radius for all sufficiently large members of the family.*

Thus a linear ordinary-CA bound uniform over gaps bounded below is impossible in this large-characteristic field class. The actual ratio M/n is not asserted to be constant. This is neither a prime-ambient-field result nor a first-order-regime tightness statement.

The prime-selection input. Put $\theta = 1/4 - \varepsilon$. The lower linear sieve and Bombieri–Vinogradov imply that there are infinitely many primes $p \equiv 41 \pmod{48}$ for which every odd prime divisor of $(p-1)/4$ is at least p^θ . Here is the specific reduction, so that the fixed progression introduces no hidden distribution hypothesis. Sift

$$\mathcal{A} = \{(p-1)/8 : x/2 < p \leq x, p \equiv 41 \pmod{48}\}.$$

For squarefree d coprime to 6, the condition $d \mid (p-1)/8$ selects one reduced residue class modulo $48d$. Hence the local density is $g(d) = 1/\varphi(d)$, with main size $X = (\text{Li}(x) - \text{Li}(x/2))/16$; set $g(2) = g(3) = 0$. Bombieri–Vinogradov bounds the sum of absolute remainders through $D = x^{1/2-\varepsilon}$ by $O_A(x/\log^A x)$ for every fixed A . With $z = x^{1/4-\varepsilon}$, the sieve parameter

$$s = \frac{\log D}{\log z} = \frac{1/2 - \varepsilon}{1/4 - \varepsilon} \in (2, 3)$$

has positive lower sieve function $f(s) = 2e^\gamma \log(s-1)/s$. Since $\prod_{\ell < z} (1 - g(\ell)) \asymp 1/\log z$, there are $\gg_\varepsilon x/\log^2 x$ unsifted primes. Their associated odd integers $(p-1)/8$ have no prime factor below $z \geq p^\theta$. This is the standard shifted-prime sieve argument; see [35, Section 5, Example 3, pp. 36–37] and [36, Theorem 2 and equation (11)].

Excluding small polynomial orbits. Let ζ be a primitive eighth root of unity. No affine polynomial agrees at three distinct points of μ_8 with

$$W_2(Y) = (1 + Y^4)/2 - Y^2$$

in characteristic $p > 17$ with $p \equiv 1 \pmod{8}$. For completeness this finite assertion can be checked integrally. Write $v_j = 1 + \zeta^{4j} - 2\zeta^{2j}$. For $0 \leq i < j < h < 8$, take the norm of

$$(\zeta^j - \zeta^i)(v_h - v_i) - (\zeta^h - \zeta^i)(v_j - v_i)$$

from $\mathbb{Q}(\zeta)$ to $\mathbb{Q}$, equivalently its resultant with $Z^4 + 1$ after reducing powers of Z . The complete distribution is

Norm	64	256	576	1088	3136	4096	6400
number of triples	12	8	8	8	4	8	8.

All 56 determinants are nonzero and their norms have prime factors only 2, 3, 5, 7, 17. Reduction at $p > 17$ therefore preserves nonvanishing. An independent exact verifier and the individual determinants are in `research/dickson_fixed_gap/verify_two_orbit.py` and its JSON output.

Proof of Theorem N.1. Choose the primes above, put $k = (p - 1)/4$ and $N = 4k$, and use $W(X) = (1 + X^{2k})/2 - X^k$ on $\mathbb{F}_p^\times$. Let M be its true maximum agreement with degree- $< k$ polynomials. Proposition 2.4 gives $M \geq 3k/2$, while the degree of $W - P$ gives $M \leq 2k$. The group $H = \mu_k$ preserves the word and acts on its nearest list by $P(X) \mapsto P(hX)$.

The size r of any such orbit divides k . Here $v_2(k) = 1$, so the roughness condition implies either $r \in \{1, 2\}$ or $r \geq p^\theta$. In the former case invariance under a subgroup of order $k/2$ forces $P(X) = a + bX^{k/2}$. Passing to $Y = X^{k/2}$, whose fibers have size $k/2$, the preceding eight-point calculation bounds its agreement by k . Such a polynomial cannot be nearest. Consequently there are at least p^θ nearest polynomials. Select $L = \lfloor p^\theta \rfloor$ of them; for large p , $L \leq k$.

This nearest agreement remains M over $\mathbb{F}_{p^2}$: a degree- $< k$ polynomial matching k base-field coordinates and values has base-field coefficients by interpolation. Choose an agreement anchor incident to $\ell \geq ML/N$ selected candidates, and divide each candidate minus the received anchor value by the anchor's linear factor. On the $N - 1$ old coordinates the resulting degree- $\leq k - 2$ quotients each have exactly $M - 1$ agreements. No polynomial of this degree has M old agreements, since restoring the anchor would contradict maximality of M .

Let $Q = p^2$ and $q = (k - 10)/3$. This is an integer because $p \equiv 41 \pmod{48}$ gives $k \equiv 1 \pmod{3}$. For each of the $Q - N$ unused field points let s_x count distinct quotient values. Pairwise root counting and Cauchy-Schwarz give average diversity at least

$$\mu = \frac{\ell(Q - N)}{Q - N + (\ell - 1)(k - 2)} \geq \ell/2,$$

where $Q \geq 2NL$. Choose the q points with largest s_x , give them direction one, and retain direction zero on the old domain. Independent uniform translations of the new value sets have expected union size

$$Q \left[1 - \prod_x (1 - s_x/Q) \right] \geq Q[1 - (1 - \mu/Q)^q] \geq q\ell/4 \geq \frac{qML}{4N}.$$

For the middle inequality, use $1 - u \leq e^{-u}$ and $q\ell/(2Q) \leq 1/4$. Every counted label has at least M agreements with a selected quotient.

If a correlated witness pair F, G of degree at most $k - 2$ has $G = 0$, its joint agreements are confined to the old domain, with maximum $M - 1$. If $G \neq 0$, there are at most $k - 2$ old zeros and at most q new agreements. Since $q \leq M - k + 1$, this again totals at most $M - 1$. Thus ordinary correlated agreement of size M is absent.

Finally,

$$K = k - 1, \quad n = N - 1 + q = \frac{13(k - 1)}{3}, \quad \frac{M - K}{n} \geq \frac{k/2 + 1}{n} > \frac{3}{26}.$$

For $k \geq 20$ we have $q \geq k/6$, so the number of nearby labels is at least

$$\frac{qML}{4N} \geq \frac{kL}{64} \geq \frac{3nL}{832} \geq \frac{nL}{300} = \Omega_\varepsilon(n^{5/4-\varepsilon}).$$

Here $p = 12n/13 + 5 > K - 1$. Also $H_p(1 - M/n) \leq 1 - M/n + 1/\log_2 p < 1 - K/n$ for large p , which proves the strict Elias assertion. $\square$

A complete small-field replay in `research/ordinary_ca_superlinear/verify.py` exhausts all 17^4 source polynomials and all 17^3 anchored-code polynomials. It certifies 27 nearby labels at threshold 6 for length 18 and dimension 3 over $\mathbb{F}_{17^2}$, with old and nonzero-direction joint-agreement bounds both equal to 5. This checks the boundary mechanism, not the asymptotic sieve theorem or the Elias condition for this small fixture.

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